



Cochrane
Library

Cochrane Database of Systematic Reviews

Manual therapy and exercise for lateral elbow pain (Review)

Wallis JA, Bourne AM, Jessup RL, Johnston RV, Frydman A, Cyril S, Buchbinder R

Wallis JA, Bourne AM, Jessup RL, Johnston RV, Frydman A, Cyril S, Buchbinder R.
Manual therapy and exercise for lateral elbow pain.
Cochrane Database of Systematic Reviews 2024, Issue 5. Art. No.: CD013042.
DOI: [10.1002/14651858.CD013042.pub2](https://doi.org/10.1002/14651858.CD013042.pub2).

www.cochranelibrary.com

Manual therapy and exercise for lateral elbow pain (Review)

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

WILEY

TABLE OF CONTENTS

ABSTRACT	1
PLAIN LANGUAGE SUMMARY	2
SUMMARY OF FINDINGS	5
BACKGROUND	9
OBJECTIVES	10
METHODS	10
RESULTS	15
Figure 1.	16
Figure 2.	21
DISCUSSION	31
Figure 3.	33
Figure 4.	34
AUTHORS' CONCLUSIONS	34
ACKNOWLEDGEMENTS	35
REFERENCES	36
CHARACTERISTICS OF STUDIES	45
DATA AND ANALYSES	132
Analysis 1.1. Comparison 1: Manual therapy versus placebo manual therapy, Outcome 1: Mean pain (0 to 10, 0 is no pain)	133
Analysis 1.2. Comparison 1: Manual therapy versus placebo manual therapy, Outcome 2: Mean disability (0 to 100, 0 is no disability)	133
Analysis 1.3. Comparison 1: Manual therapy versus placebo manual therapy, Outcome 3: Mean grip strength (lb/in2)	133
Analysis 2.1. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 1: Mean pain (0 to 10, 0 is no pain)	135
Analysis 2.2. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 2: Mean disability (0 to 100, 0 is no disability)	136
Analysis 2.3. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 3: Health-related quality of life (SMD)	136
Analysis 2.4. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 4: Participant reported treatment success	137
Analysis 2.5. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 5: Proportion of overall withdrawals	137
Analysis 2.6. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 6: Proportion with any adverse event	138
Analysis 2.7. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 7: Pain-free grip strength (SMD)	138
Analysis 2.8. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 8: Proportion on sick leave	138
Analysis 3.1. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 1: Mean pain (0 to 10, 0 is no pain)	140
Analysis 3.2. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 2: Mean disability (0 to 100, 0 is no disability)	141
Analysis 3.3. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 3: Mean health related quality of life	141
Analysis 3.4. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 4: Participant reported treatment success	142
Analysis 3.5. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 5: Proportion of overall withdrawals	142
Analysis 3.6. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 6: Proportion with any adverse event	143
Analysis 3.7. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 7: Pain free grip strength (SMD)	143
Analysis 3.8. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 8: Proportion on sick leave	143
Analysis 4.1. Comparison 4: Manual therapy versus minimal or no intervention, Outcome 1: Mean pain (0 to 10, 0 is no pain) ...	145

Analysis 4.2. Comparison 4: Manual therapy versus minimal or no intervention, Outcome 2: Mean disability (0 to 100, 0 is no disability)	145
Analysis 4.3. Comparison 4: Manual therapy versus minimal or no intervention, Outcome 3: Mean health related quality of life ..	146
Analysis 4.4. Comparison 4: Manual therapy versus minimal or no intervention, Outcome 4: Proportion of overall withdrawals ..	146
Analysis 4.5. Comparison 4: Manual therapy versus minimal or no intervention, Outcome 5: Proportion with any adverse event ..	146
Analysis 4.6. Comparison 4: Manual therapy versus minimal or no intervention, Outcome 6: Mean pain free grip strength (kg) ..	147
Analysis 5.1. Comparison 5: Prescribed exercises versus minimal or no intervention, Outcome 1: Mean pain (0 to 10, 0 is no pain)	148
Analysis 5.2. Comparison 5: Prescribed exercises versus minimal or no intervention, Outcome 2: Mean disability (0 to 100, 0 is no disability)	149
Analysis 5.3. Comparison 5: Prescribed exercises versus minimal or no intervention, Outcome 3: Mean health-related quality of life	149
Analysis 5.4. Comparison 5: Prescribed exercises versus minimal or no intervention, Outcome 4: Participant-reported treatment success	150
Analysis 5.5. Comparison 5: Prescribed exercises versus minimal or no intervention, Outcome 5: Proportion of overall withdrawals	150
Analysis 5.6. Comparison 5: Prescribed exercises versus minimal or no intervention, Outcome 6: Proportion with any adverse event	150
Analysis 5.7. Comparison 5: Prescribed exercises versus minimal or no intervention, Outcome 7: Pain-free grip strength (SMD) ..	151
Analysis 6.1. Comparison 6: Manual therapy and prescribed exercises versus prescribed exercises alone, Outcome 1: Mean pain (0 to 10, 0 is no pain)	152
Analysis 6.2. Comparison 6: Manual therapy and prescribed exercises versus prescribed exercises alone, Outcome 2: Mean disability (0 to 100, 0 is no disability)	153
Analysis 6.3. Comparison 6: Manual therapy and prescribed exercises versus prescribed exercises alone, Outcome 3: Participant-reported treatment success	153
Analysis 6.4. Comparison 6: Manual therapy and prescribed exercises versus prescribed exercises alone, Outcome 4: Proportion of overall withdrawals	154
Analysis 6.5. Comparison 6: Manual therapy and prescribed exercises versus prescribed exercises alone, Outcome 5: Mean grip strength (kg)	154
Analysis 7.1. Comparison 7: Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection alone, Outcome 1: Mean pain (0 to 10, 0 is no pain)	155
Analysis 7.2. Comparison 7: Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection alone, Outcome 2: Mean disability (0 to 100, 0 is no disability)	156
Analysis 7.3. Comparison 7: Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection alone, Outcome 3: Mean health-related quality of life (1 to 3, 1 = perfect health)	156
Analysis 7.4. Comparison 7: Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection alone, Outcome 4: Proportion of overall withdrawals	156
Analysis 7.5. Comparison 7: Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection alone, Outcome 5: Proportion with any adverse event	156
Analysis 7.6. Comparison 7: Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection alone, Outcome 6: Mean grip strength (pain-free, kg)	157
Analysis 8.1. Comparison 8: Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection alone, Outcome 1: Mean pain (0 to 10, 0 is no pain)	158
Analysis 8.2. Comparison 8: Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection alone, Outcome 2: Mean elbow disability (0 to 100, 0 is no disability)	159
Analysis 8.3. Comparison 8: Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection alone, Outcome 3: Mean health-related quality of life (EQ-5D utility score)	159
Analysis 8.4. Comparison 8: Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection alone, Outcome 4: Participant-reported treatment success	160
Analysis 8.5. Comparison 8: Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection alone, Outcome 5: Proportion of overall withdrawals	160
Analysis 8.6. Comparison 8: Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection alone, Outcome 6: Proportion with any adverse event	160
Analysis 8.7. Comparison 8: Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection alone, Outcome 7: Mean grip strength (affected side/unaffected side x 100)	161

Analysis 9.1. Comparison 9: Manual therapy, prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 1: Mean pain (0 to 10, 0 is no pain)	162
Analysis 9.2. Comparison 9: Manual therapy, prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 2: Mean elbow disability (0 to 100, 0 is no disability)	163
Analysis 9.3. Comparison 9: Manual therapy, prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 3: Mean health-related quality of life (EQ-5D utility score)	163
Analysis 9.4. Comparison 9: Manual therapy, prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 4: Participant-reported treatment success	163
Analysis 9.5. Comparison 9: Manual therapy, prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 5: Proportion of overall withdrawals	164
Analysis 9.6. Comparison 9: Manual therapy, prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 6: Proportion with any adverse event	164
Analysis 10.1. Comparison 10: Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 1: Mean pain (0 to 10, 0 is no pain)	165
Analysis 10.2. Comparison 10: Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 2: Mean elbow disability (0 to 100, 0 is no disability)	166
Analysis 10.3. Comparison 10: Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 3: Participant-reported treatment success	166
Analysis 10.4. Comparison 10: Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 4: Proportion of overall withdrawals	166
Analysis 10.5. Comparison 10: Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 5: Proportion with any adverse event	167
Analysis 10.6. Comparison 10: Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 6: Mean grip strength (affected side/unaffected side x 100)	167
Analysis 11.1. Comparison 11: Subgroup analysis short-term vs long-term symptoms, Outcome 1: Subgroup analysis: pain end of intervention	168
Analysis 11.2. Comparison 11: Subgroup analysis short-term vs long-term symptoms, Outcome 2: Subgroup analysis: disability end of intervention	168
ADDITIONAL TABLES	169
APPENDICES	188
WHAT'S NEW	194
HISTORY	194
CONTRIBUTIONS OF AUTHORS	194
DECLARATIONS OF INTEREST	195
SOURCES OF SUPPORT	195
DIFFERENCES BETWEEN PROTOCOL AND REVIEW	195
NOTES	196
INDEX TERMS	196

[Intervention Review]

Manual therapy and exercise for lateral elbow pain

Jason A Wallis¹, Allison M Bourne¹, Rebecca L Jessup¹, Renea V Johnston¹, Aviva Frydman¹, Sheila Cyril¹, Rachelle Buchbinder¹

¹School of Public Health and Preventive Medicine, Monash University, Melbourne, Australia

Contact: Jason A Wallis, jason.wallis@monash.edu.

Editorial group: Cochrane Musculoskeletal Group.

Publication status and date: Edited (no change to conclusions), published in Issue 7, 2024.

Citation: Wallis JA, Bourne AM, Jessup RL, Johnston RV, Frydman A, Cyril S, Buchbinder R. Manual therapy and exercise for lateral elbow pain. *Cochrane Database of Systematic Reviews* 2024, Issue 5. Art. No.: CD013042. DOI: [10.1002/14651858.CD013042.pub2](https://doi.org/10.1002/14651858.CD013042.pub2).

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

ABSTRACT

Background

Manual therapy and prescribed exercises are often provided together or separately in contemporary clinical practice to treat people with lateral elbow pain.

Objectives

To assess the benefits and harms of manual therapy, prescribed exercises or both for adults with lateral elbow pain.

Search methods

We searched the databases CENTRAL, MEDLINE and Embase, and trial registries until 31 January 2024, unrestricted by language or date of publication.

Selection criteria

We included randomised or quasi-randomised trials. Participants were adults with lateral elbow pain. Interventions were manual therapy, prescribed exercises or both. Primary comparators were placebo or minimal or no intervention. We also included comparisons of manual therapy and prescribed exercises with either intervention alone, with or without glucocorticoid injection. Exclusions were trials testing a single application of an intervention or comparison of different types of manual therapy or prescribed exercises.

Data collection and analysis

Two review authors independently selected studies for inclusion, extracted trial characteristics and numerical data, and assessed study risk of bias and certainty of evidence using GRADE. The main comparisons were manual therapy, prescribed exercises or both compared with placebo treatment, and with minimal or no intervention. Major outcomes were pain, disability, health-related quality of life, participant-reported treatment success, participant withdrawals, adverse events and serious adverse events. The primary endpoint was end of intervention for pain, disability, health-related quality of life and participant-reported treatment success and final time point for adverse events and withdrawals.

Main results

Twenty-three trials (1612 participants) met our inclusion criteria (mean age ranged from 38 to 52 years, 47% female, 70% dominant arm affected). One trial (23 participants) compared manual therapy to placebo manual therapy, 12 trials (1124 participants) compared manual therapy, prescribed exercises or both to minimal or no intervention, six trials (228 participants) compared manual therapy and exercise to exercise alone, one trial (60 participants) compared the addition of manual therapy to prescribed exercises and glucocorticoid injection, and four trials (177 participants) assessed the addition of manual therapy, prescribed exercises or both to glucocorticoid injection.

Twenty-one trials without placebo control were susceptible to performance and detection bias as participants were not blinded to the intervention. Other biases included selection (nine trials, 39%, including two quasi-randomised), attrition (eight trials, 35%) and selective reporting (15 trials, 65%) biases. We report the results of the main comparisons.

Manual therapy versus placebo manual therapy

Low-certainty evidence, based upon a single trial (23 participants) and downgraded due to indirectness and imprecision, indicates manual therapy may reduce pain and elbow disability at the end of two to three weeks of treatment. Mean pain at the end of treatment was 4.1 points with placebo (0 to 10 scale) and 2.0 points with manual therapy, MD -2.1 points (95% CI -4.2 to -0.1). Mean disability was 40 points with placebo (0 to 100 scale) and 15 points with manual therapy, MD -25 points (95% CI -43 to -7). There was no follow-up beyond the end of treatment to show if these effects were sustained, and no other major outcomes were reported.

Manual therapy, prescribed exercises or both versus minimal intervention

Low-certainty evidence indicates manual therapy, prescribed exercises or both may slightly reduce pain and disability at the end of treatment, but the effects were not sustained, and there may be little to no improvement in health-related quality of life or number of participants reporting treatment success. We downgraded the evidence due to increased risk of performance bias and detection bias across all the trials, and indirectness due to the multimodal nature of the interventions included in the trials.

At four weeks to three months, mean pain was 5.10 points with minimal treatment and manual therapy, prescribed exercises or both reduced pain by a MD of -0.53 points (95% CI -0.92 to -0.14, $I^2 = 43\%$; 12 trials, 1023 participants). At four weeks to three months, mean disability was 63.8 points with minimal or no treatment and manual therapy, prescribed exercises or both reduced disability by a MD of -5.00 points (95% CI -9.22 to -0.77, $I^2 = 63\%$; 10 trials, 732 participants). At four weeks to three months, mean quality of life was 73.04 points with minimal treatment on a 0 to 100 scale and prescribed exercises reduced quality of life by a MD of -5.58 points (95% CI -10.29 to -0.99; 2 trials, 113 participants). Treatment success was reported by 42% of participants with minimal or no treatment and 57.1% of participants with manual therapy, prescribed exercises or both, RR 1.36 (95% CI 0.96 to 1.93, $I^2 = 73\%$; 6 trials, 770 participants). We are uncertain if manual therapy, prescribed exercises or both results in more withdrawals or adverse events. There were 83/566 participant withdrawals (147 per 1000) from the minimal or no intervention group, and 77/581 (126 per 1000) from the manual therapy, prescribed exercises or both groups, RR 0.86 (95% CI 0.66 to 1.12, $I^2 = 0\%$; 12 trials). Adverse events were mild and transient and included pain, bruising and gastrointestinal events, and no serious adverse events were reported. Adverse events were reported by 19/224 (85 per 1000) in the minimal treatment group and 70/233 (313 per 1000) in the manual therapy, prescribed exercises or both groups, RR 3.69 (95% CI 0.98 to 13.97, $I^2 = 72\%$; 6 trials).

Authors' conclusions

Low-certainty evidence from a single trial in people with lateral elbow pain indicates that, compared with placebo, manual therapy may provide a clinically worthwhile benefit in terms of pain and disability at the end of treatment, although the 95% confidence interval also includes both an important improvement and no improvement, and the longer-term outcomes are unknown. Low-certainty evidence from 12 trials indicates that manual therapy and exercise may slightly reduce pain and disability at the end of treatment, but this may not be clinically worthwhile and these benefits are not sustained. While pain after treatment was an adverse event from manual therapy, the number of events was too small to be certain.

PLAIN LANGUAGE SUMMARY

Manual therapy and exercise for treating tennis elbow

Key messages

Compared with placebo (sham) manual therapy, manual therapy for tennis elbow may reduce pain and disability at the end of treatment. Longer-term effects are unknown.

Compared with minimal or no treatment, manual therapy, prescribed exercises, or both, may slightly reduce pain and disability, and result in little to no improvement in quality of life and treatment success. We are uncertain if more people have adverse effects from the treatments.

What is tennis elbow?

Tennis elbow (also called 'lateral elbow pain') may arise spontaneously or following overuse of the forearm. It causes pain over the outer region of the elbow, usually with movements involving lifting and gripping.

How is tennis elbow treated?

Physiotherapists and others commonly provide manual therapy and prescribe exercises for people with tennis elbow. Manual therapy involves the movement of a joint or muscle by the therapist. Prescribed exercises are performed by the patient under supervision or at home. Other common treatments include pain-killing and anti-inflammatory medicines and glucocorticoid injections.

What did we want to find out?

The benefits and harms of manual therapy and exercise for adults with tennis elbow.

What did we do?

We searched for studies of manual therapy, prescribed exercises, or both, compared with placebo or minimal treatment (e.g. advice). We also included studies that compared manual therapy and prescribed exercises with either intervention alone, with or without glucocorticoid (steroid) injection. We extracted data on pain, disability, quality of life, treatment success and adverse events, and assessed how confident we were in the results.

What did we find?

We found 23 studies involving 1612 participants. The average age ranged from 38 to 52 years and 47% of participants were female.

- 1 study (23 participants) compared manual therapy to placebo manual therapy.
- 12 studies (1124 participants) compared manual therapy, prescribed exercises, or both, to minimal treatment.
- 6 trials (228 participants) compared manual therapy and prescribed exercises to prescribed exercises alone.
- 1 trial (60 participants) compared the addition of manual therapy to prescribed exercises and glucocorticoid injection.
- 4 trials (177 participants) compared manual therapy, prescribed exercises, or both, and glucocorticoid injection to a glucocorticoid injection alone.

Main results

Manual therapy compared with placebo manual therapy:

Pain was measured on a 0- to 10-point scale (lower scores mean less pain) at the end of treatment.

- People who had manual therapy, prescribed exercises, or both, rated their pain as 2 points.
- People who had placebo manual therapy rated their pain as 4.1 points.

Disability was measured on a 0- to 100-point scale (lower scores mean less disability) at the end of treatment.

- People who had manual therapy, prescribed exercises, or both, rated their disability as 15 points.
- People who had placebo manual therapy rated their disability as 40 points.

No other outcomes were measured, and no outcomes were collected after the end of treatment.

Manual therapy, prescribed exercises, or both, compared with minimal treatment:

Pain was measured on a 0- to 10-point scale (lower scores mean less pain) at the end of treatment.

- People who had manual therapy, prescribed exercises, or both, rated their pain as 4.6 points.
- People who had minimal treatment rated their pain as 5.1 points.

Disability was measured on a 0- to 100-point scale (lower scores mean less disability) at the end of treatment.

- People who had manual therapy, prescribed exercises, or both, rated their disability as 58.8 points.
- People who had minimal treatment rated their disability as 63.8 points.

Quality of life was measured on a 0- to 100-point scale (higher scores indicate better quality of life) at the end of treatment.

- People who had manual therapy, prescribed exercises, or both, rated their quality of life as 67.5 points.
- People who had minimal treatment rated their quality of life as 73.0 points.

Treatment success was rated by participants at the end of the treatment.

- 571 out of 1000 people reported treatment success with manual therapy, prescribed exercises, or both.
- 420 out of 1000 people reported treatment success with minimal treatment.

Participant withdrawals were measured at the end of the study.

- 126 out of 1000 people receiving manual therapy, prescribed exercises, or both, withdrew from studies.
- 147 out of 1000 people receiving minimal treatment withdrew from studies.

Adverse events were measured after treatment.

- 313 out of 1000 people reported adverse events with manual therapy, prescribed exercises, or both.
- 85 out of 1000 people reported adverse events with minimal treatment.

No serious adverse events were reported.

Manual therapy and exercise for lateral elbow pain (Review)

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

What are the limitations of the evidence?

For most findings, we have low confidence due to the small number of participants or events. Manual therapy and prescribed exercises may reduce pain and disability slightly in people with tennis elbow, with little or no benefit for quality of life and treatment success. Effects may not be sustained. We are less confident of the risk of adverse events or withdrawals from treatment because of the small number of events.

How up-to-date is the evidence?

The evidence is up-to-date to 31 January 2024.

SUMMARY OF FINDINGS

Summary of findings 1. Manual therapy, prescribed exercises or both compared with placebo for lateral elbow pain

Manual therapy, prescribed exercises or both compared with placebo for lateral elbow pain

Patient or population: lateral elbow pain

Setting: community

Intervention: manual therapy

Comparison: placebo manual therapy

Timing of outcome: end of intervention for outcomes assessing benefits

Outcomes	Anticipated absolute effects* (95% CI)			Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	What happens
	Placebo manual therapy	Manual therapy	Difference				
Pain Scale: numerical rating scale (0 to 10, 0 is no pain) Follow-up: end of treatment - mean 2 to 3 weeks	Mean pain was 4.1 points^a	Mean pain was 2 points	2.1 points better (4.2 better to 0.1 better)	-	23 (1 RCT)	⊕⊕⊕⊕ LOW ^{b,c}	Manual therapy may improve pain at the end of treatment. ^d
Disability Scale: 0 to 100 (0 is no disability) Follow-up: end of treatment - mean 2 to 3 weeks	Mean disability was 40 points^a	Mean disability with manual therapy was 15 points	25 points better (42.7 better to 7.4 better)	-	23 (1 RCT)	⊕⊕⊕⊕ LOW ^{b,c}	Manual therapy may improve disability at the end of treatment. ^d
Quality of life	-	-	-	-	-	-	Quality of life was not reported.
Participant-reported treatment success	-	-	-	-	-	-	Participant-reported treatment success was not reported.
Proportion of withdrawals	-	-	-	-	-	-	Withdrawals were not reported.
Proportion with adverse events	-	-	-	-	-	-	Adverse events were not reported.
Proportion with a serious adverse event	-	-	-	-	-	-	Serious adverse events were not reported.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RCT:** randomised controlled trial

GRADE Working Group grades of evidence

High certainty: We are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect.

Very low certainty: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.

^aValues for mean pain and mean disability with placebo manual therapy were from [Nourbakhsh 2008](#) at two to three weeks.

^bDowngraded due to indirectness - the intervention was limited to manual therapy alone, specifically oscillating energy manual therapy, which may not be easily applied outside the trial setting.

^cDowngraded due to imprecision - evidence was from a single small trial, and the confidence interval includes both an important improvement and no improvement.

^dWe assumed a minimal clinically important difference (MCID) of 1.5 points on a 10-point scale for pain, and 11 points on a 100-point scale for disability. The 95% CI includes both a clinically important and clinically unimportant improvement.

Summary of findings 2. Manual therapy, prescribed exercises or both compared to minimal or no intervention for lateral elbow pain

Manual therapy, prescribed exercises or both compared to minimal or no intervention for lateral elbow pain

Patient or population: lateral elbow pain

Setting: primary care, outpatient clinics

Intervention: manual therapy, prescribed exercises or both delivered together

Comparison: minimal or no intervention

Timing of outcome: end of intervention for pain, disability, quality of life and participant-reported treatment success. Last follow-up for adverse events and withdrawals.

Outcomes	Anticipated absolute effects* (95% CI)			Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Minimal or no intervention	Manual therapy, prescribed exercises or both	Difference				
Pain Scale: VAS (0 to 10, 0 is no pain) Follow-up: range 4 weeks to 3 months ^a	Mean pain was 5.1 points ^b	Mean pain was 4.6 points	0.53 points better (0.92 better to 0.14 better)	-	1023 (12 RCTs)	⊕⊕⊕⊕ LOW ^{c,d}	Manual therapy, prescribed exercises or both may reduce pain slightly at the end of treatment, which may not be clinically important. ^e

Disability Scale: Disability questionnaires (0 to 100, 0 is no disability) Follow-up: range 4 weeks to 3 months ^a	Mean disability was 63.8 points ^b	Mean disability was 58.8 points	5.00 points better (9.22 better to 0.77 better)	-	732 (10 RCTs)	⊕⊕○○ LOW ^{c,d}	Manual therapy, prescribed exercises or both may reduce disability slightly at the end of treatment, which may not be clinically important. ^e
Health-related quality of life^f Scale: SF-36 (0 to 100, 100 indicates perfect health) Follow-up: range 4 weeks to 3 months ^a	Mean quality of life was 73.0 points ^b	Mean quality of life was 67.5 points	5.58 points worse (10.29 worse to 0.99 worse)	-	113 (2 RCTs)	⊕⊕○○ LOW ^{c,g}	Manual therapy, prescribed exercises or both may result in little to no improvement in quality of life.
Participant-reported treatment success Follow-up: range 6 weeks to 3 months ^a	420 per 1000	571 per 1000	151 more per 1000 (17 fewer to 390 more)	RR 1.36 (0.96 to 1.93)	770 (6 RCTs)	⊕⊕○○ LOW ^{c,d}	Manual therapy, prescribed exercises or both may result in little to no difference in the number of participants reporting treatment success.
Proportion of overall withdrawals	147 per 1000	126 per 1000	21 fewer per 1000 (50 fewer to 18 more)	RR 0.86 (0.66 to 1.12)	1147 (12 RCTs)	⊕○○○ VERY LOW ^{c,d,h}	We are uncertain if manual therapy, prescribed exercises or both results in more withdrawals from treatment.
Proportion with any adverse event	85 per 1000	313 per 1000	228 more (2 fewer to 1100 more)	RR 3.69 (0.98 to 13.97)	457 (6 RCTs)	⊕○○○ VERY LOW ^{c,d,h}	We are uncertain if manual therapy, prescribed exercises or both results in more people with adverse events.
Proportion with a serious adverse event	-	-	-	-	-	-	Serious adverse events were not reported.

*The risk in the intervention group (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; RCT: randomised controlled trial; RR: risk ratio; VAS: visual analogue scale

GRADE Working Group grades of evidence

High certainty: We are very confident that the true effect lies close to that of the estimate of the effect

Moderate certainty: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different

Low certainty: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect

Very low certainty: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect

^aFollow-up for pain measures - four weeks ([Rachold 2023](#)), six weeks ([Bisset 2006](#); [Blanchette 2011](#); [Olaussen 2015](#); [Smidt 2002a](#); [Soderberg 2012](#); [Struijs 2004](#)), seven weeks ([Tonks 2007](#)), eight weeks ([Vuvan 2020](#)), three months ([Haahr 2003](#); [McQueen 2021](#); [Peterson 2011](#)); disability measures - four weeks ([Rachold 2023](#)), six weeks ([Bisset 2006](#); [Blanchette 2011](#); [Olaussen 2015](#); [Smidt 2002a](#); [Struijs 2004](#)), seven weeks ([Tonks 2007](#)), eight weeks ([Vuvan 2020](#)), three months ([McQueen 2021](#); [Peterson 2011](#)); quality of life measures - four weeks ([Rachold 2023](#)), three months ([Peterson 2011](#)); treatment success - six weeks ([Bisset 2006](#); [Olaussen 2015](#); [Smidt 2002a](#); [Struijs 2004](#)), eight weeks ([Vuvan 2020](#)), three months ([Haahr 2003](#)).

^bMean pain and mean disability values in the minimal treatment group are from [Bisset 2006](#) at six weeks; mean quality of life values in the minimal treatment group are from [Rachold 2023](#) at four weeks.

^cDowngraded due to risk of bias - increased risk of performance and detection biases across all trials as participants were aware of their treatment group.

^dDowngraded due to indirectness - diversity in the interventions including placebo injection ([Olaussen 2015](#)), ergonomic advice ([Haahr 2003](#)) and ultrasound therapy ([Smidt 2002a](#); [Struijs 2004](#)); diversity in the application and nature of the minimal interventions including advice ([Bisset 2006](#); [Blanchette 2011](#); [McQueen 2021](#); [Olaussen 2015](#); [Smidt 2002a](#); [Soderberg 2012](#); [Vuvan 2020](#)), usual GP care ([Haahr 2003](#)), observation ([Tonks 2007](#)), waiting list control ([Peterson 2011](#)); and whether they were applied to both treatment groups including prescribed analgesics ([Olaussen 2015](#)) and braces ([McQueen 2021](#); [Soderberg 2012](#)).

^eWe assumed a minimal clinically important difference (MCID) of 1.5 points in a 10-point scale for pain, and 11 points on a 100-point scale for disability ([Poltaowski 2011](#)).

^fOutcome data for health-related quality of life in this meta-analysis included two scales - Gothenburg Quality of Life questionnaire (seven-point Likert scale ranging from 1 'very bad' to 7 'excellent, could not be better' from [Peterson 2011](#) and the SF-36 (0 to 100, 100 indicates perfect health) from [Rachold 2023](#)). We re-expressed the standardised mean difference (SMD) (-0.45, -0.83 to -0.08) as the difference on the SF-36 scale (0 to 100) by multiplying the SMD by the standard deviation (SD) of the control group at baseline (SD 12.41) from [Rachold 2023](#).

^gDowngraded due to imprecision for continuous outcome - evidence from two trials with large confidence intervals that include both little effect and worse outcome.

^hDowngraded due to imprecision for dichotomous outcome - a small number of events.

BACKGROUND

This review is one of two Cochrane reviews of physical therapies for lateral elbow pain (or lateral epicondylitis or tennis elbow). In keeping with our reviews of physical therapies for rotator cuff disease and adhesive capsulitis (Page 2014a; Page 2014b; Page 2016a; Page 2016b), this review evaluates the benefits and harms of manual therapy and exercise and a second review will evaluate the benefits and harms of electrotherapy modalities (Pompilio da Silva 2018).

Description of the condition

Lateral elbow pain is described by many analogous terms in the literature, including tennis elbow, lateral epicondylitis (or epicondylosis), rowing elbow, tendonitis of the common extensor origin, extensor tendinopathy and peritendinitis of the elbow. For the purposes of this review, and in keeping with previous Cochrane reviews for this condition, we will use the term lateral elbow pain.

Lateral elbow pain affects approximately 3% to 4% of adults (Sanders 2015; Shiri 2011). The annual incidence is estimated to be 3.5 per 1000 adults, with a peak incidence of 9 per 1000 adults aged between 40 and 49 years (Sanders 2015). Lateral elbow pain may be slightly more common in women (Sanders 2015), and affects up to 15% of workers, including those in office work, health care, construction, maintenance, repair and cleaning (Shiri 2011; Walker-Bone 2012; Yanai 2024). Repetitive bending and straightening the elbow for more than one hour per day at work is a risk factor for lateral elbow pain (Walker-Bone 2012), and lateral elbow pain is also associated with playing sports (Abrams 2012; Fu 2018; Hume 2006). As well as pain and disability, lateral elbow pain may also result in psychological distress (Walker-Bone 2012).

Lateral elbow pain is thought to be an overuse injury of the extensor muscles in the forearm, more commonly affecting the extensor carpi radialis brevis (ECRB) tendon inserted in the lateral epicondyle of the humerus (AAOS 2020; Johns 2020). Pain is typically felt around the lateral elbow region and may radiate to the forearm, following the extensor muscles (Seiça 2023). Lateral elbow pain is typically aggravated by repetitive movement, lifting and gripping. Examination findings include localised tenderness over the common extensor origin at the lateral epicondyle and elicitation of pain in resisting dorsiflexion of the wrist, middle finger, or both, and loss of grip strength (AAOS 2020). In most instances, the diagnosis is based upon clinical assessment alone and exclusion of other causes of elbow pain such as inflammatory arthritis or osteoarthritis affecting the elbow, or referred pain from the neck.

Histological studies indicate degenerative changes at the tendon insertion site including deterioration and disorganisation of collagen tendon fibres that have been invaded by fibroblasts and atypical vascular granulation tissue (Nirschl 1979; Regan 1992). These changes in local tendon pathology may contribute to the patient's pain experience, in association with both nociceptive and non-nociceptive pain mechanisms and motor impairments such as muscle atrophy (Seiça 2023). Clinical presentations and recovery time vary between individuals.

The acute phase of lateral elbow pain usually lasts six to 12 weeks and often results in work absence (Mallen 2009). One study estimated 29 working days lost over 12 months in 5% of workers diagnosed with lateral (or medial) elbow pain (Walker-Bone 2012).

For most, it is a self-limiting condition; one study found that 80% of participants with elbow pain already greater than four weeks' duration recovered after one year without any specific treatment (Bisset 2006). A systematic review evaluating the course of improvement of lateral elbow pain, based upon the placebo or no treatment control arms of randomised trials, found that about 90% of people with untreated lateral elbow pain achieve symptom resolution in one year (Ikonen 2022). This review also observed that the probability of resolution remains constant throughout the first year of follow-up and does not depend on previous symptom duration.

Persistent pain and disability symptoms may be associated with previous occurrences, manual jobs, high baseline levels of pain or distress, and psychological factors including anxiety, depression, pain-related fear and ineffective pain coping skills (Alizadehkhayat 2007). In addition, an ultrasound study indicated that the presence of a lateral collateral ligament tear or large (≥ 6 mm) intrasubstance tear was associated with a poorer outcome (Clarke 2010). Persistent symptoms can result in significant disability, health care utilisation, lost productivity and costs (Silverstein 2006).

Description of the intervention

There is no consensus on the optimal management of lateral elbow pain (Coombes 2015; Javed 2016; Johns 2020). Various physical, pharmacological and surgical treatments may be offered. Initially, avoiding or modifying aggravating activities may be recommended. Physical treatments include manual therapy and exercise (Coombes 2015), heat and electrotherapy (Pompilio da Silva 2018), taping techniques (Giray 2019), orthotic devices (Struijs 2002), acupuncture (Green 2002) and extracorporeal shock wave therapy (Buchbinder 2005). Pharmacological treatments include topical and oral non-steroidal anti-inflammatory drugs (Pattanittum 2013), and injection therapies including glucocorticoid (Coombes 2010) and platelet rich plasma (Karjalainen 2021). Surgery may be performed in refractory cases (Buchbinder 2011; Lo 2007), with fewer than 10% of patients with lateral elbow pain estimated to undergo surgery (Konarski 2023; Nirschl 1979). However, the utility of surgery has recently been questioned (Karjalainen 2023).

In contemporary clinical practice, manual therapy and prescribed exercises are often delivered together or separately by health professionals such as physiotherapists, occupational therapists, chiropractors and osteopaths. Manual therapy involves the movement of a joint or muscle that is applied by the health professional, and may include joint mobilisation (i.e. low-velocity movement of a joint), manipulation (i.e. high-velocity thrust of a joint) or massage techniques (APA 2015). Manual therapy treatments are repetitive, and applied for a specific duration and number of sessions. For some mobilisation techniques, the patient is actively involved. For example, a Mulligan manual therapy technique, known as Mobilisation-with-Movement (MWM), requires the patient to perform a painful activity (e.g. clenched fist) while the health professional applies the mobilisation technique (Westad 2019).

Exercise therapies are planned, structured, repetitive and targeted physical activities that aim to improve specific health outcomes (Caspersen 1985). Examples of exercises prescribed for people with lateral elbow pain include various forms of resistance training (e.g. isometric, eccentric, concentric), stretching and functional movements (APA 2015). Exercises may be performed at home or

under supervision by a health professional, and include a specific dose (e.g. number of sets, repetitions, intensity and duration), and may include various features such as load, progression and equipment.

How the intervention might work

Manual therapy and prescribed exercises may be provided for people with lateral elbow pain to reduce pain and disability, and increase joint mobility of the elbow. It is possible that increasing elbow movements might help to restore structure and function of the muscle tendon complex, and may have a role in strengthening the muscle, improving grip strength, and improving blood flow and healing. It is also possible that movement of the joint or soft tissue could stimulate peripheral mechanoreceptors, inhibit abnormal sensory input (nociception), and activate the descending pain pathways of the central nervous system, resulting in reduced levels of pain (Frank 1984).

Limited evidence from small, repeated-measure controlled studies have investigated the underlying physiological mechanisms from manual therapy and exercise treatments for people with lateral elbow pain (Coombes 2016; Paungmali 2003). For manual therapy, initial pain-relieving effects and changes in sympathetic nervous system function (e.g. increased heart rate) were shown during and immediately following the single application of a Mulligan manual therapy technique (Paungmali 2003). For the single application of an isometric exercise, there were no initial pain-relieving effects observed (Coombes 2016). These results suggest that underlying local and central physiological mechanisms from manual therapy techniques may result in changes in pain for people with lateral elbow pain. Additional mechanisms from manual therapy and exercise may include reduced fear of movement, placebo and local mechanical effects (Vicenzino 2011).

Why it is important to do this review

Previous systematic reviews have concluded that manual therapy, exercise or both result in small, short-term benefits in pain, disability and grip strength (Hoogvliet 2013; Karanasios 2020; Kim 2020). Two of these reviews used suboptimal conduct and reporting methods compared with the current recommended methods (Higgins 2022). For example, one review was restricted to studies published in English and pooled data from studies with different comparators (Kim 2020), while no synthesis was attempted in the second review (Hoogvliet 2013). Neither considered treatment harms. While the third review used more up-to-date methods, including an assessment of the certainty of the evidence using the GRADE approach, it was restricted to synthesising the benefits of exercise and there was no assessment of harms (Karanasios 2020). The focus of our review includes both manual therapy and prescribed exercise as these are both commonly applied in clinical practice, either alone or in combination. Also, as further randomised controlled trials involving manual therapy and/or prescribed exercise have been published since the most recent reviews (Karanasios 2020; Kim 2020), an up-to-date synthesis of the available evidence using rigorous Cochrane conduct and reporting methods was justified.

OBJECTIVES

To assess the benefits and harms of manual therapy, prescribed exercises or both for adults with lateral elbow pain.

METHODS

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials or controlled clinical trials with quasi-randomised methods of allocating participants to treatment (such as by alternation, date of birth or other pseudo-randomised methods). We included studies reported as full text, those published as abstracts only and unpublished data. We did not impose any language restrictions.

Types of participants

We included trials that enrolled adult participants with a diagnosis of lateral elbow pain. Studies were eligible if they included participants diagnosed based upon clinical assessment alone, with or without supportive imaging, or as otherwise defined by the trial authors. Diagnostic criteria include clinical features such as pain that is maximal over the lateral epicondyle and reproducibility of pain by two or more of the following tests: palpation of the lateral epicondyle, or the common extensor origin of the elbow, or both; gripping; and resisted wrist or second or third finger extension (dorsiflexion). Ultrasound or magnetic resonance imaging may indicate the presence of focal hypoechoic areas or frank tears or alterations in the normal fibrillary pattern in the common extensor origin. There was no restriction based upon symptom duration.

We included trials that included participants with medial elbow pain or pain at other tendon insertion sites only if the lateral elbow pain results were presented separately, or at least 90% of participants had lateral elbow pain. We excluded trials that included participants with a history of significant trauma or inflammatory or degenerative disease such as rheumatoid arthritis and osteoarthritis, respectively.

Types of interventions

We included trials comparing any manual therapy or prescribed exercises either alone or delivered together to the following comparators:

- Placebo
- Minimal or no treatment
- Manual therapy and prescribed exercises compared to either the same manual therapy or prescribed exercises alone
- Manual therapy, prescribed exercises and glucocorticoid injection compared to either the same manual therapy or prescribed exercises with glucocorticoid injection
- Manual therapy, prescribed exercise or both, provided in addition to glucocorticoid injection, compared to glucocorticoid injection alone.

We excluded trials that compared any manual therapy, prescribed exercises, or both, to a different type of manual therapy or exercise, or another intervention. We also excluded trials testing a single application of an intervention.

Manual therapy interventions could include mobilisation, manipulation or massage, or a combination of these. Prescribed exercises could be supervised, performed individually or in groups, delivered at home or in other settings, land-based or water-based,

but needed to be tailored for lateral elbow pain and were not just exercise for general fitness or strength (e.g. swimming).

Minimal or no treatment could include no explicit intervention, an undefined control group, or minimal interventions such as analgesics, bracing, information or advice. These could have been delivered to all groups or just the control group. Trials were also eligible if the manual therapy and exercise groups included additional interventions commonly used in practice, such as electrotherapy, heat applications or taping interventions, even if these were not applied to the comparison group, as it was considered unlikely these would have a sizeable additional benefit.

We excluded trials primarily evaluating the effect of electrotherapy modalities, such as therapeutic ultrasound, laser therapy, transcutaneous electrical nerve stimulation (TENS), pulsed electromagnetic field therapy, bipolar interferential current, electromyographic biofeedback, phonophoresis, iontophoresis or short-wave diathermy, as we will include them in a separate Cochrane review.

Types of outcome measures

Major outcomes

1. Mean pain or mean change in pain score on a visual analogue scale or numerical rating scale, or sub-score of a total function score, or other measure.
2. Disability as measured by an elbow disease-specific disability measure such as the Patient-Rated Tennis Elbow Evaluation (PRTEE) questionnaire (Rompe 2007), the upper limb-specific Disabilities of the Arm, Shoulder and Hand (DASH) outcome questionnaire (Hudak 1996) or other disability measures.
3. Participant-reported global assessment of treatment success, as measured by a global rating of treatment satisfaction such as the Patient Global Impression of Change (PGIC) scale or as defined in the trials (e.g. proportion without elbow pain; proportion with 25% reduction in pain or disability, etc.).
4. Health-related quality of life as measured by either generic measures (such as components of the Short Form-36 (SF-36), or disease-specific tools).
5. Proportion of overall withdrawals.
6. Proportion with any adverse event.
7. Proportion with a serious adverse event, defined as adverse events that are fatal, life-threatening or require hospitalisation.

Minor outcomes

1. Grip strength (preferably pain-free maximum grip strength).
2. Return to work.

Timing of outcome assessment

For outcomes that assess benefits of treatment (pain, disability, quality of life, treatment success), the primary time point was the end of treatment. Other time points were the first assessment, if prior to end of treatment, and last follow-up (i.e. after completion of treatment). For adverse events, withdrawals, serious adverse events and return to work, we included data up to the last follow-up.

Search methods for identification of studies

Electronic databases

We searched the following electronic databases, unrestricted by date or language, on 22 September 2022. We updated the search on 31 January 2024.

1. Cochrane Central Register of Controlled Trials (CENTRAL; 2024, Issue 1) (via EBM reviews in Ovid) (Appendix 1).
2. MEDLINE (Ovid 1946 to 2024) (Appendix 2).
3. Embase (Ovid 1947 to 2024) (Appendix 3).

Trial registries

We searched the following trial registries on 31 January 2024 for ongoing trials.

1. ClinicalTrials.gov (clinicaltrials.gov/).
2. World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP) search portal (apps.who.int/trialsearch/).

Searching other resources

We screened the reference lists of all included primary studies and systematic review articles to identify potentially relevant studies.

Data collection and analysis

Selection of studies

Pairs of review authors (JW, AB, RLJ, RJ, AF) independently reviewed the search results and identified trials that fulfilled our inclusion criteria. All articles selected by at least one of the review authors were retrieved for closer examination. The review authors were not blinded to the journal or authors. We resolved disagreements about inclusion or exclusion of individual studies by consensus or by involving a third review author (RJ or RB). We recorded the selection process in sufficient detail to complete a PRISMA flow diagram (PRISMA Statement 2020), and 'Characteristics of excluded studies' table.

Data extraction and management

Pairs of review authors working independently (JW, AB, SC) extracted the following data from the included trials using pre-piloted data extraction forms. They resolved any differences by consensus, or arbitration by a third review author (RJ), if needed.

1. Trial characteristics including type (e.g. parallel or cross-over), country, source of funding and trial registration status (with registration number recorded if available).
2. Participant characteristics, including age, sex, duration of symptoms, and inclusion and exclusion criteria.
3. Intervention characteristics, including type of manual therapy, exercise, or both, and details of the treatments, such as frequency, schedule of treatment, supervision, total number of treatment sessions, use of co-interventions and characteristics of the control intervention. We used the Consensus on Exercise Reporting Template (CERT) to extract the characteristics of exercise interventions (Slade 2016).
4. Outcomes reported, including the measurement scale, direction of the scale, the mean and standard deviation, number of participants per treatment group for continuous outcomes

(such as mean pain, disability, quality of life), and number of events and number of participants per treatment group for dichotomous outcomes (such as proportion with 30% or more pain relief, treatment success, withdrawals due to adverse events, adverse events) as outlined in 'Types of outcome measures'.

5. Risk of bias domains as outlined in 'Assessment of risk of bias in included studies'.
6. Notes: funding for trial and notable declarations of interest by trial authors.

If outcome data were not reported in a suitable form for meta-analysis, and where missing data were calculated or estimated from a graph or imputed, we noted this in the 'Characteristics of included studies' tables.

Our a priori decision rules for extracting data in the event of multiple outcome reporting in trials were as follows.

1. Where trialists reported both final values and change from baseline values for the same outcome, we extracted final values.
2. Where trialists reported both unadjusted and adjusted for baseline values for the same outcome, we extracted adjusted values.
3. Where trialists reported data analysed based on the intention-to-treat (ITT) sample and another sample (e.g. per protocol, as-treated), we extracted ITT-analysed data.
4. For cross-over RCTs, we extracted data from the first period only.

Where trials did not include a measure of overall pain but included one or more other measures of pain, for the purpose of pooling data, we combined overall pain (or similar, such as average pain in last 24 hours or other time span) with another pain measure using the following hierarchy: unspecified pain, pain with activity or worst pain, pain at night or at rest.

Where trialists reported multiple pain outcome measures, for the purposes of pooling data, we extracted one measure using the following hierarchy: visual analogue scale (VAS), numerical or categorical rating scale, McGill Pain Questionnaire or other scale.

Where trialists reported multiple measures of function or disability, for the purposes of pooling data, we extracted a single measure using the following hierarchy: disease-specific disability measures such as the Patient-Rated Tennis Elbow Evaluation (PRTEE) questionnaire (Rompe 2007), or the upper limb-specific Disabilities of the Arm, Shoulder and Hand (DASH) outcome questionnaire (Hudak 1996), or another measure. We selected overall score for disability measures in preference to sub-scores.

Where trialists reported multiple measures of participant-reported global assessment of treatment success, we extracted a global rating of treatment satisfaction such as the Patient Global Impression of Change (PGIC) scale, over other measures that might be reported in the trials (e.g. proportion without elbow pain; proportion with 25% reduction in pain or disability, etc.).

Where trialists reported multiple measures of health-related quality of life, we extracted generic measures (such as components of the Short Form-36 (SF-36), and EQ5D), over disease-specific tools.

Assessment of risk of bias in included studies

Pairs of review authors (JW, AB, SC) assessed the risk of bias for each study using the criteria outlined in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2017). We resolved any disagreements by discussion or by involving a third review author (RJ). We assessed the following methodological domains.

1. Random sequence generation (to determine if the method of generating the randomisation sequence was adequate, such as random number tables, computer-generated random numbers, minimisation, coin tossing, shuffling of cards and drawing of lots).
2. Allocation sequence concealment (to determine if adequate methods were used to conceal the random sequence, such as central randomisation and sequentially numbered, sealed, opaque envelopes).
3. Blinding of participants and personnel.
4. Blinding of outcome assessors. We considered blinding separately for subjective self-reported outcomes (pain, function, treatment success, quality of life) and assessor-reported outcomes (such as withdrawals, adverse events, grip strength). For example, for unblinded outcome assessment, risk of bias for mortality may be different from that for a participant-reported pain scale.
5. Incomplete outcome data.
6. Selective outcome reporting.
7. Other potential threats to validity, such as inappropriate analysis in cross-over trials, baseline imbalance in important factors, inappropriate or uneven application of co-interventions.

We graded each potential source of bias as high risk, low risk or unclear risk (either lack of information or uncertainty over the potential for bias). We presented the figures generated by the risk of bias tool to provide summary assessments of the risk of bias.

Assessment of bias in conducting the review

We conducted the review according to the published protocol (Frydman 2018) and reported any deviations from it in the 'Differences between protocol and review' section of the review.

Measures of treatment effect

We presented simple summary data for each intervention group: number of events and number of participants per group for dichotomous outcomes, mean and standard deviation and number of participants per group for continuous outcomes, in forest plots.

We analysed dichotomous data as risk ratios (RRs) with corresponding 95% confidence intervals (CIs) for dichotomous outcomes.

For continuous data, we presented results as mean differences (MDs) if possible, including transforming all data to a common scale for pain (0 to 10), and disability (0 to 100) to standardise the measure. However, where different scales were used to measure the same outcome or concept (e.g. health-related quality of life, grip strength), we used standardised mean differences (SMDs). For mean quality of life, we re-expressed the SMD to the difference on a typical scale (SF-36, 0 to 100) by multiplying the SMD by a typical

among-person standard deviation (SD) (e.g. the SD of the control group at baseline from the most representative trial) (Higgins 2022).

In interpreting the results, we assumed a minimal clinically important difference (MCID) of 1.5 points on a 10-point scale for pain, and 11 points on a 100-point scale for disability (Poltawski 2011).

Unit of analysis issues

If we had identified trials that applied manual therapy and exercise to both elbows, but the trialists reported outcomes per participant without accounting for the bilateral correlation, we would only have reported results from one elbow where possible. If we were unable to obtain the data for a single elbow, or adjust the outcome data, we would have included the data as reported by the trialists and commented on the validity of such analyses, and assessed the effect of including such data using sensitivity analyses.

If we had identified trials that used a cross-over design, we would have only included data from the first time point to account for any potential carry over effect.

Where multiple treatment groups were reported in a single trial, we included only the relevant groups and reported this in the 'Characteristics of included studies' table. If two comparisons (e.g. manual therapy versus minimal treatment and exercise versus minimal treatment) were combined in the same meta-analysis, we halved the control group to avoid double-counting. When a study included multiple relevant intervention groups (for example, two groups treated with different types of manual therapy combined with exercise, compared to a third group with exercise alone), we combined the two manual therapy groups to perform a single pairwise comparison, using the methods described in Chapter 6 of the *Cochrane Handbook* (Higgins 2023).

Dealing with missing data

We attempted to contact the trialists where data were missing or incomplete.

For dichotomous outcomes that measured adverse events (e.g. number of withdrawals due to adverse events), we calculated the withdrawal rate using the number of participants that received treatment as the denominator.

For dichotomous outcomes that measured benefits (e.g. proportion of participants with 30% or more reduction in pain), we calculated the proportion using the number of randomised participants as the denominator.

For continuous outcomes (e.g. mean change in pain score), we calculated the MD or SMD based on the number of participants analysed at that time point. If the number of participants analysed was not presented for each time point, we used the number of randomised participants in each group at baseline.

Where possible, we computed missing SDs from other statistics such as standard errors, CIs or P values, according to the methods recommended in the *Cochrane Handbook* (Higgins 2022). If we could not calculate SDs, we imputed them (e.g. from other studies in the meta-analysis). Means and SDs were estimated from the sample size, median and interquartile ranges using Wan 2014; we described any assumptions and imputations to handle missing data and explored the effect of imputation by sensitivity analyses.

Assessment of heterogeneity

We first assessed the included trials for clinical homogeneity in terms of participants, interventions and comparators. For studies that we judged as clinically homogeneous, we assessed and quantified the possible magnitude of inconsistency (i.e. heterogeneity) across studies, using the I^2 statistic with a rough guide for interpretation as follows: 0% to 40% might not be important; 30% to 60% may represent moderate heterogeneity; 50% to 90% may represent substantial heterogeneity; 75% to 100% considerable heterogeneity (Deeks 2022). In cases of considerable heterogeneity (defined as $I^2 \geq 75\%$), we explored the data further by comparing the characteristics of individual studies and any subgroup analyses.

Assessment of reporting biases

In order to determine whether reporting bias was present, we determined whether the protocol for the trial was published before recruitment of participants for the study, and checked if all outcome measures had results reported. For studies published after 1 July 2005, we screened ClinicalTrials.gov and the WHO ICTRP search portal as described in 'Electronic searches'. We evaluated whether selective reporting of outcomes was present (outcome reporting bias).

We compared the fixed-effect estimate against the random-effects model to assess the possible presence of small sample bias in the published literature (i.e. in which the intervention effect is more beneficial in smaller studies). In the presence of small sample bias, the random-effects estimate of the intervention is more beneficial than the fixed-effect estimate (Page 2022). We further explored the potential for small-study effects in the main outcomes of the review using funnel plots if at least 10 studies reporting pain were included in a meta-analysis (Page 2022).

Data synthesis

We pooled results of trials with similar characteristics (participants, interventions, outcome measures and timing of outcome measurement) to provide estimates of benefit and harm. Contemporary clinical practice for treating lateral elbow pain often comprises manual therapy, prescribed exercise or both. To ensure that our review provided the most relevant data to inform contemporary clinical practice, we planned seven comparisons. We considered the first two comparisons comprising manual therapy, prescribed exercise or both compared with placebo or minimal or no intervention as the main questions of the review. For comparisons 3 and 4, we considered the combination of manual therapy and prescribed exercises compared to either intervention alone. For comparison 5 and 6, we considered the combination of manual therapy and exercise added to glucocorticoid injection compared to either the same manual therapy or prescribed exercises with glucocorticoid injection. For comparison 7, we considered manual therapy and exercise added to glucocorticoid injection compared to glucocorticoid injection alone.

1. Is manual therapy, prescribed exercises or both more effective than placebo? This comparison assesses the efficacy of contemporary clinical practice compared with placebo, and pools data for the following three comparisons:
 - Is manual therapy and prescribed exercises more effective than placebo?
 - Is manual therapy more effective than placebo?

- Are prescribed exercises more effective than placebo?
2. Is manual therapy, prescribed exercises or both more effective than minimal or no intervention? This comparison assesses the effectiveness of contemporary clinical practice compared with minimal or no intervention, and pools data for the following three comparisons:
 - Is manual therapy and prescribed exercises more effective than minimal or no intervention?
 - Is manual therapy more effective than minimal or no intervention?
 - Are prescribed exercises more effective than minimal or no intervention?
 3. Is manual therapy and prescribed exercises more effective than manual therapy alone?
 4. Is manual therapy and prescribed exercises more effective than prescribed exercises alone?
 5. Is manual therapy and prescribed exercises in addition to glucocorticoid injection more effective than manual therapy with glucocorticoid injection?
 6. Is manual therapy and prescribed exercises in addition to glucocorticoid injection more effective than prescribed exercises with glucocorticoid injection?
 7. Is manual therapy, prescribed exercises or both provided in addition to glucocorticoid injection more effective than glucocorticoid injection alone? This comparison assesses the added value of manual therapy, prescribed exercises or both to glucocorticoid injection alone, and pools data for the following three comparisons:
 - Is manual therapy, prescribed exercises and glucocorticoid injection more effective than glucocorticoid injection alone?
 - Is manual therapy and glucocorticoid injection more effective than glucocorticoid injection alone?
 - Are prescribed exercises and glucocorticoid injection more effective than glucocorticoid injection alone?

We synthesised effect estimates using a random-effects meta-analysis model based on the assumption that clinical and methodological diversity is likely to exist and to have an impact on the results. Where we could not pool data, we presented effect estimates and 95% CIs for each trial.

Subgroup analysis and investigation of heterogeneity

If there were sufficient data, we carried out subgroup analysis, to assess whether pain and disability differ between participants who have acute symptoms (defined as three months or less) compared to those with symptoms for more than three months and up to six months, and to those with symptoms for more than six months. We used the formal test for subgroup interaction in Review Manager 2014 (RevMan Web 2020).

If data stratified by symptom duration were not available, but if there were sufficient continuous data from at least 10 studies, we performed meta-regression to assess whether symptom duration modifies the effect of the intervention on pain and function using the Stata statistical package (Stata 2017).

We restricted these subgroup analyses to the first two comparisons (manual therapy or exercise or both delivered together versus placebo; manual therapy or exercise or both delivered together

versus minimal or no intervention) for the outcomes pain and disability.

Sensitivity analysis

We investigated the robustness of the treatment effect (pain and disability) to potential selection bias and detection bias by performing the following sensitivity analyses.

1. Selection bias: we removed the trials that reported inadequate or unclear allocation concealment from the meta-analysis to see if this changed the overall treatment effect.
2. Detection bias: we removed the trials that reported inadequate or unclear participant blinding from the meta-analysis to see if this changed the overall treatment effect.
3. We assessed the effect of including imputed data and data based on assumptions.

We restricted these sensitivity analyses to the first two comparisons (manual therapy or exercise or both delivered together versus placebo; manual therapy or exercise or both delivered together versus minimal or no intervention) for the outcomes pain and disability.

Summary of findings and assessment of the certainty of the evidence

We followed the guidelines in the *Cochrane Handbook*, Chapters 14 and 15 for interpreting results, and were aware of distinguishing a lack of evidence of effect from a lack of effect (Schünemann 2022; Schünemann 2022a). We based our conclusions only on findings from the quantitative synthesis of studies included for this review. We avoided making recommendations for practice, and our implications for research suggest priorities for future research and outline what the remaining uncertainties are in the area.

We presented the summary of findings tables for 1) manual therapy, prescribed exercises or both versus placebo; and 2) manual therapy, prescribed exercises or both versus minimal or no intervention using the following outcomes: pain; disability; treatment success; quality of life; total withdrawals, adverse events and serious adverse events. We included the end of intervention time point for pain, disability, treatment success and quality of life, and last follow-up for total withdrawals, adverse events and serious adverse events.

Two review authors (JW, RJ) independently assessed the certainty of the evidence. We used the five GRADE considerations (study limitations, consistency of effect, imprecision, indirectness and publication bias) to assess the certainty of a body of evidence as it relates to the studies that contribute data to the meta-analyses for the prespecified outcomes, and reported the certainty of evidence as high, moderate, low or very low. We used GRADEpro software to prepare the SoF tables (GRADEpro GDT 2015). We justified all decisions to downgrade the certainty of evidence using footnotes and made comments to aid the reader's understanding of the review where necessary.

RESULTS

Description of studies

Results of the search

The search retrieved 1759 records (1607 from electronic databases, 152 from trial registries). After removal of duplicates, we screened

1085 records. After 961 records were excluded, we assessed 124 full-text articles for eligibility with 23 trials (39 records) included and 76 trials (80 records) excluded, as shown in [Figure 1](#). One record is awaiting classification ([Bartlett 2007](#)), and four records were identified as ongoing trials in clinical trial registries ([Prat 2018](#); [Sahebamei 2019](#); [Sveinall 2022](#); [Takamjani 2019](#)), with a description provided in [Table 1](#).

Figure 1.

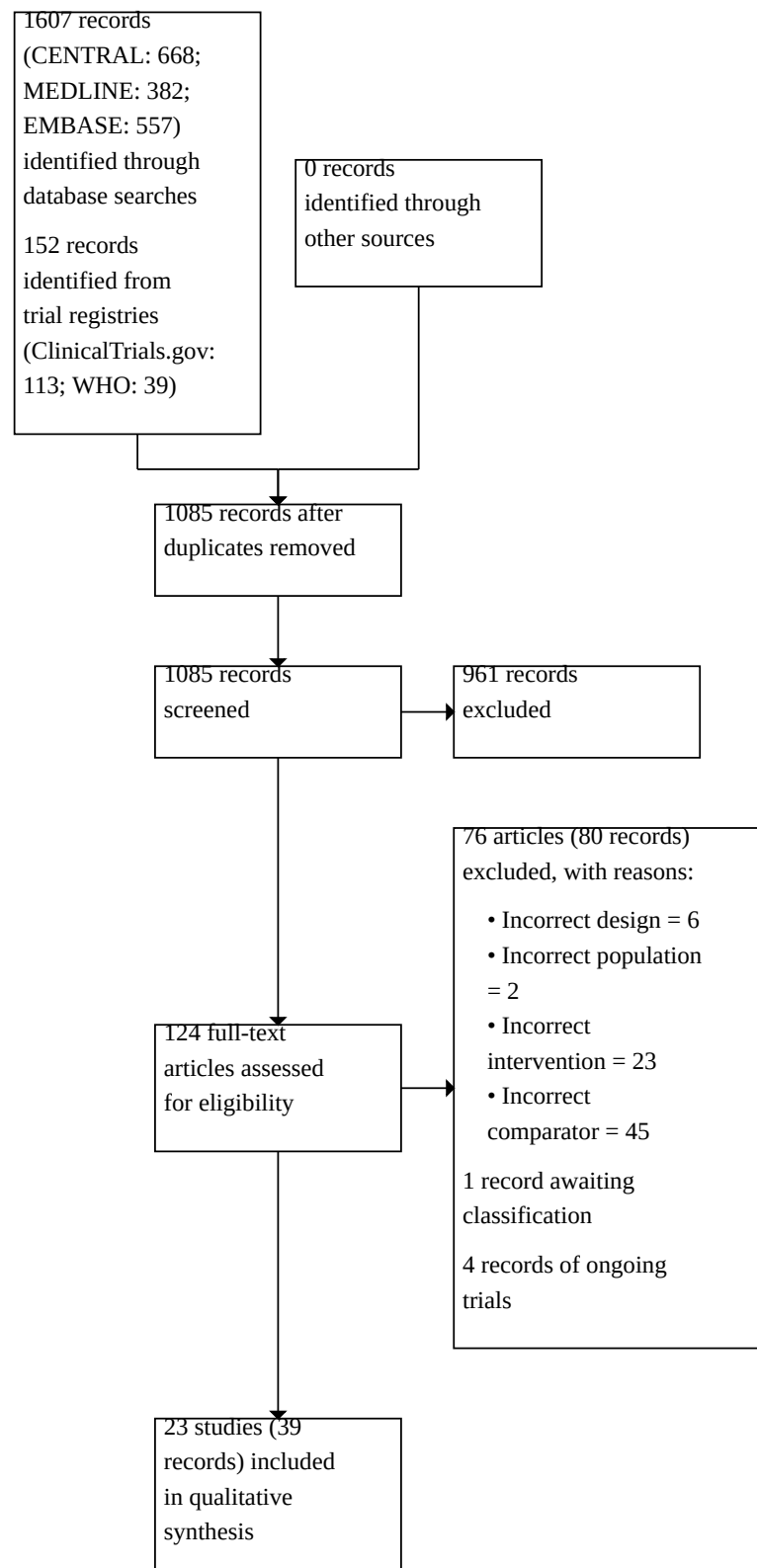
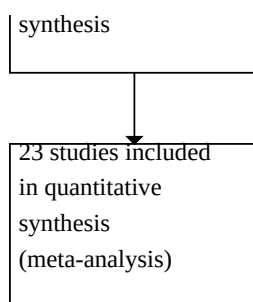


Figure 1. (Continued)



Study design and setting

All 23 included trials were parallel-group RCTs, with a description of trial characteristics provided in Table 2. Ten trials included two intervention groups (Blanchette 2011; Haahr 2003; Kochar 2002; Muralidharagopalan 2020; Nambi 2023; Nourbakhsh 2008; Peterson 2011; Reyhan 2020; Soderberg 2012; Vuvan 2020; Yilmaz 2022), eight trials included three intervention groups (Bagcaci 2023; Bisset 2006; Bostrøm 2019; Luginbuhl 2008; Olausen 2015; Rachold 2023; Smidt 2002a; Struijs 2004), and two trials (factorial design) included four intervention groups (Coombes 2013; Tonks 2007). Two trials used quasi-randomisation assignment, one with two intervention groups (Amro 2010), and one with three intervention groups (McQueen 2021).

Trials were conducted in 13 countries, including 11 high-income countries: Australia (Bisset 2006; Coombes 2013; Vuvan 2020), Netherlands (Smidt 2002a; Struijs 2004), Norway (Bostrøm 2019; Olausen 2015), Sweden (Peterson 2011; Soderberg 2012), Canada (Blanchette 2011), Denmark (Haahr 2003), Germany (Rachold 2023), Saudi Arabia (Nambi 2023), Switzerland (Luginbuhl 2008), USA (McQueen 2021; Nourbakhsh 2008) and United Kingdom (Tonks 2007). Three trials were performed in Turkey, an upper middle-income country (Bagcaci 2023; Reyhan 2020; Yilmaz 2022), one was performed in the West Bank/Gaza (Amro 2010) and two were performed in India, a lower middle-income country (Kochar 2002; Muralidharagopalan 2020).

The sample populations were recruited from primary care or community advertisements in 15 trials (Amro 2010; Bagcaci 2023; Bisset 2006; Blanchette 2011; Bostrøm 2019; Coombes 2013; Haahr 2003; Nourbakhsh 2008; Olausen 2015; Peterson 2011; Rachold 2023; Smidt 2002a; Soderberg 2012; Struijs 2004; Vuvan 2020), and from outpatient clinics in eight trials (Kochar 2002; Luginbuhl 2008; McQueen 2021; Muralidharagopalan 2020; Nambi 2023; Reyhan 2020; Tonks 2007; Yilmaz 2022).

Participant characteristics

From the 23 trials and eligible intervention groups, 1612 participants were included. The mean age of participants ranged between 38 years (Amro 2010) and 52 years (Nourbakhsh 2008), and 47% were female, ranging from 0% (Nambi 2023) to 90% (Reyhan 2020). The mean duration of symptoms ranged from 2.5 months (Smidt 2002a) to 43 months in the minimal intervention group (Blanchette 2011). The percentage of participants with their dominant elbow affected ranged from 46% (Blanchette 2011) to 80% (Smidt 2002a).

Eighteen trials required a clinical diagnosis of lateral elbow pain including positive results from physical provocation tests for inclusion, one trial required a physician diagnosis (Bagcaci 2023), and four trials did not report their diagnostic criteria (McQueen 2021; Muralidharagopalan 2020; Reyhan 2020; Yilmaz 2022). Two trials required participants to have normal X-rays (Kochar 2002; Rachold 2023). Individual trials required pain to be relieved by glucocorticoid injection (Luginbuhl 2008) or by one manual therapy session (Reyhan 2020), a medical referral (Amro 2010) or for participants to have seen their general practitioner for their condition (Olausen 2015). Thirteen trials also required participants with symptoms for a minimum duration, including symptoms for at least six weeks in seven trials (Amro 2010; Bisset 2006; Coombes 2013; Reyhan 2020; Smidt 2002a; Struijs 2004; Vuvan 2020), at least two weeks in two trials (Bostrøm 2019; Olausen 2015), at least three months in two trials (Peterson 2011; Yilmaz 2022), at least two months in one trial (Nambi 2023), and at least four weeks in one trial (Soderberg 2012).

Mean baseline pain scores varied between 1 and 7.8 out of 10 points (where a score of 10 is worst pain) in 21 trials, with data not available in two trials (Bagcaci 2023; Haahr 2003). Mean baseline disability scores varied between 29 and 77 out of 100 units (where 100 is worst disability) using a disease-specific disability questionnaire such as PRTEE, PFFQ or DASH in 14 trials (Amro 2010; Bisset 2006; Blanchette 2011; Bostrøm 2019; Coombes 2013; Olausen 2015; McQueen 2021; Muralidharagopalan 2020; Reyhan 2020; Smidt 2002a; Struijs 2004; Tonks 2007; Vuvan 2020; Yilmaz 2022). In other trials, mean disability scores were 23 and 29 points out of 100 (where 100 is worst disability) in both groups of Peterson 2011, and were 15 and 17 points using the Patient Specific Functional Scale in both groups of Nourbakhsh 2008, with data not available from five trials (Bagcaci 2023; Haahr 2003; Kochar 2002; Luginbuhl 2008; Soderberg 2012).

Interventions

One trial compared manual therapy with placebo manual therapy (Nourbakhsh 2008). There were no trials comparing manual therapy and prescribed exercises or prescribed exercises alone with placebo control.

There were 12 trials that compared manual therapy, prescribed exercises or a combination of manual therapy and prescribed exercises to minimal intervention. Of these trials, four assessed manual therapy and prescribed exercises (Bisset 2006; Olausen 2015; Smidt 2002a; Struijs 2004), two assessed manual therapy

alone (Blanchette 2011; Rachold 2023), and six assessed prescribed exercises alone (Haahr 2003; McQueen 2021; Peterson 2011; Soderberg 2012; Tonks 2007; Vuvan 2020). The minimal intervention was a single session of advice in seven trials (Bisset 2006; Blanchette 2011; McQueen 2021; Olausson 2015; Smidt 2002a; Soderberg 2012; Vuvan 2020), forearm brace and electrotherapy in one trial (Rachold 2023), forearm brace only in one trial (Struijs 2004), observation only in one trial (Tonks 2007), usual general practitioner (GP) care in one trial (Haahr 2003), and waiting list control in one trial (Peterson 2011). Co-interventions (i.e. in all intervention groups of eligible groups) included standardised advice about self-management of lateral elbow pain either verbally or via a booklet (Bisset 2006; McQueen 2021; Olausson 2015; Peterson 2011; Soderberg 2012), prescribed analgesics (Olausson 2015), braces (McQueen 2021; Soderberg 2012), and/or permitted usual treatments as needed (e.g. analgesics, heat or cold packs, Bisset 2006; Olausson 2015; Peterson 2011; Smidt 2002a). Additional intervention components that were only in the active group but not in the comparison groups included placebo injection in one trial (Olausson 2015), ultrasound therapy in two trials (Smidt 2002a; Struijs 2004), and ergonomic advice in one trial (Haahr 2003).

There were six trials that compared a combination of manual therapy and prescribed exercises with prescribed exercises alone (Amro 2010; Bagcaci 2023; Boström 2019; Kochar 2002; Reyhan 2020; Yilmaz 2022), but no trials that compared the combination to manual therapy alone. Co-interventions were diverse, including ultrasound therapy in two trials (Amro 2010; Kochar 2002), heat and massage in one trial (Amro 2010), and advice in four trials (Boström 2019; Reyhan 2020; Vuvan 2020; Yilmaz 2022).

One trial compared manual therapy, prescribed exercises and glucocorticoid injection versus sham manual therapy, prescribed exercises and glucocorticoid injection (Nambi 2023). One trial compared manual therapy, prescribed exercises and glucocorticoid injection with glucocorticoid injection alone (Coombes 2013), and three trials compared prescribed exercises and glucocorticoid injection with glucocorticoid injection alone (Luginbuhl 2008; Muralidharagopalan 2020; Tonks 2007), but no trials compared manual therapy and glucocorticoid injection to glucocorticoid injection alone. Co-interventions included advice about self-management (Coombes 2013).

Manual therapy components

Fifteen trials included manual therapy components (Table 3). Six of these trials (Amro 2010; Bagcaci 2023; Bisset 2006; Boström 2019; Coombes 2013; Kochar 2002; Reyhan 2020) applied mobilisation with movement techniques (Vicenzino 2011) based on the Mulligan concept (McDowell 2014). The mobilisation with movement techniques involved the application of a sustained mobilisation of the elbow joint by a physiotherapist (e.g. a lateral glide of the elbow as the 'mobilisation' component) followed by the participant generating an active task (e.g. gripping or making a fist as the 'movement' component). Five trials applied soft tissue techniques, including deep friction massage applied by physiotherapists in three trials (Olausson 2015; Smidt 2002a; Struijs 2004), a muscle energy technique applied by an orthopaedic clinical specialist from the USA in one trial (Nourbakhsh 2008) and a physiotherapist in one trial (Bagcaci 2023), and soft tissue massage applied by a chiropractor using an instrument in one trial (Blanchette 2011). Olausson 2015 and Nambi 2023 also included a Mills manipulation technique applied by physiotherapists, described as

a high-velocity, low-amplitude thrust into hyperextension of elbow while wrist and fingers are in full flexion, forearm fully pronated and therapist's hand over the elbow or tender spot on lateral epicondyle. One trial applied a radial nerve mobilisation technique (Yilmaz 2022).

Prescribed exercise components

Twenty trials included prescribed exercise components ranging from 2 to 12 weeks in duration (Table 4). All 20 trials reported types of resistance exercise such as concentric, eccentric, isometric or grip strengthening, with three trials reporting isometric exercises only (Haahr 2003; Luginbuhl 2008; Vuvan 2020), and five trials reporting eccentric exercises only (Boström 2019; Muralidharagopalan 2020; Olausson 2015; Soderberg 2012; Yilmaz 2022). Twelve trials included types of stretching exercises (Amro 2010; Bagcaci 2023; Bisset 2006; Coombes 2013; Haahr 2003; Kochar 2002; Nambi 2023; Olausson 2015; Reyhan 2020; Smidt 2002a; Struijs 2004; Tonks 2007). Three trials from Europe reported a four-phase series of exercises including strengthening, stretching and functional exercises as described by Pienimäki 1996 (Smidt 2002a; Struijs 2004; Tonks 2007), and two trials from Australia reported pragmatic exercise prescription consisting of strengthening, stretching and functional exercises as described by Vicenzino 2003 (Bisset 2006; Coombes 2013).

All exercise programmes were performed by participants individually in home settings. In 15 trials programmes were instructed by physiotherapists (Amro 2010; Bagcaci 2023; Bisset 2006; Boström 2019; Coombes 2013; Kochar 2002; Muralidharagopalan 2020; Nambi 2023; Olausson 2015; Reyhan 2020; Smidt 2002a; Struijs 2004; Tonks 2007; Vuvan 2020; Yilmaz 2022), in one trial the programme was instructed by an occupational therapist (McQueen 2021), in one trial the programme was instructed by an ergonomist (Haahr 2003), and three trials did not report the qualifications of the provider, or were written instructions only (Luginbuhl 2008; Peterson 2011; Soderberg 2012). Eleven trials reported using exercise equipment including resistance bands in five trials (Bisset 2006; Coombes 2013; Nambi 2023; Smidt 2002a; Struijs 2004), hand-held weights, water-filled buckets or bottles in seven trials (Bagcaci 2023; McQueen 2021; Muralidharagopalan 2020; Olausson 2015; Peterson 2011; Soderberg 2012; Vuvan 2020), sticks in three trials (Smidt 2002a; Struijs 2004; Tonks 2007) and putty in one trial (McQueen 2021).

Ten trials reported tailored decision rules for starting levels (e.g. limiting number of sets according to the severity of condition, tailored exercise selection, sport specific or work specific exercise prescription later in training phases, or starting exercise programme after pain reduction) (Bagcaci 2023; Bisset 2006; Coombes 2013; McQueen 2021; Nambi 2023; Smidt 2002a; Struijs 2004; Tonks 2007; Vuvan 2020; Yilmaz 2022), two trials reported standardised starting loads (Boström 2019; Peterson 2011), and eight trials did not provide details on starting levels (Amro 2010; Haahr 2003; Kochar 2002; Luginbuhl 2008; Muralidharagopalan 2020; Olausson 2015; Reyhan 2020; Soderberg 2012). Twelve trials reported progression strategies based on pain levels throughout training phases (Bisset 2006; Boström 2019; Coombes 2013; Kochar 2002; Nambi 2023; Olausson 2015; Smidt 2002a; Soderberg 2012; Struijs 2004; Tonks 2007; Vuvan 2020; Yilmaz 2022). Adherence strategies were reported in eight trials, including monitoring by physiotherapists at review appointments in five trials (Bisset 2006; Coombes 2013; Nambi 2023; Vuvan 2020; Yilmaz 2022),

remote monitoring by physiotherapists in two trials (Bagcaci 2023; Muralidharagopalan 2020), monitoring by the investigators at trial follow-up sessions during the intervention phase (Luginbuhl 2008; Peterson 2011), and a weekly text message reminder in one trial (Boström 2019). Three trials also used a participant diary to monitor adherence (Coombes 2013; Nambi 2023; Vuvan 2020), and one trial reported using a participant diary to capture exercise adherence, but it was unclear if it was used as a strategy to increase adherence (Soderberg 2012).

Dosages for concentric and eccentric exercises were reported in 11 trials (Bagcaci 2023; Bisset 2006; Kochar 2002; Nambi 2023; McQueen 2021; Olausson 2015; Peterson 2011; Smidt 2002a; Soderberg 2012; Struijs 2004; Yilmaz 2022), ranging between one and three sets, up to 30 repetitions, and ranging from one session daily (e.g. eccentric exercise in Boström 2019) up to six times daily in three trials that applied the programme devised by Pienimäki 1996 (Smidt 2002a; Struijs 2004; Tonks 2007). Dosages for isometric exercises were reported in four trials (Haahr 2003; Kochar 2002; Luginbuhl 2008; Vuvan 2020), and were applied for up to 45 seconds, ranging from one to three times daily. Dosages for stretching exercises were reported in nine trials (Bagcaci 2023; Bisset 2006; Haahr 2003; Kochar 2002; Nambi 2023; Olausson 2015; Reyhan 2020; Smidt 2002a; Struijs 2004), ranging from a few seconds hold ups to 40 seconds, ranging from three to five repetitions, and performed between two and six times daily.

Outcomes

The outcomes reported in each study are presented in Table 5.

Pain

All 23 trials reported pain intensity, with eight trials reporting pain intensity using a 0 to 100 scale (Bisset 2006; Blanchette 2011; Coombes 2013; Olausson 2015; Peterson 2011; Smidt 2002a; Soderberg 2012; Struijs 2004) and 12 trials using a 0 to 10 scale (Amro 2010; Bagcaci 2023; Boström 2019; Kochar 2002; McQueen 2021; Muralidharagopalan 2020; Nambi 2023; Nourbakhsh 2008; Rachold 2023; Reyhan 2020; Vuvan 2020; Yilmaz 2022). One trial combined four pain sub-scales (0 to 9; 0 = no complaints to 9 = pain as bad as could be for each sub-scale) for a total score out of 36 (Haahr 2003), one trial, Luginbuhl 2008, reported a Modified Nirschl/Pettrone score from 0 to 4 points (Stahl 1997), and one trial, Tonks 2007, reported pain as a sub-score of the Patient Rated Forearm Evaluation Questionnaire (Overend 1999).

Six trials specified the average pain intensity over the preceding week (Bisset 2006; Boström 2019; Coombes 2013; Soderberg 2012; Tonks 2007; Vuvan 2020), one trial specified average pain intensity over the previous 24 hours (Kochar 2002), one trial specified pain intensity over the previous three months (Haahr 2003), one trial specified pain intensity at the time of data collection (Amro 2010), and 14 trials did not report the time period (Bagcaci 2023; Blanchette 2011; McQueen 2021; Luginbuhl 2008; Muralidharagopalan 2020; Nambi 2023; Nourbakhsh 2008; Olausson 2015; Peterson 2011; Rachold 2023; Reyhan 2020; Smidt 2002a; Struijs 2004; Yilmaz 2022).

Seven trials reported more than one pain score using different descriptors including pain at rest, pain with activities (including a maximum voluntary muscle contraction), worst pain or pain at night (Bagcaci 2023; Coombes 2013; McQueen 2021; Peterson 2011; Reyhan 2020; Vuvan 2020; Yilmaz 2022).

Disability

Eighteen trials reported using a disease-specific disability measure, including the Patient Rated Forearm Evaluation Questionnaire (Newcomer 2005) in nine trials (Amro 2010; Bagcaci 2023; Blanchette 2011; Coombes 2013; Nambi 2023; Rachold 2023; Reyhan 2020; Tonks 2007; Vuvan 2020), the Pain Free Function Questionnaire in four trials (Bisset 2006; Olausson 2015; Smidt 2002a; Struijs 2004), and the Disability of the Arm, Shoulder and Hand questionnaire in five trials (Boström 2019; McQueen 2021; Muralidharagopalan 2020; Peterson 2011; Yilmaz 2022). One trial reported using a numeric rating scale (Nourbakhsh 2008), and four trials did not include a measure of disability (Haahr 2003; Kochar 2002; Luginbuhl 2008; Soderberg 2012).

Health-related quality of life

Five trials reported a health-related quality of life measure (Coombes 2013; Nambi 2023; Peterson 2011; Rachold 2023; Smidt 2002a), including the EQ-5D (Busija 2020) in three trials (Coombes 2013; Nambi 2023; Smidt 2002a), the SF-36 in one trial (Rachold 2023), and the Gothenburg scale (Sullivan 1993) in one trial (Peterson 2011).

Participant global assessment of treatment success

Twelve trials reported participant global assessment of treatment success (Amro 2010; Bisset 2006; Boström 2019; Coombes 2013; Haahr 2003; Luginbuhl 2008; Nambi 2023; Olausson 2015; Reyhan 2020; Smidt 2002a; Struijs 2004; Vuvan 2020). Of these, 10 trials used an ordinal scale or Likert scale ranging from 4 to 6 points to measure treatment success (e.g. ranging from 'completely recovered' to 'much improved' with treatment success defined commonly as 'completely recovered' or 'much improved') (Bisset 2006; Coombes 2013; Haahr 2003; Luginbuhl 2008; Nambi 2023; Olausson 2015; Reyhan 2020; Smidt 2002a; Struijs 2004; Vuvan 2020). One trial measured treatment success by the number of participants who perceived the treatment was effective (Amro 2010), and one trial did not report the scale used to measure treatment success (Boström 2019). Two trials also included a measure of participant satisfaction using a VAS (e.g. 0 = not satisfied at all, 10 = very satisfied) to measure treatment success (Smidt 2002a; Struijs 2004).

Adverse events

Twelve trials reported the number of participants experiencing an adverse event (Bisset 2006; Blanchette 2011; Boström 2019; Coombes 2013; Muralidharagopalan 2020; Nambi 2023; Olausson 2015; Reyhan 2020; Smidt 2002a; Tonks 2007; Vuvan 2020; Yilmaz 2022), with four trials reporting zero events (Boström 2019; Muralidharagopalan 2020; Nambi 2023; Reyhan 2020). Adverse events are presented in Table 6 and included increased pain following manual therapy or prescribed exercise interventions in six trials (Bisset 2006; Blanchette 2011; Coombes 2013; Smidt 2002a; Vuvan 2020; Yilmaz 2022), and other complications such as gastrointestinal, skin depigmentation and subcutaneous atrophy following other interventions such as glucocorticoid injections or analgesia in three trials (Coombes 2013; Olausson 2015; Tonks 2007).

One trial had a very high number of adverse events from manual therapy treatments including radiating pain into the arm (Smidt 2002a). One trial reported a high number of adverse events in the minimal intervention (wait and see) group, including increased

pain and gastrointestinal effects (Olaussen 2015). One trial did not measure adverse events in the control group (Blanchette 2011).

Grip strength

Sixteen trials measured grip strength using a dynamometer (Amro 2010; Bagcaci 2023; Bisset 2006; Blanchette 2011; Kochar 2002; Luginbuhl 2008; Nambi 2023; Nourbakhsh 2008; Olaussen 2015; Reyhan 2020; Smidt 2002a; Soderberg 2012; Struijs 2004; Tonks 2007; Vuvan 2020; Yilmaz 2022), and this was commonly measured three times, with a 30-second rest interval between each measurement, and the average score recorded. Of these 16 trials, pain-free grip strength was measured in 11 trials (Bisset 2006; Blanchette 2011; Nambi 2023; Olaussen 2015; Reyhan 2020; Smidt 2002a; Soderberg 2012; Struijs 2004; Tonks 2007; Vuvan 2020; Yilmaz 2022). Reporting of grip strength included a ratio of the affected side to the contralateral side in five trials (Bisset 2006; Luginbuhl 2008; Olaussen 2015; Smidt 2002a; Struijs 2004), or a unit of measure including kilograms (kg) in eight trials (Amro 2010; Bagcaci 2023; Blanchette 2011; Kochar 2002; Nambi 2023; Reyhan 2020; Tonks 2007; Yilmaz 2022), kilopascals (kPa) in one trial (Soderberg 2012), Newtons (N) in one trial (Vuvan 2020) and pounds per square inch (lb/in²) in one trial (Nourbakhsh 2008).

Return to work

One trial measured the number of participants on paid sick leave (Olaussen 2015).

Timing of outcome assessment

Timing of outcome assessment for the efficacy of treatment (e.g. pain) was reported at the end of intervention (primary time point) for all 23 trials, ranging from three weeks to six months. For the first time point, 11 trials reported outcomes during the intervention ranging from one to six weeks follow-up (Bisset 2006; Bostrom 2019; Coombes 2013; Kochar 2002; McQueen 2021; Muralidharagopalan 2020; Peterson 2011; Reyhan 2020; Smidt 2002a; Soderberg 2012; Yilmaz 2022). For the last time point, 13 trials reported outcomes ranging from 3 to 12 months following the end of treatment (Bisset 2006; Blanchette 2011; Bostrom 2019; Coombes 2013; Haahr 2003; McQueen 2021; Muralidharagopalan 2020; Nambi 2023; Olaussen 2015; Reyhan 2020; Smidt 2002a; Struijs 2004; Yilmaz 2022).

Excluded studies

We excluded six trials due to incorrect design, two trials due to incorrect population, 23 trials due to incorrect intervention and 45 trials due to incorrect comparisons.

Incorrect designs that were excluded were a single-group pre-post design in two trials (Abbott 2001a; Abbott 2001b), a non-randomised design in three trials (Kamal 2022; Nilsson 2007; Nilsson 2012), and a clinical commentary in one paper (Day 2019).

For incorrect intervention, we excluded trials where all participants received the same prescribed exercises or manual therapy component. For example, in Wegener 2016 both groups received identical prescribed exercises. We also excluded trials that tested the immediate effects from the single application of manual therapy or prescribed exercises in a laboratory setting (Coombes 2016; Fernandez-Carnero 2008; Paungmali 2003; Vicenzino 1996; Vicenzino 2001; Zinke 2020).

For incorrect comparison, we excluded trials that compared manual therapy or prescribed exercises to an alternative intervention such as glucocorticoid injection, acupuncture, electrotherapy, or another manual therapy or prescribed exercises. For example, Gunduz 2012 compared manual therapy (plus hot pack and ultrasound) with glucocorticoid injection or extracorporeal shock wave therapy and was excluded. In another example, Cleland 2005 compared manual therapy applied to the spine and elbow compared with manual therapy applied to the elbow alone and was also excluded.

Incorrect populations that we excluded were healthy controls with experimentally induced lateral elbow pain in one trial (Slater 2006), and handball goalie's elbow in one trial (Tyrdal 1998).

Risk of bias in included studies

Risk of bias assessment for each study is reported in the [Characteristics of included studies](#) table and summarised in [Figure 2](#).

Figure 2.

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment - self reported (detection bias)	Blinding of outcome assessment - assessor reported outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Amro 2010	+	+	+	+	+	?	?	+
Bagcaci 2023	+	?	+	+	+	+	+	+
Bisset 2006	+	+	+	+	+	+	+	+
Blanchette 2011	+	+	+	+	+	+	?	+
Bostrøm 2019	+	+	+	+	+	+	+	+
Coombes 2013	+	+	+	+	+	+	+	+
Haahr 2003	+	+	+	+	+	+	?	+
Kochar 2002	+	?	+	+	+	?	+	+
Luginbuhl 2008	?	?	+	+	+	+	+	?
McQueen 2021	+	+	+	+	+	+	+	?
Muralidharagopalan 2020	+	?	+	+	+	?	?	?
Nambi 2023	+	+	+	+	+	+	+	+
Nourbakhsh 2008	+	+	+	+	+	+	?	+
Olaussen 2015	+	+	+	+	+	+	+	+
Peterson 2011	+	?	+	+	+	+	?	+
Rachold 2023	+	+	+	+	+	+	+	+
Reyhan 2020	+	+	+	+	+	+	?	?

Figure 2. (Continued)

Reyhan 2020	+	+	-	-	-	+	-	?
Smidt 2002a	+	+	-	-	+	+	?	+
Soderberg 2012	+	+	-	-	-	+	+	+
Struijs 2004	+	?	-	-	+	+	?	+
Tonks 2007	+	+	-	-	-	-	?	+
Vuvan 2020	+	+	-	-	+	+	+	+
Yilmaz 2022	+	+	-	-	+	+	+	+

Allocation

Fourteen trials described both an adequate random sequence generation (e.g. using a computer-based or random numbers table) and adequate allocation concealment (e.g. sealed, opaque envelopes) and we judged them as being at low risk of selection bias (Bisset 2006; Blanchette 2011; Boström 2019; Coombes 2013; Haahr 2003; Nambi 2023; Nourbakhsh 2008; Rachold 2023; Reyhan 2020; Smidt 2002a; Soderberg 2012; Tonks 2007; Vuvan 2020; Yilmaz 2022).

We judged two quasi-randomised trials to have used suboptimal randomisation. Amro 2010 assigned participants alternatively to experimental or control groups and treatment allocation was not blinded, so it was at high risk of selection bias. In McQueen 2021, participants were alternatively allocated to one of two groups according to their medical registry number (odd numbers assigned to group 1 and even numbers assigned to group 2).

Seven trials did not report the method of allocation concealment and we judged them as having an unclear risk of selection bias (Bagcaci 2023; Kochar 2002; Luginbuhl 2008; McQueen 2021; Muralidharagopalan 2020; Peterson 2011; Struijs 2004). For example, Peterson 2011 described "blind" allocation and Struijs 2004 described a "minimisation strategy" without providing sufficient detail of the concealment method to judge low risk.

Olaussen 2015 was at high risk of selection bias as the authors did not conceal allocation to the group that received minimal intervention.

Blinding

We judged 21 trials to be at high risk of performance bias as participants and personnel performing the prescribed exercises or manual therapy were not blinded. We judged two trials to be at low risk of performance bias as participants were blinded to treatment allocation and likely unaware of the group they were allocated to (Nambi 2023; Nourbakhsh 2008). In Nourbakhsh 2008, the intervention involved a therapist applying an oscillating energy technique compared to a placebo technique where the therapist applied a gentle pressure to the tissue with oscillation techniques to non-tender areas to ensure participants had a sense of a treatment being provided. In Nambi 2023, the intervention involved a high-velocity thrust technique at the elbow compared to a 'sham' manipulation technique involving a low-velocity movement at the elbow that was not comprehensible to the study participants.

For patient-reported clinical outcomes (e.g. pain and disability), we judged 21 trials to be at high risk of detection bias as participants were not blinded to the intervention, with the two trials involving placebo manual therapy treatments judged as low risk (Nambi 2023; Nourbakhsh 2008).

For the assessor-reported outcome of grip strength, we judged eight trials to be at low risk of detection bias as the assessor was blinded to participant allocation (Bisset 2006; Nambi 2023; Nourbakhsh 2008; Olaussen 2015; Smidt 2002a; Struijs 2004; Vuvan 2020; Yilmaz 2022), and we judged nine trials to be at high risk of bias as the assessor was not blinded to participant allocation (Amro 2010; Bagcaci 2023; Blanchette 2011; Kochar 2002; Luginbuhl 2008; Peterson 2011; Reyhan 2020; Soderberg 2012; Tonks 2007). We judged six trials as at low risk of detection bias for this criterion as they did not include any assessor-reported outcomes in this review (Boström 2019; Coombes 2013; Haahr 2003; McQueen 2021; Muralidharagopalan 2020; Rachold 2023).

Incomplete outcome data

Fifteen trials had small amounts of participants lost to follow-up that we deemed to be a low risk for attrition bias as they were unlikely to change the results (Bagcaci 2023; Bisset 2006; Blanchette 2011; Boström 2019; Coombes 2013; Nambi 2023; Nourbakhsh 2008; Olaussen 2015; Peterson 2011; Reyhan 2020; Smidt 2002a; Soderberg 2012; Struijs 2004; Vuvan 2020; Yilmaz 2022), including two trials reporting outcomes for all participants (Bagcaci 2023; Reyhan 2020). Three trials did not report losses to follow-up and we judged them as at unclear risk of attrition bias (Amro 2010; Kochar 2002; Muralidharagopalan 2020). We judged four trials as having a high risk of attrition bias due to the attrition rates in the manual therapy or prescribed exercise groups (Haahr 2003; McQueen 2021; Rachold 2023; Tonks 2007), including one trial with a high attrition rate for the long-term outcome (12 months), with 37% of available data (Luginbuhl 2008).

Selective reporting

Ten trials had a trial registration or published protocol available (Bagcaci 2023; Bisset 2006; Boström 2019; Coombes 2013; Nambi 2023; Olaussen 2015; Rachold 2023; Soderberg 2012; Vuvan 2020; Yilmaz 2022). Of these 10 trials, eight reported outcomes consistent with the published protocol and methods and we deemed them at low risk of reporting bias (Bagcaci 2023; Bisset 2006; Nambi 2023; Olaussen 2015; Rachold 2023; Soderberg 2012; Vuvan 2020; Yilmaz 2022). We judged two of these trials as at high risk of reporting bias as not all outcomes listed in the trial registry or published protocol

were reported in the trial publication (Boström 2019; Coombes 2013). For example, grip strength was not reported in the trial publication for Coombes 2013, and a quality of life outcome was missing from the trial publication in Boström 2019.

Thirteen trials did not have a trial registry or published protocol available (Amro 2010; Blanchette 2011; Haahr 2003; Kochar 2002; Luginbuhl 2008; McQueen 2021; Muralidharagopalan 2020; Nourbakhsh 2008; Peterson 2011; Reyhan 2020; Smidt 2002a; Struijs 2004; Tonks 2007). We judged eight of these trials as at unclear risk as the reporting of results in the trial publication was consistent with the methods (Amro 2010; Blanchette 2011; Haahr 2003; Muralidharagopalan 2020; Peterson 2011; Smidt 2002a; Struijs 2004; Tonks 2007). We judged three of these trials as at high risk of bias because the trial publication did not sufficiently report results for pain measures (e.g. variance measure) to enable comparison of pain outcomes between groups (Kochar 2002; McQueen 2021; Reyhan 2020). We judged two of these trials as at high risk of reporting bias for not reporting the time point of evaluation (Luginbuhl 2008; Nourbakhsh 2008), with Luginbuhl 2008 reporting the latest follow-up only, which could have been six weeks, three months or 12 months.

Other potential sources of bias

We judged six trials to have other potential sources of bias (Blanchette 2011; Luginbuhl 2008; McQueen 2021; Muralidharagopalan 2020; Rachold 2023; Reyhan 2020). Muralidharagopalan 2020 did not adequately report baseline characteristics or co-interventions. Blanchette 2011 reported substantial baseline differences in patient characteristics between groups, with the onset of the symptoms in the minimal intervention group substantially higher than the manual therapy group. Rachold 2023 reported a baseline difference in the quality of life outcome, despite concealed allocation. Luginbuhl 2008 and McQueen 2021 did not report participant characteristic data separately for the groups. In Reyhan 2020 there was a discrepancy in the reporting of results. For example, the number of participants excluded in the trial was reported as zero in the figure, but four excluded participants were described in the results text.

Effects of interventions

See: [Summary of findings 1 Manual therapy, prescribed exercises or both compared with placebo for lateral elbow pain](#); [Summary of findings 2 Manual therapy, prescribed exercises or both compared to minimal or no intervention for lateral elbow pain](#)

See [Summary of findings 1](#) for the main comparison of manual therapy, prescribed exercises or both versus placebo.

See [Summary of findings 2](#) for the second comparison of manual therapy, prescribed exercises or both versus minimal intervention.

Manual therapy, prescribed exercises or both versus placebo

The main comparison is limited to manual therapy alone compared with placebo, as we did not identify any trials investigating manual therapy and prescribed exercises versus placebo or prescribed exercises versus placebo ([Summary of findings 1](#)).

Manual therapy versus placebo

One trial with 23 participants compared manual therapy with placebo (Nourbakhsh 2008). Outcomes were only reported at a

single time point at the end of the two- to three-week intervention period.

Major outcomes

Pain

Low-certainty evidence indicates that manual therapy may reduce pain at the end of two to three weeks of treatment. Mean pain was 4.1 points with placebo on a 0 to 10 scale (lower scores indicate less pain) and 2.0 points with manual therapy, MD -2.1 points (95% CI -4.2 to -0.1; 1 trial, 23 participants; [Analysis 1.1](#)). We downgraded the evidence due to indirectness as it may not be reflective of common clinical practice or easily applied outside the trial setting, and due to imprecision as the evidence was from a single trial and the confidence interval includes both an important improvement and no improvement. The outcome was not assessed at later time points.

Disability

Low-certainty evidence indicates that manual therapy may reduce disability at the end of two to three weeks of treatment. Mean disability was 40 points on a 0 to 100 scale (lower scores indicate less disability) with placebo and 15 points with manual therapy, MD -25 points (95% CI -43 to -7; [Analysis 1.2](#)). We downgraded the evidence due to indirectness of the intervention and imprecision. The outcome was not assessed at later time points.

Sensitivity and subgroup analyses were precluded as there was a single study.

Quality of life

This outcome was not measured.

Participant-reported treatment success

This outcome was not measured.

Participant withdrawals

This outcome was not measured.

Adverse events

This outcome was not measured.

Serious adverse events

This outcome was not measured.

Minor outcomes

Grip strength

Low-certainty evidence indicates that manual therapy may have little or no effect on grip strength. Mean grip strength was 79.2 lb/in² with placebo (higher score equals higher strength) and 73.6 lb/in² with manual therapy, MD -5.6 lb/in² (95% CI -29.8 to 18.6; 1 trial, 23 participants; [Analysis 1.3](#)). We downgraded the evidence due to indirectness of the intervention and imprecision.

Return to work

This outcome was not measured.

Manual therapy, prescribed exercises or both versus minimal or no intervention

Twelve trials were included in this comparison (Bisset 2006; Blanchette 2011; Haahr 2003; Olausson 2015; Peterson 2011; Rachold 2023; Smidt 2002a; Tonks 2007; Vuvan 2020; McQueen 2021; Soderberg 2012; Struijs 2004). The main findings are summarised in [Summary of findings 2](#). Four trials compared manual therapy and prescribed exercises with minimal intervention (Bisset 2006; Olausson 2015; Smidt 2002a; Struijs 2004); two trials compared manual therapy with minimal intervention (Blanchette 2011; Rachold 2023); and six trials compared prescribed exercises with minimal intervention (Haahr 2003; McQueen 2021; Peterson 2011; Soderberg 2012; Tonks 2007; Vuvan 2020).

Major outcomes

Pain

Low-certainty evidence indicates that manual therapy, prescribed exercises or both may reduce pain slightly at the end of the intervention, and may have little or no effect on pain prior to the completion of treatment or at later follow-up. At the end of the intervention (four weeks to three months), mean pain was 5.10 points with minimal or no intervention and 4.57 points with manual therapy, prescribed exercises or both, MD -0.53 points (95% CI -0.92 to -0.14, $I^2 = 43\%$; 12 trials, 1023 participants; [Analysis 2.1](#)). At three to six weeks, the MD was -0.33 points (95% CI -0.98 to 0.33, $I^2 = 74\%$; 5 trials, 418 participants) and, similarly, at 8 weeks to 12 months the MD was -0.20 points (95% CI -0.48 to 0.08, $I^2 = 17\%$; 8 trials, 821 participants). We downgraded the evidence due to increased risk of performance bias and detection bias across all the trials as participants were aware of their treatment group, and we also downgraded for indirectness due to the multimodal nature of the interventions included in the trials.

Sensitivity analyses indicated that the effects of manual therapy, prescribed exercises or both on pain were robust to selection bias. The effect estimate at the end of treatment was similar when restricting the analysis to the eight trials at low risk of selection bias, MD -0.66 points (95% CI -1.16 to -0.16) (Bisset 2006; Blanchette 2011; Haahr 2003; Rachold 2023; Smidt 2002a; Soderberg 2012; Tonks 2007; Vuvan 2020). As all studies were at risk of detection biases, no sensitivity analysis for detection bias could be performed.

Subgroup analysis indicated that symptom duration appeared to have little impact on pain, although the analysis was limited by the small number of studies (subgroup difference $\text{Chi}^2 = 3.70$, $\text{df} = 2$ ($P = 0.16$), $I^2 = 46\%$; [Analysis 11.1](#)).

Disability

Low-certainty evidence indicates that manual therapy, prescribed exercises or both may reduce disability slightly at the end of the intervention, and may have little or no effect on disability prior to the completion of treatment or at later follow-up. At the end of the intervention (four weeks to three months), mean disability was 63.8 points with minimal or no intervention and 58.80 points with manual therapy, prescribed exercises or both, MD -5.00 points (95% CI -9.22 to -0.77, $I^2 = 63\%$; 10 trials, 732 participants; [Analysis 2.2](#)). At three to six weeks, the MD was -2.46 points (95% CI -6.65 to 1.74, $I^2 = 11\%$; 3 trials, 302 participants) and, similarly, at 8 weeks to 12 months the MD was -1.79 points (95% CI -4.61 to 1.03, $I^2 =$

27%; 7 trials, 588 participants). We downgraded the evidence due to increased risk of performance bias and detection bias across all the trials, and we also downgraded for indirectness of the interventions.

Sensitivity analyses indicated that the effects of manual therapy, prescribed exercises or both on disability were robust to selection bias. The effect estimate was similar when restricting the analysis to the six studies with low risk of selection bias, MD -7.23 points (95% CI -12.55 to -1.91) (Bisset 2006; Blanchette 2011; Rachold 2023; Smidt 2002a; Tonks 2007; Vuvan 2020). As all studies were at risk of detection biases, no sensitivity analysis could be performed.

Subgroup analyses indicated that symptom duration appeared to have little impact on disability, although the analysis was limited by the small number of studies (subgroup difference $\text{Chi}^2 = 3.74$, $\text{df} = 2$ ($P = 0.15$), $I^2 = 46.5\%$; [Analysis 11.2](#)).

Quality of life

Low-certainty evidence indicates that manual therapy, prescribed exercises or both may have little or no effect on quality of life at end of intervention (four weeks to three months: SMD -0.45, 95% CI -0.83 to -0.08, $I^2 = 0\%$, 2 trials, 113 participants) and at 8 weeks to 12 months (SMD -0.18, 95% CI -0.82 to 0.45; 2 trials, 156 participants; [Analysis 2.3](#)). No trial reported on quality of life at first time point.

Re-expressing the SMD on a 0- to 100-point scale (higher score indicates better quality of life) gave the following results at the end of the intervention (four weeks to three months): mean quality of life was 73.04 points with minimal or no intervention and 67.46 points with prescribed exercises, a difference of 5.58 points worse (95% CI -10.29 worse to -0.99 worse; 2 trials, 113 participants; [Summary of findings 2](#)). We downgraded the evidence due to bias and imprecision as the evidence is from two trials with large confidence intervals that include both little effect and worse outcome.

Participant-reported treatment success

Low-certainty evidence indicates that manual therapy, prescribed exercises or both may have little or no effect on the number of participants reporting treatment success at end of the intervention and at final follow-up. At the end of the intervention (six weeks to three months), 162 of 386 participants (42%) with minimal treatment and 201 of 384 participants (57%) who received manual therapy, prescribed exercises or both reported treatment success (RR 1.36, 95% CI 0.96 to 1.93, $I^2 = 73\%$; 6 trials, 770 participants; [Analysis 2.4](#)). At the last follow-up (12 months), the RR was 1.02 (95% CI 0.97 to 1.08, $I^2 = 0\%$; 5 trials, 702 participants). We downgraded the evidence due to bias and indirectness of the intervention.

At first follow-up (three weeks), the RR was 1.76 (95% CI 0.98 to 3.18, $I^2 = 0\%$; 2 trials, 243 participants; [Analysis 2.4](#)). We downgraded the evidence to very low certainty due to bias, indirectness of the intervention and imprecision from a small number of events.

Participant withdrawals

We are uncertain if manual therapy, prescribed exercises or both results in more participants withdrawing, due to very low-certainty evidence ([Analysis 2.5](#)). In total, 83 of 566 participants (147 per 1000) withdrew for any reason from the minimal treatment group and 77 of 581 (126 per 1000) withdrew for any reason from the

manual therapy, prescribed exercises or both groups, RR 0.86 (95% CI 0.66 to 1.12, $I^2 = 0\%$; 12 trials, 1147 participants; [Analysis 2.5](#)). We downgraded the evidence due to bias, indirectness of the interventions and imprecision due to a small number of events.

Adverse events

We are uncertain if manual therapy, prescribed exercises or both results in more participants experiencing adverse events, due to very low-certainty evidence. Six trials reported adverse events ([Bisset 2006](#); [Blanchette 2011](#); [Olaussen 2015](#); [Smidt 2002a](#); [Tonks 2007](#); [Vuvan 2020](#)). Adverse events were mild and transient and included pain, bruising and gastrointestinal events. One trial reported 41 events from manual therapy including radiating pain into the arm ([Smidt 2002a](#)). Two trials did not report on adverse events ([Haahr 2003](#); [Peterson 2011](#)). In total, 19 of 224 (85 per 1000) participants in the minimal treatment group and 70 of 233 (313 per 1000) participants in the manual therapy, prescribed exercises or both delivered together group reported adverse events, RR 3.69 (95% CI 0.98 to 13.97, $I^2 = 72\%$; 6 trials, 457 participants; [Analysis 2.6](#)). We downgraded the evidence due to bias, indirectness of the interventions and imprecision from a small number of events.

Serious adverse events

No serious adverse events were reported.

Minor outcomes

Grip strength

Low-certainty evidence indicates that manual therapy, prescribed exercises or both may have little or no effect on grip strength at any time point. Grip strength was measured using a variety of measures. At the end of the intervention (six to eight weeks), the SMD was 0.18 (95% CI -0.11 to 0.48, $I^2 = 65\%$; 8 trials, 606 participants; [Analysis 2.7](#)). At three weeks, the SMD was 0.29 (95% CI 0.05 to 0.53, $I^2 = 0\%$; 3 trials, 280 participants) and at 12 months, the SMD was 0.18 (95% CI -0.01 to 0.36, $I^2 = 0\%$; 4 trials, 470 participants). We downgraded the evidence due to bias and indirectness of the interventions.

Return to work

We are uncertain if manual therapy, prescribed exercises or both results in fewer participants taking sick leave, due to very low-certainty evidence. Based upon one trial with 118 participants ([Olaussen 2015](#)), 30 of 60 participants who received minimal intervention were still on paid sick leave at 12 months compared with 31 of 58 participants receiving manual therapy and prescribed exercises, RR 1.07 (95% CI 0.75 to 1.52; [Analysis 2.8](#)). We downgraded the evidence due to bias, indirectness of the interventions and imprecision from a small number of events.

Manual therapy and prescribed exercises versus minimal or no intervention

Four trials were included in this comparison ([Bisset 2006](#); [Olaussen 2015](#); [Smidt 2002a](#); [Struijs 2004](#)).

Major outcomes

Pain

Low-certainty evidence indicates that manual therapy and prescribed exercises may have little or no effect on pain at any time point. At the end of the intervention (six weeks), the MD in pain (measured on a 0 to 10 scale) was -0.68 points (95% CI -1.41 to 0.04,

$I^2 = 60\%$; 4 trials, 485 participants; [Analysis 3.1](#)). At three weeks, the MD was -0.70 points (95% CI -2.12 to 0.72, $I^2 = 88\%$; 2 trials, 243 participants) and at 12 months, the MD was -0.47 points (95% CI -0.84 to -0.09, $I^2 = 0\%$; 4 trials, 470 participants). We downgraded the evidence due to performance bias and detection bias across all the trials as participants were aware of their treatment group, and indirectness due to the multimodal nature of the interventions included in the trials.

Disability

Low-certainty evidence indicates that manual therapy and prescribed exercises may have little or no effect on disability at any time point. At the end of the intervention (six weeks), there was a MD in disability on a 0 to 100 scale of -8.57 points (95% CI -13.34 to -3.79, $I^2 = 32\%$; 4 trials, 478 participants; [Analysis 3.2](#)). Similarly, at three weeks, the MD was -3.68 points (95% CI -8.58 to 1.22, $I^2 = 14\%$; 2 trials, 243 participants) and at 12 months, the MD was -4.90 points (95% CI -8.96 to -0.84, $I^2 = 0\%$; 4 trials, 469 participants). We downgraded the evidence due to bias and indirectness of the interventions.

Quality of life

No trials measured quality of life at the end of the intervention or prior to the end of the intervention. We are uncertain if manual therapy and prescribed exercises improves quality of life at the last time point, due to very low-certainty evidence. At 12 months, the MD in quality of life measured using the EQ-5D-3L utility score (value 1 equal to perfect health) was 0.01 points (95% CI -0.04 to 0.06; 1 trial, 123 participants; [Analysis 3.3](#)). We downgraded the evidence for bias, indirectness of the intervention and imprecision as the evidence was from a single trial and the confidence interval includes both a small improvement and worse outcomes.

Participant-reported treatment success

We are uncertain if manual therapy and prescribed exercises results in more participants reporting treatment success at the end of intervention and first follow-up, due to very low-certainty evidence. At the end of the intervention (six weeks), treatment success was reported in 71 of 246 participants (29%) in the minimal intervention group and 110 of 239 participants (46%) in the manual therapy and prescribed exercises group: RR 1.58 (95% CI 1.11 to 2.26, $I^2 = 52\%$; 4 trials, 485 participants; [Analysis 3.4](#)). At three weeks, the RR was 1.76 (95% CI 0.98 to 3.18, $I^2 = 0\%$; 2 trials, 243 participants). We downgraded the evidence due to bias, indirectness of the interventions and imprecision from a small number of events.

At last follow-up (12 months), the RR was 1.03 (95% CI 0.96 to 1.09, $I^2 = 0\%$; 4 trials, 470 participants). The evidence was of low certainty, downgraded for bias and indirectness of the interventions.

Participant withdrawals

We are uncertain if manual therapy and prescribed exercises results in more participant withdrawals due to very low-certainty evidence. In total, 18 of 249 participants (7%) withdrew for any reason from the minimal treatment group and 10 of 241 participants (4%) withdrew for any reason from the manual therapy and prescribed exercises groups, RR 0.59 (95% CI 0.28 to 1.25, $I^2 = 0\%$; 4 trials, 490 participants; [Analysis 3.5](#)). We downgraded the evidence due to bias, indirectness of the interventions and imprecision from a small number of events.

Adverse events

We are uncertain if manual therapy and prescribed exercises results in more participants experiencing adverse events, due to very low-certainty evidence. Three trials reported adverse events (Bisset 2006; Olausson 2015; Smidt 2002a). Adverse events were mild and transient and included pain and gastrointestinal events. In total, 19 of 181 participants (11%) in the minimal treatment group and 53 of 185 participants (29%) in the manual therapy and prescribed exercises group reported adverse events, RR 2.37 (95% CI 0.47 to 11.91, $I^2 = 82\%$; 3 trials, 366 participants; Analysis 3.6). We downgraded the evidence due to bias, indirectness of the interventions and imprecision from a small number of events.

Serious adverse events

No serious adverse events were reported.

Minor outcomes

Grip strength

Low-certainty evidence indicates manual therapy and prescribed exercises may have little or no effect on grip strength at any time point. At the end of the intervention (six weeks), the SMD was 0.20 (95% CI -0.19 to 0.60, $I^2 = 79\%$; 4 trials, 485 participants; Analysis 3.7). At three weeks, the SMD was 0.28 (95% CI 0.03 to 0.54, $I^2 = 0\%$; 2 trials, 243 participants) and at 12 months, the SMD was 0.18 (95% CI -0.01 to 0.36, $I^2 = 0\%$; 4 trials, 470 participants). We downgraded the evidence due to bias and indirectness of the interventions.

Return to work

We are uncertain if manual therapy and prescribed exercises results in fewer participants taking sick leave, due to very low-certainty evidence based upon a single trial that measured paid sick leave at 12 months (Olausson 2015). There were 30/60 participants with paid sick leave at 12 months in the no intervention group compared with 31/58 participants in the manual therapy and prescribed exercises group, RR 1.07 (95% CI 0.75 to 1.52; 1 trial, 118 participants; Analysis 3.8). We downgraded the evidence due to bias, indirectness of the interventions and imprecision from a small number of events.

Manual therapy versus minimal or no intervention

Two trials were included in this comparison (Blanchette 2011; Rachold 2023). For all the outcomes, we downgraded the evidence due to performance bias and detection bias as participants were aware of their treatment group, indirectness due to the multimodal nature of the interventions and imprecision due to large confidence intervals or small number of events.

Major outcomes

Pain

We are uncertain if manual therapy reduces pain, due to very low-certainty evidence. At the end of the intervention (four to six weeks), the MD in pain on a 0- to 10-point scale was -0.36 points (95% CI -1.25 to 0.53; 2 trials, 60 participants) and the MD at eight weeks to three months was -0.24 points (95% CI -1.10 to 0.61; 2 trials, 60 participants; Analysis 4.1). This outcome was not measured prior to the end of the intervention.

Disability

We are uncertain if manual therapy reduces disability, due to very low-certainty evidence. At the end of the intervention (four to six weeks), the MD in disability measured on a 0- to 100-point scale was -2.91 points (95% CI -16.79 to 10.96; 2 trials, 60 participants) and at eight weeks to three months it was 1.22 points (95% CI -7.17 to 9.62; 2 trials, 60 participants; Analysis 4.2). This outcome was not measured prior to the end of the intervention.

Quality of life

We are uncertain if manual therapy improves quality of life, due to very low-certainty evidence. At the end of the intervention (four weeks), the MD in quality of life measured on a 0 to 100 scale was -8.69 points (95% CI -19.79 to 2.41; 1 trial, 33 participants) and at eight weeks it was -8.51 points (95% CI -18.35 to 1.33; 1 trial, 33 participants). This outcome was not measured prior to the end of the intervention.

Participant-reported treatment success

No trials measured this outcome.

Participant withdrawals

We are uncertain if manual therapy results in more participant withdrawals due to very low-certainty evidence. Sixteen of 43 participants (37%) withdrew for any reason from the minimal treatment group and 12 of 45 withdrew for any reason from the manual therapy group, RR 0.59 (95% CI 0.13 to 2.68; Analysis 4.4).

Adverse events

We are uncertain if manual therapy results in more participants experiencing adverse events, due to very low-certainty evidence. Adverse events were mild and transient and included pain and bruises that ceased at six weeks. None of 12 participants in the minimal treatment group and 14 of 15 participants (93%) in the manual therapy group reported adverse events, RR 23.56 (95% CI 1.55 to 358.60; Analysis 4.5).

Serious adverse events

No serious adverse events were reported.

Minor outcomes

Grip strength

We are uncertain if manual therapy improves grip strength, due to very low-certainty evidence. At the end of the intervention (six weeks), the MD in pain-free grip strength was MD -1.00 kg (95% CI -11.30 to 9.30; 1 trial, 27 participants; Analysis 4.6). This outcome was only measured at the end of the intervention.

Return to work

No trials measured this outcome.

Prescribed exercises versus minimal or no intervention

Six trials were included in this comparison (Haahr 2003; McQueen 2021; Peterson 2011; Soderberg 2012; Tonks 2007; Vuvan 2020).

Major outcomes

Pain

Low-certainty evidence indicates that prescribed exercises may result in little to no effect on pain at any time point. At the end of the intervention (six weeks to three months), the MD in pain on a 0 to 10 scale was 0.49 points (95% CI -1.11 to 0.13), $I^2 = 51\%$; 6 trials, 478 participants; [Analysis 5.1](#)). At three to six weeks, the MD was 0.16 points (95% CI -0.19 to 0.50, $I^2 = 0\%$; 3 trials, 175 participants) and at 12 months, the MD was 0.10 points (95% CI -0.26 to 0.46, $I^2 = 2\%$; 2 trials, 291 participants). We downgraded the evidence due to performance bias and detection bias across all the trials as participants were aware of their treatment group, and for indirectness due to the multimodal nature of the interventions included in the trials.

Disability

Low-certainty evidence indicates that prescribed exercises may result in little to no effect on disability at the end of the intervention. At the end of the intervention (seven weeks to three months), the MD in disability measured on a 0 to 100 scale was 2.13 points (95% CI -8.91 to 4.65, $I^2 = 70\%$; 4 trials, 196 participants; [Analysis 5.2](#)). We downgraded the evidence due to bias and indirectness of the interventions.

We are uncertain if prescribed exercises reduce disability at first and last follow-up, due to very low-certainty evidence. At six weeks, the MD was 1.48 points (95% CI -6.89 to 9.85) and at 12 months it was 0.09 points (95% CI -0.40 to 0.58) (1 trial, 59 participants). We downgraded the evidence due to bias, indirectness of the interventions and imprecision as the evidence was from a single trial and the confidence intervals include both a small improvement and worse outcomes.

Quality of life

We are uncertain if prescribed exercises improve quality of life, due to very low-certainty evidence. At the end of intervention (three months), the MD in health-related quality of life (using the Gothenburg scale with a seven-point Likert scale ranging from 'very bad' to 'excellent, could not be better') was MD -0.50 points (95% CI -1.01 to 0.01; 1 trial, 80 participants; [Analysis 5.3](#)). We downgraded the evidence due to bias, indirectness of the interventions and imprecision as the evidence was from a single trial and the confidence interval included no improvement and worse outcomes. No trials measured this outcome prior to the completion of the intervention or at a later time point.

Participant-reported treatment success

We are uncertain if prescribed exercises result in more participants reporting treatment success at any time point, due to very low-certainty evidence. At the end of the intervention (six weeks to three months), 91 out of 140 participants (65%) reported treatment success compared with 91 out of 145 participants (63%) in the prescribed exercises group, RR 0.97 (95% CI 0.82 to 1.14, $I^2 = 0\%$; 2 trials, 285 participants; [Analysis 5.4](#)). At 12 months, the RR was 1.02 (95% CI 0.91 to 1.15; 1 trial, 232 participants). We downgraded the evidence due to bias, indirectness of the intervention and imprecision from a small number of events. No trials measured this outcome prior to the end of treatment.

Participant withdrawals

We are uncertain if prescribed exercises result in more participant withdrawals due to very low-certainty evidence. Forty-nine of 274 participants (18%) withdrew for any reason from the minimal treatment group and 55 of 295 participants (19%) withdrew for any reason from the prescribed exercise group, RR 0.96 (95% CI 0.62 to 1.49, $I^2 = 22\%$; 6 trials, 569 participants; [Analysis 5.5](#)). We downgraded the evidence due to bias, indirectness of the intervention and imprecision due to a small number of events.

Adverse events

We are uncertain if prescribed exercises result in more participants experiencing adverse events due to very low-certainty evidence. Two trials reported adverse events that included 'feeling worse' on the Global Rating of Change scale ([Tonks 2007](#); [Vuvan 2020](#)). None of the 31 participants in the minimal treatment group and 3 of 33 participants in the prescribed exercises group reported adverse events, RR 6.36 (95% CI 0.35 to 115.73; 2 trials, 64 participants; [Analysis 5.6](#)). We downgraded the evidence due to bias, indirectness of the intervention and imprecision from a small number of events.

Serious adverse events

No serious adverse events were reported.

Minor outcomes

Grip strength

We are uncertain if prescribed exercises improve grip strength at any time point due to very low-certainty evidence. At the end of the intervention (six to eight weeks), the SMD between groups in grip strength was 0.24 (95% CI -0.46 to 0.94, $I^2 = 0\%$; 3 trials, 94 participants; [Analysis 5.7](#)). At three weeks, the SMD was 0.34 (95% CI -0.31 to 0.99; 1 trial, 37 participants). We downgraded the evidence due to bias, indirectness of the intervention and imprecision due to a small number of events. No trials measured this outcome following the completion of treatment.

Return to work

No trials measured this outcome.

Manual therapy and prescribed exercises versus prescribed exercises alone

Six trials were included in this comparison ([Amro 2010](#); [Bagcaci 2023](#); [Boström 2019](#); [Kochar 2002](#); [Reyhan 2020](#); [Yilmaz 2022](#)).

Major outcomes

Pain

Low-certainty evidence indicates that the addition of manual therapy to prescribed exercises may slightly reduce pain at the completion of treatment, prior to its completion and at later follow-up. At the end of the intervention (four weeks to three months), the MD in pain measured on a 0- to 10-point scale was -2.56 points (95% CI -3.63 to -1.50, $I^2 = 77\%$; 5 trials, 183 participants; [Analysis 6.1](#)). At three weeks, MD was -1.91 points (95% CI -3.25 to -0.56, $I^2 = 61\%$; 3 trials, 104 participants) and at 3 to 12 months the MD was -1.81 points (95% CI -2.80 to -0.83, $I^2 = 0\%$; 2 trials, 57 participants). We downgraded the evidence due to performance bias and detection bias across all the trials as participants were

aware of their treatment group, and for indirectness due to the multimodal nature of the interventions included in the trials.

Disability

Low-certainty evidence indicates that the addition of manual therapy to prescribed exercises may slightly reduce disability at the end of the intervention and prior to its completion, but the effect was not sustained at longer follow-up. At the end of the intervention (four weeks to three months), the MD in disability was -14.57 points (95% CI -21.96 to -7.18, $I^2 = 76\%$; 5 trials, 177 participants; [Analysis 6.2](#)). At two to four weeks, the MD was -11.46 points (95% CI -20.70 to -2.22, $I^2 = 68\%$; 3 trials, 98 participants) and at 3 to 12 months, the MD was -3.79 points (95% CI -11.65 to 4.07, $I^2 = 72\%$; 3 trials, 97 participants). We downgraded the evidence due to bias and indirectness of the interventions.

Quality of life

No trials measured this outcome.

Participant-reported treatment success

We are uncertain if the addition of manual therapy to prescribed exercises results in more participants reporting treatment success at any time point due to very low-certainty evidence. At the end of the intervention (four to six weeks), 34 of 37 participants (92%) in the manual therapy and exercise group reported treatment success compared with 20 out of 37 participants (54%) who received exercise alone, RR 2.25 (95% CI 0.27 to 18.88, $I^2 = 94\%$; 2 trials, 74 participants; [Analysis 6.3](#)). At two weeks, all 20 participants from the manual therapy and prescribed exercises group reported treatment success compared with 6 out of 20 participants who received exercise alone, RR 3.15 (95% CI 1.66 to 5.99; 1 trial, 40 participants). At three months, all 20 participants in the manual therapy and exercise group reported treatment success compared with 19 out of the 20 participants in the exercise alone group, RR 1.05 (95% CI 0.92 to 1.20; 1 trial, 40 participants). We downgraded the evidence due to bias, indirectness of the intervention and imprecision due to a small number of events.

Participant withdrawals

We are uncertain if the addition of manual therapy to prescribed exercises results in more participant withdrawals due to very low-certainty evidence. Eight of 66 participants (12%) withdrew for any reason from the manual therapy and exercise group, and 8 of 82 participants (10%) withdrew for any reason from the exercise alone group, RR 0.94 (95% CI 0.30 to 2.94, $I^2 = 39\%$; 4 trials, 148 participants; [Analysis 6.4](#)). We downgraded the evidence due to bias, indirectness of the interventions and imprecision from a small number of events.

Adverse events

Three trials measured adverse events and none were recorded ([Boström 2019](#); [Reyhan 2020](#); [Yilmaz 2022](#)).

Serious adverse events

No serious adverse events were reported.

Minor outcomes

Grip strength

We are uncertain if the addition of manual therapy to prescribed exercises improves pain-free grip strength at any time point due to very low-certainty evidence. At the end of the intervention (four to six weeks), the MD in grip strength was 3.65 kg (95% CI 2.36 to 4.93, $I^2 = 0\%$; 4 trials, 154 participants; [Analysis 6.5](#)). At two to three weeks, the MD was 2.75 kg (95% CI -0.81 to 6.32, $I^2 = 8\%$; 3 trials, 121 participants) and at three months, the MD was 4.60 kg (95% CI 1.43 to 7.77, $I^2 = 0\%$; 3 trials, 120 participants). We downgraded the evidence due to bias, indirectness of the interventions and imprecision as the confidence intervals included no improvement and a large improvement.

Return to work

No trials measured this outcome.

Manual therapy and prescribed exercises versus manual therapy alone

We did not identify any trials investigating manual therapy and prescribed exercises versus manual therapy alone.

Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection alone

Very low-certainty evidence from a single trial with 60 participants was included in this comparison ([Nambi 2023](#)). For all the outcomes, we downgraded the evidence due to performance bias and detection bias as participants were aware of their treatment group, indirectness due to the multimodal nature of the interventions and imprecision due to large confidence intervals or number of events.

Major outcomes

Pain

We are uncertain if the addition of manual therapy to exercise and glucocorticoid injection reduces pain, due to very low-certainty evidence. At the end of the intervention (four weeks), the MD in pain on a 0- to 10-point scale was -1.60 points (95% CI -2.21 to -0.99; 1 trial, 60 participants), and the MD at six months was -2.00 points (95% CI -2.34 to -1.66; 1 trial, 55 participants; [Analysis 7.1](#)). This outcome was not measured prior to the end of the intervention.

Disability

We are uncertain if the addition of manual therapy to exercise and glucocorticoid injection reduces disability, due to very low-certainty evidence. At the end of the intervention (four weeks), the MD in disability on a 0- to 100-point scale was -12.90 points (95% CI -14.71 to -11.09; 1 trial, 60 participants) and the MD at six months was -17.20 points (95% CI -17.82 to -16.58; 1 trial, 55 participants; [Analysis 7.2](#)). This outcome was not measured prior to the end of the intervention.

Quality of life

We are uncertain if the addition of manual therapy to exercise and glucocorticoid injection improves quality of life, due to very low-certainty evidence. At the end of the intervention (four weeks), the MD in quality of life on a 1- to 3-point scale was -0.20 points (95% CI -0.51 to 0.11; 1 trial, 60 participants) and the MD at six

months was 1.10 points (95% CI 1.02 to 1.18; 1 trial, 55 participants; [Analysis 7.2](#)). This outcome was not measured prior to the end of the intervention.

Participant-reported treatment success

No trials measured this outcome.

Participant withdrawals

We are uncertain if the addition of manual therapy to exercise and glucocorticoid injection results in more participant withdrawals due to very low-certainty evidence. Three of 30 participants (10%) withdrew for any reason from the manual therapy, exercise and glucocorticoid injection, and 2 of 30 (7%) withdrew for any reason from the exercise and glucocorticoid injection group, RR 1.50 (95% CI 0.27 to 8.34; 1 trial, 60 participants; [Analysis 7.4](#)).

Adverse events

We are uncertain if the addition of manual therapy to exercise and glucocorticoid injection results in more participants experiencing adverse events, due to very low-certainty evidence. No participants from this single trial with 60 participants reported adverse events ([Nambi 2023](#)).

Serious adverse events

No serious adverse events were reported.

Minor outcomes

Grip strength

We are uncertain if the addition of manual therapy to exercise and glucocorticoid injection improves grip strength, due to very low-certainty evidence. At the end of the intervention (four weeks), the MD in pain-free grip strength was MD 1.70 kg (95% CI 0.38 to 3.02; 1 trial, 60 participants) and the MD at six months was 5.40 kg (95% CI 3.97 to 6.83; 1 trial, 55 participants; [Analysis 7.6](#)). This outcome was not measured prior to the end of the intervention.

Return to work

No trials measured this outcome.

Manual therapy, prescribed exercises or both and glucocorticoid injection versus glucocorticoid injection alone

Four trials were included in this comparison ([Coombes 2013](#); [Luginbuhl 2008](#); [Muralidharagopalan 2020](#); [Tonks 2007](#)). One trial compared manual therapy, prescribed exercises and glucocorticoid injection with glucocorticoid injection alone ([Coombes 2013](#)), and three trials compared prescribed exercises and glucocorticoid injection with glucocorticoid injection alone ([Luginbuhl 2008](#); [Muralidharagopalan 2020](#); [Tonks 2007](#)). There were no trials that compared manual therapy and glucocorticoid injection with glucocorticoid injection alone.

Major outcomes

Pain

Low-certainty evidence indicates that the addition of manual therapy, prescribed exercises or both to glucocorticoid injection may result in a slight reduction in pain at the end of the intervention but had no effect on pain at other time points. At the end of the intervention (six weeks to six months), the MD in pain was -0.89 points on a 0- to 10-point scale (95% CI -1.56 to -0.22, $I^2 = 0\%$; 4 trials,

171 participants; [Analysis 8.1](#)). At two to four weeks, the MD was 0.21 points (95% CI -0.37 to 0.80, $I^2 = 0\%$; 2 trials, 132 participants) and at 12 months, it was MD 0.35 points (95% CI -0.12 to 0.81, $I^2 = 0\%$). We downgraded the evidence due to performance bias and detection bias across all the trials as participants were aware of their treatment group, and indirectness due to the multimodal nature of the interventions included in the trials.

Disability

Low-certainty evidence indicates that the addition of manual therapy, prescribed exercises or both to glucocorticoid injection may have little to no effect on disability at any time point. At the end of the intervention (six weeks to six months), the MD was -0.98 points (95% CI -3.18 to 1.21, $I^2 = 0\%$; 3 trials, 151 participants; [Analysis 8.2](#)). At two to four weeks, the MD was 1.03 points (95% CI -2.51 to 4.56, $I^2 = 0\%$; 2 trials, 132 participants) and at 12 months, the MD was -1.10 points (95% CI -3.05 to 0.85, $I^2 = 0\%$; 2 trials, 132 participants). We downgraded the evidence due to bias and indirectness of the interventions.

Quality of life

We are uncertain if the addition of manual therapy, prescribed exercises or both to glucocorticoid injection improves quality of life at any time point due to very low-certainty evidence. Measured using the EQ-5D-3L utility score (1 is equal to perfect health), the MD between groups at the end of the intervention (six months) was 0.05 points (95% CI 0.02 to 0.08; 1 trial, 82 participants; [Analysis 8.3](#)). At four weeks, the MD was -0.02 points (95% CI -0.05 to 0.01) and at 12 months, the MD was -0.01 points (95% CI -0.05 to 0.03). We downgraded the evidence due to bias, indirectness of the interventions and imprecision as the evidence was from a single trial and the confidence intervals included a small improvement and worse outcomes.

Participant-reported treatment success

We are uncertain if the addition of manual therapy, prescribed exercises or both to glucocorticoid injection results in more participants reporting treatment success at any time point due to very low-certainty evidence. At the end of the intervention (three to six months), 28 of 49 participants (57%) treated with manual therapy, prescribed exercises or both and glucocorticoid injection reported treatment success compared with 29 of 53 participants (55%) who received manual therapy, prescribed exercises or both alone, RR 1.05 (95% CI 0.74 to 1.48, $I^2 = 0\%$; 2 trials, 102 participants; [Analysis 8.4](#)). At four weeks, the RR was 0.94 (95% CI 0.71 to 1.26; 1 trial, 82 participants) and at 12 months, the RR was 0.98 (95% CI 0.80 to 1.19; 1 trial, 82 participants). We downgraded the evidence due to bias, indirectness of the interventions and imprecision from a small number of events.

Participant withdrawals

We are uncertain if the addition of manual therapy, prescribed exercises or both to glucocorticoid injection results in more participant withdrawals due to very low-certainty evidence. Seven of 52 participants (13%) withdrew for any reason from the manual therapy, prescribed exercises and glucocorticoid group and 2 of 55 participants (4%) withdrew for any reason from the glucocorticoid group, RR 2.78 (95% CI 0.27 to 28.79, $I^2 = 49\%$; 2 trials, 107 participants; [Analysis 8.5](#)). We downgraded the evidence due to

bias, indirectness of the interventions and imprecision due to a small number of events.

Adverse events

We are uncertain if the addition of manual therapy, prescribed exercises or both to glucocorticoid injection results in more participants experiencing adverse events due to very low-certainty evidence. Seven of 78 participants (9%) reported adverse events in the manual therapy, prescribed exercises and glucocorticoid group and 10 of 79 participants (13%) reported adverse events in the glucocorticoid alone group, RR 0.78 (95% CI 0.33 to 1.83, $I^2 = 0\%$; 3 trials, 157 participants; [Analysis 8.6](#)). We downgraded the evidence due to bias, indirectness of the interventions and imprecision from a small number of events.

Serious adverse events

One trial with 82 participants measured serious adverse events with no adverse events recorded ([Coombes 2013](#)).

Minor outcomes

Grip strength

We are uncertain if the addition of manual therapy, prescribed exercises or both to glucocorticoid injection improves grip strength at any time point due to very low-certainty evidence. At the end of the intervention (three months), the MD in grip strength as a ratio of affected side/unaffected side was 9.00 (95% CI -2.84 to 20.84; 1 trial, 20 participants; [Analysis 8.7](#)). We downgraded the evidence due to bias, indirectness of the intervention and imprecision as the evidence was from a single trial and the confidence interval included a large improvement and worse outcomes. No trials measured grip strength prior to treatment completion or at longer follow-up.

Return to work

No trials measured this outcome.

Manual therapy, prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone

Very low-certainty evidence from one trial with 83 participants was available for this comparison ([Coombes 2013](#)). For all outcomes in this comparison, we downgraded the evidence due to performance bias and detection bias across all the trials as participants were aware of their treatment group, indirectness due to the multimodal nature of the interventions included in the trials and imprecision due to large confidence intervals or a small number of events.

Major outcomes

Pain

We are uncertain if the addition of manual therapy and prescribed exercises to glucocorticoid injection alone reduces pain at any time point. At the end of the intervention (six months), the MD in pain measured on a 0- to 10-point scale was -0.46 points (95% CI -2.11 to 1.19); 1 trial, 82 participants; [Analysis 9.1](#)). At four weeks, the MD was 0.30 (95% CI -0.32 to 0.92) and at 12 months, the MD was 0.42 points (95% CI -0.07 to 0.91).

Disability

We are uncertain if the addition of manual therapy and prescribed exercises to glucocorticoid injection alone reduces disability at any

time point. At the end of the intervention (six months), the MD in disability measured on a 0- to 100-point scale was 1.40 points (95% CI -7.40 to 4.60; 1 trial, 82 participants; [Analysis 9.2](#)). At four weeks, the MD was 1.50 points (95% CI -2.41 to 5.41) and at 12 months, the MD was -0.80 points (95% CI -3.22 to 1.62).

Quality of life

We are uncertain if the addition of manual therapy and prescribed exercises to glucocorticoid injection improves quality of life at any time point. At the end of the intervention (six months), the MD in health-related quality of life measured using the EQ-5D-3L utility score with the value 1 equal to perfect health was 0.05 points (95% CI 0.02 to 0.08; 1 trial, 82 participants; [Analysis 9.3](#)). At four weeks, it was MD -0.02 points (95% CI -0.05 to 0.01) and at 12 months, it was MD -0.01 points (95% CI -0.05 to 0.03).

Participant-reported treatment success

We are uncertain if the addition of manual therapy and prescribed exercises to glucocorticoid injection results in more participants reporting treatment success at any time point. Twenty-one of 39 participants (54%) reported treatment success at the end of the intervention period (six months) with manual therapy, prescribed exercises and glucocorticoid injection compared with 24 of 43 participants (56%) who received glucocorticoid injection alone, RR 0.96 (95% CI 0.65 to 1.43; 1 trial, 82 participants; [Analysis 9.4](#)). At four weeks, the RR was 0.94 (95% CI 0.71 to 1.26) and at 12 months it was 0.98 (95% CI 0.80 to 1.19).

Participant withdrawals

We are uncertain if the addition of manual therapy and prescribed exercise to glucocorticoid injection results in more participant withdrawals. Two of 43 participants (5%) withdrew for any reason from the manual therapy, prescribed exercises and glucocorticoid group and 2 of 40 participants (5%) withdrew for any reason from the glucocorticoid group, RR 1.07 (95% CI 0.16 to 7.27; 1 trial, 83 participants; [Analysis 9.5](#)).

Adverse events

We are uncertain if the addition of manual therapy and prescribed exercises to glucocorticoid injection results in more participants experiencing adverse events. Seven of 40 participants (18%) reported adverse events from the manual therapy, prescribed exercises and glucocorticoid group and 9 of 43 participants (21%) reported adverse events from the glucocorticoid group, RR 0.84 (95% CI 0.34 to 2.03; 1 trial, 83 participants; [Analysis 9.6](#)).

Severe adverse events

No serious adverse events were recorded.

Minor outcomes

Grip strength

No trials measured this outcome.

Return to work

No trials measured this outcome.

Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone

Three trials were included in this comparison ([Luginbuhl 2008](#); [Muralidharagopalan 2020](#); [Tonks 2007](#)).

Major outcomes

Pain

Low-certainty evidence indicates that the addition of prescribed exercises to glucocorticoid injection alone may slightly reduce pain at the end of intervention, but this is not clinically worthwhile. At the end of the intervention (six weeks to three months), there was a MD in pain on a 0- to 10-point scale of -0.97 points (95% CI -1.71 to -0.24, $I^2 = 0\%$; 3 trials, 89 participants; [Analysis 10.1](#)). We downgraded the evidence due to performance bias and detection bias across all the trials as participants were aware of their treatment group and for indirectness due to the multimodal nature of the interventions included in the trials.

We are uncertain if the addition of prescribed exercises to glucocorticoid injection alone reduces pain prior to the completion of treatment or at later follow-up due to very low-certainty evidence. The MD at two weeks was -0.45 points (95% CI -2.18 to 1.28; 1 trial, 50 participants) and at 12 months it was -0.20 points (95% CI -1.56 to 1.16; 1 trial, 50 participants). We downgraded the evidence due to bias, indirectness of the interventions and imprecision as the evidence was from a single trial and the confidence intervals included a small improvement and worse outcomes.

Disability

We are uncertain if the addition of prescribed exercises to glucocorticoid injection alone reduces disability at any time point, due to very low-certainty evidence. At the end of the intervention (six to seven weeks), the MD in disability measured on a 0- to 100-point scale was -0.92 points (95% CI -3.28 to 1.44, $I^2 = 0\%$; 2 trials, 69 participants; [Analysis 10.2](#)). At two weeks, the MD was -1.12 points (95% CI -9.44 to 7.20; 1 trial, 50 participants) and at 12 months, it was -1.66 points (95% CI -4.95 to 1.63; 1 trial, 50 participants). We downgraded the evidence due to bias, indirectness of the interventions and imprecision as the confidence intervals included a small improvement and worse outcomes.

Quality of life

No trials measured this outcome.

Participant-reported treatment success

We are uncertain if the addition of prescribed exercises to glucocorticoid injection results in more participants reporting treatment success due to very low-certainty evidence. At the end of the intervention (three months), 7 of 10 participants (70%) in the prescribed exercises and glucocorticoid injection group reported treatment success compared with 5 of 10 participants (50%) in the glucocorticoid injection only group, RR 1.40 (95% CI 0.67 to 2.94; 1 trial, 20 participants; [Analysis 10.3](#)). We downgraded the evidence due to bias, indirectness of the interventions and imprecision from a small number of events. No trials measured treatment success prior to the completion of treatment or at later follow-up.

Participant withdrawals

We are uncertain if the addition of prescribed exercises to glucocorticoid injection results in more participant withdrawals due to very low-certainty evidence. Five of 12 participants (42%) withdrew for any reason from the prescribed exercises and glucocorticoid group and none of 12 participants (0%) withdrew for any reason from the glucocorticoid group, RR 11.00, 95% CI 0.67 to 179.29; 1 trial, 24 participants; [Analysis 10.4](#)). We downgraded the evidence due to bias, indirectness of the interventions and imprecision from a small number of events.

Adverse events

We are uncertain if the addition of prescribed exercises to glucocorticoid injection results in more participants experiencing adverse events due to very low-certainty evidence. None of 38 participants (0%) reported adverse events from the prescribed exercises and glucocorticoid group and 1 of 36 participants (3%) reported adverse events from the glucocorticoid group, RR 0.33 (95% CI 0.01 to 7.45; 2 trials, 74 participants; [Analysis 10.5](#)). We downgraded the evidence due to bias, indirectness of the interventions and imprecision from a small number of events.

Serious adverse events

No serious adverse events were reported.

Minor outcomes

Grip strength

We are uncertain if the addition of prescribed exercises to glucocorticoid injection improves grip strength due to very low-certainty evidence. At the end of the intervention (three months), the MD in grip strength measured as a ratio of affected side/unaffected side was 9.00 (95% CI -2.84 to 20.84; 1 trial, 20 participants; [Analysis 8.7](#)). We downgraded the evidence due to bias, indirectness of interventions and imprecision as the evidence was from a single trial and the confidence interval included a large improvement and worse outcomes. No trials measured grip strength prior to the completion of treatment or at later follow-up.

Return to work

No trials measured this outcome.

DISCUSSION

Summary of main results

We evaluated the benefits and harms of manual therapy and exercise for 1612 participants with lateral elbow pain from 23 trials conducted in clinical practice settings from 13 countries. To answer the most clinically relevant questions, we have compared the effects of these interventions with placebo, minimal or no intervention (e.g. wait and see approaches), or in addition to glucocorticoid injection.

For the primary comparison with placebo, only one small trial of manual therapy addressed this question. Low-certainty evidence indicated that compared with placebo, manual therapy may provide a clinically worthwhile benefit in pain and disability at the end of treatment, although the 95% confidence interval also included both an important improvement and no improvement, and the longer-term outcomes are unknown. We downgraded the certainty of evidence due to indirectness and imprecision.

For the primary comparison of minimal or no intervention, 12 trials addressed this question. Low-certainty evidence indicated that manual therapy, prescribed exercises or both may slightly reduce pain and disability at the end of treatment. However, this may not be clinically worthwhile, and these benefits were not sustained over time. There was also little to no difference in quality of life and treatment success. While pain after treatment was an adverse event of manual therapy, the number of events was too small to be certain. We downgraded the certainty of evidence due to bias, indirectness and imprecision. Separately pooling data for the four trials that compared manual therapy and prescribed exercises with minimal intervention, the two trials that compared manual therapy with minimal intervention, and the six trials that compared prescribed exercises with minimal intervention were consistent with these findings of no worthwhile benefit.

There was low-certainty evidence, based upon five trials and downgraded for bias and indirectness, that adding manual therapy to prescribed exercises may result in a sustained reduction in pain although the 95% confidence interval also included no worthwhile benefit. Low-certainty evidence indicated that it may slightly reduce disability at the end of the intervention, but this effect was not sustained at longer follow-up.

There was very low-certainty evidence based on a single trial that adding manual therapy to prescribed exercises and glucocorticoid injection may slightly reduce pain and disability at the end of the intervention, but this may not be clinically worthwhile.

There was low-certainty evidence, based upon four trials and downgraded for bias and imprecision, that adding manual therapy and/or prescribed exercises to glucocorticoid injection may slightly reduce pain at the end of the intervention, but this may not be clinically worthwhile. Longer-term effects on pain were also very uncertain with evidence downgraded due to bias, indirectness and imprecision. There was very low-certainty evidence about effects on disability at any time point.

Overall completeness and applicability of evidence

A majority of trials were conducted in high-income countries. While there is no reason to suspect that the results would not be applicable to people with lateral elbow pain in low- and middle-income countries, these interventions may be less accessible in these settings.

Manual therapy and prescribed exercises for lateral elbow pain are commonly delivered together in clinical practice, but evidence assessing this multi-modal intervention with placebo was lacking.

There was a single placebo-controlled trial that assessed the effects of manual therapy. In this trial, the placebo comparison comprised manual therapy directed at the soft tissue above and below the tender points at the lateral elbow. While it could be argued that both participant groups received a form of manual therapy, differences in outcomes were observed at the end of treatment, indicating that participants may have been adequately blinded to the intervention group. In addition, all treatments were delivered by a single therapist, suggesting that there was consistent application of the intervention and placebo interventions, including verbal and non-verbal communication with the participants. Whether any differences in outcome were sustained is unknown as there was no longer-term follow-up.

The manual therapy and exercise interventions used in the remaining trials are largely reflective of clinical practice. Exercise interventions usually involved the extensor muscles of the wrist with gradually increasing resistance, including isometric, concentric and eccentric exercises. The manual therapy interventions typically involved soft tissue techniques (e.g. deep friction massage) or an approach known as 'mobilisation with movement'. For this technique, the patient performs a pain-producing movement in conjunction with the therapist applying sustained mobilisation of the elbow. The minimal to no treatment interventions across these trials included advice, usual GP care, observation and waiting list controls. Notably, the absence of clinically worthwhile benefits of manual therapy, prescribed exercises or both compared with minimal or no treatment means that higher-quality placebo-controlled trials are unnecessary as the effects would likely be even smaller.

We also included a detailed description of the interventions across trials using the Consensus on Exercise Reporting Template (CERT) (Slade 2016). While there was detailed advice on the delivery of the exercise interventions, including progression and tailoring strategies, information on how to apply the manual therapy interventions was sometimes lacking. If we had identified benefits of manual therapy and prescribed exercises, provision of more detailed advice on how to apply the manual therapy interventions, particularly around the decisions to modify the treatments, would have helped clinicians apply the results to clinical practice.

Quality of the evidence

For the primary comparison of manual therapy versus placebo manual therapy, we downgraded the evidence for pain and disability once for imprecision because the evidence was limited to a single small trial and the 95% confidence intervals included both an important improvement and no improvement. We also downgraded this evidence a second time for indirectness as the intervention (oscillating energy manual therapy technique) may not be reflective of current clinical practice or easily applied outside the trial setting. In addition, the results may differ in participants who are treated with other manual therapy techniques and exercise together.

For the second comparison of manual therapy, prescribed exercises or both versus minimal or no intervention, we downgraded the evidence for pain, disability and participant-reported treatment success due to risk of performance and detection biases, as participants were aware of group allocation and this has the potential to overestimate these self-reported outcomes. We downgraded this evidence again for indirectness due to the diversity of manual therapy and exercise interventions across the trials. This was due to potential differences in manual therapy techniques, with some requiring more advanced training (e.g. Mulligans technique), which may not be available across all settings. For quality of life, we downgraded the evidence for imprecision as the evidence was derived from a single trial with large confidence intervals that included both little effect and worse outcomes. For adverse events and participant withdrawals, we downgraded the evidence a third time due to the small number of events.

Potential biases in the review process

We applied a comprehensive search of major electronic databases and trial registries without language restrictions to reduce the risk of missing potentially eligible trials. Two independent assessors screened and selected studies and performed risk of bias judgements. None of the review authors were involved in the conduct of the included trials.

We chose the end of the intervention as the primary time point and also assessed outcomes prior to the completion of

treatment and at longer follow-up. As trials employed variable length of interventions, this could be considered a limitation. In two comparisons ([Analysis 2.1](#)), there were 11 studies allowing formal assessment of the presence of publication bias using a funnel plot ([Figure 3](#); [Figure 4](#)). The potential for publication bias was present with more small studies showing a positive result (i.e. less pain) than small negative trials (i.e. more pain) ([Figure 3](#)). However, the impact of any missing studies was small, given that improvements in pain and disability were minimal.

Figure 3.

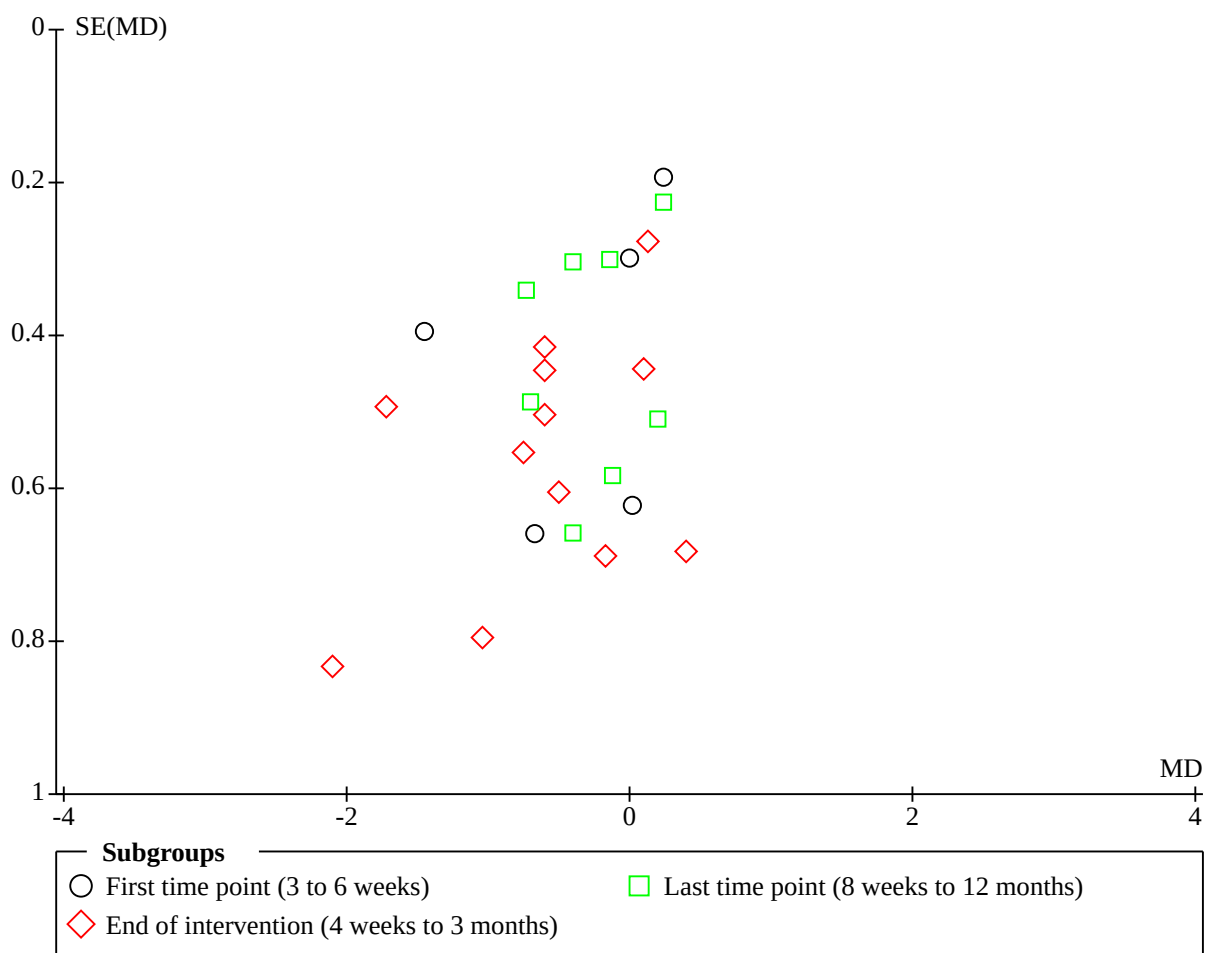
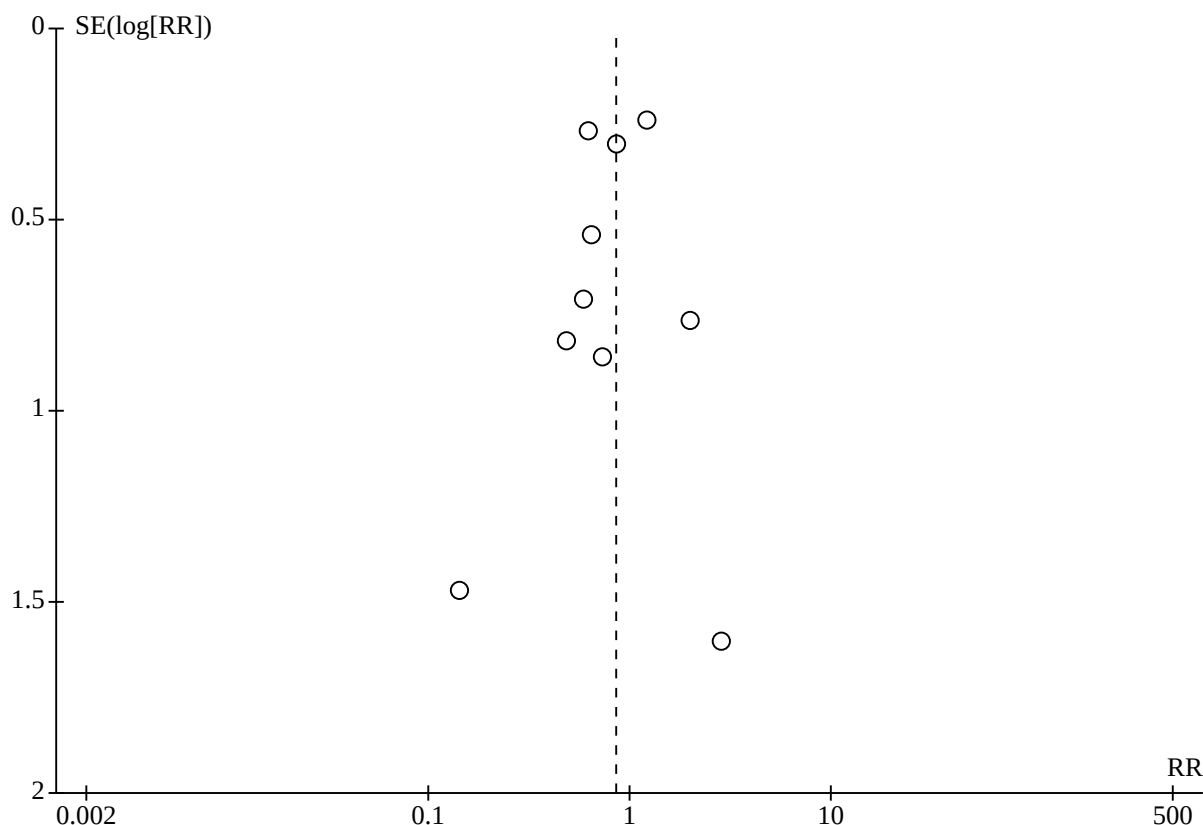


Figure 4.



For [McQueen 2021](#), no measures of variance were provided in the publication. In our calculation of SD for disability (12 months) based on the provided P value and MD, the SD appeared small compared with the other time points. In a post hoc sensitivity analysis removing this trial, the effect estimate was slightly larger, but the overall conclusion was unchanged.

Of the four trials that are currently ongoing, two trials with a planned total sample size of at least 140 participants are investigating manual therapy or exercise or both versus minimal or treatment ([Sahebjamei 2019](#); [Sveinall 2022](#)). The results of these trials are unlikely to shift the effect estimate or the precision of the estimate because there are currently 769 participants in these analyses. The other two ongoing trials are investigating manual therapy and prescribed exercises versus prescribed exercises alone ([Prat 2018](#); [Takamjani 2019](#)). Although the planned total sample sizes of these three trials is unknown, it is possible that the results may change the effect estimate and precision because there is currently a maximum of 183 participants in these analyses ([Analysis 6.1](#)).

Agreements and disagreements with other studies or reviews

We identified seven previous systematic reviews on this topic ([Bisset 2005](#); [Herd 2008](#); [Hoogvliet 2013](#); [Karanasios 2020](#); [Kim 2020](#); [Kohia 2008](#); [Smidt 2003](#)). Older systematic reviews were unable to draw conclusions due to insufficient or inconclusive evidence based on too few studies ([Bisset 2005](#); [Herd 2008](#); [Kohia 2008](#); [Smidt 2003](#)).

More recent systematic reviews largely concur with our findings of small, likely unimportant benefits in pain and disability ([Hoogvliet 2013](#); [Karanasios 2020](#); [Kim 2020](#); [Kim 2020](#)). [Karanasios 2020](#), which used more up-to-date review methods including GRADE, and included six RCTs ([Bisset 2006](#); [Olaussen 2015](#); [Peterson 2011](#); [Smidt 2002a](#); [Tonks 2007](#); [Vuvan 2020](#)), compared exercise with or without other physiotherapy interventions with a wait and see approach and found very low-certainty evidence of small benefits for pain and disability that may not be clinically worthwhile.

Our review identified five additional RCTs of prescribed exercises and/or manual therapy compared to no or minimal treatment ([Blanchette 2011](#); [Haahr 2003](#); [McQueen 2021](#); [Soderberg 2012](#); [Struijs 2004](#)) and our conclusions are largely in agreement with [Karanasios 2020](#) for these outcomes. Our review synthesised harms as well as benefits and found more adverse events following manual therapy and/or exercise (313 per 1000) compared to minimal or no treatment (85 per 1000). However, there were too few events to be certain.

AUTHORS' CONCLUSIONS

Implications for practice

Contemporary clinical practice for treating lateral elbow pain often comprises manual therapy together with prescribed exercises. However, our review questions the utility of these interventions to treat people with lateral elbow pain, although there may be other benefits of attending a physical therapist to provide education, advice and ongoing support.

Low-certainty evidence from a single trial comparing manual therapy alone with placebo provides evidence for a small improvement in pain and disability in people with lateral elbow pain at the end of treatment, although the 95% confidence interval also included both an important improvement and no improvement. Whether these effects are sustained is unknown.

Low-certainty evidence also showed that manual therapy and prescribed exercise may slightly reduce pain and disability at the end of treatment compared with minimal or no intervention, but these benefits were likely unimportant and were not sustained at longer follow-up. There were no observed benefits in health-related quality of life or treatment success. While pain after treatment was an adverse event of manual therapy, the number of events was too small to be certain.

There was low-certainty evidence that adding manual therapy to prescribed exercises may result in a sustained reduction in pain although the 95% confidence interval also included no worthwhile benefit and there were no sustained benefits to disability. Similarly, there was low-certainty evidence that adding manual therapy and/or prescribed exercises to glucocorticoid injection may not be clinically worthwhile.

People contemplating treatment from a therapist involving manual therapy and prescribed exercises should be informed about the findings of this evidence synthesis to help them make an evidence-informed decision that takes into consideration the likely benefits, harms and costs. They should also be informed that their symptoms are likely to improve over time irrespective of treatment and symptom duration.

Implications for research

Additional randomised trials that examine the benefits and harms of manual therapy, prescribed exercises or both compared with placebo in people with lateral elbow pain may increase the certainty of the evidence of a transient benefit in pain and disability at the end of treatment and clarify whether this is sustained. Further trials examining the benefits and harms of manual therapy, prescribed exercises or both compared with minimal or no intervention are unlikely to add important new information.

This review has not considered cost-effectiveness. Manual therapy interventions were applied over five weeks on average, including two sessions per week with a therapist (i.e. 10 visits). Exercise interventions are usually prescribed at a single visit for the participants to complete at home for up to 12 weeks, with one or two additional visits to progress or modify exercises, as well as materials to support adherence such as exercise diaries and equipment. The higher number of supervised sessions for manual therapy may increase healthcare costs for patients and the healthcare system, whereas a lower-cost, less resource-intensive intervention (i.e. fewer supervised sessions) for people with lateral elbow pain may achieve similar outcomes.

ACKNOWLEDGEMENTS

The review authors would like to thank the authors of the earlier version of the protocol for their contributions: Nynke Smidt, Willem JJ Assendelft, H Arola, Antti Malmivaara, Lex M Bouter and Sally Green.

REFERENCES

References to studies included in this review

Amro 2010 {published data only}

Amro A, Diener I, Bdaire WO, Hamed IM, Shalabi AI, Ilyyan DI. The effects of Mulligan mobilisation with movement and taping techniques on pain, grip strength, and function in patients with lateral epicondylitis. *Hong Kong Physiotherapy Journal* 2010;**28**(1):19-23.

Bagcaci 2023 {published data only}

* Bagcaci S, Unuvar BS, Gercek H, Ugurluc I, Sert OA, Yilmaz K. A randomized controlled trial on pain, grip strength, and functionality in lateral elbow pain: Mulligan vs muscle energy techniques. *Journal of Back and Musculoskeletal Rehabilitation* 2023;**36**:419-27.

Bisset 2006 {published data only}

* Bisset L, Beller E, Jull G, Brooks P, Darnell R, Vicenzino B. Mobilisation with movement and exercise, corticosteroid injection, or wait and see for tennis elbow: randomised trial. *BMJ* 2006;**333**:1-6. [DOI: [doi:10.1136/bmj.38961.584653.AE](https://doi.org/10.1136/bmj.38961.584653.AE)]

Bisset LM, Coppieters MW, Vicenzino B. Sensorimotor deficits remain despite resolution of symptoms using conservative treatment in patients with tennis elbow: a randomized controlled trial. *Archives of Physical Medicine & Rehabilitation* 2009;**90**(1):1-8.

Gardner J. Waiting rivals injections for tennis elbow. *Family Practice News* 2007.

NCT00110318. Trial of physiotherapy and corticosteroid injections in lateral epicondylalgia. 2002. <https://clinicaltrials.gov/ct2/show/NCT00110318>.

Blanchette 2011 {published data only}

Blanchette MA, Normand MC. Augmented soft tissue mobilization vs natural history in the treatment of lateral epicondylitis: a pilot study. *Journal of Manipulative & Physiological Therapeutics* 2011;**34**(2):123-30.

Boström 2019 {published data only}

Boström K, Mæhlum S, Cvanarova Småstuen M, Storheim KC. Clinical comparative effectiveness of acupuncture versus manual therapy treatment of lateral epicondylitis: feasibility randomized clinical trial. *BMC Pilot and Feasibility Studies* 2019;**5**(110):1-10.

NCT02321696. Physiotherapy or acupuncture for lateral epicondylitis. <https://clinicaltrials.gov/show/nct02321696> 2014.

Coombes 2013 {published data only}

ACTRN12609000051246. Optimising corticosteroid injection for lateral epicondylalgia with the addition of physiotherapy: a randomised control trial with placebo comparison. <http://www.anzctr.org.au/Trial/Registration/TrialReview.aspx?ACTRN=12609000051246>.

* Coombes BK, Bisset L, Brooks P, Khan A, Vicenzino B. Effect of corticosteroid injection, physiotherapy, or both on clinical

outcomes in patients with unilateral lateral epicondylalgia: a randomized controlled trial. *JAMA* 2013;**309**(5):461-9.

Coombes BK, Connelly L, Bisset L, Vicenzino B. Economic evaluation favours physiotherapy but not corticosteroid injection as a first-line intervention for chronic lateral epicondylalgia: evidence from a randomised clinical trial. *British Journal of Sports Medicine* 2016;**50**(22):1400-5.

Vicenzino B, Coombes BK, Bisset L, Brooks P, Khan A. Corticosteroid injection worsened recovery and recurrence of tennis elbow; physiotherapy had no effect. *Annals of Internal Medicine* 2013;**158**(10):JC8.

Haahr 2003 {published data only}

Haahr JP, Andersen JH. Prognostic factors in lateral epicondylitis: a randomized trial with one-year follow-up in 266 new cases treated with minimal occupational intervention or the usual approach in general practice. *Rheumatology* 2003;**42**(10):1216-25.

Kochar 2002 {published data only}

Kochar M, Dogra A. Effectiveness of a specific physiotherapy regimen on patients with tennis elbow: clinical study. *Physiotherapy* 2002;**88**(6):333-41.

Luginbuhl 2008 {published data only}

Luginbuhl R, Brunner F, Schneeberger AG. No effect of forearm band and extensor strengthening exercises for the treatment of tennis elbow: a prospective randomised study. *Chirurgia Degli Organi di Movimento* 2008;**91**(1):35-40.

McQueen 2021 {published data only}

* McQueen KS, Powell RK, Keener T, Whalley R, Calfee RP. Role of strengthening during nonoperative treatment of lateralepicondyle tendinopathy. *Journal of Hand Therapy* 2021;**34**:619-26.

Muralidharagopalan 2020 {published data only}

* Muralidharagopalan NR, Gopal R, Natarajan S. Addition of eccentric elbow exercises to local corticosteroid injection for tennis elbow – a prospective randomized study. *European Journal of Molecular & Clinical Medicine* 2020;**7**(8):3528-37.

Nambi 2023 {published data only}

* Nambi G, Alghadiri M, Verma A, Aldhafian OR, Alshahrani NN, Saleh AK, et al. Clinical and radiological effects of corticosteroid injection combined with deep transverse friction massage and Mill's manipulation in lateral epicondylalgia – a prospective, randomized, single-blinded, sham controlled trial. *PLOS One* 2023;**18**(2):1-16.

Nourbakhsh 2008 {published data only}

Nourbakhsh MR, Fearon FJ. The effect of oscillating-energy manual therapy on lateral epicondylitis: a randomized, placebo-control, double-blinded study. *Journal of Hand Therapy* 2008;**21**(1):4-13.

Olaussen 2015 {published data only}

NCT00826462. Treatment study of steroid injection and physical therapy for acute lateral epicondylitis. <https://clinicaltrials.gov/show/NCT00826462> 2009.

Olaussen M, Holmedal O, Lindbaek M, Brage S. Physiotherapy alone or in combination with corticosteroid injection for acute lateral epicondylitis in general practice: a protocol for a randomised, placebo-controlled study. *BMC Musculoskeletal Disorders* 2009;**10**(152):1-8.

* Olaussen M, Holmedal Ø, Mdala I, Brage S, Lindbæk M. Corticosteroid or placebo injection combined with deep transverse friction massage, Mills manipulation, stretching and eccentric exercise for acute lateral epicondylitis: a randomised controlled trial. *BMC Musculoskeletal Disorders* 2015;**16**(122):1-13.

Peterson 2011 {published data only}

Peterson M, Butler S, Eriksson M, Svardsudd K. A randomized controlled trial of exercise versus wait-list in chronic tennis elbow (lateral epicondylitis). *Upsala Journal of Medical Sciences* 2011;**116**(4):269-79.

Rachold 2023 {published data only}

Rachold W, Reis O, Berger C, Buchmann J. Therapeutic effects of manual therapy on lateral epicondylitis: a randomised controlled trial. *Physikalische Medizin, Rehabilitationsmedizin, Kurortmedizin* 2023;**33**:79-86.

Reyhan 2020 {published data only}

Reyhan AC, Sindel D, Dereli EE. The effects of Mulligan's mobilization with movement technique in patients with lateral epicondylitis. *Journal of Back and Musculoskeletal Rehabilitation* 2020;**33**(2020):99-107.

Smidt 2002a {published data only}

Hart L. Corticosteroid injections, physiotherapy, or a wait-and-see policy for lateral epicondylitis? *Clinical Journal of Sport Medicine* 2002;**12**(6):403-4.

Korthals-de Bos IB, Smidt N, van Tulder MW, Rutten-van Molken MP, Ader HJ, van der Windt DA, et al. Cost effectiveness of interventions for lateral epicondylitis: results from a randomised controlled trial in primary care. *Pharmacoeconomics* 2004;**22**(3):185-95.

Smidt N, Assendelft WJJ, Van Der Windt D, Bouter LM. Corticosteroid injections, physiotherapy or a "wait and see" policy for lateral epicondylitis: a randomised clinical trial. In: 16th Annual Meeting of The International Society of Technology Assessment in Health Care. 2000.

* Smidt N, Van Der Windt DAWM, Assendelft WJJ, Deville WLJM, Korthals-de Bos IBC, Bouter LM. Corticosteroid injections, physiotherapy, or a wait-and-see policy for lateral epicondylitis: a randomised controlled trial. *Lancet* 2002;**359**(9307):657-62.

Soderberg 2012 {published data only}

Soderberg J, Grooten W, Ang BO. Short-term effects of eccentric training on lateral epicondylalgia-a randomized controlled trial. *Physiotherapy (United Kingdom)* 2011;**1**:eS434-5.

* Soderberg J, Grooten WJ, Ang BO. Effects of eccentric training on hand strength in subjects with lateral epicondylalgia: a randomized-controlled trial. *Scandinavian Journal of Medicine & Science in Sports* 2012;**22**(6):797-803.

Struijs 2004 {published data only}

Struijs PA, Assendelft WJ, Kerkhoffs GM, Souer S, van Dijk CN. The predictive value of the extensor grip test for the effectiveness of bracing for tennis elbow. *American Journal of Sports Medicine* 2005;**33**(12):1905-9.

* Struijs PA, Kerkhoffs GM, Assendelft WJ, Van Dijk CN. Conservative treatment of lateral epicondylitis: brace versus physical therapy or a combination of both-a randomized clinical trial. *American Journal of Sports Medicine* 2004;**32**(2):462-9.

Struijs PA, Korthals-de Bos IB, van Tulder MW, van Dijk CN, Bouter LM, Assendelft WJ. Cost effectiveness of brace, physiotherapy, or both for treatment of tennis elbow. *British Journal of Sports Medicine* 2006;**40**(7):637-43.

Tonks 2007 {published data only}

* Tonks JH, Pai SK, Murali SR. Steroid injection therapy is the best conservative treatment for lateral epicondylitis: a prospective randomised controlled trial. *International Journal of Clinical Practice* 2007;**61**(2):240-6.

Vuvan 2020 {published data only}

* Vuvan V, Vincenzino B, Mellor R, Jamesheales L, Coombes BK. Unsupervised isometric exercise versus wait-and-see for lateral elbow tendinopathy. *Medicine & Science in Sports & Exercise* 2020;**52**(2):287-95.

Yilmaz 2022 {published data only}

NCT04219488. Investigating the effects of neuromobilization in lateral epicondylitis. <https://clinicaltrials.gov/ct2/show/NCT04219488> 2020.

* Yilmaz K, Bayramlar, KY, Ayhan C, Tufekci O. Investigating the effects of neuromobilization in lateralepicondylitis. *Journal of Hand Therapy* 2022;**35**:97-106.

References to studies excluded from this review
Abbott 2001a {published data only}

Abbott JH. Mobilization with movement applied to the elbow affects shoulder range of movement in subjects with lateral epicondylalgia. *Manual Therapy* 2001;**6**(3):170-7.

Abbott 2001b {published data only}

Abbott JH, Patla CE, Jensen RH. The initial effects of an elbow mobilization with movement technique on grip strength in subjects with lateral epicondylalgia. *Manual Therapy* 2001;**6**(3):163-9.

ACTRN12612000993897 {published data only}

ACTRN12612000993897. Effects of prolotherapy injections and an exercise program used singly and in combination for refractory tennis elbow. <https://www.anzctr.org.au/Trial/Registration/TrialReview.aspx?ACTRN=12612000993897>;2012.

ACTRN12617000219381 {published data only}

ACTRN12617000219381. The influence of aerobic exercise on conditioned pain modulation (CPM) and manipulation induced pain modulation (MIPM) effects in participants with tennis elbow. <https://anzctr.org.au/Trial/Registration/TrialReview.aspx?id=372232> 2017.

Altun 2018 {published data only}

Altun RD, Incel NA, Cimen O B, Sahin G. Efficacy of ESWT for lateral epicondylitis treatment: comparison with physical therapy modalities. *Journal of Musculoskeletal Research* 2018;**21**(1):1-9.

Anitha 2018 {published data only}

Anitha A, Prachi G. Effectiveness of eccentric strengthening of wrist extensors along with conventional therapy in patients with lateral epicondylitis. *Research Journal of Pharmacy and Technology* 2018;**11**(12):5340-4.

Bateman 2024 {published data only}

Bateman M, Skeggs A, Whitby E, Fletcher-Barrett V, Stephens G, Dawes M, et al. Optimising physiotherapy for people with lateral elbow tendinopathy - results of a mixed-methods pilot and feasibility randomised controlled trial (OPTimisE). *Musculoskeletal Science & Practice* 2024;**69**:102905. [DOI: [10.1016/j.msksp.2023.102905](https://doi.org/10.1016/j.msksp.2023.102905)]

Cleland 2005 {published data only}

Cleland JA, Flynn TW, Palmer JA. Incorporation of manual therapy directed at the cervicothoracic spine in patients with lateral epicondylalgia: a pilot clinical trial. *Journal of Manual and Manipulative Therapy* 2005;**13**(3):143-51.

Coombes 2016 {published data only}

Coombes BK, Wiebusch M, Heales L, Stephenson A, Vicenzino B. Isometric exercise above but not below an individual's pain threshold influences pain perception in people with lateral epicondylalgia. *Clinical Journal of Pain* 2016;**32**:1069-75.

Croisier 2007 {published data only}

Croisier JL, Foidart-Dessalle M, Tinant F, Crielaard JM, Forthomme B. An isokinetic eccentric programme for the management of chronic lateral epicondylar tendinopathy. *British Journal of Sports Medicine* 2007;**41**(4):269-75.

CTRI/2022/02/040270 {published data only}

* CTRI/2022/02/040270. Efficacy of marma therapy in comparison with physical therapy in management of tennis elbow. <http://www.ctri.nic.in/Clinicaltrials/pmaindet2.php?trialid=64678>.

Day 2019 {published data only}

Day JM, Lucado AM, Uhl TL. A comprehensive rehabilitation program for treating lateral elbow tendinopathy. *International Journal of Sports Physical Therapy* 2019;**14**(5):818-34.

Drechsler 1997 {published data only}

Drechsler WI, Knarr JF, Snyder-Mackler L. A comparison of two treatment regimens for lateral epicondylitis: a randomized trial of clinical interventions. *Journal of Sport Rehabilitation* 1997;**6**(3):226-34.

Fernandez-Carnero 2008 {published data only}

Fernandez-Carnero J, Fernandez-de-las-Penas C, Cleland JA. Immediate hypoalgesic and motor effects after a single cervical spine manipulation in subjects with lateral epicondylalgia. *Journal of Manipulative & Physiological Therapeutics* 2008;**31**(9):675-81.

Fernandez-Carnero 2011 {published data only}

Fernandez-Carnero J, Cleland JA, Arbizu RL. Examination of motor and hypoalgesic effects of cervical vs thoracic spine manipulation in patients with lateral epicondylalgia: a clinical trial. *Journal of Manipulative & Physiological Therapeutics* 2011;**34**(7):432-40.

Gunduz 2012 {published data only}

Gunduz R, Malas FU, Borman P, Kocaoglu S, Ozcakar L. Physical therapy, corticosteroid injection, and extracorporeal shock wave treatment in lateral epicondylitis. Clinical and ultrasonographical comparison. *Clinical Rheumatology* 2012;**31**(5):807-12.

Gupta 2021 {published data only}

* Gupta R, Chahal A. Comparative effect of dry needling and neural mobilization on pain, strength, range of motion, and quality of life in patients with lateral epicondylitis: protocol for randomized clinical trial. *Journal of Chiropractic Medicine* 2021;**20**(2):77-84.

Hsu 2016 {published data only}

Hsu CY, Lee KH, Huang HC, Chang ZY, Chen HY, Yang TH. Manipulation therapy relieved pain more rapidly than acupuncture among lateral epicondylalgia (tennis elbow) patients: a randomized controlled trial with 8-week follow-up. *Evidence-based Complementary and Alternative Medicine* 2016;**3079247**:1-7.

IRCT848 {published data only}

IRCT848. Effect of dry needling on radiated pain of shoulder muscles trigger points in tennis elbow patients. <https://en.irct.ir/trial/848>.

Kamal 2022 {published data only}

Kamal KC, Traistaru MR, Alexandru DO, Kamal AM, Maria DT, Kamal D. Complex rehabilitation program in patients with lateral epicondylitis. *Aging Clinical and Experimental Research* 2022;**34**(Suppl 1):S335. [DOI: [10.1007/s40520-022-02147-3](https://doi.org/10.1007/s40520-022-02147-3)]

Kaux 2022 {published data only}

Kaux JF, Forthomme F, Croisier JL, Demoulin C, Vanderthommen M. Effectiveness of tecar therapy combined with eccentric exercises in lateral elbow tendinopathy. *Aging Clinical and Experimental Research* 2022;**34**(Suppl 1):S189. [DOI: [10.1007/s40520-022-02147-3](https://doi.org/10.1007/s40520-022-02147-3)]

KCT0007378 {published data only}

* KCT0007378. Effect of blood flow restriction exercise program on auto-parts factory workers with lateral elbow tendinopathy: randomized controlled trial. <https://cris.nih.go.kr/cris/search/detailSearchEn.do?seq=22097>.

Khandaker 2014 {published data only}

Khandaker MN, Islam S, Emran A, Islam J, Ahmed SM, Khan MM, et al. The effect of stretching exercise in the management of lateral epicondylitis. *Bangladesh Medical Journal* 2014;**43**(2):61-6.

Nuruzzaman KM, Mozaffar AS, Rahman KM, Salek AKM, Emran A. Effect of stretching and strengthening exercise in the management of lateral epicondylitis. *International Journal of Rheumatic Diseases* 2017;**20**(Suppl 1):29.

Kukuksen 2013 {published data only}

Kukuksen S, Yilmaz H, Salli A, Ugurlu H. Muscle energy technique versus corticosteroid injection for management of chronic lateral epicondylitis: randomized controlled trial with 1-year follow-up. *Archives of Physical Medicine & Rehabilitation* 2013;**94**(11):2068-74.

Lee 2018 {published data only}

* Lee J, Kim T, Lim K. Effects of eccentric control exercise for wrist extensor and shoulder stabilization exercise on the pain and functions of tennis elbow. *Journal of Physical Therapy Science* 2018;**30**:590-4.

Shirzhadi Z. The effect of the shoulder muscles strengthening in treatment of patients with tennis elbow. WHO 2018.

Manias 2006 {published data only}

Manias P, Stasinopoulos D. A controlled clinical pilot trial to study the effectiveness of ice as a supplement to the exercise programme for the management of lateral elbow tendinopathy. *British Journal of Sports Medicine* 2006;**40**(1):81-5.

Martinez-Silvestrini 2005 {published data only}

Martinez-Silvestrini JA, Newcomer KL, Gay RE, Schaefer MP, Kortebein P, Arendt KW. Chronic lateral epicondylitis: comparative effectiveness of a home exercise program including stretching alone versus stretching supplemented with eccentric or concentric strengthening. *Journal of Hand Therapy* 2005;**18**(4):411-9.

McLean 2002 {published data only}

McLean S, Naish R, Reed L, Urry S, Vicenzino B. A pilot study of the manual force levels required to produce manipulation induced hypoalgesia. *Clinical Biomechanics* 2002;**17**(4):304-8.

Mostafee 2020 {published data only}

Mostafae N, Divandari A, Negahban H, Kachooei AR, Moradi A, Ebrahimzadeh MH, et al. Shoulder and scapula muscle training plus conventional physiotherapy versus conventional physiotherapy only: a randomized controlled trial of patients with lateral elbow tendinopathy. *Physiotherapy Theory and Practice* 2022;**38**(9):1153-64. [DOI: [10.1080/09593985.2020.1821417](https://doi.org/10.1080/09593985.2020.1821417)]

Mostafae N, Divandari A, Negahban H, RezaKachooei A, Moradi A, Ebrahimzadeh MH, et al. Shoulder and scapula muscle training plus conventional physiotherapy versus conventional physiotherapy only: a randomized controlled trial of patients with lateral elbow tendinopathy. *Physiotherapy Theory and Practice* 2020;**38**(9):1153-64.

Nagrale 2009 {published data only}

Nagrale AV, Herd CR, Ganvir S, Ramteke G. Cyriax physiotherapy versus phonophoresis with supervised exercise in subjects with lateral epicondylalgia: a randomized clinical trial. *Journal of Manual and Manipulative Therapy* 2009;**17**(3):171-8.

NCT00930709 {published data only}

NCT00930709. Botulinum toxin versus active strength training in the treatment of lateral epicondylitis (BooST). <https://clinicaltrials.gov/show/NCT00930709>.

NCT01273688 {published data only}

NCT01273688. Eccentric training with or without elbow brace for epicondylitis. <https://clinicaltrials.gov/show/NCT01273688>.

NCT01897259 {published data only}

NCT01897259. Comparison of conservative methods for the treatment of lateral epicondylitis: a randomized, prospective study. <https://clinicaltrials.gov/ct2/show/NCT01897259>.

NCT02052089 {published data only}

NCT02052089. Comparative study for the optimal treatment method of lateral epicondylitis. <https://clinicaltrials.gov/ct2/show/NCT02052089>.

NCT02304952 {published data only}

NCT02304952. Eccentric exercise or radiofrequency microtenotomy as treatment of lateral epicondylalgia. <https://clinicaltrials.gov/ct2/show/study/NCT02304952>.

NCT02521298 {published data only}

NCT02521298. Treatment of lateral elbow tendinopathy. <https://clinicaltrials.gov/show/NCT02521298>.

NCT02930304 {published data only}

NCT02930304. Non-surgical treatment approaches in patients with newly diagnosed lateral epicondylitis: a randomized clinical trial. <https://clinicaltrials.gov/show/NCT02930304>.

NCT02991560 {published data only}

NCT02991560. Effects of various taping techniques in lateral epicondylitis on functional and ultrasonographic outcomes. <https://clinicaltrials.gov/ct2/show/NCT02991560>.

NCT03074500 {published data only}

NCT03074500. Kinesiotaping in lateral epicondylitis. <https://clinicaltrials.gov/show/NCT03074500>.

NCT03161782 {published data only}

NCT03161782. The efficacy of different types of stretching on range of motion, pain and function in elbow limitation. <https://clinicaltrials.gov/ct2/show/NCT03161782>.

NCT03167710 {published data only}

NCT03167710. Dry needling, manipulation and stretching vs. manual therapy, exercise and ultrasound for lateral epicondylalgia. <https://clinicaltrials.gov/show/NCT03167710>.

NCT03834090 {published data only}

NCT03834090. Effectiveness of radial extracorporeal shock wave therapy and supervised exercises in lateral epicondylitis. <https://clinicaltrials.gov/ct2/show/NCT03834090>.

NCT05238090 {published data only}

* NCT05238090. Evaluation of the applying a stretching protocol to lateral epicondylitis. <https://clinicaltrials.gov/ct2/show/NCT05238090>.

NCT05696327 {published data only}

NCT05696327. Eccentric Tyler twist wrist extensor strengthening in lateral epicondylitis. <https://clinicaltrials.gov/study/NCT05696327>.

NCT06087081 {unpublished data only}

NCT06087081. Mills manipulation and Mulligan PRP affect pain, grip strength and function on lateral epicondylitis. <https://classic.clinicaltrials.gov/ct2/show/NCT06087081> 2023.

Nilsson 2007 {published data only}

* Nilsson P, Thom E, Baigi A, Marklund B, Mansson J. A prospective pilot study of a multidisciplinary home training programme for lateral epicondylitis. *Musculoskeletal Care* 2007;**5**(1):36-50.

Nilsson 2012 {published data only}

Nilsson P, Baigi A, Sward L, Moller M, Mansson J. Lateral epicondylalgia: a structured programme better than corticosteroids and NSAID. *Scandinavian Journal of Occupational Therapy* 2012;**19**(5):404-10.

Nishizuka 2017 {published data only}

Nishizuka T, Iwatsuki K, Kurimoto S, Yamamoto M, Hirata H. Efficacy of a forearm band in addition to exercises compared with exercises alone for lateral epicondylitis: a multicenter, randomized, controlled trial. *Journal of Orthopaedic Science* 2017;**22**(2):289-94.

NTR7223 {published data only}

NTR7223. The effect of US-guided percutaneous needle tenotomy and physiotherapy in lateral elbow tendinopathy (tennis elbow). <https://www.trialregister.nl/trial/6432>.

Park 2010 {published data only}

Park JY, Park HK, Choi JH, Moon ES, Kim BS, Kim WS, et al. Prospective evaluation of the effectiveness of a home-based program of isometric strengthening exercises: 12-month follow-up. *Clinics in Orthopedic Surgery* 2010;**2**(3):173-8.

Paungmali 2003 {published data only}

Paungmali A, O'Leary S, Souvlis T, Vicenzino B. Hypoalgesic and sympathoexcitatory effects of mobilization with movement for lateral epicondylalgia. *Physical Therapy* 2003;**83**(4):374-83.

Peterson 2014 {published data only}

NCT00888225. Tennis elbow trial. <https://clinicaltrials.gov/show/NCT00888225>.

* Peterson M, Butler S, Eriksson M, Svardsudd K. A randomized controlled trial of eccentric vs. concentric graded exercise in

chronic tennis elbow (lateral elbow tendinopathy). *Clinical Rehabilitation* 2014;**28**(9):862-72.

Pienimäki 1996 {published data only}

Pienimäki TT, Tarvainen TK, Siira PT, Vanharanta H. Progressive strengthening and stretching exercises and ultrasound for chronic lateral epicondylitis. *Physiotherapy* 1996;**82**(9):522-30.

Radpasand 2009 {published data only}

Radpasand M, Owens E. Combined multimodal therapies for chronic tennis elbow: pilot study to test protocols for a randomized clinical trial. *Journal of Manipulative & Physiological Therapeutics* 2009;**32**(7):571-85.

Ramteke 2020 {published data only}

Ramteke S, Samal S. To study the effect of rotator cuff exercises on tennis elbow. *Indian Journal of Public Health Research & Development* 2020;**11**(3):611-4.

Sarkar 2013 {published data only}

Sarkar B, Das PG, Equebal A, Mitra PK, Kumar R, Anwer S. Efficacy of low-energy extracorporeal shockwave therapy and a supervised clinical exercise protocol for the treatment of chronic lateral epicondylitis: a randomised controlled study. *Hong Kong Physiotherapy Journal* 2013;**31**(1):19-24.

Sethi 2018 {published data only}

Sethi K, Noohu MM. Scapular muscles strengthening on pain, functional outcome and muscle activity in chronic lateral epicondylalgia. *Journal of Orthopaedic Science* 2018;**23**:777-82.

Sevier 2015 {published data only}

Sevier TL, Stegink-Jansen CW. Astym treatment vs. eccentric exercise for lateral elbow tendinopathy: a randomized controlled clinical trial. *PeerJ* 2015;**e967**:1-26.

Sharma 2020 {published data only}

Sunita Sharma S, Kumar P, Chaudhuri P, Malhotra N, Narula H, Chahal A. Is wrist manipulation and low-level laser therapy collectively beneficial for patients with lateral epicondylitis? *Indian Journal of Public Health Research & Development* 2020;**11**(5):489-95.

Slater 2006 {published data only}

Slater H, Arendt-Nielsen L, Wright A, Graven-Nielsen T. Effects of a manual therapy technique in experimental lateral epicondylalgia. *Manual Therapy* 2006;**11**(2):107-17.

Solveborn 1997a {published data only}

Solveborn SA. Radial epicondylalgia ('tennis elbow'): treatment with stretching or forearm band. A prospective study with long-term follow-up including range-of-motion measurements. *Scandinavian Journal of Medicine & Science in Sports* 1997;**7**(4):229-37.

Stasinopoulos 2006 {published data only}

Stasinopoulos D, Stasinopoulos I. Comparison of effects of Cyriax physiotherapy, a supervised exercise programme and polarized polychromatic non-coherent light (Biopton light) for the treatment of lateral epicondylitis. *Clinical Rehabilitation* 2006;**20**(1):12-23.

Stasinopoulos 2009 {published data only}

Stasinopoulos D, Stasinopoulos I, Pantelis M, Stasinopoulou K. Comparing the effects of exercise program and low-level laser therapy with exercise program and polarized polychromatic non-coherent light (bioptron light) on the treatment of lateral elbow tendinopathy. *Photomedicine and Laser Surgery* 2009;**27**(3):513-20.

Stasinopoulos 2010 {published data only}

Stasinopoulos D, Stasinopoulos I, Pantelis M, Stasinopoulou K. Comparison of effects of a home exercise programme and a supervised exercise programme for the management of lateral elbow tendinopathy. *British Journal of Sports Medicine* 2010;**44**(8):579-83.

Stasinopoulos 2017 {published data only}

Stasinopoulos D, Stasinopoulos I. Comparison of effects of eccentric training, eccentric-concentric training, and eccentric-concentric training combined with isometric contraction in the treatment of lateral elbow tendinopathy. *Journal of Hand Therapy* 2017;**30**(1):13-9.

Struijs 2003 {published data only}

Struijs PA, Damen PJ, Bakker EW, Blankevoort L, Assendelft WJ, van Dijk CN. Manipulation of the wrist for management of lateral epicondylitis: a randomized pilot study. *Physical Therapy* 2003;**83**(7):608-16.

Svernlöv 2001a {published data only}

Svernlöv B, Adolfsson L. Non-operative treatment regime including eccentric training for lateral humeral epicondylalgia. *Scandinavian Journal of Medicine & Science in Sports* 2001;**11**(6):328-34.

TCTR20200612001 {published data only}

TCTR20200612001. Short-term effects of mindfulness deep transverse friction on tennis elbow. <https://www.thaicalclinicaltrials.org/show/TCTR20200612001>.

Tyler 2010 {published data only}

Tyler TF, Thomas GC, Nicholas SJ, McHugh MP. Addition of isolated wrist extensor eccentric exercise to standard treatment for chronic lateral epicondylitis: a prospective randomized trial. *Journal of Shoulder & Elbow Surgery* 2010;**19**(6):917-22.

Tyrdal 1998 {published data only}

Tyrdal S, Pettersen OJ. The effect of strength training on 'handball goalie's elbow'--a prospective uncontrolled clinical trial. *Scandinavian Journal of Medicine & Science in Sports* 1998;**8**(1):33-41.

Verhaar 1996 {published data only}

Verhaar JA, Walenkamp GH, van Mameren H, Kester AD, van der Linden AJ. Local corticosteroid injection versus Cyriax-type physiotherapy for tennis elbow. *Journal of Bone & Joint Surgery - British Volume* 1996;**78**(1):128-32.

Vicenzino 1996 {published data only}

Vicenzino B, Collins D, Wright A. The initial effects of a cervical spine manipulative physiotherapy treatment on the pain and dysfunction of lateral epicondylalgia. *Pain* 1996;**68**(1):69-74.

Vicenzino 2001 {published data only}

Vicenzino B, Paungmali A, Buratowski S, Wright A. Specific manipulative therapy treatment for chronic lateral epicondylalgia produces uniquely characteristic hypoalgesia. *Manual Therapy* 2001;**6**(4):205-12.

Viswas 2012 {published data only}

Viswas R, Ramachandran R, Korde Anantkumar P. Comparison of effectiveness of supervised exercise program and Cyriax physiotherapy in patients with tennis elbow (lateral epicondylitis): a randomized clinical trial. *The Scientific World Journal* 2012;**939645**:1-8.

Wegener 2016 {published data only}

Wegener RL, Brown T, O'Brien L. A randomized controlled trial of comparative effectiveness of elastic therapeutic tape, sham tape or eccentric exercises alone for lateral elbow tendinosis. *Hand Therapy* 2016;**21**(4):131-9.

Zunke 2020 {published data only}

DRKS00013964. Manual mobilization of the thoracic spine: effect on pain free grip and sympathetic nervous system activity on patients with lateral epicondylitis humeri. https://www.drks.de/drks_web/navigate.do?navigationId=trial.HTML&TRIAL_ID=DRKS00013964.

* Zunke P, Auffarth A, Hitzl W, Moursy M. The effect of manual therapy to the thoracic spine on pain-free grip and sympathetic activity in patients with lateral epicondylalgia humeri. A randomized, sample sized planned, placebo-controlled, patient-blinded monocentric trial. *BMC Musculoskeletal Disorders* 2020;**21**(186):1-11.

References to studies awaiting assessment
Bartlett 2007 {published data only}

ISRCTN89551492. An investigation of the effectiveness of a mobilisation with movement (MWM) technique for lateral epicondylalgia on pain and function in clinical practice. <http://www.isrctn.com/ISRCTN89551492> 2007.

References to ongoing studies
Prat 2018 {published data only}

Prat P I, Cibrowski D, Zuliani A, Stecco A. Efficacy of fascial manipulation and eccentric exercise for lateral elbow pain. In: *Journal of Bodywork & Movement Therapies*. Vol. 22. 2018:845-72.

Sahebamei 2019 {published data only}

IRCT20180812040773N1. The role of elbow exercises protocol in treatment of patients with lateral epicondylitis (tennis elbow). <https://en.irct.ir/trial/33707> 2019.

Sveinall 2022 {published data only}

NCT04803825. The NORwegian Tennis Elbow (NOTE) Study. <https://clinicaltrials.gov/study/NCT04803825> 2022.

Takamjani 2019 {published data only}

IRCT20190315043058N1. The effect of Mulligan's techniques along with routine physiotherapy, in comparison with routine physiotherapy alone; on pain, grip strength, and function in patients diagnosed with chronic lateral epicondylitis. <https://en.irct.ir/trial/38410> 2019.

Additional references
AAOS 2020

American Academy of Orthopaedic Surgeons. Tennis elbow (lateral epicondylitis). <https://orthoinfo.aaos.org/en/diseases--conditions/tennis-elbow-lateral-epicondylitis/> (accessed 6 May 2022).

Abrams 2012

Abrams GD, Renstrom PA, Safran MR. Epidemiology of musculoskeletal injury in the tennis player. *British Journal of Sports Medicine* 2012;**46**:492-8.

Alizadehkhayat 2007

Alizadehkhayat O, Fisher AC, Kemp GJ, Frostick SP. Pain, functional disability, and psychologic status in tennis elbow. *Clinical Journal of Pain* 2007;**23**:482-9.

APA 2015

Australian Physiotherapy Association. What is physiotherapy? www.physiotherapy.asn.au (accessed 2 April 2015).

Bisset 2005

Bisset L, Paungmali A, Vicenzino B, Beller E. A systematic review and meta-analysis of clinical trials on physical interventions for lateral epicondylalgia. *British Journal of Sports Medicine* 2005;**39**(7):411-22.

Buchbinder 2005

Buchbinder R, Green SE, Youd JM, Assendelft WJ, Barnsley L, Smidt N. Shock wave therapy for lateral elbow pain. *Cochrane Database of Systematic Reviews* 2005, Issue 4. Art. No: CD003524. [DOI: [10.1002/14651858.CD003524](https://doi.org/10.1002/14651858.CD003524)]

Buchbinder 2011

Buchbinder R, Johnston RV, Barnsley L, Assendelft WJJ, Bell SN, Smidt N. Surgery for lateral elbow pain. *Cochrane Database of Systematic Reviews* 2011, Issue 3. Art. No: CD003525. [DOI: [10.1002/14651858.CD003525.pub2](https://doi.org/10.1002/14651858.CD003525.pub2)]

Busija 2020

Busija L, Ackerman IN, Haas R, Wallis J, Nolte S, Bentley S, et al. Adult measures of general health and health-related quality of life. *Arthritis Care & Research* 2020;**72**:522-64.

Caspersen 1985

Caspersen CJ, Powell KE, Christenson GM. Physical activity, exercise, and physical fitness: definitions and distinctions for health-related research. *Public Health Reports* 1985;**100**(2):126-31.

Clarke 2010

Clarke AW, Ahmad M, Curtis M, Connell DA. Lateral elbow tendinopathy: correlation of ultrasound findings with pain and functional disability. *American Journal of Sports Medicine* 2010;**38**:1209-14.

Coombes 2010

Coombes BK, Bisset L, Vicenzino B. Efficacy and safety of corticosteroid injections and other injections for management of tendinopathy: a systematic review of randomised controlled trials. *Lancet* 2010;**376**(9754):1751-67.

Coombes 2015

Coombes BK, Bisset L, Vicenzino B. Management of lateral elbow tendinopathy: one size does not fit all. *Journal of Orthopaedic & Sports Physical Therapy* 2015;**45**(11):938-49.

Coombes 2016

Coombes BK, Wiebusch M, Heales L, Stephenson A, Vicenzino B. Isometric exercise above but not below an individual's pain threshold influences pain perception in people with lateral epicondylalgia. *Clinical Journal of Pain* 2016;**32**(12):1069-75.

Deeks 2022

Deeks JJ, Higgins JPT, Altman DG (editors). Chapter 10: Analysing data and undertaking meta-analyses. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.3 (updated February 2022). Cochrane, 2022.. Available from www.training.cochrane.org/handbook.

Frank 1984

Frank C, Akeson WH, Woo SL, Amiel D, Coutts RD. Physiology and therapeutic value of passive joint motion. *Clinical Orthopaedics and Related Research* 1984;**185**:113-25.

Fu 2018

Fu MC, Ellenbecker TS, Renstrom PA, Windler GS, Dines DM. Epidemiology of injuries in tennis players. *Current Reviews in Musculoskeletal Medicine* 2018;**11**:1-5.

Giray 2019

Giray E, Karali-Bingul D, Akyuz G. The effectiveness of kinesiotope, sham taping or exercises only in lateral epicondylitis treatment: a randomized controlled study. *PM R* 2019;**11**(7):681-93.

GRADEpro GDT 2015 [Computer program]

GRADEpro GDT. Hamilton (ON): McMaster University (developed by Evidence Prime), 2015. Available at gradepr.org.

Green 2002

Green S, Buchbinder R, Barnsley L, Hall S, White M, Smidt N, et al. Acupuncture for lateral elbow pain. *Cochrane Database of Systematic Reviews* 2002, Issue 1. Art. No: CD003527. [DOI: [10.1002/14651858.CD003527](https://doi.org/10.1002/14651858.CD003527)]

Herd 2008

Herd CR, Meserve BB. A systematic review of the effectiveness of manipulative therapy in treating lateral epicondylalgia. *Journal of Manual and Manipulative Therapy* 2008;**16**(4):225-37.

Higgins 2017

Higgins JP, Altman DG, Sterne JAC, editor(s). Chapter 8: Assessing risk of bias in included studies. In: Higgins JPT, Churchill R, Chandler J, Cumpston MS (editors), *Cochrane Handbook for Systematic Reviews of Interventions* version 5.2.0 (updated June 2017). The Cochrane Collaboration, 2017. Available from training.cochrane.org/handbook/archive/v5.2/.

Higgins 2022

Higgins JPT, Li T, Deeks JJ (editors). Chapter 6: Choosing effect measures and computing estimates of effect. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.3 (updated February 2022). Cochrane, 2022. Available from www.training.cochrane.org/handbook.

Higgins 2023

Higgins JPT, Li T, Deeks JJ (editors). Chapter 6: Choosing effect measures and computing estimates of effect. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.4 (updated August 2023). Cochrane, 2023. Available from www.training.cochrane.org/handbook.

Hoogvliet 2013

Hoogvliet P, Randsdorp MS, Dingemanse R, Koes BW, Huisstede BMA. Does effectiveness of exercise therapy and mobilisation techniques offer guidance for the treatment of lateral and medial epicondylitis? A systematic review. *British Journal of Sports Medicine* 2013;**47**:1112-9.

Hudak 1996

Hudak PL, Amadio PC, Bombardier C. Development of an upper extremity outcome measure: the DASH (disabilities of the arm, shoulder and hand). The Upper Extremity Collaborative Group (UECG). *American Journal of Industrial Medicine* 1996;**29**(6):602-8. [DOI: [10.1002/\(SICI\)1097-0274\(199606\)29:6<602::AID-AJIM4>3.0.CO;2-L](https://doi.org/10.1002/(SICI)1097-0274(199606)29:6<602::AID-AJIM4>3.0.CO;2-L)]

Hume 2006

Hume PA, Reid D, Edwards T. Epicondylar injury in sport: epidemiology, type, mechanisms, assessment, management and prevention. *Sports Medicine* 2006;**36**:151-70.

Ikonen 2022

Ikonen J, Lähdeoja T, Arden CL, Buchbinder R, Reito A, Karjalainen TV. Persistent tennis elbow symptoms have little prognostic value - the frustrating imperative to wait-and-see: a systematic review with meta-analysis. *Clinical Orthopaedics and Related Research* 2022;**480**(4):647-60.

Javed 2016

Javed M, Mustafa S, Boyle S, Scott F. Elbow pain: a guide to assessment and management in primary care. *British Journal of General Practice* 2016;**65**(640):610-2.

Johns 2020

Johns N, Shridhar V. Lateral epicondylitis: current concepts. *Australian Journal of General Practice* 2020;**49**(11):707-9.

Karanasios 2020

Karanasios S, Korakakis V, Whiteley R, Vasilogeorgis I, Woodbridge S, Giotfos G. Exercise interventions in lateral elbow tendinopathy have better outcomes than passive interventions, but the effects are small: a systematic review and meta-analysis of 2123 subjects in 30 trials. *British Journal of Sports Medicine* 2020;**0**:1-10.

Karjalainen 2021

Karjalainen TV, Silagy M, O'Bryan E, Johnston RV, Cyril S, Buchbinder R. Autologous blood and platelet-rich plasma injection therapy for lateral elbow pain. *Cochrane Database of Systematic Reviews* 2021, Issue 9. Art. No: CD010951. [DOI: [10.1002/14651858.CD010951.pub2](https://doi.org/10.1002/14651858.CD010951.pub2)]

Karjalainen 2023

Karjalainen T, Buchbinder R. Is it time to reconsider the indications for surgery in patients with tennis elbow? *The Bone & Joint Journal* 2023;**105-B**(2):109-11.

Kim 2020

Kim YJ, Wood SM, Yoon AP, Howard JC, Yang LY, Chung KC. Efficacy of nonoperative treatments for lateral epicondylitis: a systematic review and meta-analysis. *Hand/Peripheral Nerve* 2020;**147**:112-25.

Kohia 2008

Kohia M, Brackley J, Byrd K, Jennings A, Murray W, Wilfong E. Effectiveness of physical therapy treatments on lateral epicondylitis. *Journal of Sport Rehabilitation* 2008;**17**(2):119-36.

Konarski 2023

Konarski W, Poboży T. A clinical overview of the natural course and management of lateral epicondylitis. *Orthopedics* 2023;**46**(4):e210-18. [DOI: [10.3928/01477447-20230329-05](https://doi.org/10.3928/01477447-20230329-05)]

Lo 2007

Lo MY, Safran MR. Surgical treatment of lateral epicondylitis: a systematic review. *Clinical Orthopaedics and Related Research* 2007;**463**:98-106.

Mallen 2009

Mallen CD, Chesterton LS, Hay EM. Tennis elbow. *BMJ* 2009;**339**:b3180.

McDowell 2014

McDowell JM, Johnson GM, Hetherington BH. Mulligan concept manual therapy: standardizing annotation. *Manual Therapy* 2014;**19**(5):499-503.

Newcomer 2005

Newcomer KL, Martinez-Silvestrini JA, Schaefer MP, Gay R, Arendt K. Sensitivity of the patient-rated forearm evaluation questionnaire in lateral epicondylitis. *Journal of Hand Therapy* 2005;**18**:400-6.

Nirschl 1979

Nirschl RP, Pettrone FA. Tennis elbow. The surgical treatment of lateral epicondylitis. *Journal of Bone and Joint Surgery* 1979;**61**(6A):832-9.

Overend 1999

Overend T, Wouri-Fearn J, Kramer J, MacDermaid J. Reliability of a patient rated forearm evaluation questionnaire for patients with lateral epicondylitis. *Journal of Hand Therapy* 1999;**12**:31-7.

Page 2014a

Page MJ, Green S, Kramer S, Johnston RV, McBain B, Chau M, et al. Manual therapy and exercise for adhesive capsulitis (frozen shoulder). *Cochrane Database of Systematic Reviews* 2014, Issue 8. Art. No: CD011275. [DOI: [10.1002/14651858.CD011275](https://doi.org/10.1002/14651858.CD011275)]

Page 2014b

Page MJ, Green S, Kramer S, Johnston RV, McBain B, Buchbinder R. Electrotherapy modalities for adhesive capsulitis (frozen shoulder). *Cochrane Database of Systematic Reviews* 2014, Issue 10. Art. No: CD011324. [DOI: [10.1002/14651858.CD011324](https://doi.org/10.1002/14651858.CD011324)]

Page 2016a

Page MJ, Green S, McBain B, Surace SJ, Deitch J, Lyttle N, et al. Manual therapy and exercise for rotator cuff disease. *Cochrane Database of Systematic Reviews* 2016, Issue 6. Art. No: CD012224. [DOI: [10.1002/14651858.CD012224](https://doi.org/10.1002/14651858.CD012224)]

Page 2016b

Page MJ, Green S, Mrocki MA, Surace SJ, Deitch J, McBain B, et al. Electrotherapy modalities for rotator cuff disease. *Cochrane Database of Systematic Reviews* 2016, Issue 6. Art. No: CD012225. [DOI: [10.1002/14651858.CD012225](https://doi.org/10.1002/14651858.CD012225)]

Page 2022

Page MJ, Higgins JPT, Sterne JAC. Chapter 13: Assessing risk of bias due to missing results in a synthesis. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.3 (updated February 2022). Cochrane, 2022. Available from www.training.cochrane.org/handbook.

Pattanittum 2013

Pattanittum P, Turner T, Green S, Buchbinder R. Non-steroidal anti-inflammatory drugs (NSAIDs) for treating lateral elbow pain in adults. *Cochrane Database of Systematic Reviews* 2013, Issue 5. Art. No: CD003686. [DOI: [10.1002/14651858.CD003686.pub2](https://doi.org/10.1002/14651858.CD003686.pub2)]

Paungmali 2003

Paungmali A, O'Leary S, Souvlis T, Vicenzino B. Hypoalgesic and sympathoexcitatory effects of mobilization with movement for lateral epicondylalgia. *Physical Therapy* 2003;**83**(4):374-83.

Pienimäki 1996

Pienimäki T, Tarveinen T, Siira P, Vanharanta H. Progressive strengthening and stretching exercises and ultrasound for chronic lateral epicondylitis. *Physiotherapy* 1996;**82**:522-30.

Poltawski 2011

Poltawski L, Watson T. Measuring clinically important change with the Patient-rated Tennis Elbow Evaluation. *Hand Therapy* 2011;**16**(3):52-7.

Pompilio da Silva 2018

Pompilio da Silva M, Tamaoki MJS, Blumetti FC, Belloti JC, Smidt N, Buchbinder R. Electrotherapy modalities for lateral elbow pain. *Cochrane Database of Systematic Reviews* 2018, Issue 6. Art. No: CD013041. [DOI: [10.1002/14651858.CD013041](https://doi.org/10.1002/14651858.CD013041)]

PRISMA Statement 2020

Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;**372**:n71. [DOI: [10.1136/bmj.n71](https://doi.org/10.1136/bmj.n71)]

Regan 1992

Regan W, Wold LE, Coonrad R, Morrey BF. Microscopic histopathology of chronic refractory lateral epicondylitis. *American Journal of Sports Medicine* 1992;**20**(6):746-9.

RevMan Web 2020 [Computer program]

Review Manager Web (RevMan Web). Version 3.1.2. The Cochrane Collaboration, 2020. Available at revman.cochrane.org.

Rompe 2007

Rompe JD, Overend TJ, MacDermid JC. Validation of the Patient-rated Tennis Elbow Evaluation Questionnaire. *Journal of Hand Therapy* 2007;**20**:3-10.

Sanders 2015

Sanders TL, Maradit Kremers H, Bryan AJ, Ransom JE, Smith J, Morrey BF. The epidemiology and health care burden of tennis elbow: a population-based study. *American Journal of Sports Medicine* 2015;**43**:1066-107.

Schünemann 2022

Schünemann HJ, Higgins JPT, Vist GE, Glasziou P, Akl EA, Skoetz N, et al. Chapter 14: Completing 'Summary of findings' tables and grading the certainty of the evidence. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.3 (updated February 2022). Cochrane, 2022. Available from www.training.cochrane.org/handbook.

Schünemann 2022a

Schünemann HJ, Vist GE, Higgins JPT, Santesso N, Deeks JJ, Glasziou P, et al. Chapter 15: Interpreting results and drawing conclusions. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.3 (updated February 2022). Cochrane, 2022. Available from www.training.cochrane.org/handbook.

Seiça 2023

Seiça EC, Buruian A, Gameiro D, Peixoto D, Pereira C. Lateral epicondylitis: current concepts in pathology, investigation, and management. *SN Comprehensive Clinical Medicine* 2023;**5**(65):1-9.

Shiri 2011

Shiri R, Viikari-Juntura E. Lateral and medial epicondylitis: role of occupational factors. *Best Practice & Research Clinical Rheumatology* 2011;**25**:43-57.

Silverstein 2006

Silverstein BA, Adam D. Work-related disorders of the back and upper extremity in Washington State, 1989-1996. Olympia, Washington: SHARP Program, Washington State Department of Labor and Industries 2006.

Slade 2016

Slade SC, Dionne CE, Underwood M, Buchbinder R. Consensus on Exercise Reporting Template (CERT): explanation and elaboration statement. *British Journal of Sports Medicine* 2016;**50**:1428-37.

Smidt 2003

Smidt N, Assendelft WJ, Arola H, Malmivaara A, Green S, Buchbinder R, et al. Effectiveness of physiotherapy for lateral epicondylitis: a systematic review. *Annals of Medicine* 2003;**35**(1):51-62.

Stahl 1997

Stahl S, Kaufman T. The efficacy of an injection of steroids for medial epicondylitis. A prospective study of sixty elbows. *Journal of Bone and Joint Surgery* 1997;**79**:1648-52.

Stata 2017 [Computer program]

Stata. Version 15. College Station, TX, USA: StataCorp, 2017. Available at www.stata.com.

Struijs 2002

Struijs PA, Smidt N, Arola H, Dijk CN van, Buchbinder R, Assendelft WJJ. Orthotic devices for the treatment of tennis elbow. *Cochrane Database of Systematic Reviews* 2002, Issue 1. Art. No: CD001821. [DOI: [10.1002/14651858.CD001821](https://doi.org/10.1002/14651858.CD001821)]

Sullivan 1993

Sullivan M, Karlsson J, Bengtsson C, Furunes B, Lapidus L, Lissner L. 'The Goteborg Quality of Life Instrument' — a psychometric evaluation of assessments of symptoms and well-being among women in a general population. *Scandinavian Journal of Primary Health Care* 1993;**11**:267-75.

Vicenzino 2003

Vicenzino B. Lateral epicondylalgia: a musculoskeletal physiotherapy perspective. *Manual Therapy* 2003;**8**:66-79.

Vicenzino 2011

Vicenzino B, Hing W, Hall T, Rivett D. Mobilisation with Movement: the Art and the Science. Elsevier Australia, 2011.

Walker-Bone 2012

Walker-Bone K, Palmer KT, Reading I, Coggon D, Cooper C. Occupation and epicondylitis: a population-based study. *Rheumatology* 2012;**51**:305-10.

Wan 2014

Wan X, Wang W, Liu J, Tong T. Estimating the sample mean and standard deviation from the sample size, median, range and/or interquartile range. *BMC Medical Research Methodology* 2014;**14**(135):1-13.

Westad 2019

Westad K, Tjoestolvsen F, Hebron C. The effectiveness of Mulligan's mobilisation with movement (MWM) on peripheral joints in musculoskeletal (MSK) conditions: a systematic review. *Musculoskeletal Science and Practice* 2019;**39**:157-63.

Yanai 2024

Yanai K, Tajika T, Arisawa S, Hatori Y, Honda A, Hasegawa S, et al. Prevalence and factors associated with lateral epicondylitis among hospital healthcare workers. *JSES International* (in press):1-6.

References to other published versions of this review

Frydman 2018

Frydman A, Johnston RV, Smidt N, Green S, Buchbinder R. Manual therapy and exercise for lateral elbow pain. *Cochrane Database of Systematic Reviews* 2018, Issue 6. Art. No: CD013042. [DOI: [10.1002/14651858.CD013042](https://doi.org/10.1002/14651858.CD013042)]

* Indicates the major publication for the study

CHARACTERISTICS OF STUDIES

Characteristics of included studies [ordered by study ID]

Amro 2010

Study characteristics

Methods	Design: parallel, 2-arm, quasi-randomised trial (assignment by alternation)
	Setting: primary care, West Bank/Gaza, Palestine
	Timing: not reported
	Interventions: (1) manual therapy (Mulligan) and exercise; (2) exercise; 4-week intervention period
	Sample size: 34 participants were required, no sample size calculation method was reported
	Intention-to-treat analysis: not reported

Manual therapy and exercise for lateral elbow pain (Review)

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Amro 2010 (Continued)

Participants

Number of participants

- Number of participants assessed for eligibility not reported
- Number of participants excluded not reported
- Data for 34 participants available at 4-week follow-up (17 for manual therapy, taping, exercise and ultrasound, 17 for exercise and ultrasound only)

Inclusion criteria

- Medical referral received for subacute lateral elbow pain
- Positive results on 2 or more physical examination tests: active wrist extension test, Cozen's test, Thomsen test, resisted middle finger extension test, Mill's test or passive stretching

Exclusion criteria

- Lateral elbow pain due to cervical pathologies
- Acute lateral elbow pain
- Post-traumatic lateral elbow pain

Baseline characteristics

Manual therapy and exercise (n = 17 participants)

Mean age: 37.8 years (SD not reported); 4 female, 13 male

Mean (SD) pain (/100): 63 (11)

Mean (SD) disability using Patient-Rated Tennis Elbow Evaluation questionnaire (/100): 51.7 (14.5) units

Mean (SD) maximum grip strength: 35.8 (13) kg

Exercise (n = 17 participants)

Mean age: 36.8 years (SD not reported); 6 female, 11 male

Mean (SD) pain (/100): 66 (14) units

Mean (SD) disability using Patient-Rated Tennis Elbow Evaluation questionnaire (/100): 57.7 (9.7) unit

Mean (SD) maximum grip strength: 29 (16.7) kg.

Interventions

Manual therapy and exercise

Components of intervention: manual therapy - Mulligan mobilisation with movement technique that involved a physiotherapist performing a sustained lateral glide of the forearm (mobilisation component) followed by the participant making a fist (movement component); *exercise* - refer to exercise component of exercise group. Following the Mulligan technique, taping was applied (Hypofix and rigid Leuko-tape) on the origin of extensor carpi radialis with the elbow in slight flexion, forearm in pronation, and a lateral glide was applied to the extensor muscles group.

Dose: manual therapy - the technique was applied 36 times, with a brief rest period (a few seconds) after 12 repetitions. Each participant received 3 treatment sessions per week, for 4 weeks, with each session about 30 to 45 minutes; *exercise* - refer to exercise component of exercise group.

Supervision: physiotherapists delivered the interventions and received training from the researchers. Exercises performed at home.

Additional treatment: none

Exercise

Components of intervention: strengthening and stretching exercises

Dose: no dosage details were reported

Amro 2010 (Continued)

Supervision: physiotherapists delivered the interventions and received training from the researchers. Exercises performed at home.

Additional treatment: none

Co-interventions

Participants in both groups received the same combination of exercise (stretching and strengthening), plus heat therapy, massage and ultrasound from physiotherapists with 3 treatment sessions per week, for 4 weeks, with each session about 30 to 45 minutes.

Outcomes	Outcomes were reported at 4 weeks (end of intervention). Outcomes • Pain, 0 to 10 VAS (0 = no pain) • Disability, 0 to 100 using PRTEE questionnaire (0 = no disability) • Number of participants who thought the treatment was effective • Maximum grip strength (kg) measured using a hand grip dynamometer Outcomes included in this review <i>Major outcomes</i> • Mean pain • Mean disability • Participant-reported treatment success <i>Minor outcomes</i> • Mean maximum grip strength (kg)
Source of funding	Not reported
Notes	Trial registration: not registered Data analysis: we used data at 4 weeks (end of intervention) for Analysis 6 - Manual therapy and prescribed exercises versus prescribed exercises alone. Adverse events: not reported Withdrawals: not reported
Risk of bias	
Bias	Authors' judgement Support for judgement
Random sequence generation (selection bias)	High risk Participants were alternately allocated to the experimental or control arm.
Allocation concealment (selection bias)	High risk Allocation was not concealed.
Blinding of participants and personnel (performance bias) All outcomes	High risk Participants and study personnel were not blind to treatment allocation.
Blinding of outcome assessment - self reported (detection bias)	High risk Participants not blind to treatment allocation, and measures of pain, disability and global assessment were likely influenced by lack of blinding.

Amro 2010 (Continued)

Blinding of outcome assessment - assessor reported outcomes	High risk	The assessor was not blinded to treatment allocation, and measurement of grip strength was likely influenced by lack of blinding.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	We assumed all 34 patients completed the trial and were analysed, but this was not explicitly stated, so we judged there to be unclear risk of incomplete outcome data.
Selective reporting (reporting bias)	Unclear risk	There was no trial registration, or published protocol available, so we judged this as an unclear risk of selective reporting.
Other bias	Low risk	The study appears to be free of other sources of bias.

Bagcaci 2023

Study characteristics

Methods	Design: parallel group, 3-arm randomised controlled trial Setting: Department of Physiotherapy and Rehabilitation, KTO Karatay University, Konya, Turkey Interventions: (1) manual therapy (Mulligans's mobilisation with movement) and prescribed exercises; (2) manual therapy (muscle energy technique) and prescribed exercises; (3) prescribed exercises; 4-week intervention period Sample size: 14 participants per group provided 90% power to detect an effect size of 0.58 for pain, with a significance level of 5% Intention-to-treat analysis: not reported
Participants	Number of participants 48 participants assessed for eligibility 3 participants excluded (3 not meeting inclusion criteria) 45 participants randomised (15 to Mulligans's and prescribed exercises; 15 to muscle energy techniques and prescribed exercises; 15 to prescribed exercises) - Data for 45 participants (100%) available at 4-week follow-up (15 to Mulligans's and prescribed exercises; 15 to muscle energy techniques and prescribed exercises; 15 to prescribed exercises) Inclusion criteria - Physician diagnosis of chronic lateral elbow pain - Pain score of at least 3 on VAS (0 to 10) Exclusion criteria - Concomitant disorders of the wrist and forearm, such as fracture or dislocation of the elbow - Treatment with glucocorticoid injections in the previous 6 months Baseline characteristics <i>Mulligans's and prescribed exercises (n = 15 participants)</i> Mean (SD) age: 41.7 (4.4) years; 8 female, 7 male Mean (SD) duration of symptoms: 6.3 (2.7) months Mean (SD) body mass index: 23.2 (1.9) kg/m²

Bagcaci 2023 (Continued)

Academic position n (%): high school 5 participants (33); university 7 (47); Masters or PhD 3 (20)

Employment n (%): unemployed 2 participants (13); employed 11 participants (73); homemaker 2 participants (13)

Muscle energy techniques and prescribed exercises (n = 15 participants)

Mean (SD) age: 44.5 (3.96) years; 8 female, 7 male

Mean (SD) duration of symptoms: 7.3 (3.1) months

Mean (SD) body mass index: 22.5 (2.8) kg/m²

Academic position n (%): high school - 3 participants (20); university - 10 participants (67); Masters or PhD - 2 participants (13)

Employment n (%): unemployed - 1 participant (7); employed - 13 participants (87); homemaker - 1 participant (7)

Prescribed exercises (n = 15 participants)

Mean (SD) age: 42.1 (4.3) years; 8 female, 7 male

Mean (SD) duration of symptoms: 7.7 (2.98) months

Mean (SD) body mass index: 21.9 (2.8) kg/m²

Academic position n (%): high school - 4 participants (27); university - 8 participants (53); Masters or PhD - 3 participants (20)

Employment n (%): unemployed - 1 participant (7); employed - 12 participants (80); homemaker - 2 participants (13)

Interventions
Mulligans's mobilisation with movement and prescribed exercises

Components of intervention: Manual therapy - First, the pain-free angle of application was determined for each patient. The lateral condyle of the humerus was fixed by the first bar space of the physiotherapist. The elbow joint was glided until no pain was felt in the elbow joint and the hand was in the contracted position. Participants were asked to repeat the movement of the elbow joint and press the ball 10 times without pain. The same procedure was performed in three sets, with each set consisting of 10 repetitions of the exercise to control pain. The pauses between sets were 15 to 20 secs, and the interval between repetitions in each set was 30 secs; *Prescribed exercises* - refer to below.

Dose: 3 times a week for a total of 12 times over 4 weeks

Supervision: experienced and certified physical therapist

Setting: Department of Physiotherapy and Rehabilitation (University)

Muscle energy technique and prescribed exercises

Components of intervention: Manual therapy - Initially, each participant performed 5 to 7 secs of isometric muscle contraction in the barrier by passive stretching by the physical therapist until the muscle or dysfunctional joint in question tolerated it. After a short rest period of 2 to 3 secs, passive stretching was continued by the physical therapist until the second barrier was reached. This continued until muscle tension or joint dysfunction disappeared. *Prescribed exercises* - refer to below.

Dose: 3 times a week, for a total of 12 times over 4 weeks

Supervision: physical therapist

Setting: Department of Physiotherapy and Rehabilitation (University)

Prescribed exercises

Bagcaci 2023 (Continued)

Components of intervention: Stretching and strengthening (eccentrics) for the forearm extensors (extensor carpi radialis brevis) in a sitting position. Static stretching involved extension of the elbow joint, pronation of the forearm and flexion of the wrist with ulnar deviation.

Dose: daily for 4 weeks; 10 repetitions (eccentrics), 10 sec holds (stretch). The weight that patients could lift at 10 maximal repetitions for eccentric training was calculated, and they were asked to work with this weight.

Supervision: prescribed by a physical therapist, performed at home

Setting: Department of Physiotherapy and Rehabilitation (University)

Outcomes

Outcomes were reported at 4 weeks (end of intervention).

Outcomes

- Change from baseline in pain at rest, 0 to 10 on VAS (0 = no pain)
- Change from baseline in pain during activity, 0 to 10 on VAS (0 = no pain)
- Change from baseline in disability, 0 to 100 using the Turkish version of the PRTEE (0 = no disability)
- Change from baseline in maximum grip strength on extension (kg) using handheld hydraulic dynamometer
- Change from baseline in maximum grip strength on flexion (kg) using handheld hydraulic dynamometer
- Change from baseline in finger strength (kg) using mechanical pinch teacher
- Participant overall withdrawals

Outcomes used in this review

Major outcomes

- Mean pain during activity (change scores)
- Mean disability (change scores)
- Proportion of overall withdrawals

Minor outcomes

- Mean maximum grip strength on extension (change scores)

Source of funding

None

Notes

Trial registration: [www.clinicaltrials.gov](https://clinicaltrials.gov/ct2/show/NCT04505215) (NCT04505215). Comparison of Mulligan technique and muscle energy technique in lateral epicondylitis. <https://clinicaltrials.gov/ct2/show/NCT04505215> 2019.

Data analysis: we used data at 4 weeks (end of intervention) for Analysis 6 - Manual therapy and prescribed exercises versus prescribed exercises. Unable to report baseline outcome scores as they were not reported. We combined Mulligan's mobilisation and muscle energy technique groups in the same meta-analysis using the formulae in Chapter 6 of the *Cochrane Handbook* to form a single pairwise comparison of manual therapy plus prescribed exercise versus prescribed exercise alone.

Adverse events: not reported

Withdrawals: in all groups, 0 participants (0%) did not complete the trial

Risk of bias

Bias

Authors' judgement

Support for judgement

Random sequence generation (selection bias)

Low risk

Block randomisation using a computer-assisted randomisation program (www.randomizer.org/) stratified by sex and age.

Bagcaci 2023 (Continued)

Allocation concealment (selection bias)	Unclear risk	Not explicitly reported. Although it is implied it was done centrally, it is not clear who had access to the randomisation sequence.
Blinding of participants and personnel (performance bias) All outcomes	High risk	While the authors reported that the researchers who performed the assessment and treatment, as well as the participants were blinded to the groups, all participants and personnel were likely aware of group allocation due to the nature of interventions delivered.
Blinding of outcome assessment - self reported (detection bias)	High risk	Measures of pain and disability were likely influenced by lack of blinding of participants and personnel.
Blinding of outcome assessment - assessor reported outcomes	High risk	The investigator who assessed grip strength was unlikely to be blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No missing data were observed.
Selective reporting (reporting bias)	Low risk	Trial registration was consistent with the reporting of outcomes measures.
Other bias	Low risk	The trial appears to be free of other sources of bias.

Bisset 2006

Study characteristics

Methods	Design: parallel group, 3-arm randomised controlled trial Setting: primary care, Brisbane, Australia Timing: March 2002 to June 2005 Interventions: (1) manual therapy and exercise; (2) glucocorticoid injection; (3) advice; 6-week intervention period Sample size: 60 participants per group provided 80% power to detect a difference of at least 25% of participant-reported global improvement between groups, assuming the minimum success rate to be 68% at 52 weeks, with a significance level at 5% Intention-to-treat analysis: yes
Participants	Number of participants • 497 participants assessed for eligibility • 299 participants excluded (96 did not meet inclusion criteria, 148 met exclusion criteria, 55 declined to participate) • 198 participants randomised (66 to manual therapy and exercise, 65 to glucocorticoid injection and 67 to advice) • Data for 184 participants (93%) available at 3-week follow-up (64 for manual therapy and exercise, 63 for glucocorticoid injection and 57 for advice) • Data for 188 participants (95%) available at 6-week follow-up (63 for manual therapy and exercise, 65 for glucocorticoid injection and 60 for advice) • Data for 182 participants (92%) available at 3-month follow-up (58 for manual therapy and exercise, 65 for glucocorticoid injection and 59 for advice)

Bisset 2006 (Continued)

- Data for 181 participants (91%) available at 6-month follow-up (59 for manual therapy and exercise, 64 for glucocorticoid injection and 58 for advice)
- Data for 190 participants (96%) available at 12-month follow-up (63 for manual therapy and exercise, 65 for glucocorticoid injection and 62 for advice)

Inclusion criteria

- Age 18 to 65 years
- Pain over the lateral elbow that increased on palpation of the lateral epicondyle, gripping, resisted extension of the wrist, second or third finger
- Pain of at least 6 weeks duration

Exclusion criteria

- Any treatment of the elbow pain by a healthcare practitioner within the preceding 6 months
- Bilateral elbow symptoms
- Cervical radiculopathy
- Any other elbow joint pathology
- Peripheral nerve involvement
- Previous surgery to the elbow
- A history of dislocation, fracture of the elbow or tendon ruptures
- Systemic or neurological disorders
- Shoulder, wrist and hand pathology
- Contraindications to corticosteroids

Baseline characteristics

Manual therapy and exercise (n = 66 participants)

Mean (SD) age: 47.9 (7.2) years; 21 female, 45 male

Median (IQR) duration of symptoms: 16 (11 to 35) weeks

Dominant elbow affected: 42 participants (64%)

Previous episodes of lateral elbow pain: 14 participants (21%)

Alleged cause: overuse (usual activities) - 2 participants (3%); overuse (unusual activities) - 18 participants (27%); other (such as sport, unexpected movement) - 12 (18%); unknown - 34 (52%)

Employment: unemployed - 9 participants (14%); non-manual work - 36 participants (66%); manual work - 21 participants (32%)

Mean (SD) pain severity in previous week (/100): 57.5 (25.0) units

Mean (SD) disability using Pain Free Function questionnaire (/100): 75.4 (18.9) units

Mean (SD) assessor rating of severity (/100): 51.6 (19.3) units

Mean (SD) pain-free grip force ratio (affected side/unaffected side x 100): 42.4 (20.8) units

Glucocorticoid injection (n = 65 participants)

Mean (SD) age: 47.8 (8.2) years; 25 female, 40 male

Median (IQR) duration of symptoms: 26 (12 to 42) weeks

Dominant elbow affected: 49 participants (75%)

Previous episodes of lateral elbow pain: 19 participants (29%)

Bisset 2006 (Continued)

Alleged cause: overuse (usual activities) - 2 participants (3%); overuse (unusual activities) - 12 participants (18%); other (such as sport, unexpected movement) - 11 participants (17%); unknown - 40 participants (62%)

Employment: unemployed - 12 participants (18%); non-manual work - 32 participants (50%); manual work - 21 participants (32%)

Mean (SD) pain severity in previous week (/100): 53.5 (23.0) units

Mean (SD) disability using Pain Free Function questionnaire (/100): 80.8 (16.1) units

Mean (SD) assessor rating of severity (/100): 58.0 (17.3) units

Mean (SD) pain-free grip force ratio (affected side/unaffected side x 100): 38.4 (17.1) units

Advice (n = 67 participants)

Mean (SD) age: 47.3 (8.1) years; 24 female, 43 male

Median (IQR) duration of symptoms: 26 (10 to 42) weeks

Dominant elbow affected: 42 participants (63%)

Previous episodes of lateral elbow pain: 22 participants (33%)

Alleged cause: overuse (usual activities) - 4 participants (6%); overuse (unusual activities) - 8 participants (12%); other (such as sport, unexpected movement) - 16 participants (24%); unknown - 39 participants (58%)

Employment: unemployed - 10 participants (15%); non-manual work - 30 participants (45%); manual work: 27 participants (40%)

Mean (SD) pain severity in the previous week (/100): 61.3 (22.6) units

Mean (SD) disability using Pain Free Function questionnaire (/100): 76.7 (21.3) units

Mean (SD) assessor rating of severity (/100): 51.9 (19.5) units

Mean (SD) pain-free grip force ratio (affected side/unaffected side x 100): 48.0 (21.5) units

Interventions

Manual therapy and exercise

Components of intervention: manual therapy - as described by [Vincenzino 2003](#), mobilisation with movement techniques. A common technique described was a lateral glide across the elbow joint that was performed and sustained by the therapist (mobilisation component) while the participant performed an impaired action such as gripping (movement component). Spinal manual therapy techniques may also have been used. Participants were also taught manual therapy techniques they could apply to themselves at home; *exercise* - strength training and stretching of the forearm muscles and upper limb, and correction of postural alignment. All exercises were performed at home, and on both affected and unaffected arm. All exercises were prescribed by a physiotherapist and progressed as appropriate. Participants also received home exercise equipment (resistant exercise band) and an exercise instruction booklet. Other - Taping techniques were used aiming to maintaining pain relief outside the clinical visits. Additional advice included rest from activities that aggravate the pain in the initial stages of rehabilitation (including treatments used in rehabilitation) and advice to avoid lifting objects with the hand facing the floor as this subjects the extensor muscles of the forearm to increased loads that frequently exacerbate pain.

Dose: manual therapy - 8 treatments of 30 minutes duration over 6 weeks; *exercise* - strength training included 3 sets, 8 to 12 repetitions, stretching included 3 to 5 repetitions, 20 to 30 sec holds for each muscle group, programme completed 2 to 3 times a day for 6 weeks

Supervision: one of six post graduate qualified physiotherapists trained in the treatment protocol administered the treatment. Exercises performed at home.

Additional treatment: none

Manual therapy and exercise for lateral elbow pain (Review)

Bisset 2006 (Continued)

Glucocorticoid injection

Components of intervention: glucocorticoid injections delivered to painful elbow points. A second injection was permitted after 2 weeks if deemed necessary by the medical practitioners. Participants were advised to return gradually to normal activities.

Dose: 1 mL quantity of 1% lidocaine with 10 mg of triamcinolone acetonide in 1 mL

Supervision: 1 of 2 medical practitioners administered the treatment

Additional treatment: none

Advice

Participants were given reassurance that the condition would settle, and were given instruction on modifying daily activities to avoid pain and use analgesic drugs, heat, cold or braces as needed.

Co-interventions

Participants in all the groups received an information booklet outlining the disease process and included practical advice on self-management and ergonomics on entering the study. Additional treatments were discouraged during the intervention period (except for the use of analgesics as needed).

Outcomes	Outcomes were reported at 3 weeks (first time point), 6 weeks (end of intervention), 3 months, 6 months and 12 months follow-up (last time point). Outcomes - Pain in the previous 7 days, 0 to 100 VAS (0 = no pain) - Disability, 0 to 100 scale using Pain Free Function questionnaire (0 = no disability) - Global improvement reported by the participant on a 6-point scale (1 = completely recovered; 2 = much improved; 3 = little improved; 4 = not changed; 5 = little worse; 6 = much worse) - Adverse events and serious adverse events from treatments - Participant overall withdrawals - Pain-free grip strength (kg) using a digital grip dynamometer, reported as a ratio of affected side to unaffected side x 100 (based on the mean of 3 efforts with intervening 30-second rest intervals) - Assessor's rating of severity of the elbow complaint on a 0 to 100 VAS (0 = no severity, 100 = maximum severity) Outcomes included in this review <i>Major outcomes</i> - Mean pain - Mean disability - Participant-reported treatment success (measured as 'completely recovered' or 'much improved') - Proportion of overall withdrawals - Proportion with any adverse event - Proportion with a serious adverse event <i>Minor outcomes</i> - Mean pain-free grip strength (ratio of affected side/unaffected side x 100)
Source of funding	University of Queensland and the National Health and Medical Research Council, Primary Health Care Project Grant, Australia
Notes	Trial registration: ClinicalTrials.gov (NCT00110318) Data analysis: we used data at 3 weeks (first time point), 6 weeks (end of intervention) and 12 months (last time point) for Analysis 2 - Manual therapy, prescribed exercises or both versus minimal or no intervention and Analysis 3 - Manual therapy and prescribed exercises versus minimal or no intervention.

Bisset 2006 (Continued)

We did not use data at 3 months or 6 months. We did not use data from the glucocorticoid injection arm. We transformed pain to 0 to 10 (0 = no pain).

Adverse events: 20 participants experienced an adverse event from treatments (7 manual therapy and exercise; 13 glucocorticoid injection; 0 advice). Pain after treatment was the most commonly reported side effect (7 manual therapy and exercise; 12 glucocorticoid injection). One participant in each group (manual therapy and exercise, glucocorticoid injection) reported pain lasting 7 days or longer. Two participants reported loss of skin pigment, and one also had atrophy of subcutaneous tissue after receiving the glucocorticoid injection.

Withdrawals: in the manual therapy and exercise group, 3 participants (4%) did not complete trial; in the glucocorticoid injection group, 0 participants (0%) did not complete trial; in the advice group, 5 participants (7%) did not complete trial.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation sequence was kept off site and drawn up by a computerised random number generator in permuted blocks of 6.
Allocation concealment (selection bias)	Low risk	They randomised participants by telephone via concealed allocation to the 3 groups. Group allocation was kept concealed from all study personnel throughout the entire study, including the data analysis phase, except for the administrative assistant responsible for contacting participants.
Blinding of participants and personnel (performance bias) All outcomes	High risk	While all personnel were blind to treatment allocation (except the administrative assistant responsible for contacting participants), the participants were not blind to treatment allocation.
Blinding of outcome assessment - self reported (detection bias)	High risk	Participants were not blind to treatment allocation. Pain, disability, global assessment and adverse events were likely to be influenced by a lack of blinding.
Blinding of outcome assessment - assessor reported outcomes	Low risk	A blinded assessor measured grip strength.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Results showed 5/67 participants from the wait and see group, 0/65 participants from the injection group and 3/66 participants from the manual therapy and exercise group did not complete the trial and were not included in the final analysis. The per protocol analysis only minimally changed the results compared with the intention-to-treat analysis.
Selective reporting (reporting bias)	Low risk	Adverse events were not reported in the wait and see group. However, data were reported for all outcomes listed in the methods, and all relevant outcomes reported were listed in the trial protocol.
Other bias	Low risk	The study appears to be free of other sources of bias.

Blanchette 2011**Study characteristics**

Methods **Design:** parallel-group, 2-arm randomised controlled trial

Blanchette 2011 (Continued)

Setting: university biomechanics laboratory, Quebec, Canada

Timing: February 2007 to April 2007

Interventions: (1) manual therapy; (2) advice; 5-week intervention period

Sample size: 58 participants per group provided 80% power to detect a clinically significant difference (figure not reported) in function between groups at 6 weeks, using the Patient-Rated Tennis Elbow Evaluation questionnaire, with a significance level set at 5% (sample size was determined after completion of the study).

Intention-to-treat analysis: not reported

Participants

Number of participants

- 35 participants assessed for eligibility
- 5 participants excluded (2 did not meet inclusion criteria; 3 refused to participate)
- 30 participants randomised (15 to manual therapy; 15 to advice)
- Data for 27 participants (90%) available at 6-week follow-up (15 for manual therapy, 12 for advice)

Inclusion criteria

- Age 18 years or older
- Diagnosis of lateral elbow pain using a clinical examination (chiropractor) including Cozen test (resisted extension of the wrist) and Mill test (passive extension of the elbow with the wrist flexed)

Exclusion criteria

- Bleeding disorders (haemophilia)
- Taking anticoagulants (aspirin)
- History of thrombosis or thrombophlebitis
- A condition that weakens the skin (e.g. hives)
- Diabetes
- Kidney disease
- Uncontrolled hypertension
- The presence of an infection at the time of the evaluation
- History of elbow surgery or diffuse pain syndrome (e.g. fibromyalgia)
- Rheumatoid arthritis, or other inflammatory arthritis affecting the wrist or the elbow
- Carpal tunnel syndrome
- Radiculopathy from the neck
- Fracture of the upper limb with residual deformity
- Pending litigation settlement (other than the Work Compensation Board) for the elbow problem
- Corticosteroid injections within 30 days preceding the study

Baseline characteristics

Manual therapy (n = 15 participants)

Mean (SD) age: 47 (10) years; 9 female, 6 male

Elbow affected: dominant - 7 (46%); non-dominant - 8 participants (54%); bilateral - 0 participants (0%)

Mean (SD) onset of condition: 22 (25) months

Mean (SD) pain severity (/100): 46 (23) units

Mean (SD) function on Patient-Rated Tennis Elbow Evaluation questionnaire (/100): 37 (19) units

Mean (SD) pain-free grip strength: 25 (14) kg

Advice (n = 12 participants, data for 3 participants not reported)

Blanchette 2011 (Continued)

Mean (SD) age: 46 (10) years; 6 female, 6 male

Elbow affected: dominant - 6 participants (50%); non-dominant - 3 participants (25%); bilateral - 3 participants (25%)

Mean (SD) onset of condition: 43 (50) months

Mean (SD) pain severity (/100): 39 (29) units

Mean (SD) function on Patient-Rated Tennis Elbow Evaluation questionnaire (/100): 30 (18) units

Mean (SD) pain-free grip strength: 26 (15) kg

Interventions
Manual therapy

Components of intervention: augmented soft tissue mobilisation - the therapist used an instrument manufactured by Graston Technique (USA), specifically developed to increase the capacity of the therapist to carry out the technique. The instrument is first moved along the longitudinal axis of musculo-tendinous structures, and then in various directions around bony prominence. With the patient seated, the instrument moves along the skin of the wrist, forearm, epicondyle, ulnar pit and distal portion of the triceps.

Dose: 10 treatments, with 2 treatments per week over 5 weeks

Supervision: 1 chiropractor administered the treatment

Setting: a university biomechanics laboratory

Additional treatment: none reported

Advice

Participants received a single session of advice, from the same chiropractor as the manual therapy group, about the natural history of the condition (e.g. a self-limiting pathology, the pain decreases the function to protect the insertion of the extensor muscles from too large mechanical stress during the healing process), advice about computer station ergonomics, stretches and analgesia (ice, generic anti-inflammatory medication) and a 2-page summary information sheet.

Co-interventions

None reported

Outcomes

Outcomes were reported at 6 weeks (end of intervention) and 3 months follow-up (last time point).

Outcomes

- Pain, 0 to 100 VAS (0 = no pain)
- Disability, 0 to 100 using the PRTEE (0 = no disability)
- Adverse events
- Participant overall withdrawals
- Pain-free grip strength (kg) measured with a digital grip dynamometer

Outcomes included in this review
Major outcomes

- Mean pain
- Mean disability
- Proportion of overall withdrawals
- Proportion with any adverse event.

Minor outcomes

Blanchette 2011 (Continued)

- Mean pain-free grip strength (kg)

Source of funding	Fondation de Recherche Chiropratique du Québec
Notes	Trial registration: not reported Data analysis: we used data at 6 weeks (end of intervention) and 3 months (last time point) for Analysis 2 - Manual therapy, prescribed exercises or both versus minimal or no intervention; and Analysis 4 - Manual therapy versus minimal or no intervention. We transformed pain to 0 to 10 (0 = no pain). Adverse effects: in the manual therapy group, 14/15 (93%) of participants experienced aches and bruises following the treatment, and these effects had ceased at the 6-week follow-up. No data were reported for the advice group. Withdrawals: in the manual therapy group, 0/15 participants (0%) did not complete the trial; in the advice group, 3/15 participants (20%) did not complete the trial.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Sequence generation method: they were randomised into 2 groups, with a number assigned to each participant and placed in an opaque urn. We assumed this was an adequate method.
Allocation concealment (selection bias)	Low risk	A research assistant with no further involvement in this study drew from the opaque urn to form the 2 groups.
Blinding of participants and personnel (performance bias) All outcomes	High risk	The participants were not blinded, and one chiropractor performed all the manual therapy treatments for the experimental group, delivered all the advice to the control group, collected and managed the data, and performed the statistical analysis. Therefore, participants and study personnel were not blind.
Blinding of outcome assessment - self reported (detection bias)	High risk	The participants were not blind to treatment allocation, and pain, function and adverse events are likely to be influenced by lack of blinding.
Blinding of outcome assessment - assessor reported outcomes	High risk	The chiropractor who assessed grip strength was not blinded, and also delivered the intervention.
Incomplete outcome data (attrition bias) All outcomes	Low risk	There were no missing outcome data. The 3 participants in the control group who could not return after their group assignment were excluded from the analysis.
Selective reporting (reporting bias)	Unclear risk	There was no trial registry or protocol available for this trial. The data reported in the results were consistent with the outcomes listed in the methods. However, adverse events were not reported in the minimal intervention group.
Other bias	High risk	There were baseline differences between the groups, with the average onset of the symptoms (in months) significantly higher in the advice group than the manual therapy group.

Bostrøm 2019

Study characteristics

Manual therapy and exercise for lateral elbow pain (Review)

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Boström 2019 (Continued)

Methods

Design: parallel-group, 3-arm, randomised controlled trial

Setting: institute of sports medicine and rehabilitation, Oslo, Norway

Timing: April 2015 to May 2016

Interventions: (1) manual therapy and exercise; (2) acupuncture and exercise; (3) exercise; 3-month intervention period

Sample size: calculation not performed (as data in this study were obtained to generate preliminary estimates of treatment effectiveness)

Intention-to-treat analysis: yes

Participants

Number of participants

- 50 participants assessed for eligibility
- 14 participants excluded (12 did not meet the inclusion criteria, 2 declined to participate)
- 36 patients were randomised (12 to manual therapy and exercise, 13 patients to treatment with acupuncture, 11 to exercise)
- Data for 32 participants (100%) were available at 4-week follow-up (11 for manual therapy and exercise, 12 for acupuncture and exercise, 9 for exercise)
- Data for 32 participants (100%) were available at 12-week follow-up (10 for manual therapy and exercise, 12 for acupuncture and exercise, 10 for exercise)
- Data for 30 participants (100%) were available at 2-year follow-up (11 for manual therapy and exercise, 11 for acupuncture and exercise, 8 for exercise)

Inclusion criteria

- Report pain with an intensity of 4 or higher on a numeric rating scale (NRS; 0 to 10)
- Pain on palpation
- Increased pain on resisted dorsiflexion of the wrist with the elbow extended and the fingers flexed
- Resisted extension of the third finger

Exclusion criteria

- Patients with symptom duration of less than 2 weeks
- Treatment with corticosteroid injection within the last 4 weeks
- Bilateral symptoms
- Radio-ulna or radio-humeral osteoarthritis
- Neck or shoulder problems
- Inflammatory rheumatic disease of the central or peripheral nervous system
- Unwillingness to participate in the study

Baseline characteristics

Manual therapy and exercise (n = 12 participants)

Mean (SD) age: 49 (2) years; 3 female, 9 male

Employment: manual work - 2 participants (16%); office work - 10 participants (83%)

Higher education: 11 participants (92%)

Mean (SD) pain (/10): 4.1 (1.3) units

Mean (SD) disability using Quick-DASH (/100): 31.7 (15.2)

Acupuncture and exercise (n = 13 participants)

Mean (SD) age: 51 (4) years; 6 female, 7 male

Boström 2019 (Continued)

Employment: manual work - 5 participants (38%); office work - 8 participants (62%)

Higher education: 10 participants (76%)

Mean (SD) pain (/10). 0 = no pain, 10 = worst pain. Unadjusted mean: 4.1 (1.3).

Mean (SD) disability using Quick-DASH (/100): 30.5 (14.2)

Exercise (n = 11 participants)

Mean (SD) age: 47 (3) years; 6 female, 6 male

Employment: manual work - 5 participants (45%); office work - 6 participants (55%)

Higher education: 8 participants (73%)

Mean (SD) pain (/10): 4.8 (1.3)

Mean (SD) disability using Quick-DASH (/100): 29.2 (14.4)

Interventions

Manual therapy

Components of intervention: mobilisation with movement techniques, with the therapists performing a lateral glide with gripping, a posterior-anterior glide on the radial head with supination of the radio-ulnar joint, and a lateral gapping manipulation technique.

Dose: during a period of 12 weeks, participants attended a minimum of 3 and a maximum of 8 treatment sessions; exercise - daily for 12 weeks following enrolment.

Supervision: 2 physiotherapists with specific manual therapy qualifications and long clinical experience performed all the manual therapy sessions.

Additional treatment: nil

Exercise

Components of intervention: eccentric exercise programme to strengthen the extensor muscles and tendon

Dose: daily basis for the 12 weeks, with advice to increase the load once a week by 10% of the starting weight, or less if their pain intensified. Written instructions on how to perform the exercises at home was given. A text message was sent weekly as a reminder to do their daily exercise.

Supervision: nil, exercises performed at home

Additional treatment: nil

Acupuncture

Components of intervention: generalised treatment, consisting of selected local points LI11 and LI10 over the muscular origin of the lateral extensor group of the forearm and LU5 in the cubital region. As distal points, they selected LI4 and TE5 for the treatment of pain in the upper limb, GB34 for treatment of tendinitis in general, and ST36 for treatment of pain. The acupuncturist inserted the needles down to the musculature, approximately 15 mm in depth, to obtain a De Qi sensation. All the points except ST36 were manipulated with a reducing technique to obtain pain relief.

Dose: acupuncture needles remained in situ for 20 mins. During a period of 12 weeks, participants attended a minimum of 3 and a maximum of 8 treatment sessions.

Supervision: acupuncturist with 12 years of clinical experience

Additional treatment: nil

Co-interventions

Bostrom 2019 (Continued)

All groups received the same exercise, information and advice, including details about the natural course of the condition and expected duration of symptoms. Patients were encouraged to use their arm normally, but to avoid carrying heavy loads and pain-provoking activities such as gripping and repetitive wrist movement. Patients were allowed other kinds of treatments during the trial, except corticosteroid injections.

Outcomes

Outcomes were reported at 1 week, 2 weeks, 3 weeks (first time point for pain severity), 4 weeks (first time point for disability), 3 months (end of intervention) and 12 months follow-up (last time point).

Outcomes

- Pain, 0 to 10 numerical rating scale (0 = no pain)
- Disability, 0 to 100 scale using Quick-DASH (0 = no disability)
- Participant satisfaction at 3 months follow-up
- Overall withdrawals
- Adverse events

Outcomes included in this review

Major outcomes

- Mean pain
- Mean disability
- Proportion of overall withdrawals
- Proportion with any adverse event

Minor outcomes

- Nil

Source of funding

No funding was obtained

Notes

Trial registration: ClinicalTrials.gov (NCT02321696)

Data analysis: we used data at 3 weeks (first follow-up for pain severity), 4 weeks (first follow-up for disability), 3 months (end of intervention) and 12 months (last follow-up) for Analysis 6 - Manual therapy and prescribed exercises versus prescribed exercises alone. We did not use data from the group that included acupuncture; 95% confidence intervals were converted to SDs; for the measurement of treatment success we were unable to include patient satisfaction data due to insufficient data reporting.

Adverse events: no participants reported adverse events or harms.

Withdrawals: in the manual therapy and exercise group 4/12 participants (33%) did not complete the trial; in the acupuncture and exercise group 4/13 participants (31%) did not complete the trial; in the exercise group 6/11 (55%) did not complete the trial.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation was organised in blocks of 6 with a 1:1:1 ratio.
Allocation concealment (selection bias)	Low risk	Participants drew a sealed, opaque envelope containing disclosure of group allocation from a collection of at least 6 envelopes. To prevent possible manipulation of group assignment, additional randomisation envelopes were constantly added to avoid ending up with only one envelope left at the end of a block.

Boström 2019 (Continued)

Blinding of participants and personnel (performance bias) All outcomes	High risk	The participants and study personnel were not blind.
Blinding of outcome assessment - self reported (detection bias)	High risk	The participants were not blind to treatment allocation, and pain and function assessments were likely influenced by lack of blinding.
Blinding of outcome assessment - assessor reported outcomes	Low risk	There were no assessor reported outcomes included in this review.
Incomplete outcome data (attrition bias) All outcomes	Low risk	All 36 participants in this trial were analysed using intention-to-treat analysis.
Selective reporting (reporting bias)	High risk	Data were not reported for all outcomes listed in the published protocol (e.g. quality of life).
Other bias	Low risk	The study appears to be free of other sources of bias.

Coombes 2013

Study characteristics

Methods	Design: parallel-group, 4-arm, factorial design, randomised controlled trial Setting: university research centre and primary care, Brisbane, Australia Timing: July 2008 to May 2010 Interventions: (1) manual therapy, exercise and glucocorticoid injection; (2) manual therapy, exercise and placebo injection; (3) glucocorticoid injection alone; (4) placebo injection alone; 8 week intervention period Sample size: initially, 30 participants per group provided 80% power to detect a difference of at least 25% of participant-reported global improvement for the 2 factorial (at margin) comparisons (manual therapy and exercise versus no manual therapy and exercise; glucocorticoid vs placebo), with a significance level set at 5%. Before recruitment ended, the sample size required was increased to 41 participants per group to provide adequate power for the following a priori pairwise comparisons: glucocorticoid injection versus placebo injection; manual therapy and exercise, plus glucocorticoid injection versus manual therapy and exercise, plus placebo injection; manual therapy and exercise, plus placebo injection versus placebo injection; and manual therapy and exercise, plus glucocorticoid injection versus glucocorticoid injection; and to account for loss to follow-up. Intention-to-treat analysis: yes
Participants	Number of participants 956 participants assessed for eligibility 791 participants excluded (657 did not meet inclusion criteria; 106 refused to participate; 28 did not attend testing) 165 participants randomised (40 for manual therapy and exercise, plus glucocorticoid injection, 41 for manual therapy and exercise, plus placebo injection, 43 for glucocorticoid injection, 41 for placebo injection)

Coombes 2013 (Continued)

- Data for 164 participants (99%) available at 4-week follow-up (40 for manual therapy and exercise, plus glucocorticoid injection, 41 for manual therapy and exercise, plus placebo injection, 42 for glucocorticoid injection, 41 for placebo injection)
- Data for 163 participants (99%) available at 26-week follow-up (39 for manual therapy and exercise, plus glucocorticoid injection, 41 for manual therapy and exercise, plus placebo injection, 43 for glucocorticoid injection, 40 for placebo injection)
- Data for 163 participants (99%) available at 52-week follow-up (39 for manual therapy and exercise, plus glucocorticoid injection, 41 for manual therapy and exercise, plus placebo injection, 42 for glucocorticoid injection, 40 for placebo injection)

Inclusion criteria

- Aged 18 years or older
- Unilateral lateral elbow pain of longer than 6 weeks' duration
- Pain over the lateral humeral epicondyle with pain severity greater than 30 mm on a 100 mm VAS, provoked by at least 2 of the following tests: gripping, palpation, resisted wrist or middle finger extension, or stretching of forearm extensor muscles with reduced pain-free grip

Exclusion criteria:

- Receipt of injection during preceding 6 months
- Receipt of a course of physiotherapy during preceding 3 months
- Concomitant neck or other arm pain necessitating treatment or preventing participation in usual work or recreational activities during preceding 6 months
- Symptoms suggesting radicular, neurological or systemic arthritic conditions
- Pregnancy
- Breastfeeding
- Contraindication to injection

Baseline characteristics

Manual therapy, exercise and glucocorticoid injection (n = 40 participants)

Mean (SD) age: 50.8 (8.5) years; 15 female, 25 male

Median (IQR) duration of symptoms: 15 (10 to 26) weeks

Mean (SD) worst pain severity (/100): 59.0 (15.8) units

Mean (SD) disability on the Patient-Rated Tennis Elbow Evaluation questionnaire (/100): 38.1 (13.8) units

Mean (SD) health-related quality of life (EQ-5D): 0.74 (0.09)

Manual therapy, exercise and placebo injection (n = 41 participants)

Mean (SD) age: 48.7 (7.7) years; 15 female, 26 male

Median (IQR) duration of symptoms: 16 (8 to 24) weeks

Mean (SD) worst pain severity (/100): 63.2 (18.0) units

Mean (SD) disability on the Patient-Rated Tennis Elbow Evaluation questionnaire (/100): 36.4 (13.3) units

Mean (SD) health-related quality of life (EQ-5D): 0.74 (0.12)

Glucocorticoid injection (n = 43 participants)

Mean (SD) age: 49.3 (8.9) years; 16 female, 27 male

Median (IQR) duration of symptoms: 16 (10 to 27) weeks

Coombes 2013 (Continued)

Mean (SD) worst pain severity (/100): 62.0 (20.3) units

Mean (SD) disability on the Patient-Rated Tennis Elbow Evaluation questionnaire (/100): 42.0 (14.4) units

Mean (SD) health-related quality of life (EQ-5D): 0.68 (0.20)

Placebo injection (n = 41 participants)

Mean (SD) age: 49.9 (7.4) years; 17 female, 21 male

Median (IQR) duration of symptoms: 16 (8 to 32) weeks

Mean (SD) worst pain severity (/100): 62.4 (19.8) units

Mean (SD) disability on the Patient-Rated Tennis Elbow Evaluation questionnaire (/100): 41.6 (14.4) units

Mean (SD) health-related quality of life (EQ-5D): 0.74 (0.13)

Interventions

Manual therapy and exercise

Components of intervention: manual therapy - as described by [Vincenzino 2003](#), Mulligan mobilisation with movement techniques including sustained lateral elbow glides, posterior to anterior radioulnar glides and de-loading the common extensor origin (mobilisation component), while the participants arm is supported in elbow extension and pronation, and the participant slowly performs a pain-free grip over approximately 6 seconds (movement component). If the technique was successful, participants were taught self-treatment techniques to perform at home. Taping techniques also applied.

Exercise - as described by [Vincenzino 2003](#), involved retraining of gripping and forearm movements and posture correction, progressive resistance training for the wrist extensors using theraband resistance, and general arm strengthening exercises with free weights or weightbearing antigravity resistance. The home programme was regularly reviewed and exercise diaries were monitored to facilitate programme adherence.

Dose: manual therapy - 8 treatments of 30 minutes duration over 8 weeks, techniques were repeated 6 to 10 times during a single treatment session if substantial improvement in pain-free grip force was noted with the glide application; *Dose: exercise* - programme involved retraining of gripping and forearm movements and posture correction was performed twice daily. The dosage of progressive resistance exercise and general arm strengthening not reported.

Supervision: 1 of 11 post graduate qualified physiotherapists trained in the treatment protocol administered the manual therapy and exercise treatments

Additional treatment: none

Glucocorticoid injection

Components of intervention: injection of corticosteroid and local anaesthetic (10 mg/mL of triamcinolone acetonide in a 1 mL injection plus 1 mL of 1% lignocaine) was applied to the site of greatest palpable tenderness at the common extensor origin.

Dose: the single injection was given within 10 days of randomisation.

Supervision: 1 of 5 medical practitioners delivered the injection.

Additional treatment: none

Placebo injection

Components of intervention: injection of placebo (0.5 mL of 0.9% isotonic saline) that was applied to the site of greatest palpable tenderness at the common extensor origin

Dose: the single injection was given within 10 days of randomisation

Coombes 2013 (Continued)

Supervision: 1 of 5 medical practitioners delivered the injection

Additional treatment: none

Co-interventions

Components of intervention: participants in all groups received advice to avoid activities that caused or provoked pain and to refrain from performing strenuous activity for 2 weeks after receipt of injection. Following the 2-week period, a gradual return to normal activities was encouraged (even if substantial initial relief was obtained) to minimise potential recurrence. Participants could use an analgesic or anti-inflammatory medication, heat or cold pack, or braces as needed, but were discouraged from seeking treatments other than those assigned.

Outcomes

Outcomes were reported at 4 weeks (first time point), 6 months (end of intervention) and 12 months follow-up (last time point).

Outcomes

- Pain (worst and resting), 0 to 100 VAS (0 = no pain)
- Disability, 0 to 100 using PRTEE questionnaire (0 = no disability)
- Health-related quality of life using EQ-5D-3L utility score (a value of 1 = perfect health)
- Global rating of change measured on a 6-point scale, ranging from 1 = complete recovery to 6 = much worse
- Adverse events and serious adverse events from treatments
- Participant overall withdrawals

Outcomes included in this review
Major outcomes

- Mean pain (worst pain)
- Mean disability
- Mean health-related quality of life
- Participant-reported treatment success (measured as 'complete recovery' or 'much improvement')
- Proportion of overall withdrawals
- Proportion with any adverse event
- Proportion with a serious adverse event

Minor outcomes

- Nil

Source of funding

Australian National Health and Medical Research Council grant 511238

Notes

Trial registration: trial registered at Australian New Zealand Clinical trials Registry, AC-TRN12609000051246

Data analysis: we used data at 4 weeks (first time point), 6 months (end of intervention) and 12 months (last time point) for Analysis 8 - Manual therapy, prescribed exercises or both and glucocorticoid injection versus glucocorticoid injection alone; and Analysis 9 - Manual therapy, prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone; for pain, disability and quality of life measures we transformed the median and interquartile range values to means and SDs using Wan 2014; we did not use data from (2) manual therapy, exercise and placebo injection and (4) placebo injection alone; we transformed pain to 0 to 10 (0 = no pain).

Adverse events: 43 participants experienced an adverse event from treatments (7 manual therapy, exercise and glucocorticoid injection; 16 manual therapy, exercise and placebo injection; 9 glucocorticoid injection, 11 placebo injection). Adverse events were reported as minor, transient and similar between injection and physiotherapy treatments. Skin depigmentation (4/83) and subcutaneous atrophy (3/83) occurred exclusively in patients receiving glucocorticoid injection, showed a delayed onset (evident on examinations at 8 weeks or 3 months) and was resolved by 6 months.

Coombes 2013 (Continued)

Withdrawals: in manual therapy, exercise and glucocorticoid injection group 2/40 participants (5%) did not complete the trial; in manual therapy, exercise and placebo injection group, 0/41 participants (0%) did not complete the trial; in the glucocorticoid injection group, 2/43 participants (5%) did not complete the trial; in the placebo injection group, 3/41 participants (7%) did not complete the trial.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The sequence was a computer-generated schedule developed by the Queensland Clinical Trials Center - an independent, offsite organisation.
Allocation concealment (selection bias)	Low risk	Allocation was concealed by having a research assistant, who was not involved in data collection or analysis, administer the randomisation schedule and arrange all study appointments.
Blinding of participants and personnel (performance bias) All outcomes	High risk	While the statistician was blinded to treatment allocation, neither the participants nor the clinician were blind to the treatment groups due to the nature of the intervention. This increases the risk of performance bias.
Blinding of outcome assessment - self reported (detection bias)	High risk	Participants were not blind to treatment allocation by the physiotherapist who delivered treatments, and the measures of pain, disability and treatment success were likely to be influenced by lack of blinding.
Blinding of outcome assessment - assessor reported outcomes	Low risk	This criterion is not applicable, as all outcomes were patients' reported outcomes.
Incomplete outcome data (attrition bias) All outcomes	Low risk	There was a very small amount of missing outcome data in this trial.
Selective reporting (reporting bias)	High risk	Data were not reported for all outcomes listed in the published protocol (e.g. pain-free grip force). Also, the outcome data were not reported for the 8-week and 3-month time points.
Other bias	Low risk	The study appears to be free of other sources of bias.

Haahr 2003

Study characteristics

Methods	Design: parallel-group, 2-arm randomised controlled trial Setting: primary care (general practice clinics), Ringkjøbing County, Denmark Timing: May 1998 to May 2000 Interventions: (1) exercise; (2) usual GP care; intervention period not reported Sample size: 250 participants in total provided 80% power to detect a difference of 15% (outcome not reported) between groups Intention-to-treat analysis: yes
Participants	Number of participants

Haahr 2003 (Continued)

- 360 participants assessed for eligibility
- 71 participants excluded (reasons for exclusion not reported)
- 289 participants randomised (148 to exercise, 141 to usual GP care)
- Data for 245 participants (85%) available at 3-month follow-up (124 to exercise, 121 to usual GP care)
- Data for 239 participants (83%) available at 6-month follow-up (120 to exercise, 119 to usual GP care)
- Data for 232 participants (80%) available at 12-month follow-up (116 to exercise, 116 to usual GP care)

Inclusion criteria

- Fulfilled diagnostic criteria for lateral epicondylitis including presence of pain in the elbow region and direct and indirect tenderness at or within 2 cm of the lateral humeral epicondyle on resisted extension of the wrist and/or the third finger

Exclusion criteria

- History of elbow pain for more than 1 year
- Previous elbow operation
- Known inflammatory rheumatic disorder

Baseline characteristics

Exercise (n = 141 participants)

Age > 40 years: 82 participants (58%); 80 female, 61 male

Dominant elbow affected: 102 participants (72%)

Body mass index > 25 kg/m²: 76 participants (54%)

Physically active < 4 hours per week: 94 participants (68%)

Employment: potentially strainful work - 84 participants (60%)

Pain in shoulder or forearm/hand in the last 3 months: 121 participants (98%)

Baseline pain score in upper quartile: 44 participants (32%)

Usual GP care (n = 125 participants)

Age > 40 years: 81 participants (65%); 64 female, 61 male

Dominant elbow affected: 102 participants (72%)

Body mass index > 25 kg/m²: 63 participants (50%)

Physically active < 4 hours per week: 70 participants (56%)

Employment: potentially strainful work - 65 participants (52%)

Pain in shoulder or forearm/hand in the last 3 months: 111 participants (89%)

Baseline pain score in upper quartile: 29 participants (24%)

Interventions

Exercise

Components of intervention: exercise - an ergonomist provided instructions including an illustrated written programme for performing a graded exercise programme. The home programme was described by [Solveborn 1997a](#) and included isometric contraction (pressing back of the hand for 10 secs against resistance with the elbow at right-angle) and stretching extensors and flexors for 15 to 20 secs.

Dose: 2 daily sessions, with exercises performed 3 to 5 times per session. The exercise was to be performed as long as symptoms persisted.

Supervision: nil for home exercises after the initial instruction with ergonomist

Haahr 2003 (Continued)

Setting: occupational medicine department and home

Additional treatment: a single session of information including lateral elbow pain being a self-limiting condition with a favourable prognosis, though the pain often varies in intensity. Participants were informed that no specific treatment seems to improve the overall long-term prognosis. Advice was given against complete rest and maintaining activity was encouraged whilst avoiding activities found to exaggerate the pain. In cases with a history of strenuous job tasks, the participant was encouraged to adjust work conditions if possible; participants were allowed to use analgesics available over the counter without prescription and elbow bandages at their own discretion; In cases with continuation or worsening of symptoms, the participants were allowed to consult their GP again, and no restrictions were imposed on subsequently selected treatments.

Usual GP care

Participants received treatment as preferred and agreed upon by the participant and the participant's GP.

Additional treatment: nil

Co-interventions

None

Outcomes

Outcomes were reported at 3 months (end of intervention), 6 months and 12 months follow-up (last time point).

Outcomes

- Pain score (4 sub-scales) measured pain in the elbow region in the past 3 months ranging from 0 (no complaints at all) to 9 (pain as bad as could be), indicating (i) severity of pain at worst, (ii) average pain within the last 3 months, (iii) severity of impairment in daily activities within the last 3 months due to elbow pain, and (iv) average level of pain within the last 7 days). These 4 scores were summed to give a range of 0 to 36.
- Participant self-reported overall development of the condition ('How do you evaluate the overall development of your lateral elbow pain since the first contact with your doctor 3 months/6 months/1 yr ago?', with response categories of 'much better', 'better', 'neither better nor worse', 'worse' and 'much worse').
- Participant overall withdrawals

Outcomes included in this review
Major outcomes

- Mean pain
- Participant-reported treatment success (rated as 'much better' or 'better')
- Proportion of overall withdrawals

Minor outcomes

- Nil

Source of funding

Funding was from The Research Council of the National Working Environment Authority, The Danish Insurance Association and The Medical Research Unit of Ringkjøbing County.

Notes

Trial registration: not registered

Data analysis: we used data at 3 months (end of intervention) and 12 months (last time point) for Analysis 2 - Manual therapy, prescribed exercises or both versus minimal or no intervention; and Analysis 5 - Prescribed exercises versus minimal or no intervention. We extracted the overall combined pain score from Fig 2; the 95% confidence intervals were also extracted from Fig 2 and then converted to SDs; the data were then transformed from 0 to 36 scale to 0 to 10 scale to standardise the measure for the meta-analyses; we did not use data at 6 months follow-up.

Haahr 2003 (Continued)

Adverse events: not reported

Withdrawals: in the exercise group, 32/148 participants (22%) did not complete trial; in the usual GP care group, 25/141 (18%) did not complete trial

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation to intervention or control groups was performed in advance by the project co-ordinator in blocks of 4 according to a predetermined random sequence of numbers, and was supplied to each general practice in sealed envelopes.
Allocation concealment (selection bias)	Low risk	Allocation was concealed using opaque envelopes, which were opened by the general practitioner after eligible cases had signed an informed consent form.
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding of participants or personnel was carried out for this trial due to the nature of the interventions.
Blinding of outcome assessment - self reported (detection bias)	High risk	The participants were not blind to the treatment groups, and pain outcome was likely to be influenced by a lack of blinding.
Blinding of outcome assessment - assessor reported outcomes	Low risk	There were no assessor-reported outcomes included in the trial.
Incomplete outcome data (attrition bias) All outcomes	High risk	There were high rates of attrition in the eccentric exercise group, and less attrition in the usual care group.
Selective reporting (reporting bias)	Unclear risk	There was no trial protocol published or trial registration available, so we judged this as an unclear risk of selective reporting.
Other bias	Low risk	The study appears to be free of other sources of bias.

Kochar 2002
Study characteristics

Methods	Design: parallel-group, 2-arm randomised controlled trial Setting: physiotherapy unit in orthopaedic department of medical institute, New Delhi, India Timing: not reported Interventions: (1) manual therapy and exercise; (2) exercise; 3-month intervention period Sample size: calculation method not reported Intention-to-treat analysis: not reported
Participants	Number of participants Number of participants assessed for eligibility not reported

Kochar 2002 (Continued)

- Number of participants excluded not reported
- 46 participants randomised (23 to manual therapy and exercise, 23 to exercise)
- Data for 46 participants (100%) available at 3-week follow-up (23 for manual therapy and exercise, 23 for exercise)
- Data for 46 participants (100%) available at 12-week follow-up (23 for manual therapy and exercise, 23 for exercise)

Inclusion criteria

- Patients with lesions involving the tenoperiosteal junction of extensor carpi radialis brevis confirmed by:
 - A typical history of localised pain and tenderness at the antero-inferior aspect of the lateral epicondyle aggravated by gripping and wrist extension activities, and relieved by rest
 - A positive Cozen's test or resisted wrist extension test
 - A positive Mill's test or passive stretching
 - A positive tender point at the tenoperiosteal junction
 - Full elbow range of motion and a normal X-ray

Exclusion criteria

- Cervical spine dysfunction
- Neuromeningeal involvement
- Radial tunnel syndrome
- Posterior interosseous nerve syndrome
- Arthritis (local/generalised polyarthritis)
- Radiohumeral bursitis
- Ligamentous sprain
- Bilateral tennis elbow
- Recurrent tennis elbow
- Golfer's elbow
- Painful shoulder
- Prior arm fractures
- Neurological abnormalities
- Pregnancy
- Infection
- Malignancy
- Received treatment in the preceding 3 weeks
- Corticosteroid injection in the preceding 6 months

Baseline characteristics

Manual therapy and exercise (n = 23 participants)

Mean (SD) age: 40 (9) years; 11 female, 12 male

Elbow affected: dominant - 16 (70%); non-dominant - 7 participants (30%)

Mean (SD) pain in last 24 hours, 0 to 10 VAS: 6 (1.3) units

Mean (SD) grip strength: 22.7 (8) kg

Exercise (n = 23 participants)

Mean (SD) age: 45 (12) years; 10 female, 13 male

Elbow affected: dominant - 18 (78%); non-dominant - 5 participants (22%)

Mean (SD) pain in last 24 hours, 0 to 10 VAS: 5.8 (1.3) units

Kochar 2002 (Continued)

Mean (SD) grip strength: 23.5 (10) kg

Interventions

Manual therapy and exercise

Components of intervention: manual therapy - mobilisation with movement technique with the participant performing the pain-producing movement in conjunction with sustained glide by the therapist. Participants were positioned supine with their elbow extended and forearm pronated. The therapist put the belt around their own shoulder and participant's forearm. The distal part of the forearm was stabilised. The participants then lifted the magnitude of weight that caused them pain as determined by the weight test. As the participants lifted the weights, the therapist applied lateral glide using their body weight. The belt was then removed and the participants were again encouraged to lift the weights. As the participants' pain lessened, they lifted heavier weights during the procedure; *exercise* - refer to exercise group.

Dose: manual therapy - 10 MWMs were repeated three times a session, for 10 sessions over 3 weeks on alternative days; *exercise* - started after initial 3 weeks of manual therapy treatment - refer to exercise group.

Supervision: 1 physiotherapist performed all treatments and reviewed participants fortnightly in order to progress the exercises and motivate the participants to comply with the exercise programme. Exercises performed at home.

Additional treatment: none

Exercise

Components of intervention: stretching and progressive resistance exercise within pain-free limits. Stretching involved the forearm pronated and elbow extended, the wrist is palmar flexed using the other hand of the patient and held for few seconds, then released. Progressive resistance exercise included isometric contractions with the elbow flexed to 90°, the hand of the unaffected arm applies manual resistance over the dorsum of the supinated hand of the affected arm. Pain-free isometric contractions of the wrist extensors are initiated and held for 5 to 10 seconds. Progression includes forearm pronation as the starting position and increasing resistance. Progressive resistance exercise also included active concentric and eccentric exercises introduced as pain allows, with the forearm rested on a table with elbow flexed to 90°, the hand hanging over the edge of the table with the wrist flexed and relaxed. Concentric extension of the wrist is then carried out until the wrist is in neutral, held for five seconds, then slow eccentric work of extensors lowers the hand back to a relaxed starting position. This procedure is then increased in range of movement and resistance. Participants increased the resistance by using heavier weights as pain allowed.

Dose: exercise - 10 repetitions, 3 times a day for 9 weeks

Supervision: physiotherapists reviewed participants fortnightly in order to progress the exercises and motivate the participants to comply with the exercise programme at home.

Additional treatment: none

Co-interventions

Ultrasound involved the head kept in motion and in skin contact at the palpated point at the tenoperiosteal junction of the extensor carpi radialis brevis. The area of the transducer head was 4 cm². This was applied for 5 minutes at a dosage of 3 MHz, 1.5 W/cm² and pulsed mode of 1 ms on and 5 ms off and was applied for the 10 sessions over 3 weeks on alternative days.

Outcomes

Outcomes were reported at 1 week, 2 weeks, 3 weeks (first time point) and 3 months follow-up (end of intervention).

Outcomes

- Pain in last 24 hours on a 0 to 10 VAS (0 = no pain)
- Weight test ability to lift only the weight of their own hand, and then 1 kg, 2 kg and 3 kg weights, with the wrist in extension and the elbow fully extended and forearm pronated, without pain. The forearm was supported on a table.

Kochar 2002 (Continued)

- Grip strength (kg) using a dynamometer with the average of 3 repetitions

Outcomes used in this review
Major outcomes

- Mean pain

Minor outcomes

- Mean grip strength (kg)

Source of funding	Not reported
Notes	Trial registration: not reported Data analysis: we used data at 3 weeks (first time point) and 3 months (end of intervention) for Analysis 6 - Manual therapy and prescribed exercises versus prescribed exercise alone. We extracted data for first follow-up and end of intervention from Fig 1 and 3. Adverse events: not reported Withdrawals: not reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Participants were randomised into 2 treatment groups by a random draw method.
Allocation concealment (selection bias)	Unclear risk	Allocation concealment was not described.
Blinding of participants and personnel (performance bias) All outcomes	High risk	There was no blinding of participants and personnel due to the nature of the interventions.
Blinding of outcome assessment - self reported (detection bias)	High risk	Participants were not blind to the treatment received, and the pain outcome was likely to be influenced by a lack of blinding.
Blinding of outcome assessment - assessor reported outcomes	High risk	The outcome assessor was not blind to treatment, and the grip strength outcome is likely to be influenced by a lack of blinding.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	We assumed that no patients were lost to follow-up, although there was insufficient reporting of attrition.
Selective reporting (reporting bias)	High risk	While there was no trial registration or published protocol available, the results for one of the pain outcomes that was measured on a Likert scale was not reported in the results.
Other bias	Low risk	The study appears to be free of other sources of bias.

Luginbuhl 2008

Study characteristics

Methods

Study design: parallel-group, 3-arm randomised controlled trial

Setting: outpatient clinic of University, Zurich, Switzerland

Timing: February 1999 to August 2001

Interventions: (1) exercise, glucocorticoid injection, brace; (2) glucocorticoid injection, brace; (3) exercise and glucocorticoid injection; 3-month intervention period

Sample size: calculation method not reported

Intention-to-treat analysis: not reported

Participants

Number of participants

- 75 participants assessed for eligibility
- 40 participants excluded (reasons not reported)
- 35 participants randomised (10 to exercise, brace and glucocorticoid injection, 10 to brace and glucocorticoid injection, 9 to exercise and glucocorticoid injection, 6 lost to follow-up were not reported)
- Data for 29 participants (83%) available at 6-week follow-up (10 for exercise, brace and glucocorticoid injection, 10 for brace and glucocorticoid injection, 9 for exercise and glucocorticoid injection)
- Data for 29 participants (83%) available at 3-month follow-up (10 for exercise, brace and glucocorticoid injection, 10 for brace and glucocorticoid injection, 9 for exercise and glucocorticoid injection)
- Data for 13 participants (37%) available at 12-month follow-up (5 for exercise, forearm brace and glucocorticoid injection, 4 for brace and glucocorticoid injection, 4 for exercise and glucocorticoid injection)

Inclusion criteria

- Local tenderness at the lateral humeral epicondyle
- Increasing pain at the lateral epicondyle with resisted dorsiflexion of the wrist
- Relief of the pain by at least 50% of intensity after injection of local anaesthetics (Scandicain 1%, Mepivacain Astra Zeneca, Zug, Switzerland) at the lateral epicondyle

Exclusion criteria

- Prior surgery at the elbow
- Major trauma such as fractures
- Tendon or ligament tears
- Instability
- More than 3 cortisone injections at the lateral epicondyle during the last 6 months
- Degenerative or inflammatory arthritis
- Prior conservative treatment with a forearm support band or with a strengthening programme
- Disapproval of the patients of the conservative treatment protocol

Baseline characteristics

Characteristics were reported for the whole population.

Mean (SD) age: 47 (9) years; 16 female, 13 male

Elbow affected: dominant - 22 participants (76%); non-dominant - 7 participants (24%)

Mean duration of symptoms: 10 (11) months

Employment: strenuous manual labour - 3 participants; moderate manual work - 14 participants; light work - 13 participants; workers compensation - 2 participants; permanent working disability - 1 participant

Luginbuhl 2008 (Continued)

Exercise, brace and glucocorticoid injection (10 participants)

Mean (SD) pain, 0 to 4 scale: 3.1 (0.7) units

Mean (SD) grip strength: 81 (24) kg

Brace and glucocorticoid injection (10 participants)

Mean (SD) pain, 0 to 4 scale: 3.7 (0.7) units

Mean (SD) grip strength: 83 (31) kg

Exercise and glucocorticoid injection (n = 9 participants)

Mean (SD) pain, 0 to 4 scale: 3.4 (0.7) units

Mean (SD) grip strength: 80 (33) kg

Interventions
Exercise

Components of intervention: grip strengthening exercise using a tennis ball was prescribed to start 1 week after glucocorticoid injection (co-treatment) and an isometric resisted wrist extension exercise was added 3 weeks after the glucocorticoid injection. The exercises were instructed to be performed gently, and not cause relevant pain.

Dose: exercise - isometric grip strength exercise was performed for 20 seconds, 20 times, twice a day, for at least 3 months. Isometric resisted wrist extension exercise was performed for 20 seconds, 20 times, twice a day, for at least 2.5 months.

Supervision: no exercise supervision; participants provided an instruction sheet to perform the exercises at home

Additional treatment: none reported

Glucocorticoid injection

Components of intervention: 1 week prior to the exercise and forearm band interventions, a glucocorticoid injection was delivered at the site of the lateral epicondyle of about 2 to 3 mL of a mixture of 5 mL of Scandicain 1% and 1 mL of Kenacort 40 mg (Triamcinolon, Bristol-Myers Squibb, Baar, Switzerland).

Brace

Components of intervention: a Chamois Tennis Elbow Wrap (Scott Specialties Inc., Belleville, KS, USA) was worn about 4 cm distally to the lateral epicondyle in a slightly tight manner without causing swelling or paresthesiae at the hand.

Dose: 3 months, recommended to be worn for the whole day, especially at work or during stressful activities, and not at night during sleep. This was prescribed 1 week after glucocorticoid injection.

Co-interventions

Glucocorticoid injection. All participants were encouraged to rest, to avoid painful activities and to apply ice for 15 min several times a day for 1 week.

Outcomes

Outcomes were reported at the latest follow-up (6 weeks, 3 months or 12 months) without specifying the time point.

Outcomes

- Pain, 0 to 4 scale measured by the Modified Nirschl/Pettrone score (0 = full activity, no pain; 1 = no pain during normal daily activities, mild pain during sports/occupational activity; 2 = occasional pain during normal daily activities, moderate pain during sports/occupational activity; 3 = mild to moderate pain during normal daily activities, severe pain during sports/occupational activity; 4 = pain at rest)
- Participant global assessment of treatment success on a 4-point scale of 'much better', 'better', 'same' or 'worse'

Luginbuhl 2008 (Continued)

- Grip strength measured using a dynamometer and reported as a ratio of affected side to the contralateral side x 100
- Number treated with surgical lateral extensor release due to failure of non-operative treatment

Outcomes used in this review
Major outcomes

- Mean pain, 0 to 4 measured by the Modified Nirschl/Pettrone score (0 = full activity, no pain; 1 = no pain during normal daily activities, mild pain during sports/occupational activity; 2 = occasional pain during normal daily activities, moderate pain during sports/occupational activity; 3 = mild to moderate pain during normal daily activities, severe pain during sports/occupational activity; 4 = pain at rest)
- Participant-reported treatment success (measured as 'much better' or 'better')

Minor outcomes

- Mean grip strength (ratio of affected side/unaffected side x 100)

Source of funding	Not reported
Notes	Trial registration: not reported Data analysis: we assumed the 3-month follow-up represented 'end of intervention' as the majority of participants were lost to follow-up at 12 months. Therefore, we used data (end of intervention) for Analysis 8 - Manual therapy, prescribed exercises or both and glucocorticoid injection versus glucocorticoid injection alone; Analysis 10 - Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone. We used data from the groups that received the brace as a co-intervention to balance the effects of the brace, and did not use data from the exercise and glucocorticoid injection arm. For the pain outcome, we transformed the data to a 0 to 10 scale (0 = no pain) to standardise the measure for the meta-analyses. Adverse events: not reported. Withdrawals: 6 participants did not complete the trial without specifying groups

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	The patients were randomised into 3 groups. However, there was no information about the randomisation process, so we judged this as an unclear risk of selection bias.
Allocation concealment (selection bias)	Unclear risk	Concealment was not reported.
Blinding of participants and personnel (performance bias) All outcomes	High risk	There was no blinding of participants and personnel due to the nature of the interventions.
Blinding of outcome assessment - self reported (detection bias)	High risk	The participants were not blind to treatment, and the pain outcome was likely to be influenced by a lack of blinding.
Blinding of outcome assessment - assessor reported outcomes	High risk	The outcome assessor was not blind to treatment, and the grip strength outcome is likely to be influenced by a lack of blinding.
Incomplete outcome data (attrition bias)	High risk	There were incomplete outcome data at the 12-month time point, with data available for 13 participants (37%).

Manual therapy and exercise for lateral elbow pain (Review)

Luginbuhl 2008 (Continued)

All outcomes

Selective reporting (reporting bias)	High risk	While there was no trial registration or published protocol available, the outcomes were reported for one time point (latest) so we judged this as a high risk of selective reporting.
Other bias	Unclear risk	They did not report baseline characteristics data separately by groups.

McQueen 2021
Study characteristics

Methods	Design: parallel-group, 3-arm, quasi-randomised controlled trial Setting: Washington University School of Medicine, Program of Occupational Therapy, Milliken Hand Rehabilitation Center, Saint Louis, MO, USA Timing: not reported Interventions: (1) exercise; (2) advice, wrist orthosis; (3) no treatment; 16-week intervention period Sample size: based on 3 groups, sample size calculation indicated that 35 participants would be required for each group to demonstrate any true differences between the groups. No further details. Intention-to-treat analysis: not reported
Participants	Number of participants - Number assessed for eligibility not reported - Number of participants excluded not reported 94 participants were randomised (54 to exercise, 40 to advice, 0 to control group) - Data for 59 participants (63%) available at 12-month follow-up (38 for exercise, 21 for advice, 0 to no treatment) Inclusion criteria - Patients diagnosed with lateral epicondylalgia Exclusion criteria - Patients who have received an injection for lateral epicondylalgia within the past 6 months - Patients who do not speak English as their primary language - Patients younger than 18 years - Patients with cognitive impairment - Patients with comorbidities in the ipsilateral upper extremity Baseline characteristics <i>Exercise (n = 38 participants)</i> Mean (SD) pain at rest (/10): 1.0 (more reported) units Mean (SD) pain with activity (/10): 6.69 (not reported) units Mean (SD) disability using quick DASH (/100): 30.84 (not reported) units <i>Advice, orthosis (n = 21 participants)</i> Mean (SD) pain at rest (/10): 1.29 (not reported) units

McQueen 2021 (Continued)

Mean (SD) pain with activity (/10): 6.69 (not reported) units

Mean (SD) disability using quick DASH (/100): 29.33 (not reported) units

No treatment (n = 0 participants)

Whole group demographics

Occupation n (%): manual labour 20 participants (22%), office work 61 (66%), retired 11 (12%)

Race: White 80 (86%), Black 10 (9%), Hispanic 3 (3%), Other 1 (1%)

Age groups n (%): 30 to 39 = 11 participants (12%), 40 to 49 = 32 (34%), 50 to 59 = 33 (35%), 60 to 69 = 16 (17%), 70 to 79 = 2 (2%)

Gender: 59 female, 35 male.

Education in years n (%): 8 to 12 = 7 participants (7%), 13 to 16 = 35 (37%), 17 to 19 = 25 (27%), 20+ = 10 (11%), unknown = 17 (18%)

Symptom duration months, n (%): 0 to 3 = 59 participants (63%), 4 to 6 = 14 (15%), 7 to 12 = 14 (15%), 12+ = 4 (4%), unknown = 0 (0%)

Interventions
Exercise

Components of intervention: strengthening exercises of eccentric wrist extension and TheraPutty grip, performed once daily. Eccentric exercises were initiated at the second therapy session with a one-pound hand weight at 10 repetitions; patients were coached to progress to 30 repetitions, after which a two-pound weight was used for the same progression. Once these were mastered, grip strengthening with soft putty was introduced to the home programme, beginning with gripping with the elbow flexed and progressing to gripping with elbow extension. Gripping was practised in 3 positions: supination, forearm neutral and pronation and progressed from 3 to 5 repetitions at whichever level of the elbow position the patient was able to use while performing the exercise pain-free.

Dose: 10 repetitions once daily; patients were coached to progress to 30 repetitions, after which a two-pound weight was used for the same progression.

Supervision: therapist (likely occupational therapist)

Additional treatment: none

Advice, orthosis

Advice on pathoanatomy, activity modification, pain relief techniques of icing and stretches; wrist cock-up orthosis to wear with all activities for 2 weeks and thereafter as pain symptoms dictated

Co-interventions

Advice, wrist orthosis

Outcomes

Outcomes were reported at 6 weeks (first time point), 3 months (end of intervention), 6 months and 12 months (last time point)

Outcomes

- Pain (rest and activity), 0 to 10 VAS (0 = no pain)
- Disability, 0 to 100 using Quick DASH (0 = no disability)

Outcomes included in this review
Major outcomes

- Mean pain (rest)
- Mean disability

McQueen 2021 (Continued)

Minor outcomes

- Nil

Source of funding	Not reported
Notes	Trial registration: not reported Data analysis: we used data at 6 weeks (first time point), 3 months (end of intervention) and 12 months (last time point) for Analysis 2 - Manual therapy, prescribed exercises or both versus minimal or no intervention; and Analysis 5 - Prescribed exercises versus minimal or no intervention. We calculated SDs from P values. Adverse events: not reported Withdrawals: in the exercise and brace group, 16/54 participants (30%) did not complete the trial; in the brace group, 19/40 (48%) did not complete the trial.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	Participants were randomised to one of two groups according to their medical registry number, with odd-numbered registrants enrolled in group 1 and even-numbered registrants in group 2. However, we consider this a quasi-randomisation method (alternate allocation) and it is predictable.
Allocation concealment (selection bias)	High risk	Allocated by front desk staff who were unaware of the study. However, it is not concealed because the method was quasi-randomised.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were likely aware of the group due to the nature of the interventions.
Blinding of outcome assessment - self reported (detection bias)	High risk	Participants were likely aware of group allocation.
Blinding of outcome assessment - assessor reported outcomes	Low risk	No assessor-reported outcomes were measured in this trial.
Incomplete outcome data (attrition bias) All outcomes	High risk	There was a high proportion of dropouts throughout the trial and these were unevenly distributed between groups: 16/54 (30%) from the exercise group vs 19/40 (48%) from the control group. More dropped out of the control group for unknown reasons and lack of compliance.
Selective reporting (reporting bias)	High risk	There was no published trial protocol or trial registry. It was also unclear if adverse events were measured, and no measures of variance were reported for pain and disability outcomes.
Other bias	Unclear risk	The trial did not report participant characteristic data separately by group. For excluded participants, this appears to be due to unplanned reasons, such as low compliance or another intervention.

Muralidharagopalan 2020

Study characteristics

Methods

Design: parallel-group, 2-arm randomised controlled trial

Setting: tertiary care medical college hospital, Chennai, India

Timing: Jun 2015 to May 2020

Interventions: (1) exercise and glucocorticoid injection; (2) glucocorticoid injection; 2-week intervention period

Sample size: 45 participants in total required, method unclear

Intention-to-treat analysis: not reported

Participants

Number of participants

- Number assessed for eligibility not reported
- Number of participants excluded not reported
- 50 participants were randomised (26 to exercise and glucocorticoid injection, 24 to glucocorticoid injection)
- Data for 50 participants (100%) available at 2-week follow-up (26 for exercise and glucocorticoid injection, 24 for glucocorticoid injection)
- Data for 50 participants (100%) available at 6-week follow-up (26 for exercise and glucocorticoid injection, 24 for glucocorticoid injection)
- Data for 50 participants (100%) available at 3-month follow-up (26 for exercise and glucocorticoid injection, 24 for glucocorticoid injection)
- Data for 50 participants (100%) available at 6-month follow-up (26 for exercise and glucocorticoid injection, 24 for glucocorticoid injection)
- Data for 50 participants (100%) available at 12-month follow-up (26 for exercise and glucocorticoid injection, 24 for glucocorticoid injection)

Inclusion criteria

- 18 to 60 years of age
- Unilateral epicondylitis
- Willing to enrol in the study and present for adequate follow-up

Exclusion criteria

- < 18 years or > 60 years
- Pregnancy
- Bilateral involvement simultaneously
- Congenital deformities of upper limb
- Previous surgery of upper limb
- Inflammatory joint disorders
- Not willing to enrol in the study or follow-up

Baseline characteristics

Exercise and glucocorticoid injection (n = 26 participants)

Mean (SD) age: 37.6 (6.9) years; 15 females, 11 males

Mean (SD) duration of symptoms: 7.3 (1.3) weeks

Mean (SD) pain (/10): 7.07 (0.82) units

Mean (SD) disability using DASH: 34.99 (7.19) units

Glucocorticoid injection (n = 24 participants)

Muralidharagopalan 2020 (Continued)

Mean (SD) age: 37.5 (7.2) years; 14 females, 10 males

Mean (SD) duration of symptoms: 7.3 (1.6) weeks

Mean (SD) pain (/10): 7.08 (0.95) units

Mean (SD) disability using DASH: 33.4 (7.36) units

Interventions	Exercise <i>Components of intervention:</i> eccentric elbow strengthening exercises <i>Dose:</i> 2 weeks <i>Supervision:</i> monitored by physiotherapist remotely for 2 weeks <i>Additional treatment:</i> none Glucocorticoid injection <i>Components of intervention:</i> injection of 1 mL of triamcinolone acetonide (ABBOTT) (40 mg/mL) mixed with 2 cc of 2% lignocaine and injected into the lateral epicondyle and the most tender areas surrounding it <i>Dose:</i> single injection <i>Supervision:</i> no information <i>Additional treatment:</i> none Co-intervention Glucocorticoid injection
Outcomes	Outcomes were reported at 2 weeks (first time point), 6 weeks (end of intervention), 3 months, 6 months and 12 months (last time point). Outcomes • Pain, 0 to 10 VAS (0 = no pain) • Disability, 0 to 100 using DASH (0 = no disability) • Adverse events from treatments • Checked for tenderness on resisted dorsiflexion of wrist and middle finger (on a 3-point scale: no tenderness, some tenderness, definitive tenderness) Outcomes included in this review <i>Major outcomes</i> • Mean pain • Mean disability • Proportion with any adverse event <i>Minor outcomes</i> • Nil.
Source of funding	Not reported
Notes	Trial registration: not reported Data analysis: we used data at 2 weeks (first time point), 6 weeks (end of intervention) and 12 months (last time point) for Analysis 8 - manual therapy, prescribed exercises or both and glucocorticoid injection versus glucocorticoid injection alone; and Analysis 10 - Prescribed exercises and glucocorticoid in-

Muralidharagopalan 2020 (Continued)

jection versus glucocorticoid injection alone. We imputed SDs for (1) exercise and glucocorticoid injection, from the injection only group at 12 months as no SDs for this group were reported.

Adverse events: no adverse effects of exercise intervention or glucocorticoid injection were reported.

Withdrawals: not reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Patients were randomised into groups using a computer random number generator.
Allocation concealment (selection bias)	Unclear risk	Allocation was not described.
Blinding of participants and personnel (performance bias) All outcomes	High risk	The participants and personnel were likely aware of group allocation due to the nature of the intervention.
Blinding of outcome assessment - self reported (detection bias)	High risk	The participants were likely aware of group allocation.
Blinding of outcome assessment - assessor reported outcomes	Low risk	No assessor-reported outcomes were used in this trial.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	There were no data reported on participant follow-up or dropouts, so we judged there to be an unclear risk of bias.
Selective reporting (reporting bias)	Unclear risk	There was no published protocol or registration available, so we judged this as an unclear risk of selective reporting. While tenderness was measured on a 3-point categorical scale, this was not reported categorically, and was not included in the outcomes for this review.
Other bias	Unclear risk	It was unclear if co-interventions were used during the study period.

Nambi 2023
Study characteristics

Methods	Design: parallel-group, 2-arm, randomised, sham-controlled trial Setting: Physical Therapy Department, College of Applied Medical Sciences, Prince Sattam bin Abdulaziz University, Saudi Arabia Interventions: (1) manual therapy, prescribed exercises and glucocorticoid injection; (2) sham manual therapy, prescribed exercises and glucocorticoid injection Sample size: 27 participants per group provided 80% power to detect a mean difference of 1.5 points and a standard deviation of 4, with a significance level of 5% Intention-to-treat analysis: yes
---------	---

Nambi 2023 (Continued)

Participants

Number of participants

- 115 participants assessed for eligibility
- 55 participants excluded (19 previous physical therapy interventions, 13 other systemic problems, 6 elbow joint problems, 5 undergone surgeries, 12 not willing to participate)
- 60 participants randomised (30 to manual therapy, prescribed exercises and glucocorticoid injection, 30 to sham manual therapy, prescribed exercises and glucocorticoid injection)
- Data for 60 participants (100%) were available at 4-week follow-up (30 for manual therapy, prescribed exercises and glucocorticoid injection, 30 for sham manual therapy, prescribed exercises and glucocorticoid injection)
- Data for 57 participants (95%) were available at 8-week follow-up (28 for manual therapy, prescribed exercises and glucocorticoid injection, 29 for sham manual therapy, prescribed exercises and glucocorticoid injection)
- Data for 55 participants (92%) were available at 6-month follow-up (27 for manual therapy, prescribed exercises and glucocorticoid injection, 28 for sham manual therapy, prescribed exercises and glucocorticoid injection)

Inclusion criteria

- Male
- Aged between 18 and 60 years
- Clinical diagnosis of chronic lateral elbow pain (> 2 months of pain)
- Pain intensity between 3 and 8 on VAS

Exclusion criteria

- Participants with prior steroid injection therapy
- Associated neck or arm pain
- Severe musculoskeletal, neural, somatic and psychiatric conditions
- Waiting for surgery
- Having alcohol or drug abuse
- Involved in other weight training programmes
- Red flags for manipulation

Baseline characteristics

Manual therapy, prescribed exercises and glucocorticoid injection (n = 30 participants)

Mean (SD) age: 48.4 (3.8) years; 0 females, 30 males

Mean (SD) duration of symptoms: 6.2 (2.3) months

Mean (SD) pain (0 to 10): 7.6 (1.4) points

Mean (SD) disability using PRTEE (0 to 100): 74.1 (5.1) points

Dominant elbow n (%): 21 (70)

Manual work n (%): 21 (70)

Employment n (%): manual work - 21 participants (70); non-manual work - 6 participants (20); not working - 3 participants (10)

Sham manual therapy, prescribed exercises and glucocorticoid injection (n = 30 participants)

Mean (SD) age: 47.1 (3.5) years; 0 females, 30 males

Mean (SD) duration of symptoms: 5.6 (2.7) months

Mean (SD) pain (0 to 10): 7.4 (1.3) units

Mean (SD) disability using PRTEE (0 to 100): 74.8 (4.9) points

Nambi 2023 (Continued)

Dominant elbow n (%): 20 (67)

Manual work n (%): 20 (67)

Employment n (%): manual work - 20 participants (67); non-manual work - 6 participants (20); not working - 4 participants (13)

Interventions
Manual therapy

Components of intervention: deep transverse friction massage and Mill's manipulation

Dose: 3 sessions per week for 4 weeks, each session lasted for 30 to 40 minutes

Supervision: physiotherapist

Setting: physiotherapy department in a university hospital

Sham manual therapy

Components of intervention: low-velocity, high-amplitude movement performed at the elbow joint, which was not comprehensible to the study participants

Dose: 3 sessions per week for 4 weeks, each session lasted for 30 to 40 minutes

Supervision: physiotherapist

Setting: physiotherapy department in a university hospital

Exercise

Components of intervention: progressive resistance exercises using therabands. Initially, the painful movements were trained with minimal resistance and then progressed to the next level of resistance for the other joint movements. In the later phase, it included specific rehabilitation activity.

Dose: home exercises were performed daily for 4 weeks: consisting of eccentric exercise (3 times 30 repetitions) and isolated stretching of radial wrist extensors (3 times daily for 30 seconds).

Supervision: prescribed by a physio, completed at home without supervision

Setting: physiotherapy department in a university hospital

Glucocorticoid injection

Components of intervention: 1 mL triamcinolone acetonide (10 mg/mL) (Kenacort- A 10) with 1 mL lignocaine (1%) was administered into the most palpably tender point in the region of the lateral epicondyle.

Co-interventions

Exercise; glucocorticoid injection

Outcomes

Outcomes were reported at 4 weeks (end of intervention), 8 weeks and 6 months (last time point).

Outcomes

- Pain, 0 to 10 on VAS (0 = no pain)
- Disability, 0 to 100 using PRTEE (0 = no disability)
- Global perceived improvement reported by the participants on a 6-point Likert scale
- Health-related quality of life, EQ-5D-3L (scored 1 to 3 where 1 represents perfect health)
- Participant overall withdrawals
- Adverse events
- Depression, Hospital Anxiety and Depression Scale (HADS)
- Maximum grip strength using handheld dynamometer
- MRI imaging

Nambi 2023 (Continued)

- Ultrasound imaging

Outcomes used in the review

Major outcomes

- Mean pain
- Mean disability
- Mean health-related quality of life
- Proportion of overall withdrawals
- Proportion with any adverse event

Minor outcomes

- Mean maximum grip strength

Source of funding	Source of funding: this study is supported via funding from Prince Sattam bin Abdulaziz University project number (PSAU/2023/R/1444).
Notes	Trial registration: https://trialsearch.who.int/Trial2.aspx?TrialID=CTRI/2020/05/025135 Data analysis: we used data at 4 weeks (end of intervention) and 6 months (final time point) for Analysis 7 - Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection alone. Unable to use data for global perceived improvement as results were reported on a continuous scale. While the SDs for pain, disability, quality of life and grip strength appeared small, we contacted the author who confirmed the reported results were means and SDs. Adverse events: in both groups, no participants (0%) reported an adverse event. Withdrawals: in the manual therapy, prescribed exercises and glucocorticoid injection group, 3 participants (10%) did not complete the trial; in the sham manual therapy, prescribed exercises and glucocorticoid injection group, 2 participants (7%) did not complete the trial.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Participants were randomised using a computer-generated randomisation method.
Allocation concealment (selection bias)	Low risk	Sealed, opaque envelopes were used to conceal group allocation with participants selecting their group from a sealed, opaque envelope given by the consulting physiotherapist.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	The participant's group allocation was known to the physical therapists who provided the treatments. However, the participants were blind to the manual therapy component (sham versus active).
Blinding of outcome assessment - self reported (detection bias)	Low risk	Participants were blind to the manual therapy component (sham versus active), reducing the risk of detection bias for the outcomes of pain, disability, quality of life and treatment success.
Blinding of outcome assessment - assessor reported outcomes	Low risk	The therapist who was measuring grip strength was blinded to group allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Only a very small amount of missing data was observed, and it was equal across groups.

Manual therapy and exercise for lateral elbow pain (Review)

Nambi 2023 (Continued)

Selective reporting (reporting bias)	Low risk	Data were reported for all outcomes listed in protocol and methods, so we judged this as a low risk of selective reporting. However, we were unable to extract data for measuring treatment success as it was reported as a continuous score, and the reporting of quality of life using EQ-5D did not use a utility score.
Other bias	Low risk	This trial appears to be free of other sources of bias.

Nourbakhsh 2008

Study characteristics

Methods	Design: parallel-group, 2-arm randomised controlled trial Setting: physical therapy department of University, North-East Georgia, USA Timing: not reported Interventions: (1) manual therapy; (2) placebo manual therapy; 2- to 3-week intervention period Sample size: calculation method not reported Intention-to-treat analysis: not reported
Participants	Number of participants - Number assessed for eligibility not reported - Number of participants excluded not reported 23 participants were randomised (11 to manual therapy, 12 to placebo manual therapy) - Data for 23 participants (100%) available at 6-week follow-up (11 for manual therapy, 12 for placebo manual therapy) Inclusion criteria - Area of pain to the origin of the wrist and hand extensors on the lateral epicondyle (distal radiation of symptoms into the forearm was acceptable) - Diagnosis of lateral elbow pain confirmed by 3 pain provocation tests - Cozen's, Mill's and third finger extension tests Exclusion criteria - Proximal upper extremity or neck symptoms - History of cervical pathology nerve entrapment syndromes - Non-union fractures - Surgical treatments for lateral elbow pain - Steroid injection for elbow pain during the last 6 months Baseline characteristics <i>Manual therapy (n = 11 participants)</i> Mean (SD) age: 51.7 (9.4) years; 5 females, 6 males Mean (SD) duration of symptoms: 17.1 (17.7) months Mean (SD) pain (/10): 5.1 (2.6) units Mean (SD) disability (0/100): 52 (28) units Mean (SD) grip strength: 60 (40) lb/in²

Nourbakhsh 2008 (Continued)

Placebo manual therapy (n = 12 participants)

Mean (SD) age: 52.4 (7.2) years; 4 females, 8 males

Mean (SD) duration of symptoms: 20.3 (22.1) months

Mean (SD) pain (/10): 4.6 (2.3) units

Mean (SD) disability (/100): 61 (22) units

Mean (SD) grip strength: 80 (30) lb/in²
Interventions
Manual therapy

Components of intervention: oscillating energy manual therapy - the participant was seated comfortably on a chair with the affected arm resting on the treatment table. First a tender point was located through palpation. Then, the therapist placed the index and middle fingers of one hand, spread in a V-shape, around the detected tender point, and placed the index finger of the other hand in the medial side of the elbow, diagonally across the located tender point. After hand placement, the therapist applied a very gentle pressure, via his fingertips, to the tissue alternatively from the medial and lateral sides a few times to initiate the back-and-forth impulse (energy) oscillation between the medial and lateral hands. At this point (upon start of the oscillating movement), the therapist stopped pressing on the tissue and allowed the oscillatory impulse to continue, with no further efforts from the therapist part, between the two points of contact on the participant's elbow. At this point, the therapist could feel continued back-and-forth oscillation of the impulse (energy) between the two points of contact on the limb. The therapist focused the direction of the oscillating energy on the tender point by slightly moving the index finger on the medial side of the elbow. Treatment was repeated a few times until the treated tender point was not tender with palpation. The same procedure was repeated for all identified tender points during each session.

Dose: all participants received 6 treatment sessions in a 2- to 3-week period. Each treatment session, depending on the number of tender points explored and treated, took about 20 to 30 minutes. Treatment duration varied from point to point, ranging from 30 seconds to 2 minutes.

Supervision: orthopaedic clinical specialist, trained in craniosacral therapy

Additional treatment: none reported

Placebo manual therapy

Components of intervention: participants underwent the exact procedure explained above. The only difference was that after locating the point of greatest tenderness, the clinician placed their index and middle finger, spread in V-shape, away from the located tender point and no oscillating energy was directed to the affected area. For participants to have a sense of a treatment being provided, the therapist applied a very gentle pressure to the tissue and imparted periodic short-duration (about 5 seconds) oscillation to the non-affected areas.

Dose: all participants received 6 treatment sessions in a 2- to 3-week period. Each treatment session, depending on the number of tender points explored and treated, took about 20 to 30 minutes.

Supervision: orthopaedic clinical specialist, trained in craniosacral therapy

Additional treatment: none reported

Co-interventions

None reported

Outcomes

Outcomes were reported at 2 to 3 weeks (end of intervention) and 6 months follow-up (last time point).

Outcomes

- Pain severity in the previous 24 hours on a 0 to 10 NRS (0 = no pain)
- Disability in the previous 24 hours on a 0 to 10 NRS (10 = no disability)

Nourbakhsh 2008 (Continued)

- Disability using the PSFS (30 = no disability, participants listed 3 of their activities most affected by the injury on a 0 to 10 scale with scores a combined total)
- Grip strength (lb/in²) using a Jamar dynamometer (based on the mean of 3 efforts)

Outcomes used in this review

Major outcomes

- Mean pain severity
- Mean disability

Minor outcomes

- Mean grip strength (lb/in²)

Source of funding	Not reported
Notes	Trial registration: not reported Data analysis: we used data at 2 to 3 weeks (end of intervention) for Analysis 1 - Manual therapy versus placebo manual therapy; for the disability outcome we used the NRS and transformed the scale to 0 to 100 (0 = no disability) to standardise the analysis. We did not use data for 6 months as data were only reported for the manual therapy group. Adverse events: not reported Withdrawals: not reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Participants were randomly assigned into treatment or placebo groups by the treating therapist with a sequence generated by drawing cards (A or B).
Allocation concealment (selection bias)	Low risk	The method of allocation was likely concealed as participants drew a card out of a set of cards marked as "Group A" or "Group B" and it was likely unknown which group was A or B.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Placebo manual therapy applied by a therapist not blinded. The participants were blinded to treatment allocation and likely unaware of the group they were allocated to.
Blinding of outcome assessment - self reported (detection bias)	Low risk	Participants were blind to treatment allocation, reducing bias in the measurement of pain and disability.
Blinding of outcome assessment - assessor reported outcomes	Low risk	The outcome assessor for grip strength was blinded to the participants' group assignment.
Incomplete outcome data (attrition bias) All outcomes	Low risk	There were no missing outcome data.
Selective reporting (reporting bias)	Unclear risk	There was no trial registry or protocol available, so we judged this as an unclear risk of selective reporting. We were also unsure if other outcomes were measured, including adverse events and withdrawals.

Nourbakhsh 2008 (Continued)

Other bias	Low risk	While grip strength was observed to be higher in the placebo group at baseline, there were no baseline differences observed for pain or disability.
------------	----------	---

Olaussen 2015

Study characteristics

Methods	Design: parallel-group, 3-arm randomised controlled trial Setting: primary care, Oslo, Norway Timing: April 2009 to June 2012 Interventions: (1) manual therapy, exercise and glucocorticoid injection; (2) manual therapy, exercise and placebo injection; (3) advice; 6 week intervention period Sample size: 60 participants per group provided 80% power to detect a difference of at least 25% of participant-reported success rate at 3 months between groups, based on an estimated success rate of 55%, with a significance level at 5% Intention-to-treat analysis: yes
---------	--

Participants	Number of participants • 256 participants assessed for eligibility • 79 participants excluded (69 did not meet inclusion criteria, 10 participants declined to participate) • 177 participants randomised (59 to manual therapy, exercise and glucocorticoid injection, 58 to manual therapy, exercise and placebo injection, 60 to advice) • Data for 171 participants (97%) available at 6-week follow-up (54 for manual therapy, exercise and glucocorticoid injection, 57 for manual therapy, exercise and placebo injection, 60 for advice) • Data for 163 participants (94%) available at 3-month follow-up (53 for manual therapy, exercise and glucocorticoid injection, 54 for manual therapy, exercise and placebo injection, 56 for advice) • Data for 160 participants (90%) available at 6-month follow-up (52 for manual therapy, exercise and glucocorticoid injection, 53 for manual therapy, exercise and placebo injection, 55 for advice) • Data for 157 participants (89%) available at 12-month follow-up (52 for manual therapy, exercise and glucocorticoid injection, 53 for manual therapy, exercise and placebo injection, 52 for advice) Inclusion criteria • Aged 18 to 70 years • Seeing their general practitioner with pain of recent onset from the lateral part of the elbow were eligible for inclusion • Pain increase on resisted dorsiflexion of the wrist with the elbow extended, and the fingers flexed, resisted radial deviation of the wrist or resisted extension of the third finger Exclusion criteria • Less than 2 weeks symptom duration • Tenderness within the muscle body itself • Treatment with corticosteroid injection or physiotherapy within the last 12 months • Bilateral symptoms • Previous surgical treatment for lateral epicondylitis • Deformities of the elbow • Cervical radiculopathy • Referred pain from neck or shoulder • Previous fractures or tendon ruptures in the elbow
--------------	---

Olaussen 2015 (Continued)

- Systemic musculoskeletal disease
- Previous allergic reactions or contraindications to corticosteroids, lidocaine or NSAIDs
- Pregnancy or breastfeeding
- Fertile females not on effective birth control
- Psycho-social or other reasons for not being able to participate throughout the study

Baseline characteristics

Manual therapy, exercise and glucocorticoid injection (n = 59 participants)

Mean (SD) age: 47.9 (9.6) years; 26 female, 33 male

Median (IQR) duration of symptoms: 6 (4 to 10) weeks

Dominant elbow affected: 43 participants (73%)

Exercises regularly: 24 participants (41%)

Employment involves manual labour: 35 participants (59%)

Mean (SD) pain score (/100): 56 (19.7) units

Mean (SD) disability using Pain Free Function Index: 5.5 (1.7) units

Mean (SD) quality of life using overall complaints scale: 69 (17.1) units

Mean (SD) pain-free grip strength ratio: 37 (27) units

Manual therapy, exercise and placebo injection (n = 58 participants)

Mean (SD) age: 48.8 (9.4) years; 20 female, 38 male

Median (IQR) duration of symptoms: 8 (4 to 10) weeks

Dominant elbow affected: 42 participants (72%)

Exercises regularly: 32 participants (55%)

Employment involves manual labour: 29 participants (50%)

Mean (SD) pain score (/100): 53 (19.3) units

Mean (SD) disability using Pain Free Function Index: 5.3 (1.8) units

Mean (SD) quality of life using overall complaints scale: 62 (19.3) units

Mean (SD) pain-free grip strength ratio: 38 (27) units

Advice (n = 60 participants)

Mean (SD) age: 44.0 (9.7) years; 25 female, 35 male

Median (IQR) duration of symptoms: 7.5 (4 to 10) weeks

Dominant elbow affected: 41 participants (68%)

Exercises regularly: 30 participants (50%)

Employment involves manual labour: 36 participants (60%)

Mean (SD) pain score (/100): 48 (21.5) units

Mean (SD) disability using Pain Free Function Index: 5.4 (1.6) units

Mean (SD) quality of life using overall complaints scale: 62 (20.5) units

Olaussen 2015 (Continued)

Mean (SD) pain-free grip strength ratio: 35 (27) units

Interventions
Manual therapy and exercise

Components of intervention: manual therapy - (1) deep transverse friction massage at the tendon origin; (2) Mill's manipulation and (3) stretching of the radial wrist extensors; *exercise* - participants received oral and written instructions for home exercises including eccentric exercise and isolated stretching of radial wrist.

Dose: manual therapy - deep transverse friction massages were provided for 15 minutes per session, Mill's manipulation was provided once per session with 2 sessions per week for 6 weeks; *exercise* - performed daily for 6 weeks with eccentric exercise (3 times 30 repetitions) and isolated stretching of radial wrist extensors (3 times daily for 40 seconds), eccentric exercise was done within tolerable pain level during the exercise by using a 500 mL bottle filled with increasing volume of water or sand as patients improved; *injections* - twice (at baseline and after 3 weeks with the second injection was not given if there had been adverse reactions or increase in symptoms).

Supervision: physiotherapist performed manual therapy; exercise unsupervised performed at home

Additional treatment: none reported

Glucocorticoid injection

Components of intervention - participants received injections of 10 mg triamcinolone acetonide (1 mL of 10 mg/mL) and 0.5 mL of 2% lidocaine. The injection was placed into the point of maximal tenderness down to the level of the bone and then pulled back 1 to 2 mm, leaving several small depots at the surface of the tendon.

Placebo injection

Components of intervention: participants received an injection with 1 mL of 0.9 % isotonic saline. The injection was placed into the point of maximal tenderness down to the level of the bone and then pulled back 1 to 2 mm, leaving several small depots at the surface of the tendon. The second injection was not given if there had been adverse reactions or increase in symptoms.

Advice

Single session of advice to 'wait and see' including the natural course of the condition and expected duration of symptoms, and encouragement to use their arm as usual, but avoid pain-provoking activities; naproxen 500 mg twice daily for 2 weeks; paracetamol could be taken at the participant's own discretion up to 4 g daily.

Co-interventions

Advice

Outcomes

Outcomes were reported at 6 weeks (end of intervention), 3 months, 6 months and 12 months follow-up (last time point).

Outcomes

- Pain, 0 to 100 VAS (0 = no pain)
- Pain-free isometric activities (dorsiflexion of wrist; extension of third finger) on 3 point Likert scale (1 = no pain, 2 = some pain, 3 = strong pain)
- Disability, 0 to 100 VAS (0 = no disability)
- Disability, 0 to 8 scale using Pain Free Function Index (0 = no disability)
- Global improvement reported by the participant on a 6-point Likert scale (1 much worse; 2 worse; 3 a little worse; 4 some improvement; 5 much improvement; 6 completely recovered)
- Overall complaints, 0 to 100 VAS (0 = no complaints)
- Number of participant overall withdrawals
- Number of participants experiencing an adverse event
- Number of participants on sick leave

Olaussen 2015 (Continued)

- Pain-free and maximum grip strength (ratio affected/unaffected arm x 100)

Outcomes used in this review
Major outcomes

- Mean pain
- Mean disability using the 0 to 8 scale, Pain Free Function Index (0 = no disability)
- Participant-reported treatment success (measured as 'completely recovered' or 'much improvement')
- Proportion of overall withdrawals
- Proportion with any adverse event

Minor outcomes

- Mean pain-free grip strength (ratio affected/unaffected arm x 100)
- Proportion on sick leave

Source of funding	Fund for General Practice Research, The Norwegian Medical Association, Oslo, Norway Naproxen and lidocaine were provided by Roche Norge AS, Oslo, Norway and AstraZeneca AS, Oslo, Norway, respectively.
Notes	Trial registration: ClinicalTrials.gov (NCT00826462) Data analysis: we used data at 6 weeks (end of intervention) and 12 months (last time point) for Analysis 2 - Manual therapy, prescribed exercise or both versus minimal or no intervention; and Analysis 3 - Manual therapy and prescribed exercise versus minimal or no intervention. For the pain outcome we transformed the data to a 0 to 10 scale, and for the disability outcome we transformed the data to a 0 to 100 scale to standardise the measure for the meta-analyses. We did not use data at 3 months and 6 months follow-up. We did not use data from the manual therapy, exercise and glucocorticoid group. Adverse events: in the manual therapy, exercise and glucocorticoid injection group, 6 participants (11%) reported an adverse event; in the manual therapy, exercise and placebo injection group, 5 participants (9%) reported an adverse event; in the no intervention group, 9 participants (16%) reported an adverse event. Mild adverse events included gastrointestinal side effects and increased pain. There were no serious adverse events. Withdrawals: in the manual therapy, exercise and glucocorticoid injection group, 7 participants (12%) did not complete the trial; in the manual therapy, exercise and placebo injection group, 5 participants (9%) did not complete the trial; in the no intervention group, 8 participants (13%) did not complete the trial.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Participants were allocated to groups based on a computer-generated random sequence with numeric blocks of variable size.
Allocation concealment (selection bias)	High risk	While allocation to steroid or placebo injections was concealed by using an independent research assistant to prepare syringes, allocation to the minimal treatment group was not concealed.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Neither the participants nor the clinician were blind to the manual therapy and exercise treatments due to the nature of the intervention. This increases the risk of performance bias. However, participants were blinded to the injection type (glucocorticoid or placebo).

Olaussen 2015 (Continued)

Blinding of outcome assessment - self reported (detection bias)	High risk	Participants were not blind to treatment allocation, and pain, disability and treatment success were likely to be influenced by lack of blinding.
Blinding of outcome assessment - assessor reported outcomes	Low risk	The outcome assessors were blinded to group allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Only a very small amount of missing data was observed, and it was equal across groups.
Selective reporting (reporting bias)	Low risk	Data were reported for all outcomes listed in the protocol and methods, so we judged it as a low risk of selective reporting.
Other bias	Low risk	The trial appears to be free of other sources of bias.

Peterson 2011
Study characteristics

Methods	Design: parallel-group, 2-arm randomised controlled trial Setting: primary care, Uppsala, Sweden Timing: December 2004 to October 2006 Interventions: (1) exercise; (2) advice; 3 month intervention period Sample size: calculation method not performed Intention-to-treat analysis: yes
Participants	Number of participants • 111 participants assessed for eligibility • 30 participants excluded (16 did not meet inclusion criteria, 14 had one or more exclusion criteria) • 81 participants randomised (40 to exercise, 41 to no intervention) • Data for 79 participants (98%) available at 4-week follow-up (38 for exercise, 41 for advice) • Data for 81 participants (100%) available at 8-week follow-up (40 for exercise, 41 for advice) • Data for 80 participants (99%) available at 3-month follow-up (40 for exercise, 40 for advice) Inclusion criteria • Aged 20 to 75 years • Symptoms for more than 3 months • A verified diagnosis of lateral elbow pain using pain on palpation and a positive outcome for one or more of the following: stretching (Mill's test), loading (maximum voluntary contraction (MVC)) and Maudsley's middle finger test Exclusion criteria • Concomitant supinator syndrome • Compartment syndrome of the anconeus muscle • Rhizopathy • Inflammatory joint disease • Fibromyalgia

Peterson 2011 (Continued)

- Previous elbow surgery
- Inability to understand Swedish

Baseline characteristics

Exercise (n = 40 participants)

Mean (SD) age: 49.8 (8.1); 16 female, 24 male

Mean (SD) duration of symptoms: 106.6 (192.7) weeks

Mean (SD) number of previous episodes: 1.3 (3.91)

Mean (SD) time since last episode: 76.2 (202.14) weeks

Education level: compulsory education only - 3 participants (7.5%); vocational training - 5 participants (12.5%); upper secondary school - 12 participants (30%); college or university - 20 participants (50%)

Marital status: never married - 2 participants (5%); married or cohabiting - 35 participants (87.5%); divorced or widowed - 3 participants (7.5%)

Smoking habit: never smoked - 25 participants (62.5%); ex-smoker - 12 participants (30%); current smoker - 3 participants (7.5%)

Mean (SD) pain level (/100): 42.2 (26.5) units.

Mean (SD) disability (/100): 28.7 (7.4) units

Quality of life using Gothenburg quality of life instrument sub-scales

- Mean (SD) activity score: 29.1 (7.4) units
- Mean (SD) well-being: 5.4 (1.3) units
- Mean (SD) complaint score: 6.1 (4.1) units

No intervention (n = 41 participants)

Mean (SD) age: 47.4 (8.6); 18 female, 21 male

Mean (SD) duration of symptoms: 95.6 (118.8) weeks

Mean (SD) number of previous episodes: 0.8 (2.05)

Mean (SD) time since last episodes: 44.6 (142.34) weeks

Education level: compulsory education only - 2 participants (4.9%); vocational training - 8 participants (19.5%); upper secondary school - 12 participants (29.3%); college or university - 19 participants (46.3%)

Marital status: never married - 3 participants (7.3%); married or cohabiting - 36 (87.8%); divorced or widowed - 2 (4.9%)

Smoking habit: never smoked - 20 participants (46.8%); ex-smoker - 19 participants (46.3%); current smoker - 2 participants (4.9%)

Mean (SD) pain level (/100): 33.9 (29.3) units

Mean (SD) disability (/100): 24.6 (14.7) units

Quality of life using Gothenburg quality of life instrument sub-scales

- Mean (SD) activity score: 28.0 (6.7) units
- Mean (SD) well-being score: 5.7 (1.3) units
- Mean (SD) complaint score: 5.0 (5.2) units

Interventions

Exercise

Peterson 2011 (Continued)

Components of intervention: progressive resistance exercise for extensor muscles of the affected forearm, including eccentric and concentric exercises. The loading equipment consisted of plastic water containers with a handle. The initial load was standardised to 1 kg (1 litre of water) for women and 2 kg for men. Participants sat in a chair and supported the forearm on the arm rest or on an adjacent table. Holding the handle of the plastic water container with a clenched fist in pronation and the container hanging freely in front of the armchair or below the table-top, the load was lifted or lowered.

Dose: 3 sets of 15 repetitions per session, once daily for 3 months

Setting: home

Supervision: monitoring at 4-week and 8-week follow-up

Additional treatment: none

Advice

Single session of advice that the condition was painful but harmless, that the arm should be used in ordinary daily activities, and advised to 'wait and see'. Participants were permitted to use paracetamol as needed (not other pain relieving or anti-inflammatory medication).

Co-interventions

Advice

Outcomes

Outcomes were reported at 4 weeks (first time point), 8 weeks and 3 months (end of intervention).

Outcomes

- Pain during maximum voluntary muscle contraction, 0 to 100 VAS (0 = no pain)
- Pain during maximum muscle elongation with a load, 0 to 100 VAS (0 = no pain)
- Disability, 0 to 100 scale using DASH questionnaire (0 = no disability)
- Quality of life using 3 sub-scales of the Gothenburg Quality of Life questionnaire - complaint score (lower score = lesser tendency to complain); activity score (higher score = more active lifestyle) and well-being score (7-point Likert scale ranging from 1 = 'very bad' to 7 'excellent, could not be better')
- Participant overall withdrawals
- Muscle strength (Newtons) of forearm extensors measured with a hand-held dynamometer

Outcomes used in review
Major outcomes

- Mean pain (during maximum voluntary muscle contraction)
- Mean disability
- Mean quality of life using well-being score from Gothenburg Quality of Life questionnaire (7-point Likert scale ranging from 'very bad' to 'excellent, could not be better')
- Proportion of overall withdrawals

Minor outcomes

- None

Source of funding

This study was supported by grants from the Swedish Research Council (grant no. K2005-27X-15293-01A), The Amersham Fund at Uppsala University, The Research Fund at Uppsala County Council, The Family Medicine Foundation and Uppsala University.

Notes

Trial registration: not reported

Data analysis: we used data at 4 weeks (first follow-up, pain scores only) and 3 months (end of intervention) for Analysis 2 - Manual therapy, prescribed exercises or both versus minimal or no intervention; and Analysis 5 - Prescribed exercises versus minimal or no intervention. We did not use data from

Peterson 2011 (Continued)

8 weeks follow-up. For the pain outcome, we transformed the data to a 0 to 10 scale to standardise the measure for the meta-analyses.

Adverse events: not reported

Withdrawals: in the exercise group, 0/40 participants (0%) did not complete the trial; in the advice group, 0/41 participants (0%) did not complete the trial

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Participants were randomly assigned using a computerised random block design.
Allocation concealment (selection bias)	Unclear risk	Participants were "blindly assigned" by the author. However, it is unclear whether this was concealed, so we judged this as an unclear risk of bias.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants and personnel were not blind to treatment allocation.
Blinding of outcome assessment - self reported (detection bias)	High risk	Participants were not blind to treatment allocation, and the outcomes of pain and disability were likely to be influenced by lack of blinding.
Blinding of outcome assessment - assessor reported outcomes	High risk	It was not reported if the assessment of grip strength was measured by an assessor blinded to treatment allocation, and measurement of grip strength was likely influenced by a lack of blinding.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Only 2 participants at the 1-month time point in the exercise group, and 1 participant at 3-month time point were missing, with no participants excluded from the analysis.
Selective reporting (reporting bias)	Unclear risk	While data were reported for all outcomes listed in the methods, there was no trial protocol or trial registry reported, so we judged this as an unclear risk of selective reporting.
Other bias	Low risk	The study appears to be free of other sources of bias.

Rachold 2023

Study characteristics

Methods	Design: parallel-group, 3-arm randomised controlled trial Setting: single centre, Germany Interventions: (1) manual therapy; (2) forearm brace and electrotherapy; (3) manual therapy, forearm brace, electrotherapy. 3-week intervention period. Sample size: target 52 participants in total provided 80% power to detect an effect size of 0.25, with a significance level of 5% Intention-to-treat analysis: yes
Participants	Number of participants

Rachold 2023 (Continued)

- 92 participants assessed for eligibility
- 13 participants excluded (13 did not meet inclusion criteria)
- 79 participants randomised (30 to manual therapy, 28 to forearm brace, electrotherapy, 21 to manual therapy, forearm brace, electrotherapy)
- Data for 52 participants (66%) available at 4-week follow-up (18 for manual therapy, 15 for forearm brace and electrotherapy, 19 for manual therapy, forearm brace and electrotherapy)
- Data for 52 participants (66%) available at 8-week follow-up (18 for manual therapy, 15 for forearm brace and electrotherapy, 19 for manual therapy, forearm brace and electrotherapy)

Inclusion criteria

- Age 18 to 65 years
- Criteria for the diagnosis of lateral epicondylitis needed to be fulfilled: (1) pain was reported for the lateral region of the elbow, (2) an extension of the hand joint provoked significant pain (provocation test), and (3) other diseases or injuries were excluded using X-ray imaging

Exclusion criteria

- Injuries to the arm or cervical spine within 6 months before presentation
- Acute or chronic infection, rheumatic diseases, osteochondritis dissecans, advanced osteoarthritis of the elbow, pregnancy, diabetes, malignancy, malformation in the upper extremities, circulatory disorder in the upper extremities, radicular syndrome at C6, chronic pain syndrome, psychiatric disorders, neuropathy, myopathy, the use of beta-blockers, statins, or non-steroidal anti-inflammatory drugs or incomplete data.

Baseline characteristics

Manual therapy (n = 18 participants)

Mean (SD) age: 48.1 (9.7) years; 12 female, 6 male

Mean (SD) pain (0 to 10): 5.22 (1.83) points

Mean (SD) disability (0 to 10) using PRTEE: 41.03 (17.5) points

Mean health-related quality of life (0 to 100) using SF-36: 61.6 (17.8) points

Forearm brace and electrotherapy (n = 15 participants)

Mean (SD) age: 45.7 (9.7) years; 10 female, 5 male

Mean (SD) pain (0 to 10): 5.87 (1.54) points

Mean (SD) disability (0 to 10) using PRTEE: 39.7 (16.3) points

Mean health-related quality of life (0 to 100) using SF-36: 70.7 (12.4) points

Manual therapy, forearm brace, electrotherapy (n = 19 participants)

Mean (SD) age: 45.9 (9.4) years; 9 female, 10 male

Mean (SD) pain (0 to 10): 5.45 (1.09) points

Mean (SD) disability (0 to 10) using PRTEE: 42.3 (15.2) points

Mean health-related quality of life (0 to 100) using SF-36: 65.1 (15.1) points

Interventions

Manual therapy

Components: involved manipulating the joints, muscles, or fascia by a specialised therapist who received correspondence from the primary physician regarding functional chains and trigger points

Dose: ~20 min twice per week over 3 weeks

Rachold 2023 (Continued)

Forearm brace and electrotherapy

Components: an extensor forearm brace was used to provide constant mechanical therapy. Biphasic TENS mode at 300 ms and 50 Hz for analgesia and detensioning, respectively for ~20 min twice per week over 3 weeks.

Co-interventions

None

Outcomes

Outcomes were measured at 4 weeks (end of intervention) and 8 weeks (last time point).

Outcomes

- Pain, 0 to 10 VAS (0 = no pain)
- Disability, 0 to 100 using PRTEE (0 = no disability)
- Health-related quality of life, 0 to 100 using Short form 36 questionnaire
- Participant overall withdrawals
- Range of motion of the elbow joint in degrees

Outcomes used in the review

Major outcomes

- Mean pain
- Mean disability
- Mean health-related quality of life
- Proportion of overall withdrawals.

Minor outcomes

- None

Source of funding

Funding: institutional budget, no external funding

Notes

Trial registration: DRKS00021137. Therapeutic effects of manual therapy, transcutaneous electric nerve stimulation and forearm (extensor) brace on lateral epicondylitis - a randomised controlled trial. https://www.drks.de/drks_web/navigate.do?navigationId=trial.HTML&TRIAL_ID=DRKS00021137

Data analysis: we used data at 4 weeks (end of intervention) and 8 weeks (last time point) in Analysis 2 - Manual therapy, prescribed exercises or both versus minimal intervention (forearm brace, electrotherapy) and Analysis 4 - Manual therapy versus minimal intervention.

Adverse events: not reported

Withdrawals: in the manual therapy group, 12/30 participants (40%) did not complete the trial; in the forearm brace and electrotherapy group, 13/28 participants (46%) did not complete trial; in the manual therapy, forearm brace, electrotherapy group, 2/21 participants (10%) did not complete the trial.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Participants were randomised into one of 3 treatment arms using a randomiser software developed by Medsharing (Paris, France) https://www.med-sharing.fr .
Allocation concealment (selection bias)	Low risk	The authors reported concealed allocation.

Rachold 2023 (Continued)

Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants and personnel were not blind to treatment allocation.
Blinding of outcome assessment - self reported (detection bias)	High risk	Participants were not blind to treatment allocation, and pain, disability and quality of life were self-reported outcomes.
Blinding of outcome assessment - assessor reported outcomes	Low risk	No assessor-reported reported outcomes collected in our review were measured in this trial.
Incomplete outcome data (attrition bias) All outcomes	High risk	A high proportion of participants in 2 of the 3 groups were lost to follow-up without any outcomes reported.
Selective reporting (reporting bias)	Low risk	The study was retrospectively registered as a clinical trial, however results for all prespecified outcomes were reported.
Other bias	High risk	Despite concealed allocation at randomisation, participants had slightly worse quality of life at baseline, which may have resulted in larger improvement in the minimal treatment group.

Reyhan 2020

Study characteristics

Methods	Design: parallel-group, 2-arm randomised controlled trial Setting: physical therapy and rehabilitation department of Istanbul University, Turkey Timing: not reported Interventions: (1) manual therapy and exercise; (2) exercise; 2-week intervention period Sample size: 13 participants per group provided 80% power to detect a difference in resting pain of 1.4 units (0 to 10 VAS) between groups, with a significance level at 5% Intention-to-treat analysis: not reported
Participants	Number of participants 44 participants assessed for eligibility 4 participants excluded (4 did not meet inclusion criteria) 40 participants randomised (20 to manual therapy and exercise, 20 to exercise) - Data for 40 participants (100%) available at 2-week follow-up (20 for manual therapy and exercise, 20 for exercise) - Data for 40 participants (100%) available at 6-week follow-up (20 for manual therapy and exercise, 20 for exercise) - Data for 40 participants (100%) available at 3-month follow-up (20 for manual therapy and exercise, 20 for exercise) Inclusion criteria - Patients with chronic lateral elbow pain diagnosis (for > 6 weeks) whose pain reduced in the first mobilisation-with-movement session

Reyhan 2020 (Continued)

- No dysfunction related to the shoulder, neck or thoracic region
- Did not receive any therapy within the last 6 months
- Willing and suitable for continuing the programme

Exclusion criteria

- Diagnosis of bilateral lateral elbow pain, major trauma (such as fracture) or entrapment neuropathy
- History of tendon or ligament tear
- More than 3 cortisone injections in the last 6 months
- Lateral elbow surgery within the last 2 months
- History of rheumatic diseases, neurological disorders such as stroke or brain trauma

Baseline characteristics

Manual therapy and exercise (n = 20 participants)

Mean (SD) age: 43.4 (10.6) years; 15 female, 5 male

Mean (SD) duration of symptoms: 4.05 (0.63) months

Mean (SD) pain at rest (/10): 1.20 (1.01) units

Mean (SD) pain during activity (/10): 5.10 (1.52) units

Mean pain at night (/10): 2.20 (1.40) units

Mean pain-free grip strength: 21.79 (10.67) kg

Mean maximum grip strength: 27.18 (10.02) kg

Exercise (n = 20 participants)

Mean (SD) age: 41.35 (9.39) years; 18 female, 2 male

Mean (SD) duration of symptoms: 3.98 (0.91) months

Mean (SD) pain at rest (/10): 1.60 (1.27) units

Mean (SD) pain during activity (/10): 5.20 (2.14) units

Mean pain at night (/10): 2.85 (0.75) units

Mean pain-free grip strength: 25.22 (8.93) kg

Mean maximum grip strength: 30.15 (8.83) kg

Interventions

Manual therapy and exercise

Components of intervention: manual therapy - a mobilisation with movement technique using a lateral glide was applied with the participant in the supine position. First, the angle for pain-free application was determined for each patient. The distal humerus was fixed by the therapist holding it down with their web space positioned just above the elbow joint on the lateral humeral condyle; *exercise* - refer to exercise component of exercise group.

Dose: participants were treated 5 days per week for 2 weeks; *manual therapy* - repeated 10 times for 3 sets, with an interval of 15 to 20 seconds between sets; *exercise* - refer to exercise component of exercise group

Supervision: 1 physiotherapist

Setting: physiotherapy clinic and home setting

Additional treatment: participants were trained in manual therapy techniques to perform at home

Exercise

Reyhan 2020 (Continued)

Components of intervention: exercise - stretching the wrist extensors and flexors with the elbow extended and the forearm pronated; strengthening wrist flexors and extensors; active elbow flexion and extension; isometric wrist extension; isometric handgrip using a tennis ball with elbow at full extension and pronation, and wrist in extension

Dose: exercise - stretching for 30 seconds, followed by 30 seconds rest, repeated 3 times; active elbow movements for 20 repetitions; resistance for 3 sets, 20 repetitions, 1-minute rest between sets; resistance (isometrics) for 20 seconds, 20 repetitions

Supervision: none

Setting: home

Additional treatment: none

Co-interventions

All participants were advised to use cold application to minimise potential symptomatic pain caused by stretching and strengthening exercises. Additionally, both groups were informed about the movements they should avoid during treatment, such as heavy lifting, knitting, handwriting, driving, using a screw-driver, opening a jar, waving the hand and using a mouse.

Outcomes	Outcomes were reported at 2 weeks (first time point), 6 weeks (end of intervention) and 3 months (last time point). Outcomes - Pain at rest, during movement and at night, 0 to 10 VAS (0 = no pain) - Disability, 0 to 100 scale using PRTEE questionnaire (0 = no disability) - Global improvement reported by the participant on a 6-point scale (1 = complete recovery; 2 = significant recovery; 3 = mild recovery; 4 = no change; 5 = slight worsening; 6 = worse) - Participant overall withdrawals - Pain-free and maximum grip strength (kg) using a dynamometer (based on the mean of 3 efforts) Outcomes included in this review <i>Major outcomes</i> - Mean change in disability - Participant-reported treatment success measured as 'complete recovery' or 'significant recovery' - Proportion of overall withdrawals <i>Minor outcomes</i> - Mean pain-free grip strength (kg)
Source of funding	Not reported
Notes	Trial registration: not reported Data analysis: we used data at 2 weeks (first time point), 6 weeks (end of intervention) and 3 months (last time point) for Analysis 6 - Manual therapy and prescribed exercises versus prescribed exercises alone. For the disability outcome we estimated the mean change score and SD values from the median change score and range values using Wan 2014. We were unable to use data for the pain outcomes in the meta-analyses due to insufficient data reporting. Adverse events: no adverse events Withdrawals: in the manual therapy and exercise group, 0/20 participants (0%) did not complete the trial; in the exercise group 0/20 participants (0%) did not complete the trial

Risk of bias

Reyhan 2020 (Continued)

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The sequence was randomly generated via a computer-based system.
Allocation concealment (selection bias)	Low risk	Allocation using closed envelopes was performed by a therapist who was not included in the treatment sessions.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants and the physiotherapist were not blind to treatment allocation.
Blinding of outcome assessment - self reported (detection bias)	High risk	Participants were not blind to treatment allocation, and the measures of pain, disability and global assessment were likely influenced by lack of blinding.
Blinding of outcome assessment - assessor reported outcomes	High risk	The assessor was not reported as blinded to treatment allocation, and measurement of grip strength was likely influenced by lack of blinding.
Incomplete outcome data (attrition bias) All outcomes	Low risk	There was no loss to follow-up, and no participants were excluded from the analysis.
Selective reporting (reporting bias)	High risk	No trial protocol or trial registry was reported. For pain outcomes, sufficient data were not available to compare this outcome between groups, so we judged it as a high risk of selective reporting.
Other bias	Unclear risk	There are some reporting errors. For example, 44 patients volunteered for the study with 4 participants excluded. However, in Fig. 1, only 40 participants that completed all sessions of treatment were reported to have been screened for eligibility, and no participants were excluded. Also, the identical means and SDs for the exercise group were questionable.

Smidt 2002a
Study characteristics

Methods	Design: parallel-group, 3-arm randomised controlled trial Setting: primary care, Amsterdam, Netherlands Timing: September 1997 to October 1998 Interventions: (1) manual therapy and exercise; (2) glucocorticoid injection; (3) advice; 6 week intervention period Sample size: 60 participants per group provided 80% power to detect a clinically important difference of 25% in success rate between groups, assuming a success rate of 40% in the least successful group at 6 weeks, with a significance level at 5% Intention-to-treat analysis: yes
Participants	Number of participants 259 participants assessed for eligibility

Smidt 2002a (Continued)

- 74 participants excluded (5 did not meet inclusion criteria, 52 met exclusion criteria, 17 gave no informed consent)
- 185 participants randomised (64 to manual therapy and exercise, 62 to glucocorticoid injection and 59 to advice)
- Data for 184 participants (99%) available at 3-week follow-up (64 for manual therapy and exercise, 62 for glucocorticoid injection and 58 for advice)
- Data for 185 participants (100%) available at 6-week follow-up (64 for manual therapy and exercise, 62 for glucocorticoid injection and 59 for advice)
- Data for 183 participants (99%) available at 3-month follow-up (64 for manual therapy and exercise, 60 for glucocorticoid injection and 59 for advice)
- Data for 183 participants (99%) available at 6-month follow-up (64 for manual therapy and exercise, 60 for glucocorticoid injection and 59 for advice)
- Data for 183 participants (99%) available at 12-month follow-up (64 for manual therapy and exercise, 60 for glucocorticoid injection and 59 for advice)

Inclusion criteria

- Pain at the lateral side of the elbow, increasing with pressure on the lateral epicondyle and with resisted dorsiflexion of the wrist
- Age 18 to 70 years
- Ability to complete questionnaires in Dutch

Exclusion criteria

- Treatment of elbow complaints with physiotherapy or injections during the preceding 6 months
- Bilateral elbow symptoms
- Duration less than 6 weeks
- Presence of signs and symptoms suggestive of another cause of elbow pain (e.g. cervical radiculopathy)
- Congenital or acquired deformities of the elbow
- Surgery of the elbow
- Dislocation, tendon ruptures or fractures in the area in the preceding 12 months
- Systemic musculoskeletal or neurological disorders
- Contraindications for corticosteroids

Baseline characteristics

Manual therapy and exercise (n = 64 participants)

Median (IQR) age: 48 (41 to 52) years; 28 female, 36 male

Median (IQR) duration of current episode: 11 (8 to 21) weeks

Acute onset: 17 participants (27%)

Use of analgesia during past week: 8 participants (13%)

Dominant elbow affected: 51 participants (80%)

Concomitant neck disorders: 9 participants (14%)

Previous episodes of lateral elbow pain: 16 participants (25%)

Alleged cause: overuse (usual activities) - 21 participants (33%); overuse (unusual activities) - 15 participants (23%); other (such as sport, unexpected movement) - 12 participants (19%); unknown - 16 participants (25%)

Median (IQR) pain during day (/100): 60 (40 to 70) units

Median (IQR) elbow disability (/100): 48 (38 to 60) units

Smidt 2002a (Continued)

Median (IQR) pain-free grip strength: 27 (18 to 43) kg

Glucocorticoid injection (n = 62 participants)

Median (IQR) age: 47 (41 to 54) years; 34 female, 28 male

Median (IQR) duration of current episode: 11 (8 to 16) weeks.

Acute onset: 16 participants (26%).

Use of analgesia during past week: 10 participants (16%)

Dominant elbow affected: 48 participants (77%)

Concomitant neck disorders: 18 participants (29%)

Previous episodes of lateral elbow pain: 25 participants (40%)

Alleged cause: overuse (usual activities) - 16 participants (26%); overuse (unusual activities) - 15 participants (24%); other (such as sport, unexpected movement) - 11 participants (18%); unknown - 20 participants (32%)

Median (IQR) pain during day (/100): 60 (40 to 70) units

Median (IQR) elbow disability (/100): 48 (38 to 61) units

Median (IQR) pain-free grip strength: 30 (22 to 50) kg

Advice (n = 59 participants)

Median (IQR) age: 46 (42 to 54) years; 31 female, 28 male

Median (IQR) duration of current episode: 11 (8 to 21) weeks

Acute onset: 13 participants (22%)

Use of analgesia during past week: 9 participants (15%)

Dominant elbow affected: 46 participants (78%)

Concomitant neck disorders: 12 participants (20%)

Previous episodes of lateral elbow pain: 13 participants (31%)

Alleged cause: overuse (usual activities) - 22 participants (37%); overuse (unusual activities) - 15 participants (25%); other (such as sport, unexpected movement) - 8 participants (14%); unknown - 14 participants (24%)

Median (IQR) pain during day (/100): 60 (30 to 70) units

Median (IQR) elbow disability (/100): 45 (35 to 55) units

Median (IQR) pain-free grip strength: 27 (17 to 48) kg

Interventions

Manual therapy and exercise

Components of intervention: manual therapy - deep friction massage; *exercise* - as devised by [Pienimäki 1996](#), involved progressive, slow, repetitive wrist and forearm stretching, muscle conditioning and occupational exercises. A home exercise equipment with an instruction book was provided.

Dose: manual therapy - 9 sessions; *exercise* - 6 weeks (started after pain reduction)

Supervision: 72 physiotherapists (31 practices) received special training for techniques required for the intervention

Smidt 2002a (Continued)

Additional treatment: pulsed ultrasound (20% duty cycle, intensity of 2 W/cm² for 7.5 minutes per session)

Glucocorticoid injection

Components of intervention: participants received a single glucocorticoid injection with every tender spot identified with the needle and injected until the patient was free of pain during resisted dorsiflexion. The amount of fluid remaining in the syringe was recorded.

Dose: 1 mL triamcinolone acetonide (10 mg/mL) and 1 mL lidocaine 2%. During the 6-week intervention period, a maximum of 3 injections was recommended.

Supervision: family doctors administered the treatment, who received training in the injection technique before the trial.

Additional treatment: patients were asked to avoid pain-provoking activities, although absolute rest of the arm was not necessary.

Advice

Components of intervention: advice about practical solutions, ergonomic advice for activities that provoked pain, encouragement to await further spontaneous improvement

Dose: one visit to family doctor during the intervention period of 6 weeks

Additional treatment: none

Co-interventions

Advice. If necessary, paracetamol (2000 to 4000 mg daily) or non-steroidal anti-inflammatory drugs (NSAIDs, naproxen 1000 mg daily) were prescribed. Additional treatments were discouraged during the 6-week intervention period.

Outcomes

Outcomes were reported at 3 weeks (first time point), 6 weeks (end of intervention), 3 months, 6 months and 12 months (last time point).

Outcomes

- Pain severity during the day, 0 to 100 VAS (100 = no pain)
- Severity of the main complaint, 0 to 100 VAS (100 = no pain)
- Disability using modified pain-free function questionnaire, 0 to 40 scale and reported as 0 to 100 scale (100 = no disability)
- Global improvement measured by the participant on a 6-point scale ranging from 'completely recovered' to 'much worse'
- Quality of life using EQ-5D-3L utility score (higher utility score = better quality of life, with a score of 1 representing perfect health and a negative utility score representing a health state worse than death)
- Number of adverse events from treatments
- Number of participant withdrawals
- Pain-free and maximum grip strength (kg) measured using a hand dynamometer and reported as a ratio of maximum grip strength of unaffected side x 100 (based on the mean of 3 efforts with 20-second rest intervals)
- Pressure pain threshold measured using algometer and reported as a ratio of the pain-free pressure of the unaffected side
- Participant satisfaction measured on 0 to 10 VAS (0 = not satisfied at all, 10 = very satisfied)
- Participant inconvenience on 0 to 10 VAS (lower score = less inconvenience)
- Assessor's rating of the severity of the elbow complaint on a 0 to 10 VAS (0 = no complaints, 10 = very severe complaints)

Outcomes included in this review

Major outcomes

Smidt 2002a (Continued)

- Mean pain
- Mean disability
- Mean quality of life
- Participant-reported treatment success (measured as 'completely recovered' or 'much improved')
- Proportion of overall withdrawals
- Proportion with any adverse event
- Proportion with a serious adverse event

Minor outcomes

- Mean pain-free grip strength (ratio of affected side/unaffected side x 100)

Source of funding	Health Insurance Council (Fund for Investigative Medicine) and the Netherlands Organisation for Scientific Research	
Notes	Trial registration: NCT00826462 Data analysis: we used data at 3 weeks (first time point), 6 weeks (end of intervention) and 12 months (last time point) for Analysis 2 - manual therapy, prescribed exercises or both versus minimal or no intervention; and Analysis 3 - manual therapy and prescribed exercises versus minimal or no intervention. For pain scores we transformed to a 0 to 10 scale where 0 = no pain; for disability scores we transformed to a 0 to 100 scale where 0 = disability. We did not use data from 3 months or 6 months follow-up. We did not use data from the glucocorticoid injection arm. We extracted the first follow-up (3 weeks) data from Figure 2. Adverse events: 87 participants (47%) experienced an adverse event from treatments, including 41 participants (64%) in the manual therapy and exercise group; 36 participants (58%) in the glucocorticoid injection group; 10 participants (17%) in the advice group. Radiating pain to forearm or upper arm was the most commonly reported side effect, including 27 participants (42%) in the manual therapy and exercise group; 17 participants (27%) in the glucocorticoid injection group; 7 participants (12%) in the advice group. No serious adverse events were reported. Withdrawals: in the manual therapy and exercise group, 0 participants (0%) did not complete the trial; in the glucocorticoid injection group, 2 participants (3%) did not complete the trial; in the advice group, 0 participants (0%) did not complete the trial.	
Risk of bias		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A computerised random number generator was used to draw up an allocation schedule.
Allocation concealment (selection bias)	Low risk	An administrative assistant allocated interventions via opaque, sealed envelopes marked according to the allocation schedule.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blind to treatment allocation, and outcomes were likely to be influenced by lack of blinding. Assessors were blinded to treatment allocation.
Blinding of outcome assessment - self reported (detection bias)	High risk	Participants were not blind to treatment allocation, and outcomes of pain, disability, quality of life and global assessment were likely to be influenced by lack of blinding.
Blinding of outcome assessment - assessor reported outcomes	Low risk	Outcome assessors were blinded to treatment allocation for measurement of grip strength.

Smidt 2002a (Continued)

Incomplete outcome data (attrition bias) All outcomes	Low risk	This trial had only 2/185 participants withdraw.
Selective reporting (reporting bias)	Unclear risk	While data were reported for all outcomes listed in the methods, there was no trial protocol or trial registry reported, so we judged this as an unclear risk of selective reporting.
Other bias	Low risk	There was sufficient reporting of baseline characteristics and co-interventions without any major differences between groups. Except there were some additional treatments between 6 weeks and 12 months follow-up (e.g. physiotherapy in the manual therapy, exercise group; and analgesia in the no intervention group).

Soderberg 2012
Study characteristics

Methods	Design: parallel-group, 2-arm randomised control trial Setting: community, Stockholm, Sweden Timing: not reported Interventions: (1) exercise; (2) advice, forearm band; 6-week intervention period Sample size: 20 participants per group provided 80% power to detect a 50% reduction in lateral epicondylalgia cases in the exercise group, compared with a 10% reduction in controls with a significance level at 5%. Intention-to-treat analysis: no, per protocol
Participants	Number of participants • 58 participants assessed for eligibility • 16 participants excluded (10 did not meet inclusion criteria, 6 met exclusion criteria) • 42 participants randomised (20 to exercise and brace, 22 to brace) • Data for 37 participants (88%) available at 3-week follow-up (19 for exercise, 18 for advice) • Data for 37 participants (88%) available at 6-week follow-up (19 for exercise, 18 for advice) Inclusion criteria • History of pain around the lateral epicondyle for at least 1 month • Pain at palpation of the lateral epicondyle of humerus • Positive results in 2 of the following 3 pain-provocation tests: middle-finger test, resisted extension of the wrist and vigorimeter test Exclusion criteria • Rheumatoid arthritis • Fibromyalgia • Previous surgery in the elbow region • Neck dysfunction suggesting possible cervical rhizopathy • Entrapment of radial nerve • Ongoing treatment • Previous treatment < 3 months before enrolment into the study

Soderberg 2012 (Continued)

Baseline characteristics:
Exercise and brace (n = 20 participants)

Mean (SD) age: 48 (12.4) years; 9 female, 11 male

Duration of symptoms > 6 months: 12 participants (60%)

Alleged cause: sport - 6 participants (30%)

Number received previous steroid injection: 6 participants (30%)

Number of smokers: 6 participants (30%)

Number performing regular exercise twice a week, > 30 minutes per occasion: 17 participants (85%)

Median (IQR) pain (/100): 24 (12 to 42)

Mean (SD) pain-free grip strength: 46 (11) kPa

Forearm band (n = 22 participants)

Mean (SD) age: 50 (10.8) years; 15 female, 7 male

Duration of symptoms > 6 months: 14 participants (64%)

Alleged cause: sport - 5 participants (23%)

Number received previous steroid injection: 6 participants (27%)

Number of smokers: 4 participants (18%)

Number performing regular exercise twice a week, > 30 minutes per occasion: 16 participants (73%)

Median (IQR) pain (/100): 31 (15 to 40)

Mean (SD) pain-free grip strength: 50 (11) kPa

Interventions
Exercise

Components of intervention: eccentric exercise for forearm extensors. Initially, they were shown how and manually guided to perform accurate eccentric action with the affected arm. With the elbow flexed, the participants were instructed to sit on a chair with the affected forearm pronated and resting on a table with the wrist and hand over the edge, holding a bucket of water as a training weight. They placed the non-affected hand over the one holding the bucket and slowly lifted it with the non-affected hand, thus avoiding the concentric phase in the affected arm. With the affected hand extended, the participant removed the unaffected hand slowly and then, counting to three, lowered the hand to cause a flexion hand movement.

Dose: 2 sets, 8 to 12 repetitions daily for 1 week; 2 sets, 8 to 12 repetitions 2 times daily for next 2 weeks; 3 sets, 8 to 12 repetitions, 2 times daily for next 3 weeks (given this instruction at 3 week follow-up). The advice was to adjust their resisted weight so that they were able to perform the prescribed repetitions during pain-free intervals (less than or equal to 2 on Borg's scale) throughout the training period.

Supervision: unsupervised, performed at home. Instruction at 3-week follow-up to progress exercises

Additional treatment: none

Forearm band, advice

Components of intervention: participants were instructed to wear a forearm band (Epi Forsa 4061 Re-band, Sweden) during all daily activities and to perform warm-up exercises for the wrist extensors: flexion, extension, abduction, adduction and circumduction, for 1 minute twice a day. They were also instructed to avoid activities that increased their pain.

Co-interventions

Soderberg 2012 (Continued)

Forearm band, advice

Outcomes

Outcomes were reported at 3 weeks (first time point) and 6 weeks (end of intervention).

Outcomes

- Pain during the previous week, 0 to 100 VAS (0 = no pain, 100 = worst imaginable pain)
- Pain-free grip strength (kPa) using a vigorimeter
- Pain-free isometric wrist extensor strength (kg) using with myometer
- Change in the proportion of cases with lateral elbow pain measured as having symptoms or not at 6 weeks
- Number of participant overall withdrawals

Outcomes used in this review
Major outcomes

- Mean pain
- Proportion of overall withdrawals

Minor outcomes

- Mean pain-free grip strength (kPa)

Source of funding

Not reported

Notes
Trial registration: ClinicalTrials.gov (NCT00975442)

Data analysis: we used data at 3 weeks (first time point) and 6 weeks (end of intervention) for Analysis 2 - Manual therapy, prescribed exercises or both versus minimal or no intervention; and Analysis 5 - Prescribed exercises versus minimal or no intervention. For the pain outcome, we extrapolated median and interquartile range values from Fig 5. and then calculated the mean and SD using Wan 2014; we transformed pain to 0 to 10 (0 = no pain) to standardise the meta-analysis. For pain-free grip strength we extracted the mean from Fig 3. and the SD from the P value.

Adverse events: not reported

Withdrawals: in the exercise group 2 participants (10%) did not complete the trial; in the forearm brace group 3 participants (14%) did not complete the trial

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A randomised block design was used with stratification for duration of elbow pain and whether the participant was referred from a medical caregiver.
Allocation concealment (selection bias)	Low risk	Participants were randomised by each participant picking from a container one of two 'folded' notes showing either a 'one' (exercise group) or a 'two' (control group).
Blinding of participants and personnel (performance bias) All outcomes	High risk	No participants or personnel were blinded.
Blinding of outcome assessment - self reported (detection bias)	High risk	The participants were not blind to treatment allocation, and the measurement of pain was likely to be influenced by lack of blinding.

Soderberg 2012 (Continued)

Blinding of outcome assessment - assessor reported outcomes	High risk	Outcomes assessors were not reported as blind to treatment allocation, and outcomes of grip strength were potentially influenced by lack of blinding.
Incomplete outcome data (attrition bias) All outcomes	Low risk	There was a small loss to follow-up, with 2/20 participants from the exercise group and 3/22 participants from the minimal treatment group lost to follow-up, and analysis conducted as per protocol.
Selective reporting (reporting bias)	Low risk	Data were reported for all outcomes listed in the trial registration and methods.
Other bias	Low risk	There were no major differences in baseline characteristics observed between groups, and the participants reported compliance of over 70% regarding the exercise programme.

Struijs 2004
Study characteristics

Methods	Design: parallel-group, 3-arm randomised controlled trial Setting: orthopaedic outpatient clinic, Amsterdam, Netherlands Timing: January 1999 to May 2000 Intervention: (1) manual therapy, exercise, forearm band; (2) manual therapy, exercise; (3) forearm band; 6-week intervention period Sample size: the sample size calculation (number of participants per group not reported) provided 80% power to detect a difference in success rate of at least 25% of participant-reported global improvement between groups at 6 weeks, with the least effective treatment, with a significance level at 5% Intention-to-treat analysis: yes
Participants	Number of participants • 221 participants assessed for eligibility • 41 participants excluded (28 did not meet inclusion criteria, 13 declined to participate) • 180 participants randomised (56 to manual therapy, exercise, band, 56 to manual therapy and exercise, 68 to band) • Data for 176 participants (98%) available at 6-week follow-up (55 for manual therapy, exercise, band, 54 for manual therapy, exercise, 67 to band) • Data for 172 participants (96%) available at 6-month follow-up (54 for manual therapy, exercise, band, 54 for manual therapy, exercise, 64 to band) • Data for 170 participants (94%) available at 12-month follow-up (54 for manual therapy, exercise, band, 53 for manual therapy, exercise, 63 to band) Inclusion criteria • Clinically diagnosed lateral epicondylitis via pain on the lateral side of the elbow, which was aggravated with both pressure on the lateral epicondyle of the humerus and resisted dorsiflexion of the wrist • Complaints for at least 6 weeks Exclusion criteria: • Bilateral complaints • Clear decrease of pain in the previous 2 weeks • Any treatment for the lateral epicondylitis episode in the last 6 months before inclusion

Struijs 2004 (Continued)

- Unable to fill out questionnaires

Baseline characteristics

Manual therapy, exercise and brace (n = 56 participants)

Mean (SD) age: 47 (9) years; 28 female, 28 male

Mean (SD) duration of symptoms: 21 (37) weeks

Dominant elbow affected: 40 participants (71%)

Neck/shoulder complaints: 10 participants (18%)

Mean (SD) pain most important complaint (/100): 72 (15) units

Mean Pain-Free Function Questionnaire (/100): 52 (16) units

Mean (SD) pain-free grip strength, ratio of non-injured arm x 100: 42 (29) units

Mean (SD) pressure pain, ratio of non-injured arm x 100: 39 (20) units

Manual therapy and exercise (n = 56 participants)

Mean (SD) age: 43 (8) years; 29 female, 27 male

Mean (SD) duration of symptoms: 16 (16) weeks

Dominant elbow affected: 43 participants (77%)

Neck/shoulder complaints: 10 participants (18%)

Mean (SD) severity of complaint (/100): 44 (18) units

Mean (SD) pain most important complaint (/100): 72 (20) units

Mean (SD) Pain-free Function Questionnaire (/100): 48 (16) units

Mean (SD) pain-free grip strength, ratio of non-injured arm x 100: 45 (25) units

Mean (SD) pressure pain, ratio of non-injured arm x 100: 51 (24) units

Brace (n = 68 participants)

Mean (SD) age (SD): 46 (11); 32 female, 36 male

Mean (SD) duration of symptoms: 23 (30) weeks

Dominant elbow affected: 50 participants (74%)

Neck/shoulder complaints: 17 participants (25%)

Mean (SD) pain most important complaint (/100): 74 (18) units

Mean (SD) Pain-free Function Questionnaire (/100): 51 (17) units

Mean (SD) pain-free grip strength, ratio of non-injured arm x 100: 45 (27) units

Mean (SD) pressure pain, ratio of non-injured arm x 100: 48 (23) units

Interventions
Manual therapy and exercise

Components of intervention: manual therapy - deep friction massage; *exercise* - when pain subsided, patients were instructed on a strengthening and stretching protocol by the therapist as devised by [Pienimäki 1996](#). All patients were provided an exercise diary in which the therapist described the number and type of exercises they were to do and in which they noted their compliance with this instructed

Struijs 2004 (Continued)

programme. The exercises were done in 4 steps. When a patient was able to perform an exercise step, he or she was allowed to perform the next step.

Dose: manual therapy - 5 to 10 minutes per session for 9 sessions over 6 weeks (3, 2, 1, 1, 1 and 1 session(s) per week, respectively), unless complaints had resolved before the end of these 9 sessions; *exercise* - 10 repetitions in 2 or 3 sets, performed 4 to 6 times daily at home for 6 weeks.

Supervision: physiotherapist at outpatient clinic and unsupervised at home

Additional treatment: every session consisted of 7.5 minutes of pulsed ultrasound

Forearm band

A forearm band was provided brace immediately after randomisation, with instructions on use, and activities causing pain despite the use of the brace were discouraged. Worn continuously during daytime for the 6-week intervention period.

Outcomes

Outcomes were reported 6 weeks (end of intervention), 6 months and 12 months (last time point).

Outcomes

- Pain severity of the most important complaint, 0 to 100 VAS (0 = no pain)
- Severity of complaints, 0 to 100 VAS (0 = no complaints)
- Disability using Pain-Free Function Questionnaire, 0 to 100 (0 = no disability)
- Global improvement reported by the participant on a 6-point Likert scale of 1 completely recovered; 2 much improved; 3 little improved; 4 not changed; 5 little worse; 6 much worse
- Participant satisfaction with treatment, 0 to 100 VAS (0 = not satisfied, 100 = very satisfied)
- Inconvenience during daily activities, 0 to 100 VAS (lower score indicating less inconvenience)
- Pain-free and maximum grip strength (kg), using a Jamar hand dynamometer and reported as a ratio of affected side to unaffected side x 100
- Pressure pain at the lateral epicondyle using a Pressure Threshold Meter (kg per cm²)

Outcomes used in this review
Major outcomes

- Pain
- Disability
- Participant-reported treatment success (measured as 'completely recovered' or 'much improved')
- Proportion of overall withdrawals

Minor outcomes

- Pain-free grip strength (ratio of affected side to unaffected side x 100)
- Pressure pain at the lateral epicondyle using a Pressure Threshold Meter (kg per cm²)

Source of funding

Bauerfeind, manufacturer of orthotic devices

Notes

Trial registration: not reported

Data analysis: we used data at 6 weeks (end of intervention) and 12 months (last time point) for Analysis 2 - Manual therapy, prescribed exercises or both versus minimal or no intervention; and Analysis 3 - Manual therapy and prescribed exercises versus minimal or no intervention. We used data from the groups that included forearm band as a co-intervention to balance the effects; we did not use data from manual therapy and exercise arm. For pain and disability measures we transformed the data to 0 = no pain/disability as we assumed these data were incorrectly reported in the article. We did not use data for the 6-month follow-up. We transformed pain to 0 to 10 scale (0 = no pain) to standardise the analysis.

Adverse events: not reported

Struijs 2004 (Continued)

Withdrawals: in the manual therapy, exercise and forearm brace group, 2/56 participants (4%) did not complete the trial; in the manual therapy and exercise group 3/56 participants (5%) did not complete the trial; in the forearm brace group, 5/68 participants (7%) did not complete the trial.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Participants were randomised by a researcher using a computer program with minimisation strategy for the duration of complaints (≤ 3 months, 3 to 6 months and ≥ 6 months).
Allocation concealment (selection bias)	Unclear risk	Participants were included in the trial by the medical doctor. While the authors reported that "selection bias was prevented by randomisation using a minimisation strategy, which guarantees concealment of allocation", we have judged an unclear risk of allocation bias.
Blinding of participants and personnel (performance bias) All outcomes	High risk	The participants and therapists were not blind to treatment allocation.
Blinding of outcome assessment - self reported (detection bias)	High risk	The participants were not blind to treatment allocation, and measures of pain, disability and global assessment were likely to be influenced by lack of blinding.
Blinding of outcome assessment - assessor reported outcomes	Low risk	The outcome assessor was blinded to group allocation, and conducted blinded assessment of grip strength.
Incomplete outcome data (attrition bias) All outcomes	Low risk	There was a small loss to follow-up for each group, and similar results for the per protocol analysis compared to the intention-to-treat analysis.
Selective reporting (reporting bias)	Unclear risk	While reporting of the results was consistent with the trial methods, there was no trial registry or protocol available, so we judged this as an unclear risk of selective reporting.
Other bias	Low risk	There were no observed baseline differences between the groups, and the trial appears to be free of other sources of bias.

Tonks 2007

Study characteristics

Methods	Design: parallel-group, 4-arm, factorial design, randomised controlled trial Setting: orthopaedic outpatient clinic, Wigan, United Kingdom Timing: May 2003 to February 2005 Interventions: (1) exercise and glucocorticoid injection; (2) glucocorticoid injection; (3) exercise; (4) observation; 6-week intervention period Sample size: calculation method not reported Intention-to-treat analysis: not reported
---------	---

Tonks 2007 (Continued)

Participants

Number of participants

- 142 participants were assessed for eligibility
- 94 participants excluded (80 did not meet inclusion criteria, 14 declined participation)
- 48 were randomised (12 to exercise and glucocorticoid injection, 12 to glucocorticoid injection, 12 to exercise, 12 to observation)
- Data for 37 participants (77%) available at 7-week follow-up (7 for exercise and glucocorticoid injection, 12 for glucocorticoid injection, 8 for exercise, 10 for observation)

Inclusion criteria

- Symptoms of lateral elbow pain who had not had treatment for this complaint in the preceding 6 months
- Pain reproduced on palpation of the common extensor origin
- Pain reproduced on resisted extension of the wrist with the elbow extended
- Age over 18 years

Exclusion criteria

- Trauma to the affected elbow in the preceding 6 weeks
- Patients with a past history of elbow instability
- Previous elbow surgery
- Bilateral symptoms
- Any other pathology involving the affected upper limb
- Coexisting cervical spine pathology
- Physiotherapy or steroid injection for the presenting condition within the previous 6 months
- Patients already on oral/systemic steroids
- Patients with contraindications to injection therapy
- Patients with bleeding diatheses or on anticoagulant therapy
- Local or systemic infection
- Past history of hypersensitivity to local anaesthetics
- Poorly controlled diabetics and other immunosuppressed individuals
- Pregnant or breastfeeding patients
- Uncooperative patients with an underlying psychiatric diagnosis
- Prosthetic elbow joint

Baseline characteristics

Exercise and glucocorticoid injection (n = 12 participants)

Mean (SD) age: 41.9 (7.4) years; gender not reported

Mean (SD) pain score component of the Patient Related Forearm Evaluation Questionnaire: 4.7 (2.3)

Mean (SD) function score component of the Patient Related Forearm Evaluation Questionnaire: 3.6 (2.3)

Mean (SD) total score of the Patient Related Forearm Evaluation Questionnaire: 4.0 (2.2)

Mean (SD) pain-free grip strength: 17.6 (7.2) kg

Glucocorticoid Injection (n = 12 participants)

Mean (SD) age: 48.2 (6.5) years; gender not reported

Mean (SD) pain score component of the Patient Related Forearm Evaluation Questionnaire: 5.0 (1.4) units

Mean (SD) function score component of the Patient Related Forearm Evaluation Questionnaire: 5.0 (2.7) units

Tonks 2007 (Continued)

Mean (SD) total score of the Patient Related Forearm Evaluation Questionnaire: 5.0 (2.2) units

Mean (SD) pain-free grip strength: 14.5 (7.5) kg

Exercise (n = 12 participants)

Mean (SD) age: 43.8 (7.5) years; gender not reported

Mean (SD) pain score component of the Patient Related Forearm Evaluation Questionnaire: 4.6 (1.8) units

Mean (SD) function score component of the Patient Related Forearm Evaluation Questionnaire: 3.7 (2.0) units

Mean (SD) total score of the Patient Related Forearm Evaluation Questionnaire: 4.0 (1.1) units

Mean (SD) pain-free grip strength: 19.4 (11.4) kg

No intervention (n = 12 participants)

Mean (SD) age: 43.4 (7.1) years; gender not reported

Mean (SD) pain score component of the Patient Related Forearm Evaluation Questionnaire: 5.1 (1.2) units

Mean (SD) function score component of the Patient Related Forearm Evaluation Questionnaire: 4.9 (1.3) units

Mean (SD) total score of the Patient Related Forearm Evaluation Questionnaire: 5.0 (1.1) units

Mean (SD) pain-free grip strength: 22.3 (10.8) kg

Interventions
Exercise

Components of intervention: as devised by [Pienimäki 1996](#), comprised progressive slow, repetitive wrist and forearm stretching and muscle conditioning that was intensified over 4 phases

Dose: participants performed the exercise programme 4 to 6 times daily at home for 6 weeks. Each programme included 10 repetitions in 2 or 3 series for each exercise. Every exercise period ended with stretching for at least 30 seconds in both flexion and extension and each individual exercise movement was done slowly while the patient counted to 8.

Supervision: not reported

Additional treatment: none

Glucocorticoid injection

Components of intervention: glucocorticoid injection - single injection into the symptomatically tender region of the common extensor origin

Dose - single injection of 10 mg of triamcinolone acetonide and of 2% lignocaine hydrochloride made up to a volume of 1 mL injected

Additional treatment: none

Minimal intervention

Observation

Outcomes

Outcomes were reported at 7 weeks (end of intervention).

Outcomes

- Pain subscore component of the PRFEQ (0 = no pain) change from baseline scores
- Function subscore component of the PRFEQ (0 = no pain) change from baseline scores

Tonks 2007 (Continued)

- Disability, 0 to 10 using the PRFEQ (0 = no disability)
- Number of participant overall withdrawals
- Number of participants experiencing an adverse event (complications of treatment)
- Pain-free grip strength (kg) using a hand dynamometer
- Extensor weight strength (kg) - the maximum weight lifted using a floor mounted spring weight balance, without pain

Outcomes included in this review
Major outcomes

- Mean change in pain
- Mean change in disability
- Proportion of overall withdrawals
- Proportion with any adverse event

Minor outcomes

- Mean change in pain-free grip strength (kg)

Source of funding	Not reported
Notes	Trial registration: not reported Data analysis: we used data at 7 weeks follow-up (end of intervention) for Analysis 2 - Manual therapy, prescribed exercises or both versus minimal or no intervention; Analysis 5 - Prescribed exercises versus minimal or no intervention; Analysis 8 - Manual therapy, prescribed exercises or both and glucocorticoid injection versus glucocorticoid injection alone; and Analysis 10 - Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone. For the PRFEQ we used total score as a disability measure and pain subscore for pain outcome; we also transformed this disability outcome data to a 0 to 100 scale to standardise the measure for the meta-analyses. Adverse events: 1 adverse event was reported in the glucocorticoid injection group (depigmentation and atrophy). Withdrawals: in the exercise and glucocorticoid injection group, 5 (42%) did not complete the trial; in the glucocorticoid injection group, 0 participants (0%) did not complete the trial; in the exercise group, 4 participants (33%) did not complete the trial; in the observation group, 2 participants (17%) did not complete the trial.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A random numbers table in a block size of 24 was used so that each block consisted of the 4 treatment groups in equal proportions.
Allocation concealment (selection bias)	Low risk	Sealed, opaque numbered envelopes were used to conceal allocation.
Blinding of participants and personnel (performance bias) All outcomes	High risk	The participants and study personnel were not blind to group allocation.
Blinding of outcome assessment - self reported (detection bias)	High risk	The participants were not blind to treatment allocation, and measures of pain, disability and global assessment were likely influenced by lack of blinding.

Tonks 2007 (Continued)

Blinding of outcome assessment - assessor reported outcomes	High risk	The outcome assessor was not blinded to treatment allocation, and measurement of grip strength could be influenced by lack of blinding.
Incomplete outcome data (attrition bias) All outcomes	High risk	There was a high dropout in the groups allocated the exercise intervention, with 4/12 participants dropping out in one group, and 5/12 participants in the other group.
Selective reporting (reporting bias)	Unclear risk	While the data were reported for all outcomes listed in the methods, there was no published trial protocol or trial registry, so we judged this as an unclear risk of selective reporting.
Other bias	Low risk	There were no major baseline differences observed, and the trial appears to be free of other sources of bias.

Vuvan 2020
Study characteristics

Methods	Design: parallel group, 2-arm randomised controlled trial Setting: primary care, Brisbane, Australia Timing: September 2015 to July 2017 Interventions: (1) exercise; (2) advice; 8-week intervention period Sample size: 17 participants per group provided 80% power to detect a difference of at least 1 unit on the Patient-Rated Tennis Elbow Evaluation questionnaire between groups, with a significance level at 5% Intention-to-treat analysis: yes
Participants	Number of participants 533 participants assessed for eligibility 493 participants excluded (415 did not meet selection criteria, 78 declined to participate) 40 participants randomised (21 to exercise, 19 to advice) - Data for 39 participants (98%) available at 8-week follow-up (20 for exercise, 19 for advice) Inclusion criteria - Aged 18 to 70 years - Unilateral lateral elbow pain $\geq$ 6 weeks duration - Average pain severity during the past week $\geq$ 2 on an 11-point numerical rating scale (NRS, 0 = no pain, 10 = worst pain imaginable) - Provoked by at least 2 of: gripping, palpation of the lateral epicondyle, stretching of forearm extensor muscles or resisted wrist, second or third finger extension - Reduced pain-free grip strength Exclusion criteria - Other primary sources of elbow pain (e.g. exacerbation of elbow pain with neck movements or manual palpation, pain localised over the radiohumeral joint, abnormal findings on neural examination) - Concomitant musculoskeletal pain conditions reported by participants to be their predominant complaint - Major neurological, inflammatory or systemic conditions

Vuvan 2020 (Continued)

- Treatment by a healthcare practitioner within the preceding 3 months
- Injections within the preceding 6 months
- Major trauma, fracture or surgery in the last year

Baseline characteristics
Exercise (n = 21 participants)

Mean (SD) age: 48.0 (7.9) years; 7 female, 14 male

Median (IQR) duration of symptoms: 6 (2 to 8) months

Dominant elbow affected: 14 participants (67%)

Gripping sport: 15 participants (71.4%)

Repetitive/manual tasks as work: 17 participants (81%)

Mean (SD) pain severity (rest) in previous week (/10): 1.6 (1.1) units

Mean (SD) pain severity (worst) in previous week (/10): 6.8 (1.8) units

Mean (SD) disability using PRTEE questionnaire (/100): 33.4 (12.1) units

Mean (SD) health related quality of life using EQ-5D index (/1): 0.78 (0.06) units

Mean (SD) pain-free grip strength (affected side): 98.7 (56.7) Newtons

No intervention (n = 19 participants)

Mean (SD) age: 49.0 (10.2) years; 4 female, 15 male

Median (IQR) duration of symptoms: 3 (2 to 6) months

Dominant elbow affected: 10 participants (53%)

Gripping sport: 15 participants (79%)

Repetitive/manual tasks as work: 16 participants (84%)

Mean (SD) pain severity (rest) in previous week (/10): 2.0 (1.9) units

Mean (SD) pain severity (worst) in previous week (/10): 7.8 (1.4) units

Mean (SD) disability using PRTEE questionnaire (/100): 37.6 (13.9) units

Mean (SD) health related quality of life using EQ-5D index (/1): 0.77 (0.13) units

Mean (SD) pain-free grip strength (affected side): 111.8 (72.6) Newtons

Interventions
Exercise

Components of intervention: progressive isometric wrist extension exercise, performed at home, using a container of water with a handle as resistance and progressed weekly by adding water to the container or increasing the duration of exercise, according to standardised criteria. Participants were instructed to hold the water-filled container in the affected arm, with the forearm resting on the edge of a table (upper body in sound postural alignment, elbow approximately 90° flexion, forearm pronated, wrist 30° extension). The other hand was used to lower the water container to the floor between sets.

Dose: exercise - 1 daily session for 8 weeks; starting load was equivalent to 20% MVC of the unaffected side with 5% MVC added each fortnight; week 1, 3, 5, 7 - 30 sec holds, 4 reps, 120 secs total time under tension; week 2, 4, 6, 8: 45 sec holds, 3 reps, 135 secs total time under tension.

Supervision: single supervised instruction session with physiotherapist with postgraduate degree qualification who provided a 20-minute individualised session of information and advice to complete the exercise intervention at home

Vuvan 2020 (Continued)

Additional treatment: none

Advice

All participants were provided written and verbal general advice from the same physiotherapist regarding self-management and ergonomics

Co-interventions

Advice

Outcomes	Outcomes were reported at 8 weeks (end of intervention). Outcomes • Pain severity (at rest and worst pain) in the past week, 0 to 10 NRS (0 = no pain) • Disability, 0 to 100 scale using PRTEE questionnaire (0 = no disability) • Global rating of change reported by the participant on a 6-point scale (1 = much worse; 2 = worse; 3 = same; 4 = improved; 5 = much improved; 6 = completely recovered) • Adverse events from treatments • Participant overall withdrawals • Pain-free grip strength (N) using a digital grip dynamometer with triplicate measures collected with 30 sec rest intervals, and mean values were used for analysis • Pressure pain thresholds (kPa) • Cold and heat pain thresholds (°C) Outcomes included in this review <i>Major outcomes</i> • Mean pain (worst pain) • Mean disability • Participant-reported treatment success (measured as 'completely recovered' or 'much improved') • Proportion of overall withdrawals • Proportion with any adverse event • Proportion with a serious adverse event <i>Minor outcomes</i> • Mean pain-free grip strength (N)
Source of funding	Funding was provided by a programme grant from the National Health and Medical Research Council of Australia (ID631717).
Notes	Trial registration: Australian New Zealand Clinical Trial Registry (ACTRN12615001136594) Data analysis: we used data at 8 weeks (end of intervention) for Analysis 2 - Manual therapy, prescribed exercises or both versus minimal or no intervention; and Analysis 5 - Prescribed exercises versus minimal or no intervention. Adverse events: 3 of 21 participants in the exercise group only reported feeling worse on the global rating of change scale and were classified as adverse events. No participants reported a serious adverse event. Withdrawals: in the exercise group, 1 participant (5%) did not complete the trial; in the advice group, 0 participants (0%) did not complete the trial.
Risk of bias	
Bias	Authors' judgement Support for judgement

Vuvan 2020 (Continued)

Random sequence generation (selection bias)	Low risk	A computer-generated randomisation sequence was performed.
Allocation concealment (selection bias)	Low risk	Participants were randomly allocated using concealed, opaque envelopes to either group.
Blinding of participants and personnel (performance bias) All outcomes	High risk	The participants were not blind to treatment allocation.
Blinding of outcome assessment - self reported (detection bias)	High risk	The participants were not blind to treatment allocation, and measures of pain, disability and global improvement were likely to be influenced by lack of blinding.
Blinding of outcome assessment - assessor reported outcomes	Low risk	The assessor was blinded to treatment allocation for measurement of grip strength.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Only 1 participant from the exercise group was lost to follow-up.
Selective reporting (reporting bias)	Low risk	The reporting of results was consistent with the registered trial protocol.
Other bias	Low risk	This study appears to be free of other sources of bias.

Yilmaz 2022
Study characteristics

Methods	Design: parallel-group, 2-arm randomised controlled trial Settings: Hacettepe University and affiliated hospital units - Physical Therapy Rehabilitation, Hand Surgery Rehabilitation, Ankara, Turkey Timing: May 2014 to December 2014 Interventions: (1) manual therapy and exercise; (2) exercise; 6-week intervention period Sample size: 20 participants per group provided 80% power with a significance level at 5%, without reporting the required difference between groups Intention-to-treat analysis: not reported
Participants	Number of participants 40 participants assessed for eligibility 0 participants excluded 40 participants randomised (20 to exercise, 20 to no intervention) - Data for 35 participants (88%) available at 3-week follow-up (17 for manual therapy and exercise, 18 for exercise) - Data for 35 participants (88%) available at 6-week follow-up (17 for manual therapy and exercise, 18 for exercise) - Data for 34 participants (85%) available at 3-month follow-up (16 for manual therapy and exercise, 18 for exercise)

Yilmaz 2022 (Continued)

Inclusion criteria

- Aged > 18 years
- Symptom duration > 3 months

Exclusion criteria

- Bilateral symptoms
- Rheumatologic diseases affecting the elbow and wrist
- Musculoskeletal disorder due to connective tissue diseases
- Diffuse pain syndrome
- Cervical radiculopathy
- Nerve compression syndromes involving upper extremity
- Undergone surgery on the affected arm
- Received treatment for condition in last 6 months
- Inability to perform the exercises

Baseline characteristics

Manual therapy and exercise (n = 20 participants)

Mean (SD) age: 42.7 (7.57) years; 14 female, 6 male

Mean (SD) body mass index: 27.6 (4.97) kg/m²

Elbow affected: right - 15 (75%); left - 5 participants (25%)

Mean (SD) duration of symptoms: 30.5 (34.6) weeks

Employment: homemaker - 7 participants (35%); technician - 2 participants (10%); servant - 4 participants (20%); others - 7 participants (35%)

Education status: primary school - 11 participants (55%); secondary/high school - 4 participants (20%); university - 4 participants (20%); post-graduate 1 participant (5%)

Mean (SD) pain severity at rest (/10): 4.50 (1.85) units

Mean (SD) pain severity with activity (/10): 7.23 (1.74) units

Mean (SD) pain severity at night (/10): 4.70 (3.11) units

Mean (SD) disability using DASH questionnaire (/100): 39.2 (13.71) units

Mean (SD) pain-free grip strength: 29.2 (10.6) kg

Mean (SD) maximum grip strength: 32.1 (11.1) kg

Exercise (n = 20 participants)

Mean (SD) age: 42.9 (10.3) years; 12 female, 8 male

Mean (SD) body mass index: 29.1 (5.61) kg/m²

Elbow affected: right - 17 (85%); left - 3 participants (15%)

Mean (SD) duration of symptoms: 29.3 (32.8) weeks

Employment: homemaker - 8 participants (40%); technician - 2 participants (10%); servant - 2 participants (10%); others - 8 participants (40%)

Education status: primary school - 10 participants (50%); secondary/high school - 7 participants (35%); university - 1 participant (5%); post-graduate 1 participant (10%)

Mean (SD) pain severity at rest (/10): 2.78 (2.42) units

Yilmaz 2022 (Continued)

Mean (SD) pain severity with activity (/10): 6.03 (1.94) units

Mean (SD) pain severity at night (/10): 3.58 (2.55) units

Mean (SD) disability using DASH questionnaire (/100): 32.8 (14.1) units

Mean (SD) pain-free grip strength: 29.6 (12.3) kg

Mean (SD) maximum grip strength: 33.5 (13.4) kg

Interventions

Manual therapy and exercise

Components of intervention: manual therapy - radial nerve mobilisation with the participant in supine, the therapist holds the participant's arm and wrist, depresses the shoulder, extends the elbow and internally rotates the arm. The participant's wrist, thumb and fingers were all flexed, and wrist ulnar deviated. The tension was adjusted with rotation or lateral flexion movements of the head. The nerve was relaxed after each repetition; exercise - refer to exercise component of exercise group.

Dose: 3 days a week for 3 weeks; each position was maintained for 3 seconds with 3 sets of 10 repetitions during each treatment session with 2 minutes rest between sets.

Supervision: physiotherapist at physiotherapy clinic

Additional treatment: participants performed radial nerve mobilisation exercises at home for 6 weeks. These were performed standing, with the elbow extended, wrist in ulnar deviation, depressing the shoulder, arm rotated internally and then turning their head towards their palm and staring at it for at least 3 seconds before returning to their starting position.

Exercise

Components of intervention: eccentric wrist extension exercises performed in sitting with the elbow extended, forearm pronated, and the wrist fully extended. Holding a weight, participants fully extended their wrist, then slowly moved their wrist from full wrist extension to full wrist flexion, then passively extended their wrist with the help of their other hand to the starting position. Participants were advised to stop the exercises if they felt pain at any time during the exercises or continue exercises without weights. The concept of 10 repetition maximum was used to determine resistance and progression.

Dose: 3 times daily for 6 weeks, 3 sets of 10 repetitions with 1-minute rest interval between sets

Supervision: exercises were performed unsupervised at home; participants visited a physiotherapist every second week for follow-up examination and to receive a progression of exercises.

Additional treatment: none

Co-interventions

Advice to avoid exacerbating their condition and engaging in forceful activities, and to rest intermittently during periods of intense pain. Participants were instructed not to take any medication that provided pain relief throughout the study period.

Outcomes

Outcomes were reported at 3 weeks (first time point), 6 weeks (end of intervention) and 3 months (last time point).

Outcomes

- Pain severity (at rest, night and daily activities), 0 to 10 VAS (0 = no pain)
- Disability, 0 to 100 scale using DASH questionnaire (0 = no disability)
- Participant overall withdrawals
- Pain-free grip strength (kg) and maximum grip strength (kg) using a hydraulic hand dynamometer (Jamar, Bolingbrook IL) with an average of 3 trials recorded
- Tip pinch (kg) and key pinch (kg) using a Baseline Mechanical Pinch Gauge (FEI, White Plains, NY) with an average of 3 trials recorded

Yilmaz 2022 (Continued)

- Wrist active range of movement (degrees) for flexion, extension, radial and ulnar deviation using a goniometer

Outcomes included in this review
Major outcomes

- Mean pain severity (daily activities)
- Mean disability
- Proportion of overall withdrawals

Minor outcomes

- Mean pain-free grip strength (kg)

Source of funding	Not reported
Notes	Trial registration: Clinicaltrials.gov (NCT04219488) Data analysis: we used data at 3 weeks (first time point), 6 weeks (end of intervention) and 3 months (last time point) for Analysis 6 - Manual therapy and prescribed exercises versus prescribed exercises alone. Adverse events: 1 patient from the manual therapy and exercise group did not attend final follow-up due to pain. Withdrawals: in the manual therapy and exercise group, 4 participants (20%) did not complete the trial (did not complete assessments in final time point); in the exercise group, 2 participants (10%) did not complete the trial (did not complete assessments in final time point).

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A randomised sampling method with equal chance was used.
Allocation concealment (selection bias)	Low risk	Sealed, opaque envelopes were used.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants could not be blinded, increasing the risk of performance bias.
Blinding of outcome assessment - self reported (detection bias)	High risk	Participants could not be blinded, and measures of pain and disability were likely to be influenced by lack of blinding.
Blinding of outcome assessment - assessor reported outcomes	Low risk	The assessor was blinded to group allocation, reducing the risk of detection bias for grip strength measurements.
Incomplete outcome data (attrition bias) All outcomes	Low risk	There was minimal loss to follow-up in the manual therapy and exercise group (4 participants at final follow-up).
Selective reporting (reporting bias)	Low risk	The outcomes reported were consistent with the registered protocol and trial methods.

Yilmaz 2022 (Continued)

Other bias	Low risk	This study appears to be free of other sources of bias.
------------	----------	---

DASH: Disabilities of the Arm, Shoulder and Hand
 EQ-5D-3L: EuroQol-5 Dimension-3 Likert
 IQR: interquartile range
 MRI: magnetic resonance imaging
 MVC: maximum voluntary contraction
 MWM: Mobilisation-with-Movement
 NRS: numeric rating scale
 NSAID: non-steroidal anti-inflammatory drug
 PRTEE: Patient-Rated Tennis Elbow Evaluation
 PRFEQ: Patient-rated Forearm Evaluation Questionnaire
 PSFS: Patient Specific Functional Scale
 SD: standard deviation
 SF-36: Short Form 36
 TENS: transcutaneous electrical nerve stimulation
 VAS: visual analogue scale

Characteristics of excluded studies [ordered by study ID]

Study	Reason for exclusion
Abbott 2001a	Incorrect design
Abbott 2001b	Incorrect design
ACTRN12612000993897	Incorrect comparator
ACTRN12617000219381	Incorrect intervention
Altun 2018	Incorrect intervention
Anitha 2018	Incorrect comparison
Bateman 2024	Incorrect comparator
Cleland 2005	Incorrect comparator
Coombes 2016	Incorrect intervention
Croisier 2007	Incorrect intervention
CTRI/2022/02/040270	Incorrect comparator
Day 2019	Incorrect design
Drechsler 1997	Incorrect intervention
Fernandez-Carnero 2008	Incorrect intervention
Fernandez-Carnero 2011	Incorrect intervention
Gunduz 2012	Incorrect comparator
Gupta 2021	Incorrect intervention

Study	Reason for exclusion
Hsu 2016	Incorrect comparator
IRCT848	Incorrect intervention
Kamal 2022	Incorrect design
Kaux 2022	Incorrect comparator
KCT0007378	Incorrect comparator
Khandaker 2014	Incorrect comparator
Kucuksen 2013	Incorrect comparator
Lee 2018	Incorrect comparator
Manias 2006	Incorrect intervention
Martinez-Silvestrini 2005	Incorrect comparator
McLean 2002	Incorrect intervention
Mostafee 2020	Incorrect comparison
Nagrale 2009	Incorrect comparator
NCT00930709	Incorrect comparator
NCT01273688	Incorrect intervention
NCT01897259	Incorrect comparator
NCT02052089	Incorrect comparator
NCT02304952	Incorrect comparator
NCT02521298	Incorrect intervention
NCT02930304	Incorrect intervention
NCT02991560	Incorrect comparator
NCT03074500	Incorrect intervention
NCT03161782	Incorrect comparator
NCT03167710	Incorrect comparator
NCT03834090	Incorrect comparison
NCT05238090	Incorrect comparator
NCT05696327	Incorrect comparator
NCT06087081	Incorrect comparator

Study	Reason for exclusion
Nilsson 2007	Incorrect design
Nilsson 2012	Incorrect design
Nishizuka 2017	Incorrect intervention
NTR7223	Incorrect intervention
Park 2010	Incorrect comparator
Paungmali 2003	Incorrect intervention
Peterson 2014	Incorrect comparator
Pienimaki 1996	Incorrect comparator
Radpasand 2009	Incorrect comparator
Ramteke 2020	Incorrect comparison
Sarkar 2013	Incorrect comparator
Sethi 2018	Incorrect comparison
Sevier 2015	Incorrect comparator
Sharma 2020	Incorrect comparison
Slater 2006	Incorrect population
Solveborn 1997a	Incorrect comparator
Stasinopoulos 2006	Incorrect comparator
Stasinopoulos 2009	Incorrect comparator
Stasinopoulos 2010	Incorrect comparator
Stasinopoulos 2017	Incorrect comparator
Struijs 2003	Incorrect comparator
Svernlöv 2001a	Incorrect comparator
TCTR20200612001	Incorrect intervention
Tyler 2010	Incorrect comparator
Tyrdal 1998	Incorrect population
Verhaar 1996	Incorrect comparator
Vicenzino 1996	Incorrect intervention
Vicenzino 2001	Incorrect intervention

Study	Reason for exclusion
Viswas 2012	Incorrect comparator
Wegener 2016	Incorrect intervention
Zunke 2020	Incorrect intervention

Characteristics of studies awaiting classification [ordered by study ID]

Bartlett 2007

Methods	Design: parallel-group, 2-arm randomised controlled trial Setting: Physiotherapy Department at Solihull Hospital, UK Interventions: (1) manual therapy; (2) no intervention Sample size: not reported
Participants	Inclusion criteria - Chronic lateral elbow pain diagnosed using criteria: pain over the lateral side of the elbow provoked by palpation of the lateral epicondyle region and gripping tasks, pain over the lateral epicondyle during either resisted static contraction or stretching of the forearm extensor muscles - Symptoms of greater than 6 weeks duration - Both male and female patients - Age 18+ years Exclusion criteria - Not reported
Interventions	Manual therapy <i>Components of intervention:</i> Mobilisation with Movement <i>Dose:</i> twice a week for 3 weeks <i>Supervision:</i> physiotherapy <i>Setting:</i> physiotherapy department in a hospital <i>Additional treatment:</i> none No intervention Simply be given an appointment to attend for follow-up 3 weeks later Co-interventions None
Outcomes	Outcomes were to be reported at 3 weeks. Outcomes - Patient-Rated Forearm Evaluation Questionnaire (PRFEQ) - Pain-free grip strength
Notes	Trial registration: https://doi.org/10.1186/ISRCTN89551492

Bartlett 2007 (Continued)

Source of funding: Heart of England NHS Foundation Trust (UK)

Contact: Alison Bartlett, Physiotherapy Department, Solihull B91 2JL United Kingdom

Attempted to contact author but no response to date 14 March 2023.

Editorial note from trial registration site <https://doi.org/10.1186/ISRCTN89551492>: "18/05/2017: No publications found in PubMed, verifying study status with principal investigator."

TENS: transcutaneous electrical nerve stimulation

Characteristics of ongoing studies [ordered by study ID]

Prat 2018

Study name	Efficacy of fascial manipulation and eccentric exercise for lateral elbow pain
Methods	Design: parallel-group, 2-arm randomised controlled trial Setting: unknown Interventions: (1) manual therapy and exercise; (2) exercise Sample size: 29 participants
Participants	Inclusion criteria - Clinical diagnosis of lateral elbow pain Exclusion criteria - Details not available Baseline characteristics <i>Manual therapy and exercise (n = 17 participants)</i> <i>Exercise (n = 12 participants)</i> Mean age across groups: 47 years (SD not reported); 14 females;15 males
Interventions	Manual therapy and exercise <i>Components: manual therapy</i> - fascial manipulation involving gliding through deep friction over specific points of the deep fascia; <i>exercise</i> - refer to exercise component of exercise group. <i>Dose: manual therapy</i> - 3 sessions for 4 weeks Exercise <i>Components:</i> eccentric exercise <i>Dose:</i> daily for 4 weeks Co-interventions Exercise
Outcomes	Outcomes to be measured at 4 weeks, 8 weeks and 4 months follow-up. Outcomes - Disability, 0 to 100 scale using DASH questionnaire (lower score = less disability) - Pain-free and maximum grip strength (units not reported) using hydraulic hand dynamometer

Manual therapy and exercise for lateral elbow pain (Review)

Prat 2018 (Continued)

- Pain pressure threshold, measured with a pressure algometer on the common extensor tendon

Starting date	Unknown
Contact information	Name: Dr Antonio Stecco Address: RUSK Rehabilitation, New York University, School of Medicine, USA Email: antonio.stecco@nyumc.org
Notes	Trial registration: unknown Funding: unknown Data analysis: we plan to use data at 4 weeks, 8 weeks and 4 months follow-up for comparison - manual therapy and prescribed exercises versus prescribed exercises

Sahebamei 2019

Study name	The role of elbow exercises protocol in treatment of patients with lateral epycodylitis (tennis elbow)
Methods	Design: parallel-group, 2-arm randomised controlled trial Setting: orthopaedic clinic, Gorgan, Iran Interventions: (1) manual therapy, exercise and brace; (2) brace Sample size: target 40 participants
Participants	Inclusion criteria • Presence of lateral pain in the elbow for at least 4 to 6 weeks • At least 2 positive tests from the following 4 tests: 1) lateral epicondylar tenderness; 2) elbow pain during extension of the wrist against resistance; 3) elbow pain during tough fingers; 4) elbow pain during passive flexion Exclusion criteria • History of trauma to the elbow, or infected elbow surgery • Infusion or physiotherapy of the affected elbow in the past 6 months
Interventions	Manual therapy, exercise <i>Components:</i> manual therapy - massage; exercise - stretching, isometrics, active range of motion <i>Dose:</i> 4 weeks Brace <i>Components:</i> elbow brace <i>Dose:</i> 4 weeks Co-interventions Brace; advice to moderate activity; Mobic 15 mg twice daily
Outcomes	Outcomes to be measured at 4 weeks and 8 weeks follow-up. Outcomes

Sahebjamei 2019 (Continued)

- Pain severity, VAS
- Disability, using VAS and Tennis Elbow Function test

Starting date	23 September 2017
Contact information	Name: Associate Professor Afshin Sahebjamei Address: Gorgan University of Medical Sciences, Dezziani Clinic of Orthopaedi, Shahid Beheshti Str, Gorgan, Golestan, Iran Email: sahebjameiafshin@yahoo.com
Notes	Trial registration: IRCT20180812040773N1 Funding: Institution - Gorgan University of Medical Sciences. Data analysis: we plan to use data at 4 weeks and 8 weeks follow-up for comparison - manual therapy and prescribed exercises versus minimal treatment.

Sveinall 2022

Study name	The feasibility of heavy-slow resistance training, radial extracorporeal shockwave therapy (rESWT) and information and advice. The NORwegian Tennis Elbow (NOTE) study.
Methods	Design: parallel-group, 3-arm randomised controlled trial Setting: Håkon Sveinall, Oslo University Hospital Interventions: (1) exercise (heavy slow resistance training); (2) radial extracorporeal shock wave therapy; (3) advice Duration of experimental period: 12 weeks Sample size: 100 participants
Participants	Inclusion criteria • Ability to fill in the questionnaires (understand oral and written Norwegian) • Clinically diagnosed tennis elbow provoked by at least 2 of: pain on pressure of the common extensor origin, pain during power grip, resisted wrist extension (Tomsen test), resisted third finger extension (Maudsley's test), stretching of the extensors (Mill's test) Exclusion criteria • Patients with significant language issues (not understanding oral or written) • Norwegian contraindications to ESWT (pregnancy, coagulation disturbance, connective tissue diseases, epilepsy, use of pacemaker) • Suspicion of other serious illness
Interventions	Exercise <i>Components:</i> heavy slow resistance exercise. One physiotherapist will supervise and instruct the exercise programme on an individual basis. If the patient doesn't have access to a dumbbell, it will be offered free rental from the physiotherapy ward. Follow-up on the exercises will be scheduled as needed and tele-rehabilitation will be offered as an option. Participants will also be instructed to perform daily stretches of the forearm. <i>Dose:</i> a dumbbell as heavy as possible within the given repetitions and sets. Number of repetitions: week 1: 15 reps, Week 2 to 3: 12 reps, Week 4 to 5: 10 reps, Week 6 to 8: 8 reps, Week 9 to 12: 6 reps. Number of sets: 3. Rest between sets: 2 to 3 min. Fractional and temporal distribution of the con-

Sveinall 2022 (Continued)

traction modes per repetition and duration of one repetition: 3s concentric, 2s isometric: while change concentric/eccentric phase, 3s eccentric. No rest in-between repetitions. Full range of motion. Intervention frequency: 3 times a week, with minimum 48 h recovery time in between.

Radial extracorporeal shock wave therapy

Components of intervention: the treatment will be given by a physiotherapist trained in using the intervention and applied to the extensor carpi radialis brevis tendon insertion area.

Dose: once a week for 3 sessions; 2000 impulses of shock waves are applied to the painful tendon, with a pressure between 1.5 and 3.0 bar (depending on what the patient tolerates). Using a power handpiece that provides energy of 0.01 to 0.35 mJ/mm².

Information and advice

The information and advice group is given a single face-to-face session with a physiotherapist, lasting up to 60 minutes. Information and advice are based on assessment and will comprise standardised oral and written information. Participants will be educated regarding pain during and after activities and encouraged to be physically active. With only one session, there is great reliance on self-management. Thus, information and advice are made available through a handout.

Outcomes	Outcomes to be measured at 12 weeks and 6 months. - Disability, 0 to 100 scale using the Patient-Rated Tennis Elbow Evaluation - Disability, 0 to 100 scale using the Disabilities of the Arm, Shoulder and Hand Outcome Measure (Quick-DASH) - Quality of life using the 5-level EQ-5D (EQ-5D-5L) and VAS (EQ VAS) - Global improvement - Pain-free and maximum grip strength - Pain-free grip strength (units not reported) using hydraulic hand dynamometer - Pain pressure threshold, measured with a pressure algometer on the common extensor tendon
Starting date	18 March 2021
Contact information	Name: Håkon Sveinall Address: Oslo University Hospital Recruiting Oslo, Norway, 0450 Email: hasvei@ous-hf.no
Notes	Trial registration: NCT04803825 Funding: unknown Data analysis: we plan to use data at 3 months and 6 months follow-up for comparison - prescribed exercises versus minimal intervention (advice).

Takamjani 2019

Study name	The effect of Mulligan's techniques along with routine physiotherapy, in comparison with routine physiotherapy alone; on pain, grip strength, and function in patients diagnosed with chronic lateral epicondylitis
Methods	Design: parallel group, 3-arm randomised controlled trial Setting: physiotherapy clinic, Iran Faculty of Rehabilitation, Tehran Interventions: (1) manual therapy and exercise; (2) exercise

Takamjani 2019 (Continued)

Sample size: target 34 participants

Participants	Inclusion criteria: • Patients with chronic lateral epicondylitis symptoms lasting more than 3 months • Age 35 to 55 years • Referred by the physician • Pain intensity of at least 3 on NPRS, in the lateral epicondyle of the humerus bone, which demonstrates an increase of 2 units by gripping and active wrist extension, and decreases with rest • Positive Mill's test (passive stretching): the physiotherapist touches the lateral epicondyle with one hand, while with the other hand pronates the patient's forearm and flexes the wrist to the maximum flexion and extends the elbow. Reproduction of pain indicates that the test is positive • Positive Cozen's test (wrist resistive extension test): the wrist resistive extension and radial deviation and forearm pronation tests are performed. In case the test triggers pain, the test is considered positive. • Positive Maudsley's test: a positive test is indicated by presence of pain over the lateral epicondyle of the affected hand during the third finger resistive extension test • Presence of tenderness at the tenoperiosteal junction or at the junction of the extensor carpi radialis brevis muscle to the lateral humeral epicondyle Exclusion criteria: • Pain in the lateral humeral epicondyle, due to neck involvement • Lateral humeral epicondyle acute inflammation • Pain in the lateral humeral epicondyle, caused by acute trauma of elbow region • Radial tunnel syndrome • Elbow joint inflammation Injuries to the internal and external ligaments of the elbow • Medial humeral epicondylitis • History of elbow joint bone fractures • Presence of deformity or congenital anomalies of the elbow joint • Pregnancy • Infection of the elbow joint • Malignancies • Haemophilia • Generalised hypermobility • Presence of swelling in the elbow joint • Presence of myositis ossification in radiographic x-ray • Dissatisfaction of participants • Trigger points in the wrist and finger extensor muscles
Interventions	Manual therapy and exercise <i>Components: manual therapy</i> - Mulligan techniques; <i>exercise</i> - refer to exercise component of exercise group. <i>Dose:</i> 12 sessions over 4 weeks Exercise <i>Components:</i> strengthening and stretching exercises <i>Dose:</i> 12 sessions over 4 weeks Co-interventions <i>Components:</i> ultrasound with frequency of 3 MHz and intensity of 1.5 w/cm² in pulsed mode of operation and the pulse is 1 m/s on and 5 m/s off

Takamjani 2019 (Continued)

Outcomes	Outcomes measured at 4 weeks follow-up
	Outcomes - Pain average severity at rest in past 24 hours, 0 to 100 VAS - Disability, using Patient-Rated Elbow Evaluation Questionnaire - Maximum grip strength (no units reported) using digital hydraulic dynamometer
Starting date	19 April 2019
Contact information	Name: Professor Ismail Ebrahimi Takamjani Address: Iran University of Medical Sciences, No. 195, Khalili Alley, Saba Blvd., Rumi Bridge, Shariati Street, Tehran Email: Ebrahimitakamjani.e@iums.ac.ir
Notes	Trial registration: IRCT20190315043058N1 Funding: not reported Data analysis: we plan to use data at 4 weeks follow-up for comparison - manual therapy and prescribed exercises versus prescribed exercises.

DASH: Disabilities of the Arm, Shoulder and Hand

ESWT: extracorporeal shock wave therapy

NPRS: numerical pain rating scale

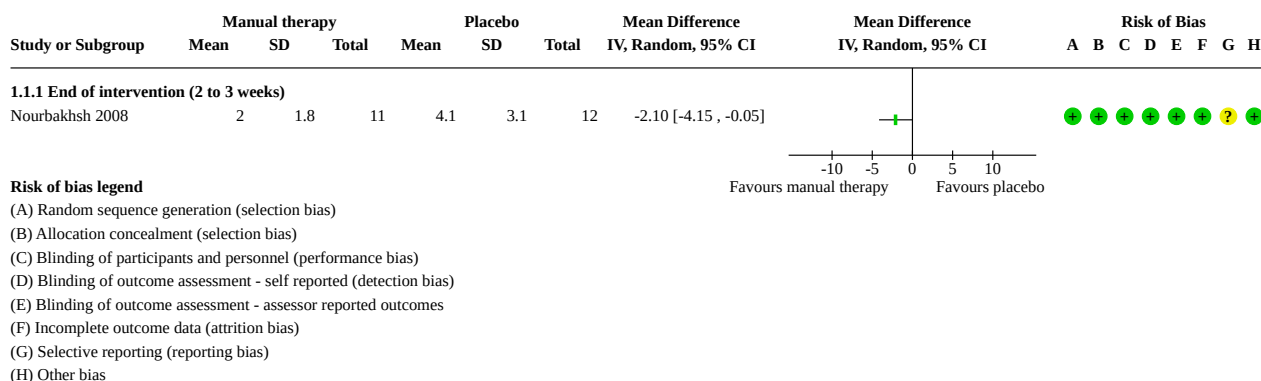
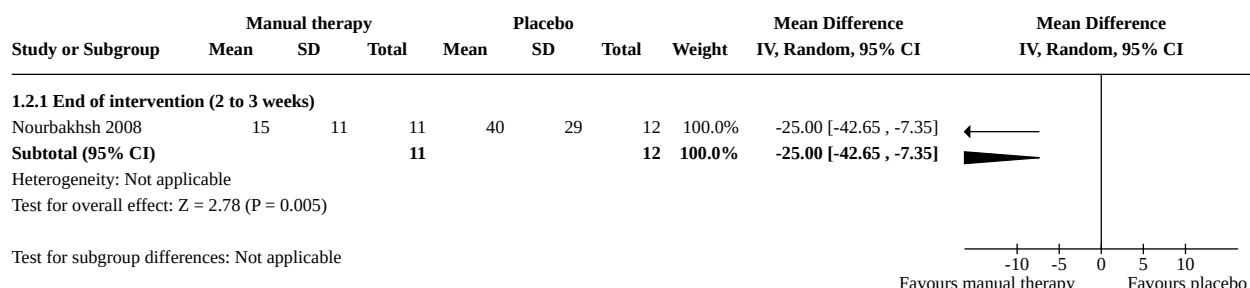
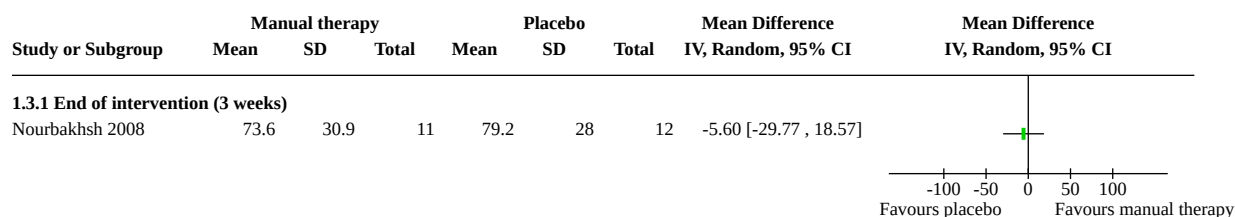
SD: standard deviation

VAS: visual analogue scale

DATA AND ANALYSES

Comparison 1. Manual therapy versus placebo manual therapy

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.1 Mean pain (0 to 10, 0 is no pain)	1		Mean Difference (IV, Random, 95% CI)	Totals not selected
1.1.1 End of intervention (2 to 3 weeks)	1		Mean Difference (IV, Random, 95% CI)	Totals not selected
1.2 Mean disability (0 to 100, 0 is no disability)	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
1.2.1 End of intervention (2 to 3 weeks)	1	23	Mean Difference (IV, Random, 95% CI)	-25.00 [-42.65, -7.35]
1.3 Mean grip strength (lb/in ²)	1		Mean Difference (IV, Random, 95% CI)	Totals not selected
1.3.1 End of intervention (3 weeks)	1		Mean Difference (IV, Random, 95% CI)	Totals not selected

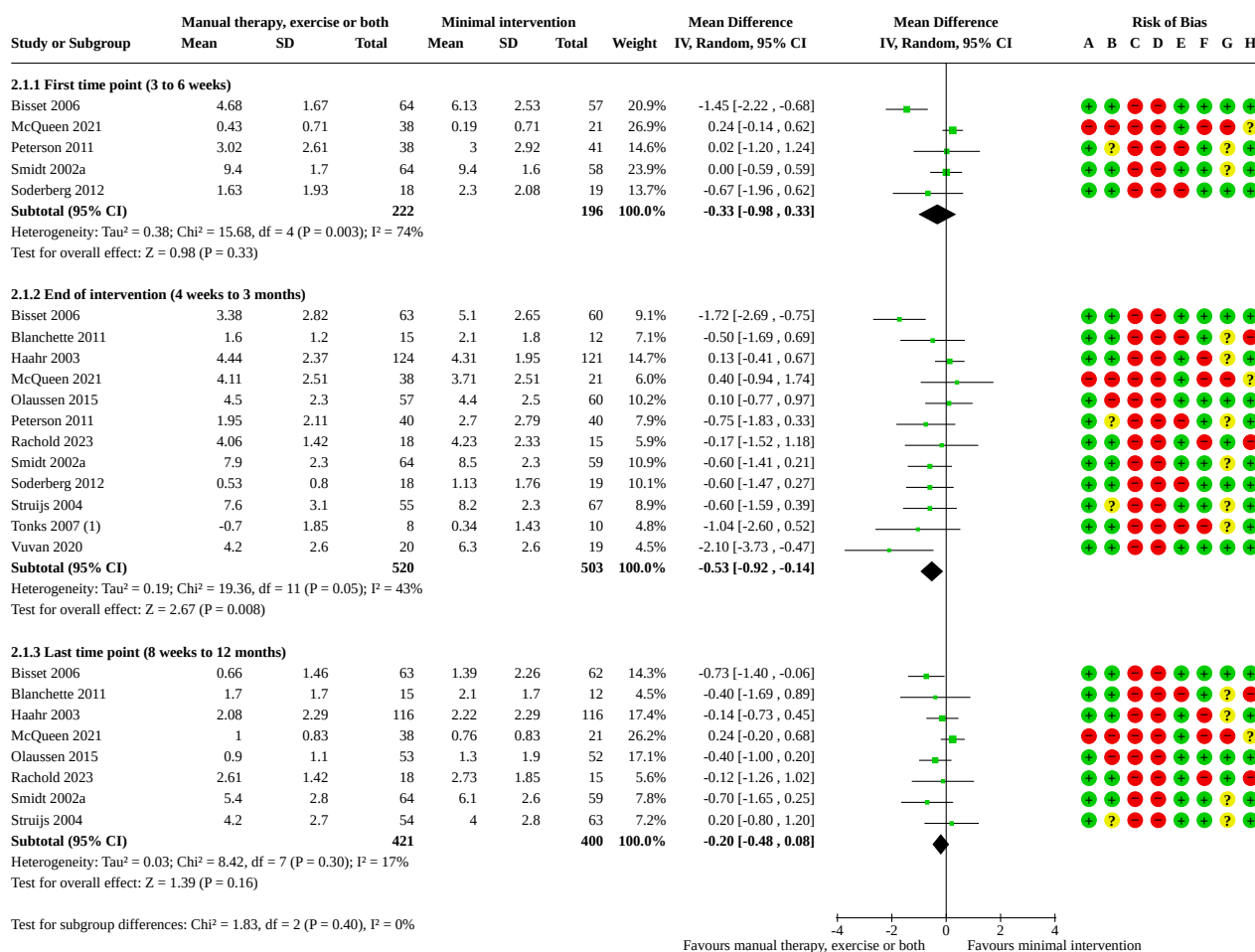
Analysis 1.1. Comparison 1: Manual therapy versus placebo manual therapy, Outcome 1: Mean pain (0 to 10, 0 is no pain)**Analysis 1.2. Comparison 1: Manual therapy versus placebo manual therapy, Outcome 2: Mean disability (0 to 100, 0 is no disability)****Analysis 1.3. Comparison 1: Manual therapy versus placebo manual therapy, Outcome 3: Mean grip strength (lb/in²)****Comparison 2. Manual therapy, prescribed exercises or both versus minimal or no intervention**

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.1 Mean pain (0 to 10, 0 is no pain)	12		Mean Difference (IV, Random, 95% CI)	Subtotals only
2.1.1 First time point (3 to 6 weeks)	5	418	Mean Difference (IV, Random, 95% CI)	-0.33 [-0.98, 0.33]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.1.2 End of intervention (4 weeks to 3 months)	12	1023	Mean Difference (IV, Random, 95% CI)	-0.53 [-0.92, -0.14]
2.1.3 Last time point (8 weeks to 12 months)	8	821	Mean Difference (IV, Random, 95% CI)	-0.20 [-0.48, 0.08]
2.2 Mean disability (0 to 100, 0 is no disability)	10		Mean Difference (IV, Random, 95% CI)	Subtotals only
2.2.1 First time point (3 to 6 weeks)	3	302	Mean Difference (IV, Random, 95% CI)	-2.46 [-6.65, 1.74]
2.2.2 End of intervention (4 weeks to 3 months)	10	732	Mean Difference (IV, Random, 95% CI)	-5.00 [-9.22, -0.77]
2.2.3 Last time point (8 weeks to 12 months)	7	588	Mean Difference (IV, Random, 95% CI)	-1.79 [-4.61, 1.03]
2.3 Health-related quality of life (SMD)	3		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
2.3.1 End of intervention (4 weeks to 3 months)	2	113	Std. Mean Difference (IV, Random, 95% CI)	-0.45 [-0.83, -0.08]
2.3.2 Last time point (8 weeks to 12 months)	2	156	Std. Mean Difference (IV, Random, 95% CI)	-0.18 [-0.82, 0.45]
2.4 Participant reported treatment success	6		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
2.4.1 First time point (3 weeks)	2	243	Risk Ratio (M-H, Random, 95% CI)	1.76 [0.98, 3.18]
2.4.2 End of intervention (6 weeks to 3 months)	6	770	Risk Ratio (M-H, Random, 95% CI)	1.36 [0.96, 1.93]
2.4.3 Last time point (12 months)	5	702	Risk Ratio (M-H, Random, 95% CI)	1.02 [0.97, 1.08]
2.5 Proportion of overall withdrawals	12	1147	Risk Ratio (M-H, Random, 95% CI)	0.86 [0.66, 1.12]
2.6 Proportion with any adverse event	5	433	Risk Ratio (M-H, Random, 95% CI)	3.69 [0.98, 13.97]
2.7 Pain-free grip strength (SMD)	8		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
2.7.1 First time point (3 weeks)	3	280	Std. Mean Difference (IV, Random, 95% CI)	0.29 [0.05, 0.53]
2.7.2 End of intervention (6 to 8 weeks)	8	606	Std. Mean Difference (IV, Random, 95% CI)	0.18 [-0.11, 0.48]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.7.3 Last time point (12 months)	4	470	Std. Mean Difference (IV, Random, 95% CI)	0.18 [-0.01, 0.36]
2.8 Proportion on sick leave	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected

Analysis 2.1. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 1: Mean pain (0 to 10, 0 is no pain)

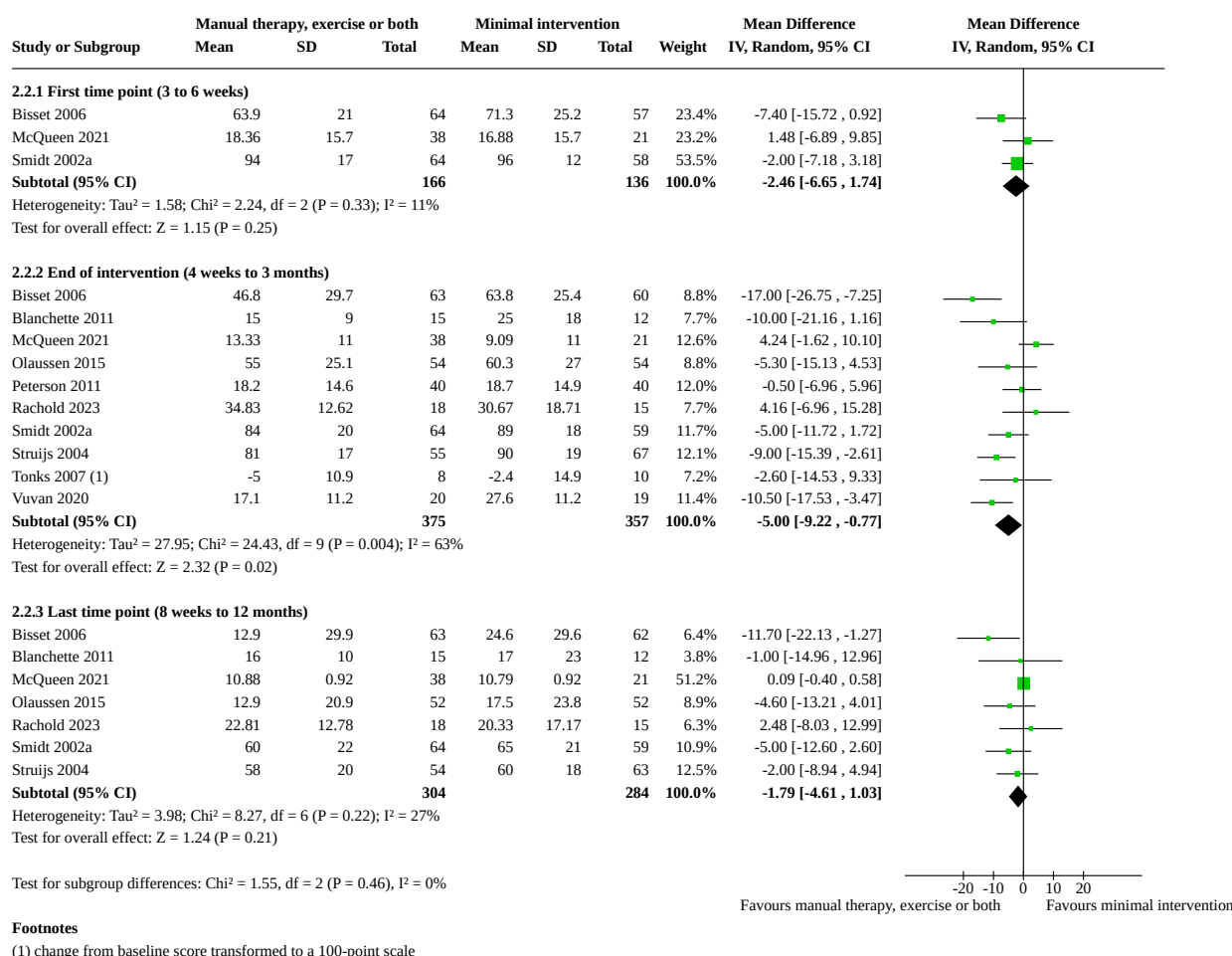
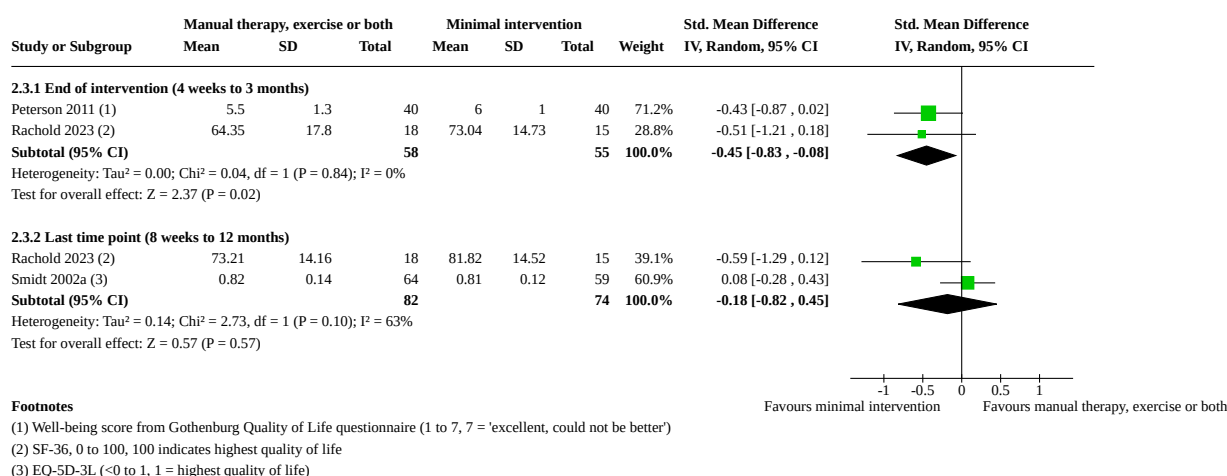


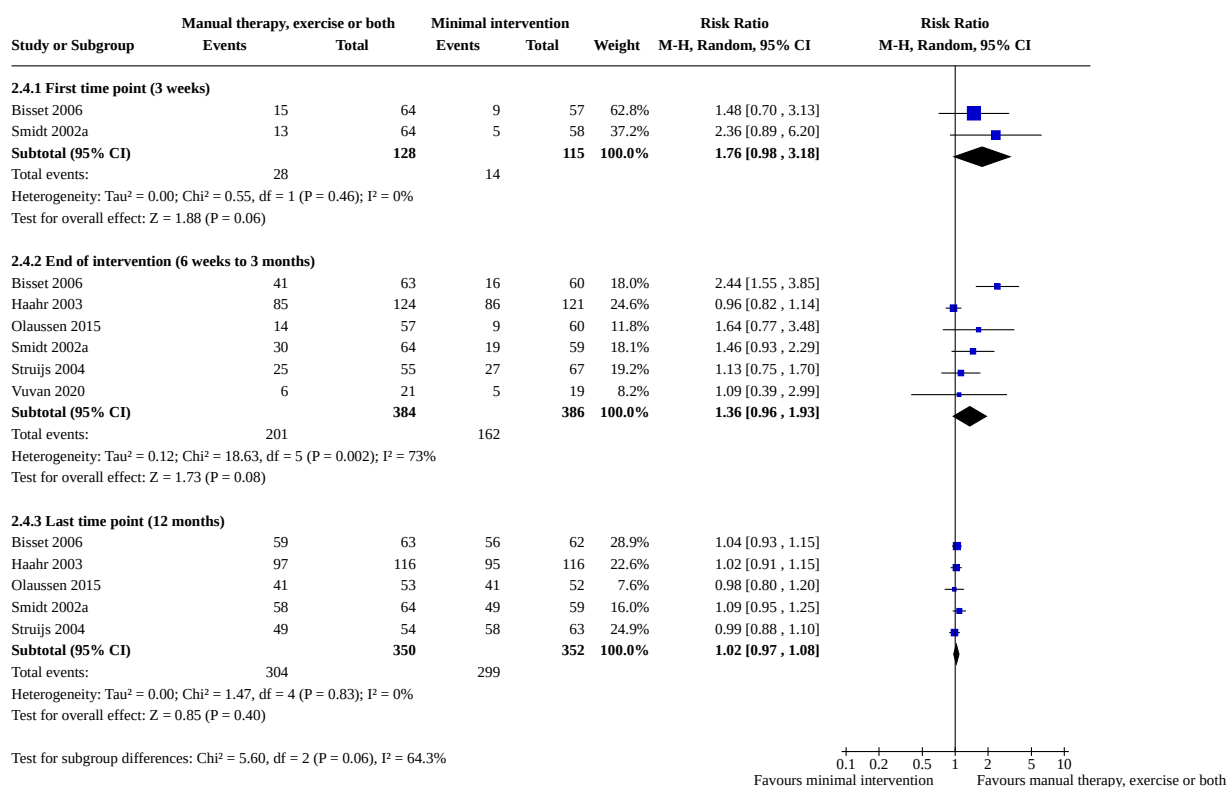
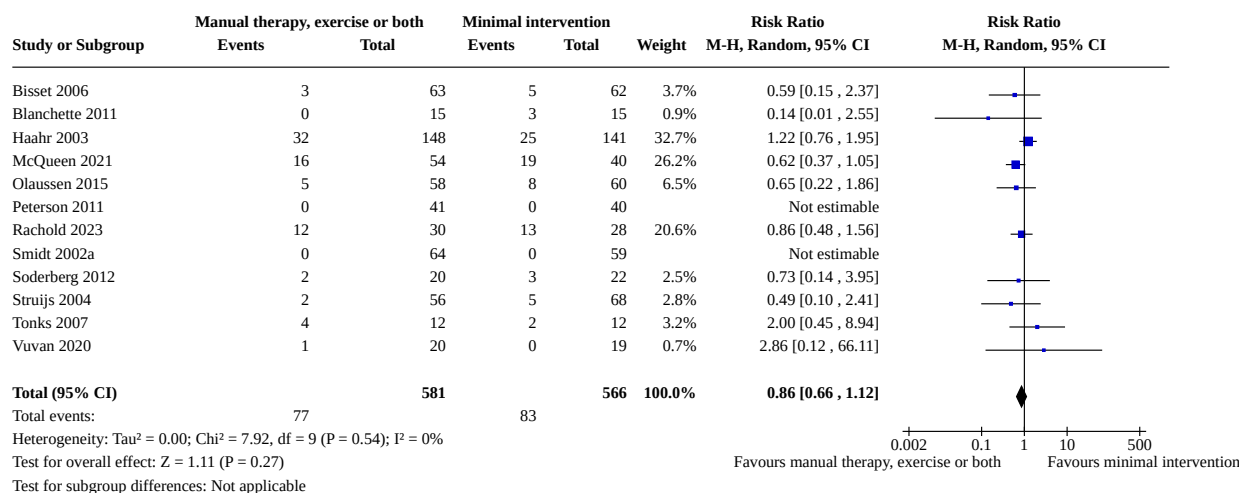
Footnotes

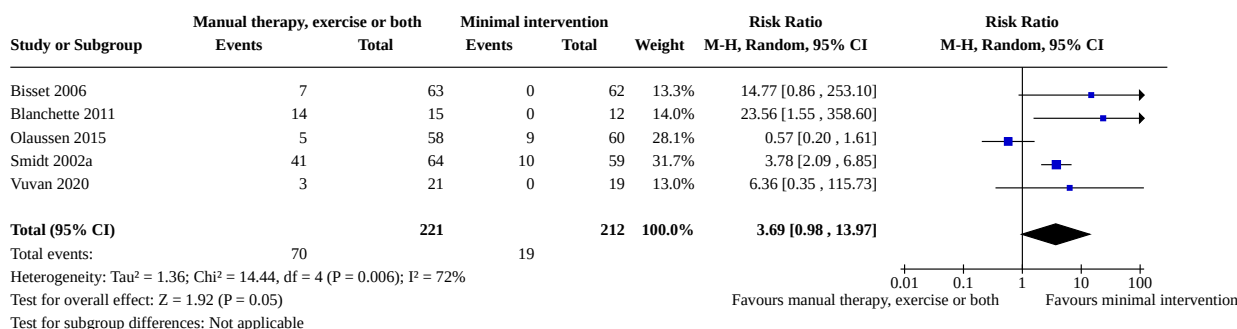
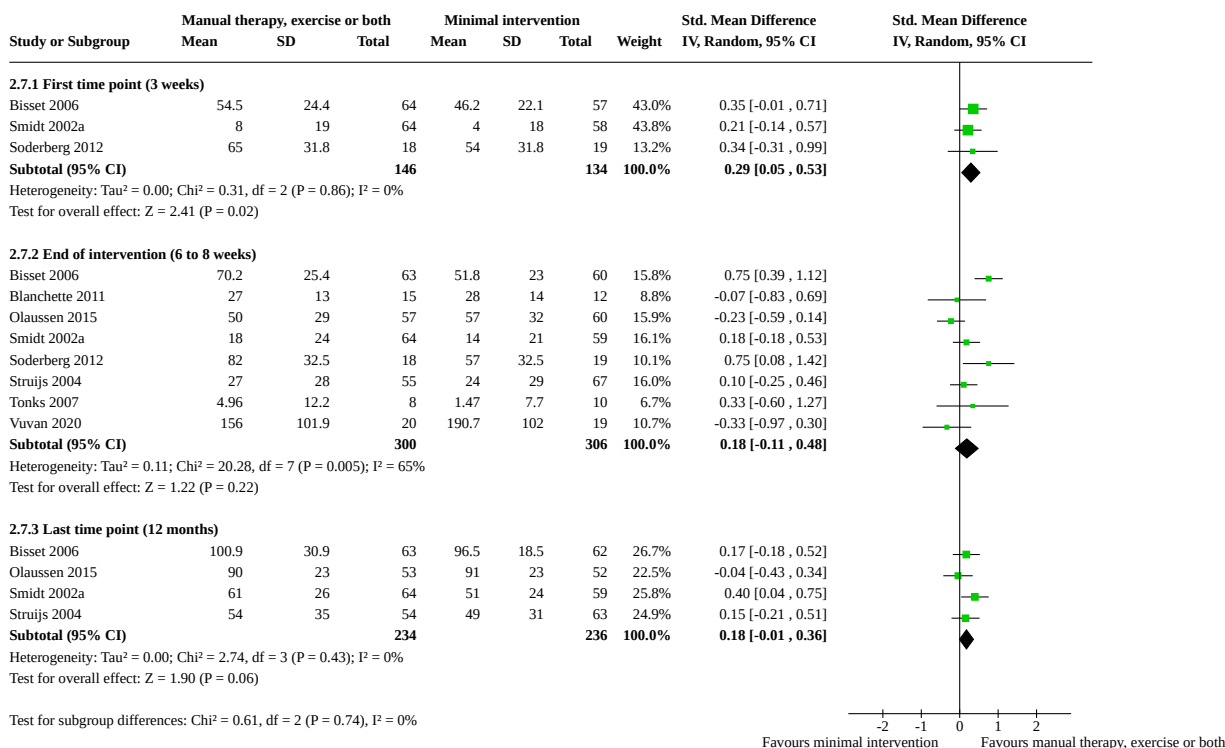
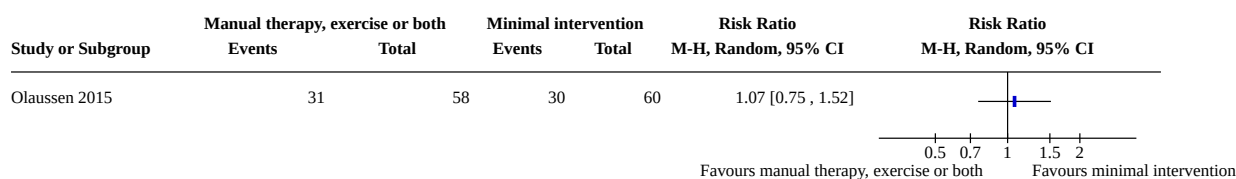
(1) change from baseline score

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment - self reported (detection bias)
- (E) Blinding of outcome assessment - assessor reported outcomes
- (F) Incomplete outcome data (attrition bias)
- (G) Selective reporting (reporting bias)
- (H) Other bias

Analysis 2.2. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 2: Mean disability (0 to 100, 0 is no disability)**Analysis 2.3. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 3: Health-related quality of life (SMD)**

Analysis 2.4. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 4: Participant reported treatment success**Analysis 2.5. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 5: Proportion of overall withdrawals**

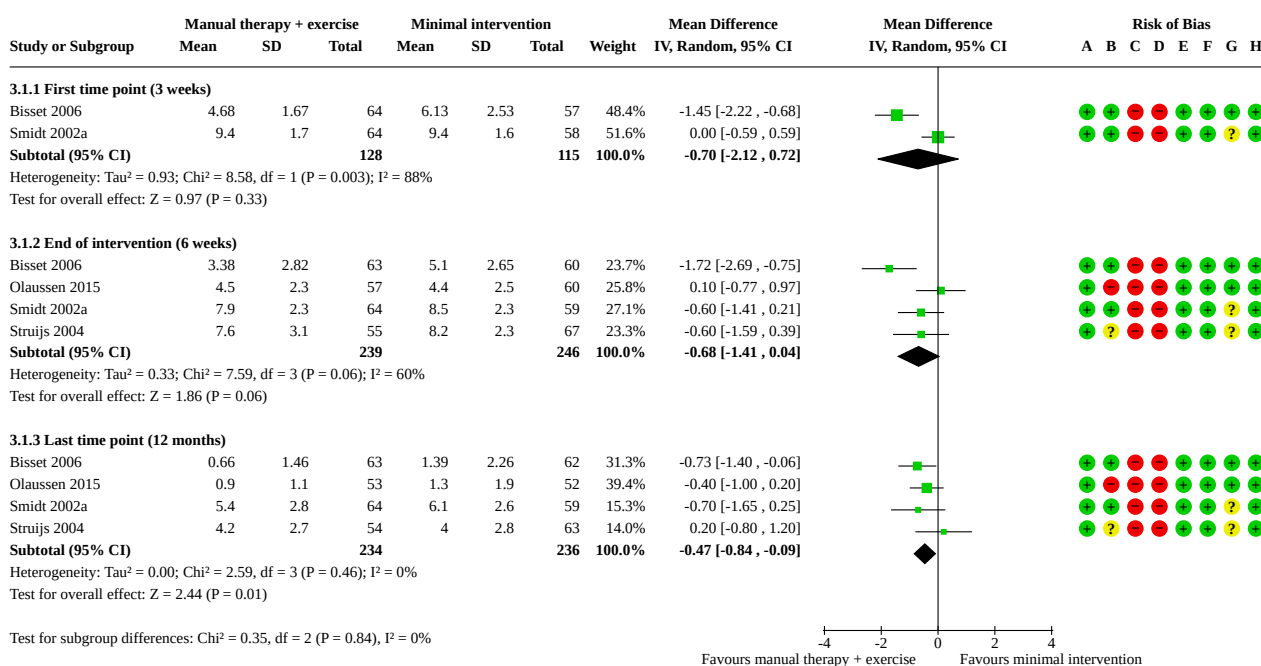
Analysis 2.6. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 6: Proportion with any adverse event**Analysis 2.7. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 7: Pain-free grip strength (SMD)****Analysis 2.8. Comparison 2: Manual therapy, prescribed exercises or both versus minimal or no intervention, Outcome 8: Proportion on sick leave**

Comparison 3. Manual therapy and prescribed exercises versus minimal or no intervention

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
3.1 Mean pain (0 to 10, 0 is no pain)	4		Mean Difference (IV, Random, 95% CI)	Subtotals only
3.1.1 First time point (3 weeks)	2	243	Mean Difference (IV, Random, 95% CI)	-0.70 [-2.12, 0.72]
3.1.2 End of intervention (6 weeks)	4	485	Mean Difference (IV, Random, 95% CI)	-0.68 [-1.41, 0.04]
3.1.3 Last time point (12 months)	4	470	Mean Difference (IV, Random, 95% CI)	-0.47 [-0.84, -0.09]
3.2 Mean disability (0 to 100, 0 is no disability)	4		Mean Difference (IV, Random, 95% CI)	Subtotals only
3.2.1 First time point (3 weeks)	2	243	Mean Difference (IV, Random, 95% CI)	-3.68 [-8.58, 1.22]
3.2.2 End of intervention (6 weeks)	4	476	Mean Difference (IV, Random, 95% CI)	-8.57 [-13.34, -3.79]
3.2.3 Last time point (12 months)	4	469	Mean Difference (IV, Random, 95% CI)	-4.90 [-8.96, -0.84]
3.3 Mean health related quality of life	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
3.3.1 Last time point (12 months)	1	123	Mean Difference (IV, Random, 95% CI)	0.01 [-0.04, 0.06]
3.4 Participant reported treatment success	4		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
3.4.1 First time point (3 weeks)	2	243	Risk Ratio (M-H, Random, 95% CI)	1.76 [0.98, 3.18]
3.4.2 End of intervention (6 weeks)	4	485	Risk Ratio (M-H, Random, 95% CI)	1.58 [1.11, 2.26]
3.4.3 Last time point (12 months)	4	470	Risk Ratio (M-H, Random, 95% CI)	1.03 [0.96, 1.09]
3.5 Proportion of overall withdrawals	4	490	Risk Ratio (M-H, Random, 95% CI)	0.59 [0.28, 1.25]
3.6 Proportion with any adverse event	3	366	Risk Ratio (M-H, Random, 95% CI)	2.37 [0.47, 11.91]
3.7 Pain free grip strength (SMD)	4		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
3.7.1 First time point (3 weeks)	2	243	Std. Mean Difference (IV, Random, 95% CI)	0.28 [0.03, 0.54]

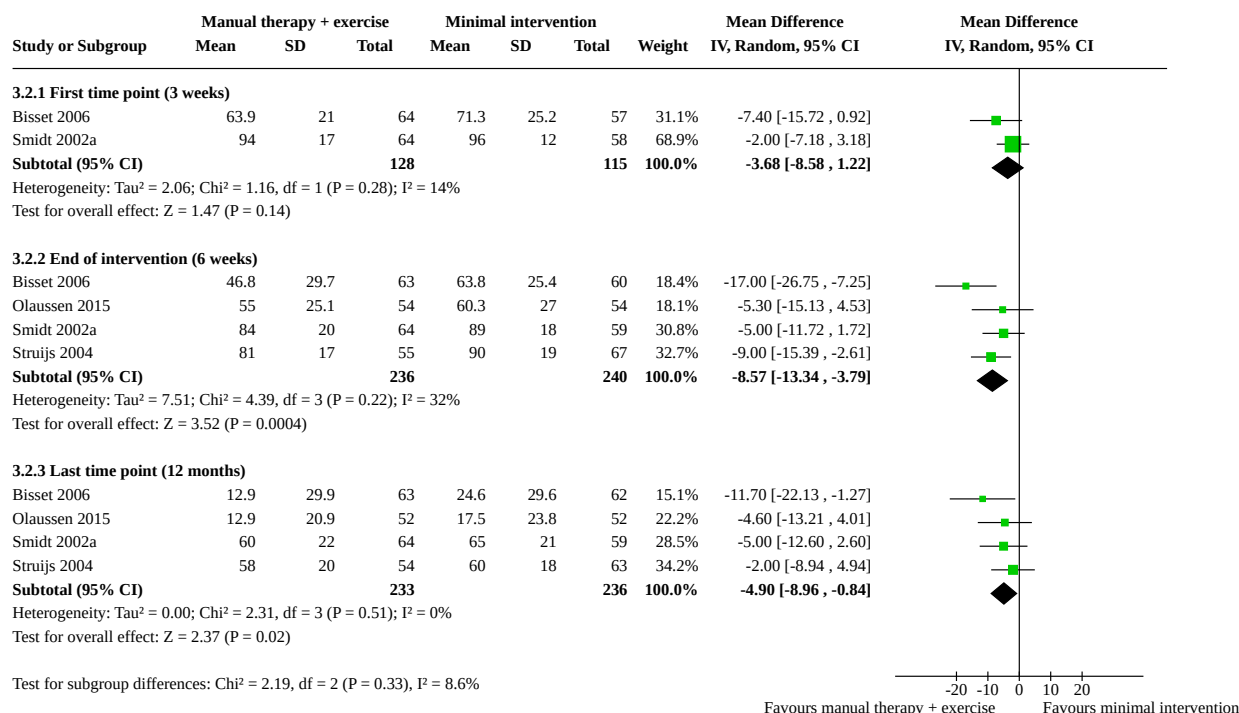
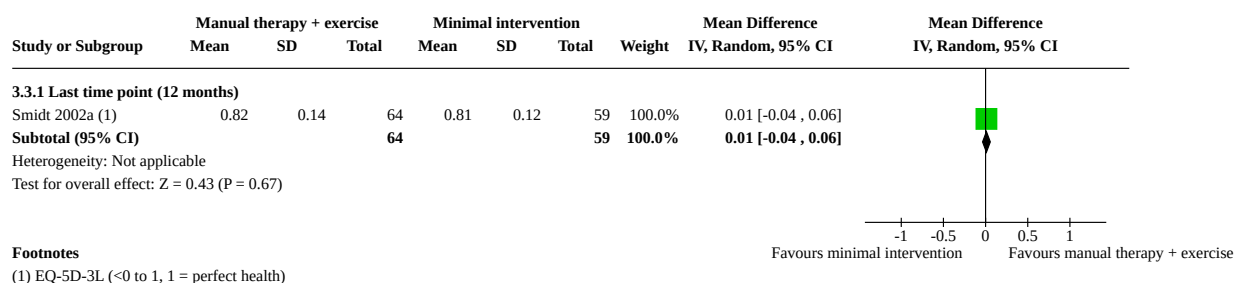
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
3.7.2 End of intervention (6 weeks)	4	485	Std. Mean Difference (IV, Random, 95% CI)	0.20 [-0.19, 0.60]
3.7.3 Last time point (12 months)	4	470	Std. Mean Difference (IV, Random, 95% CI)	0.18 [-0.01, 0.36]
3.8 Proportion on sick leave	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected

Analysis 3.1. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 1: Mean pain (0 to 10, 0 is no pain)

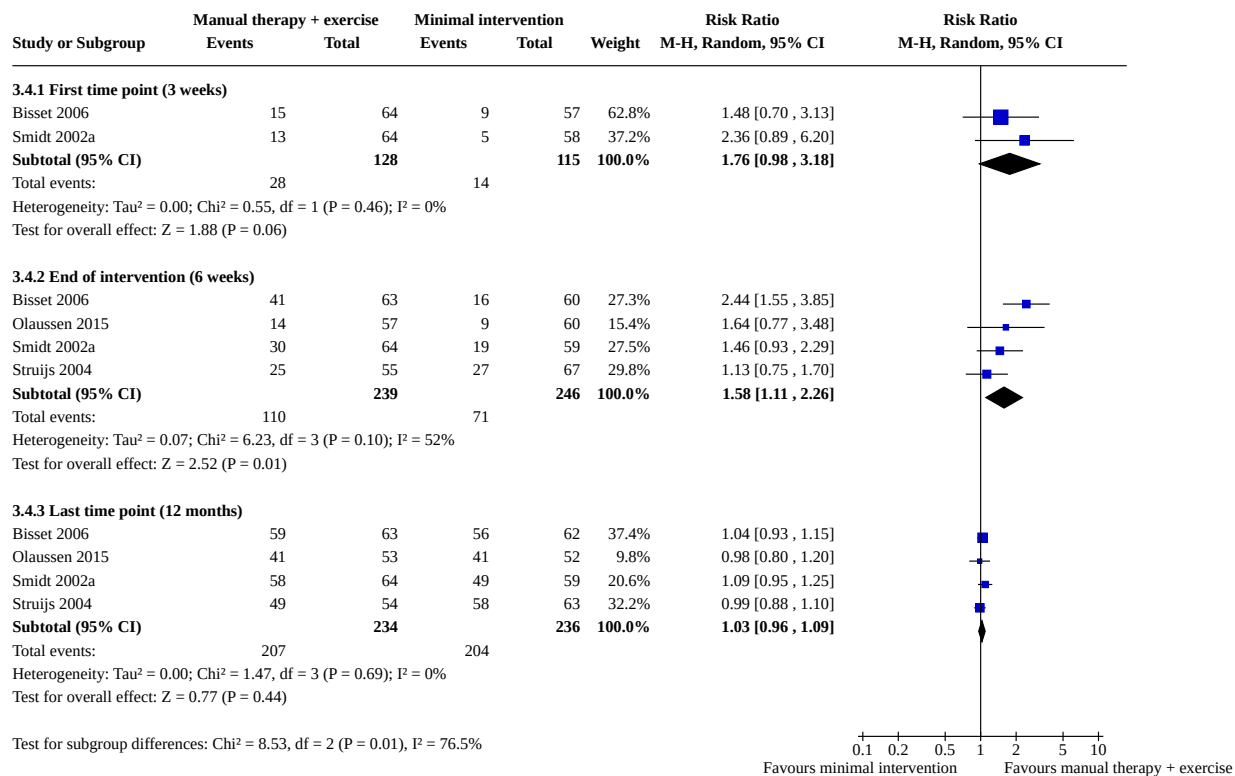
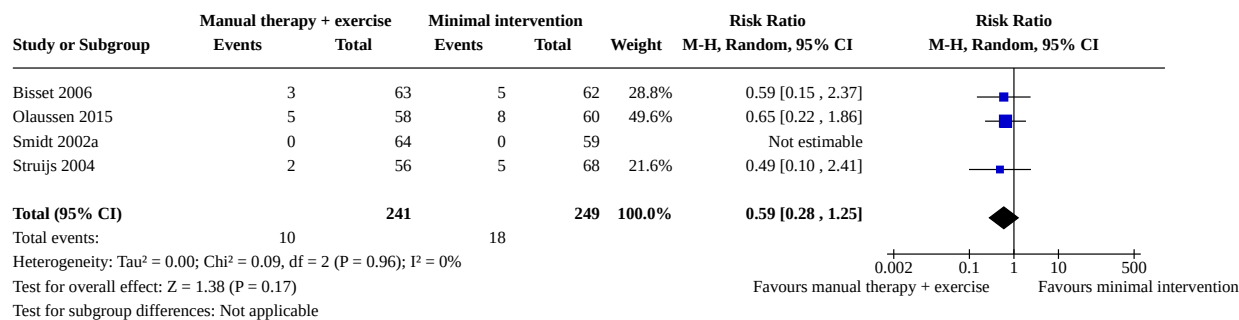


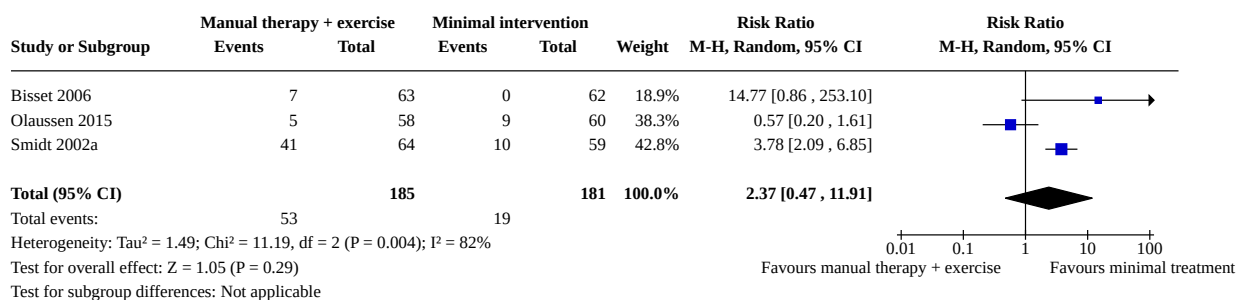
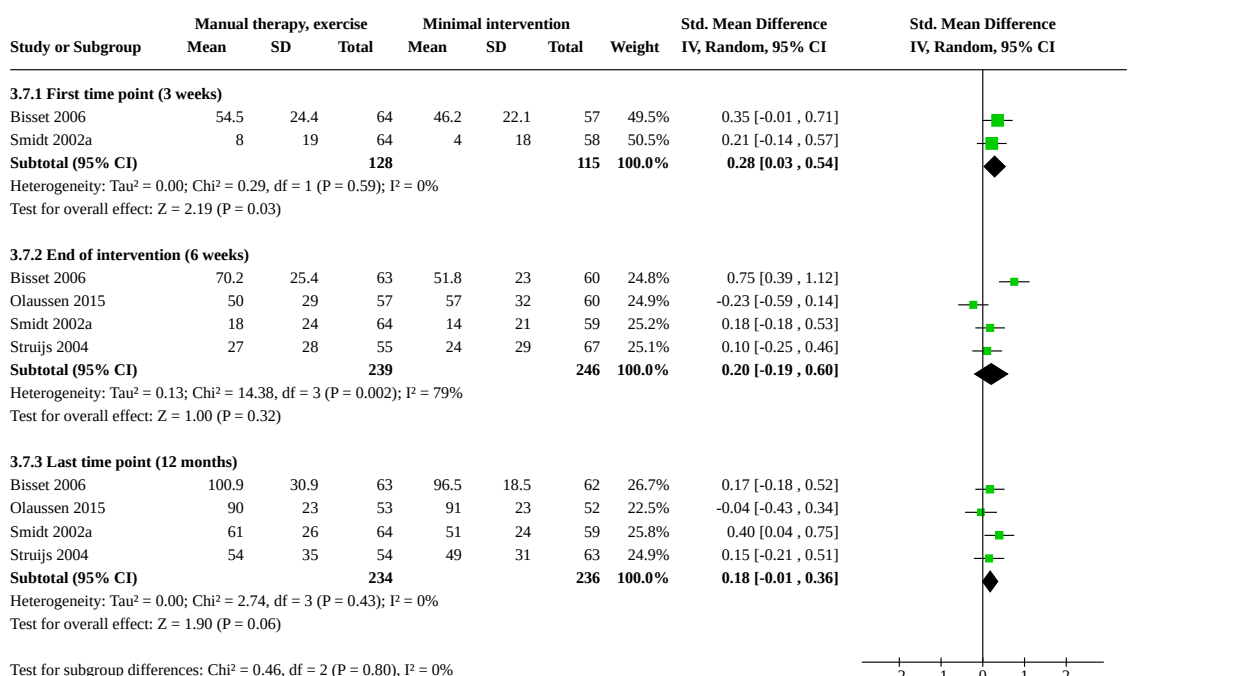
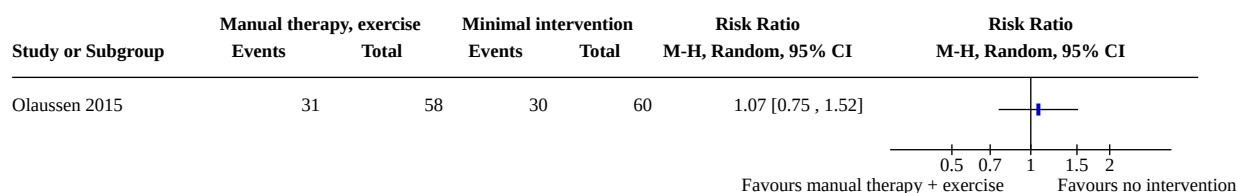
Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment - self reported (detection bias)
- (E) Blinding of outcome assessment - assessor reported outcomes
- (F) Incomplete outcome data (attrition bias)
- (G) Selective reporting (reporting bias)
- (H) Other bias

Analysis 3.2. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 2: Mean disability (0 to 100, 0 is no disability)**Analysis 3.3. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 3: Mean health related quality of life****Footnotes**

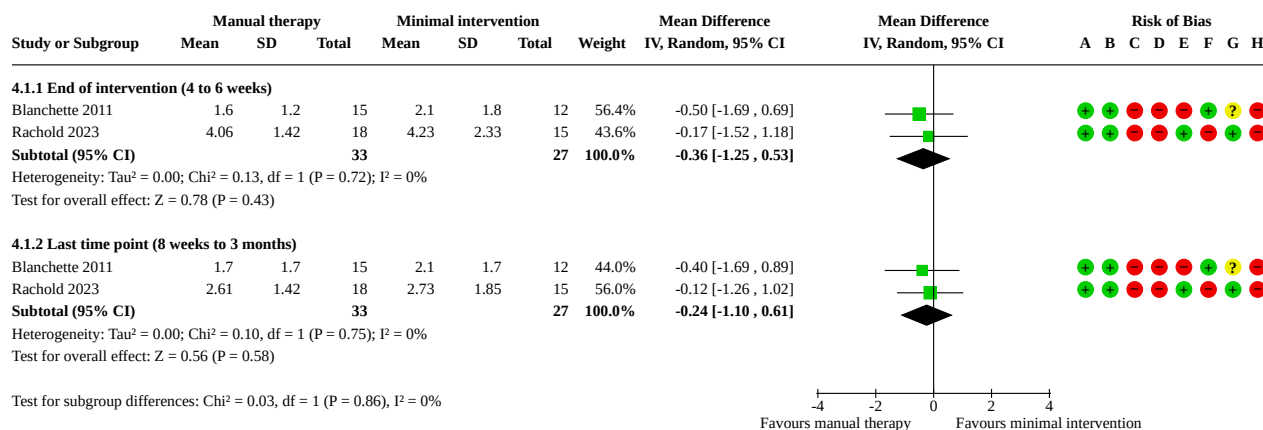
(1) EQ-5D-3L (<0 to 1, 1 = perfect health)

Analysis 3.4. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 4: Participant reported treatment success**Analysis 3.5. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 5: Proportion of overall withdrawals**

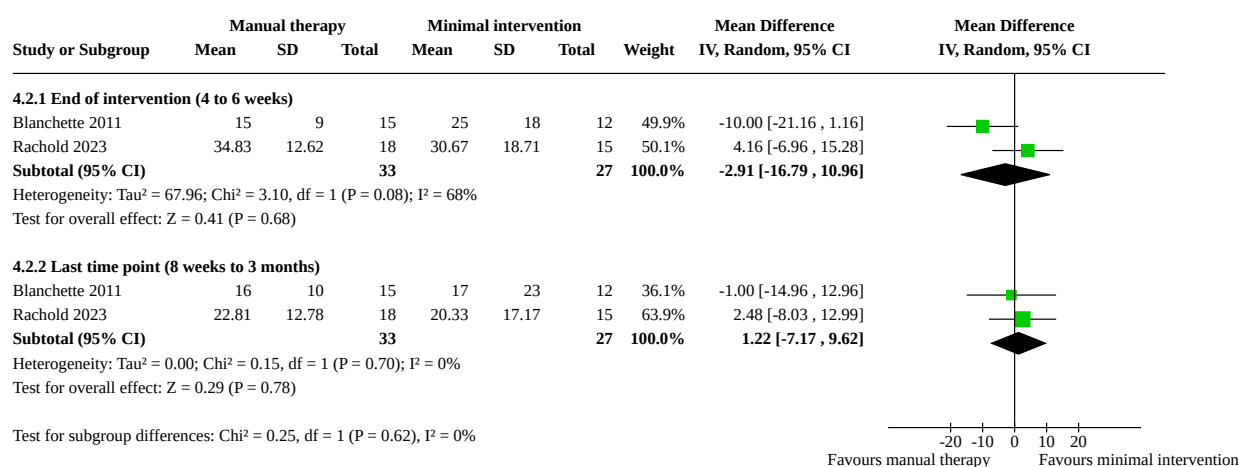
Analysis 3.6. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 6: Proportion with any adverse event**Analysis 3.7. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 7: Pain free grip strength (SMD)****Analysis 3.8. Comparison 3: Manual therapy and prescribed exercises versus minimal or no intervention, Outcome 8: Proportion on sick leave**

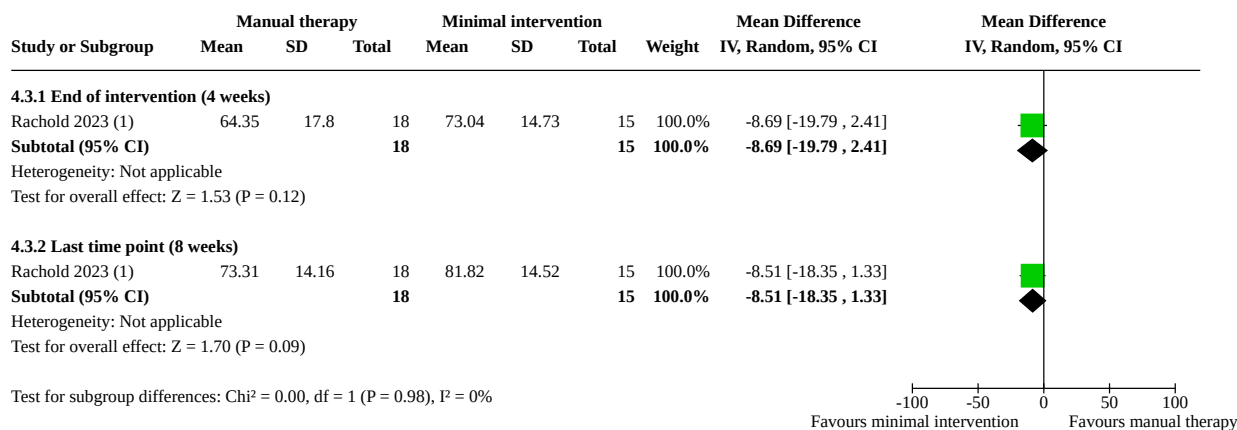
Comparison 4. Manual therapy versus minimal or no intervention

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
4.1 Mean pain (0 to 10, 0 is no pain)	2		Mean Difference (IV, Random, 95% CI)	Subtotals only
4.1.1 End of intervention (4 to 6 weeks)	2	60	Mean Difference (IV, Random, 95% CI)	-0.36 [-1.25, 0.53]
4.1.2 Last time point (8 weeks to 3 months)	2	60	Mean Difference (IV, Random, 95% CI)	-0.24 [-1.10, 0.61]
4.2 Mean disability (0 to 100, 0 is no disability)	2		Mean Difference (IV, Random, 95% CI)	Subtotals only
4.2.1 End of intervention (4 to 6 weeks)	2	60	Mean Difference (IV, Random, 95% CI)	-2.91 [-16.79, 10.96]
4.2.2 Last time point (8 weeks to 3 months)	2	60	Mean Difference (IV, Random, 95% CI)	1.22 [-7.17, 9.62]
4.3 Mean health related quality of life	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
4.3.1 End of intervention (4 weeks)	1	33	Mean Difference (IV, Random, 95% CI)	-8.69 [-19.79, 2.41]
4.3.2 Last time point (8 weeks)	1	33	Mean Difference (IV, Random, 95% CI)	-8.51 [-18.35, 1.33]
4.4 Proportion of overall withdrawals	2	88	Risk Ratio (M-H, Random, 95% CI)	0.59 [0.13, 2.68]
4.5 Proportion with any adverse event	1	27	Risk Ratio (M-H, Random, 95% CI)	23.56 [1.55, 358.60]
4.6 Mean pain free grip strength (kg)	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
4.6.1 End of intervention (6 weeks)	1	27	Mean Difference (IV, Random, 95% CI)	-1.00 [-11.30, 9.30]

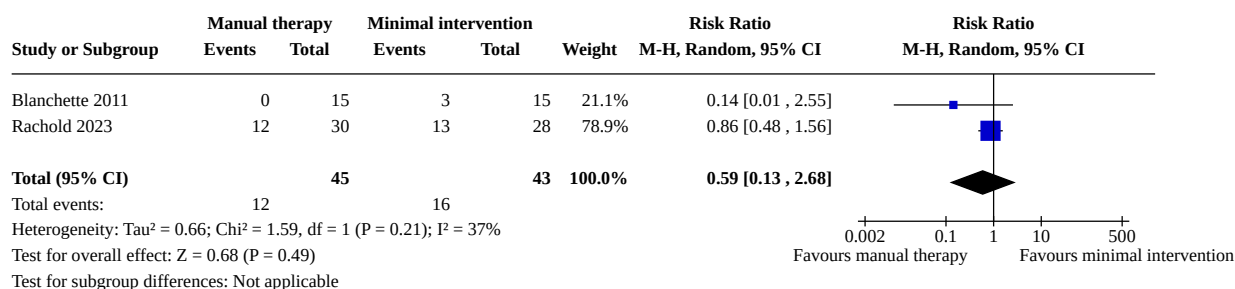
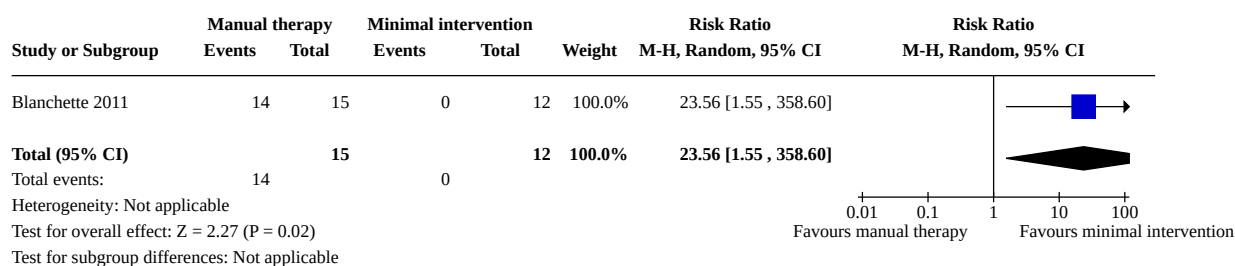
Analysis 4.1. Comparison 4: Manual therapy versus minimal or no intervention, Outcome 1: Mean pain (0 to 10, 0 is no pain)**Risk of bias legend**

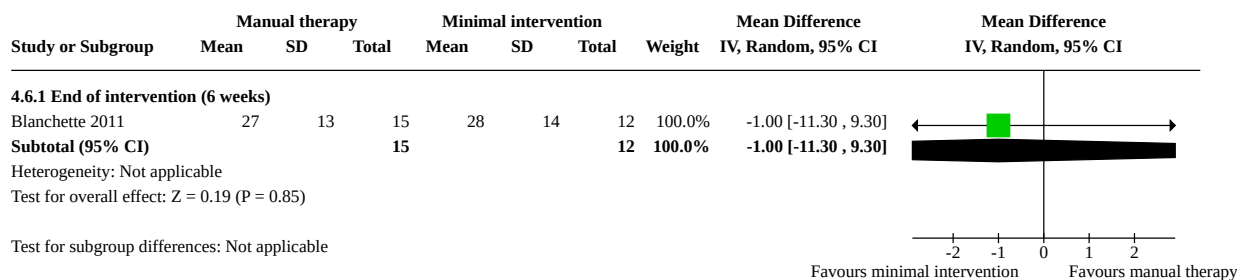
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment - self reported (detection bias)
- (E) Blinding of outcome assessment - assessor reported outcomes
- (F) Incomplete outcome data (attrition bias)
- (G) Selective reporting (reporting bias)
- (H) Other bias

Analysis 4.2. Comparison 4: Manual therapy versus minimal or no intervention, Outcome 2: Mean disability (0 to 100, 0 is no disability)

Analysis 4.3. Comparison 4: Manual therapy versus minimal or no intervention, Outcome 3: Mean health related quality of life**Footnotes**

(1) SF-36, 0 to 100, 100 indicates highest quality of life

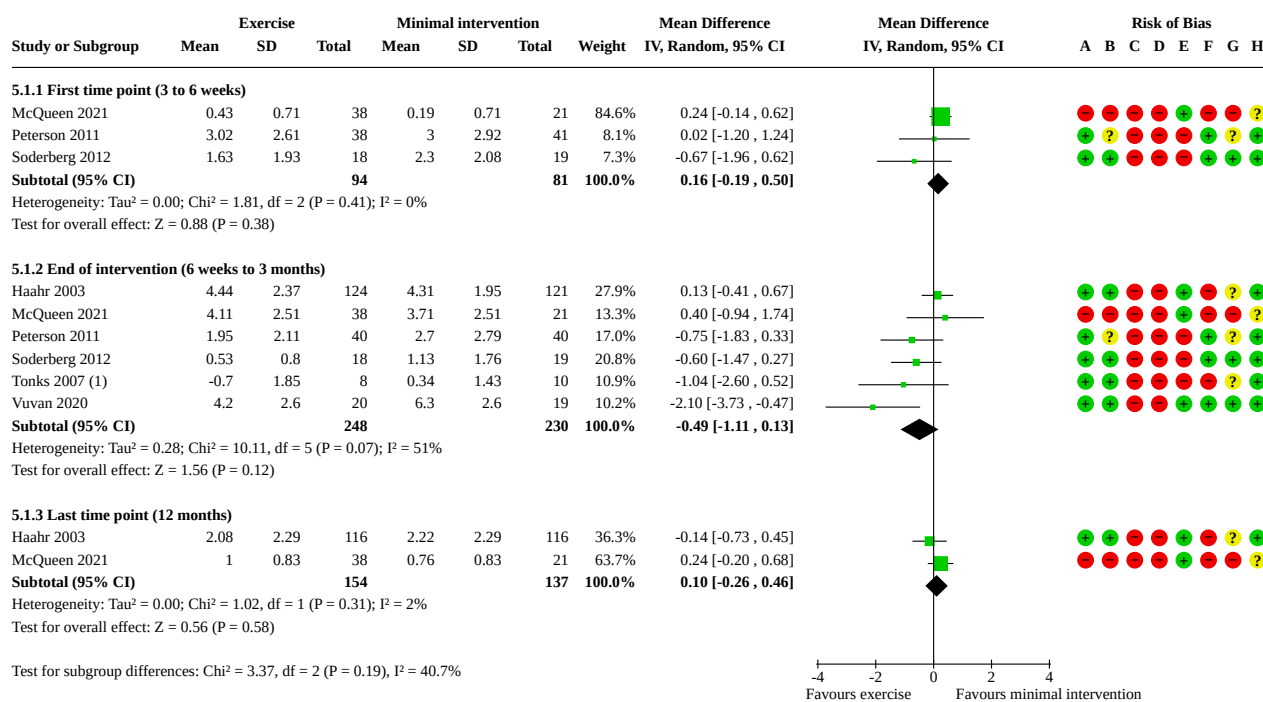
Analysis 4.4. Comparison 4: Manual therapy versus minimal or no intervention, Outcome 4: Proportion of overall withdrawals**Analysis 4.5. Comparison 4: Manual therapy versus minimal or no intervention, Outcome 5: Proportion with any adverse event**

Analysis 4.6. Comparison 4: Manual therapy versus minimal or no intervention, Outcome 6: Mean pain free grip strength (kg)**Comparison 5. Prescribed exercises versus minimal or no intervention**

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
5.1 Mean pain (0 to 10, 0 is no pain)	6		Mean Difference (IV, Random, 95% CI)	Subtotals only
5.1.1 First time point (3 to 6 weeks)	3	175	Mean Difference (IV, Random, 95% CI)	0.16 [-0.19, 0.50]
5.1.2 End of intervention (6 weeks to 3 months)	6	478	Mean Difference (IV, Random, 95% CI)	-0.49 [-1.11, 0.13]
5.1.3 Last time point (12 months)	2	291	Mean Difference (IV, Random, 95% CI)	0.10 [-0.26, 0.46]
5.2 Mean disability (0 to 100, 0 is no disability)	4		Mean Difference (IV, Random, 95% CI)	Subtotals only
5.2.1 First time point (6 weeks)	1	59	Mean Difference (IV, Random, 95% CI)	1.48 [-6.89, 9.85]
5.2.2 End of intervention (7 weeks to 3 months)	4	196	Mean Difference (IV, Random, 95% CI)	-2.13 [-8.91, 4.65]
5.2.3 Last time point (12 months)	1	59	Mean Difference (IV, Random, 95% CI)	0.09 [-0.40, 0.58]
5.3 Mean health-related quality of life	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
5.3.1 End of intervention (3 months)	1	80	Mean Difference (IV, Random, 95% CI)	-0.50 [-1.01, 0.01]
5.4 Participant-reported treatment success	2		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
5.4.1 End of intervention (8 weeks to 3 months)	2	285	Risk Ratio (M-H, Random, 95% CI)	0.97 [0.82, 1.14]
5.4.2 Last time point (12 months)	1	232	Risk Ratio (M-H, Random, 95% CI)	1.02 [0.91, 1.15]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
5.5 Proportion of overall withdrawals	6	569	Risk Ratio (M-H, Random, 95% CI)	0.96 [0.62, 1.49]
5.6 Proportion with any adverse event	1	40	Risk Ratio (M-H, Random, 95% CI)	6.36 [0.35, 115.73]
5.7 Pain-free grip strength (SMD)	3		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
5.7.1 First time point (3 weeks)	1	37	Std. Mean Difference (IV, Random, 95% CI)	0.34 [-0.31, 0.99]
5.7.2 End of intervention (6 to 8 weeks)	3	94	Std. Mean Difference (IV, Random, 95% CI)	0.24 [-0.46, 0.94]

Analysis 5.1. Comparison 5: Prescribed exercises versus minimal or no intervention, Outcome 1: Mean pain (0 to 10, 0 is no pain)

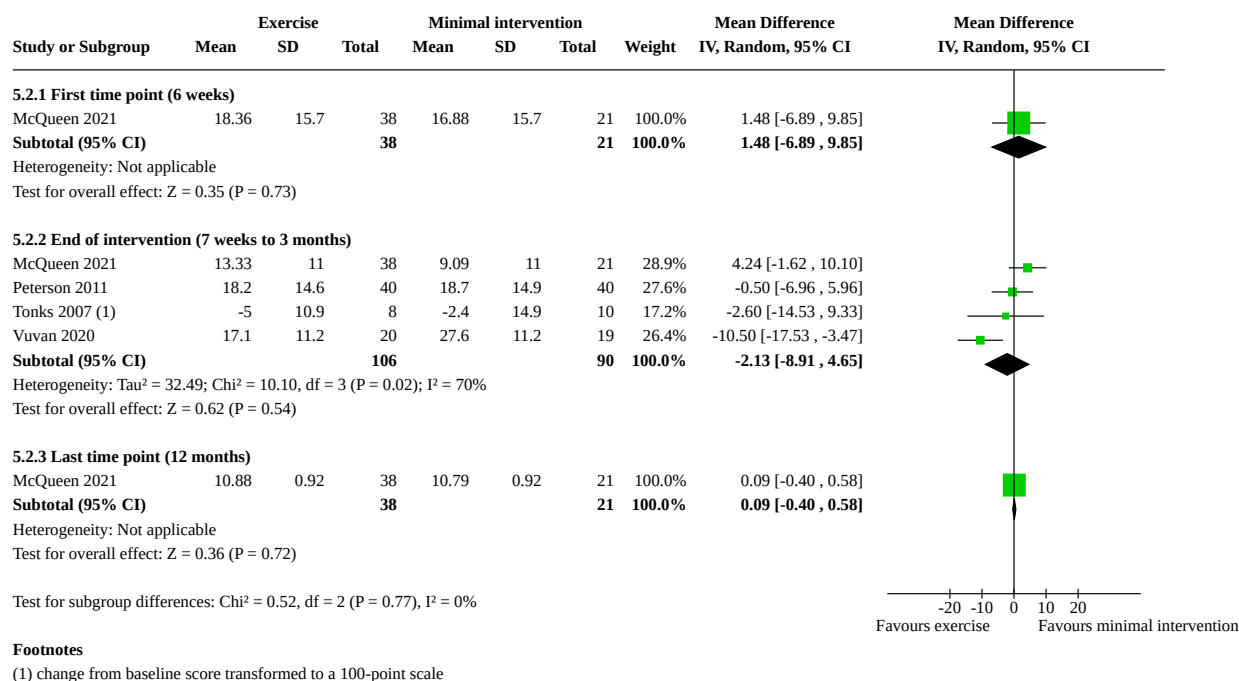
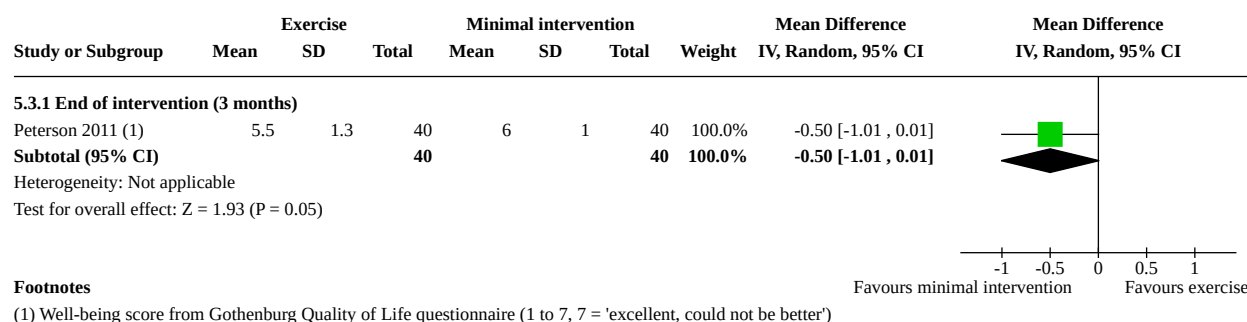


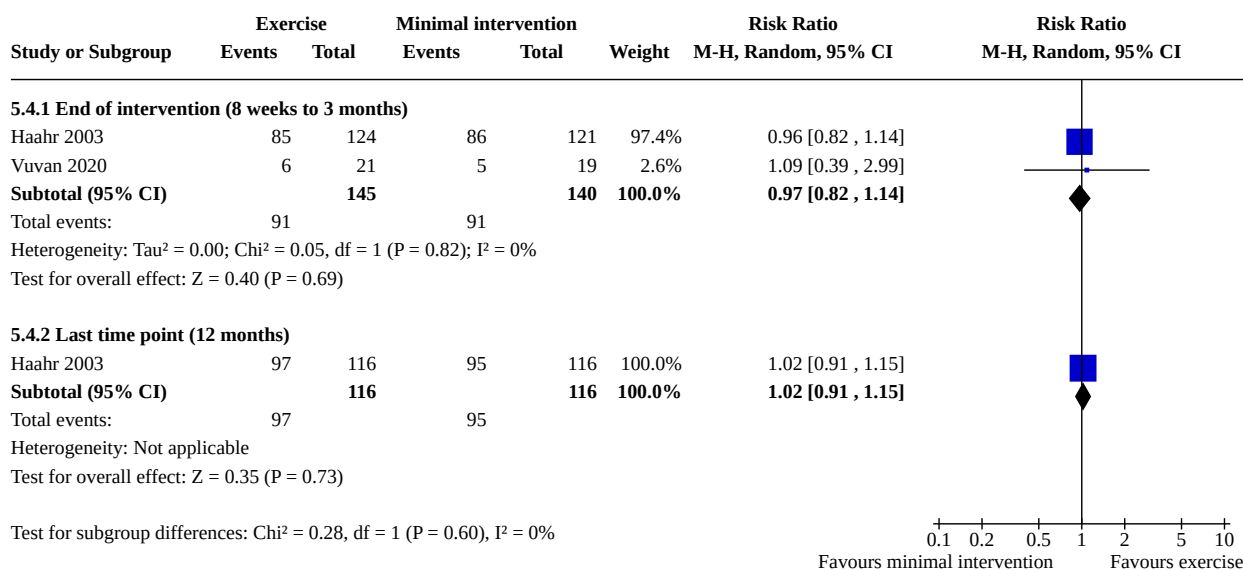
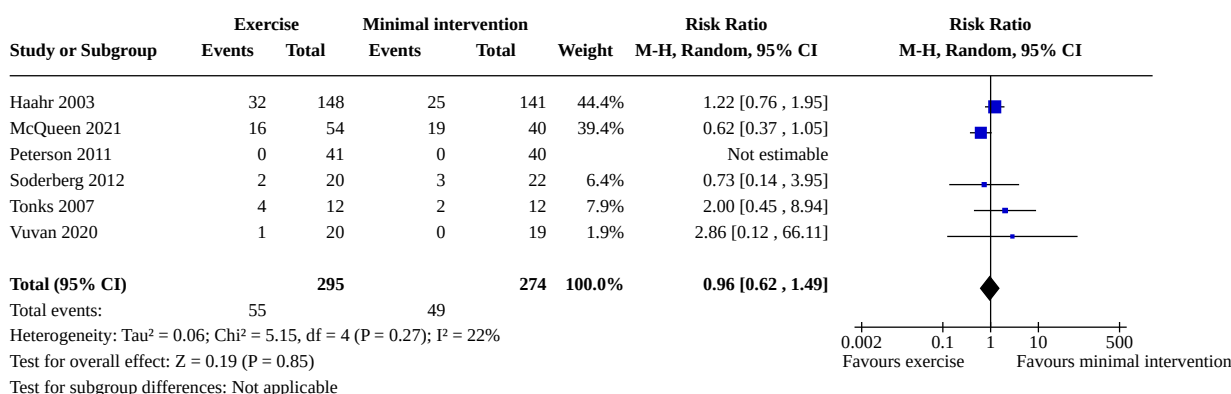
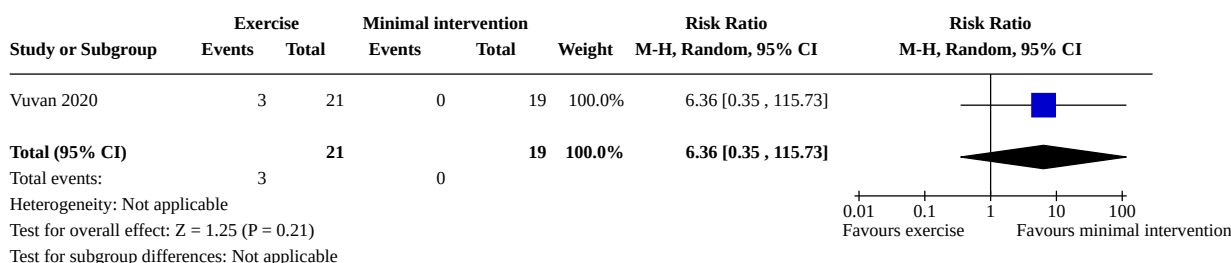
Footnotes

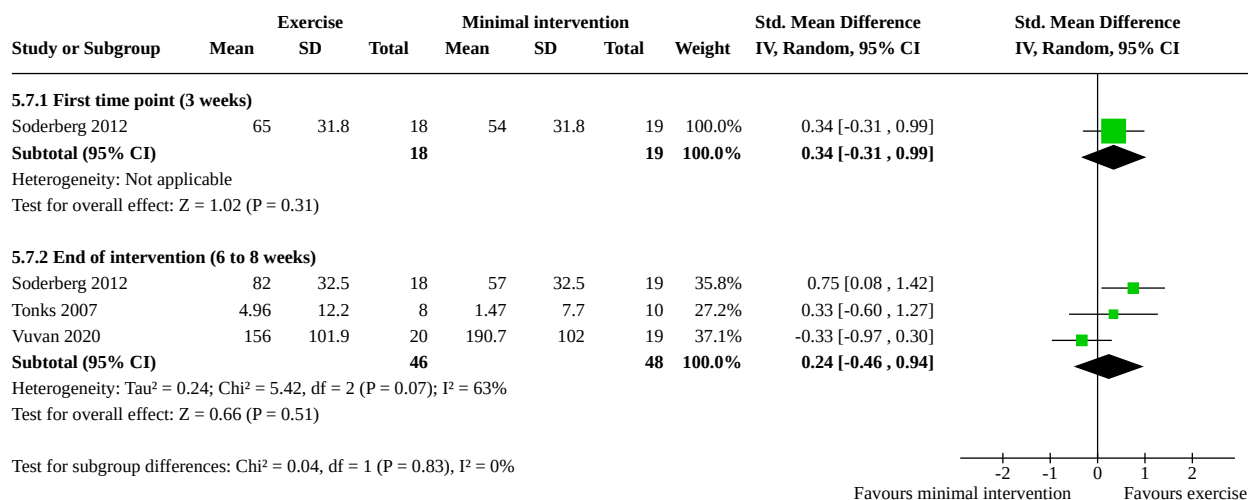
(1) change from baseline score

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment - self reported (detection bias)
- (E) Blinding of outcome assessment - assessor reported outcomes
- (F) Incomplete outcome data (attrition bias)
- (G) Selective reporting (reporting bias)
- (H) Other bias

Analysis 5.2. Comparison 5: Prescribed exercises versus minimal or no intervention, Outcome 2: Mean disability (0 to 100, 0 is no disability)**Analysis 5.3. Comparison 5: Prescribed exercises versus minimal or no intervention, Outcome 3: Mean health-related quality of life**

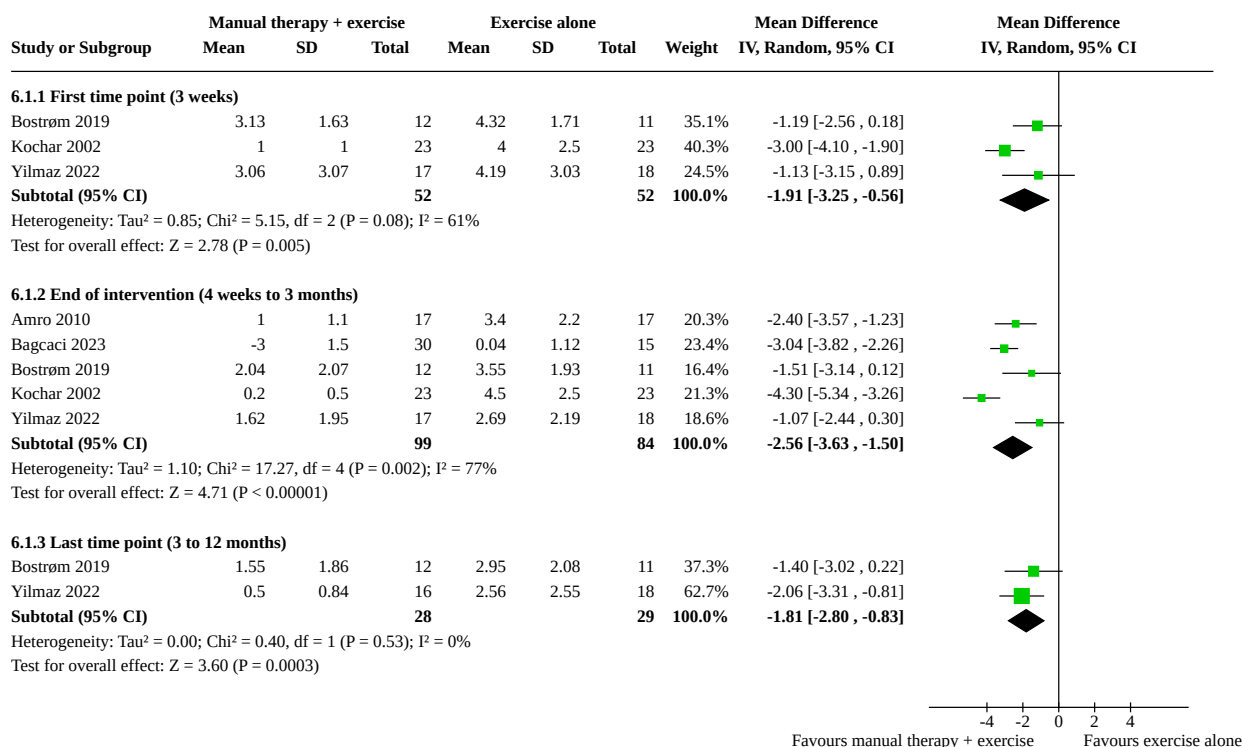
Analysis 5.4. Comparison 5: Prescribed exercises versus minimal or no intervention, Outcome 4: Participant-reported treatment success**Analysis 5.5. Comparison 5: Prescribed exercises versus minimal or no intervention, Outcome 5: Proportion of overall withdrawals****Analysis 5.6. Comparison 5: Prescribed exercises versus minimal or no intervention, Outcome 6: Proportion with any adverse event**

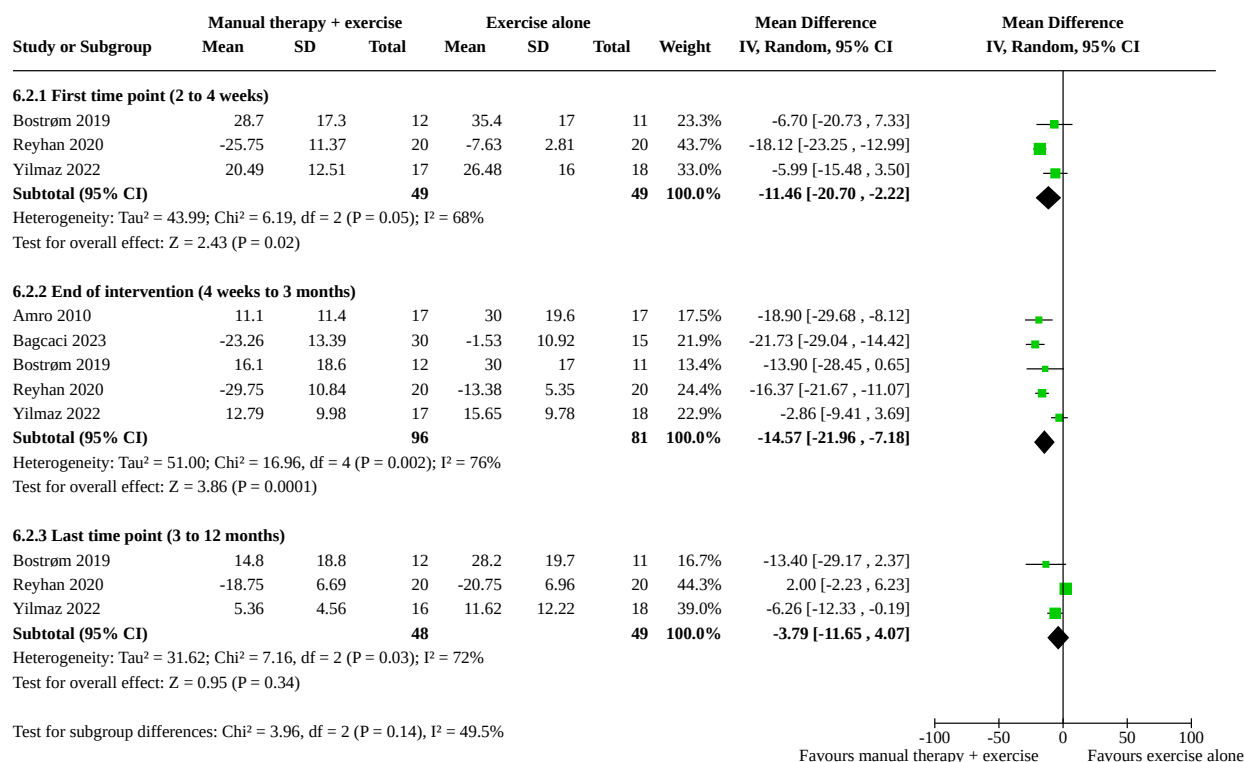
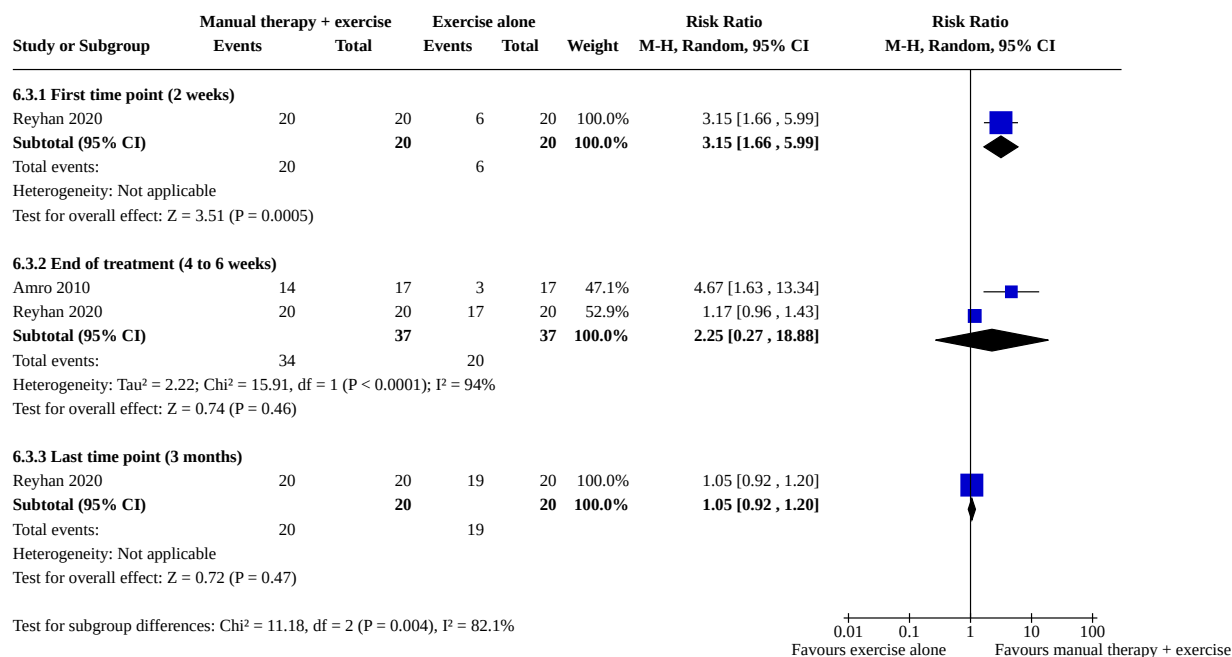
Analysis 5.7. Comparison 5: Prescribed exercises versus minimal or no intervention, Outcome 7: Pain-free grip strength (SMD)**Comparison 6. Manual therapy and prescribed exercises versus prescribed exercises alone**

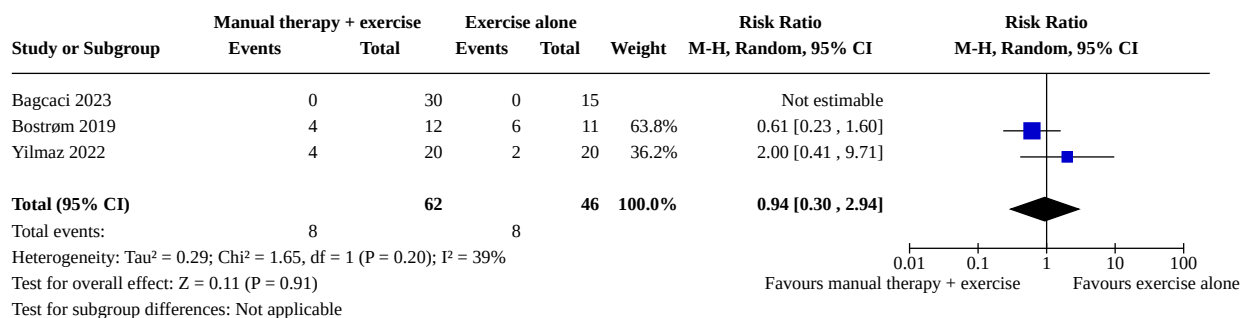
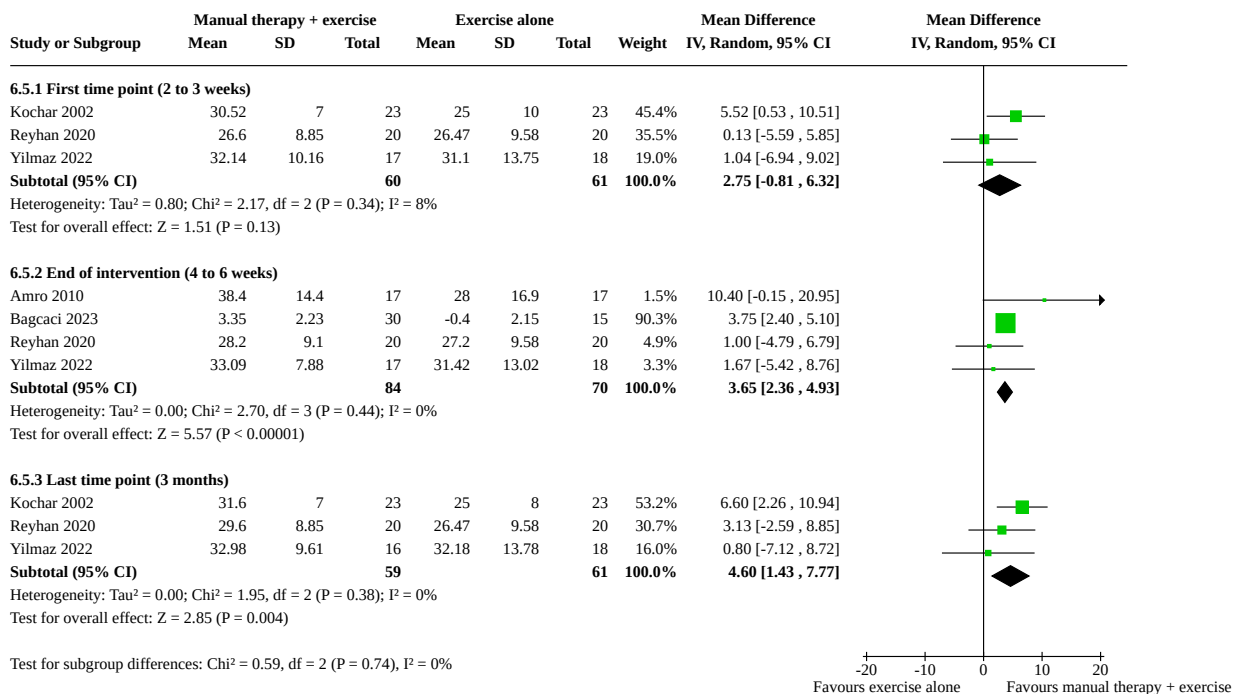
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
6.1 Mean pain (0 to 10, 0 is no pain)	5		Mean Difference (IV, Random, 95% CI)	Subtotals only
6.1.1 First time point (3 weeks)	3	104	Mean Difference (IV, Random, 95% CI)	-1.91 [-3.25, -0.56]
6.1.2 End of intervention (4 weeks to 3 months)	5	183	Mean Difference (IV, Random, 95% CI)	-2.56 [-3.63, -1.50]
6.1.3 Last time point (3 to 12 months)	2	57	Mean Difference (IV, Random, 95% CI)	-1.81 [-2.80, -0.83]
6.2 Mean disability (0 to 100, 0 is no disability)	5		Mean Difference (IV, Random, 95% CI)	Subtotals only
6.2.1 First time point (2 to 4 weeks)	3	98	Mean Difference (IV, Random, 95% CI)	-11.46 [-20.70, -2.22]
6.2.2 End of intervention (4 weeks to 3 months)	5	177	Mean Difference (IV, Random, 95% CI)	-14.57 [-21.96, -7.18]
6.2.3 Last time point (3 to 12 months)	3	97	Mean Difference (IV, Random, 95% CI)	-3.79 [-11.65, 4.07]
6.3 Participant-reported treatment success	2		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
6.3.1 First time point (2 weeks)	1	40	Risk Ratio (M-H, Random, 95% CI)	3.15 [1.66, 5.99]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
6.3.2 End of treatment (4 to 6 weeks)	2	74	Risk Ratio (M-H, Random, 95% CI)	2.25 [0.27, 18.88]
6.3.3 Last time point (3 months)	1	40	Risk Ratio (M-H, Random, 95% CI)	1.05 [0.92, 1.20]
6.4 Proportion of overall withdrawals	3	108	Risk Ratio (M-H, Random, 95% CI)	0.94 [0.30, 2.94]
6.5 Mean grip strength (kg)	5		Mean Difference (IV, Random, 95% CI)	Subtotals only
6.5.1 First time point (2 to 3 weeks)	3	121	Mean Difference (IV, Random, 95% CI)	2.75 [-0.81, 6.32]
6.5.2 End of intervention (4 to 6 weeks)	4	154	Mean Difference (IV, Random, 95% CI)	3.65 [2.36, 4.93]
6.5.3 Last time point (3 months)	3	120	Mean Difference (IV, Random, 95% CI)	4.60 [1.43, 7.77]

Analysis 6.1. Comparison 6: Manual therapy and prescribed exercises versus prescribed exercises alone, Outcome 1: Mean pain (0 to 10, 0 is no pain)



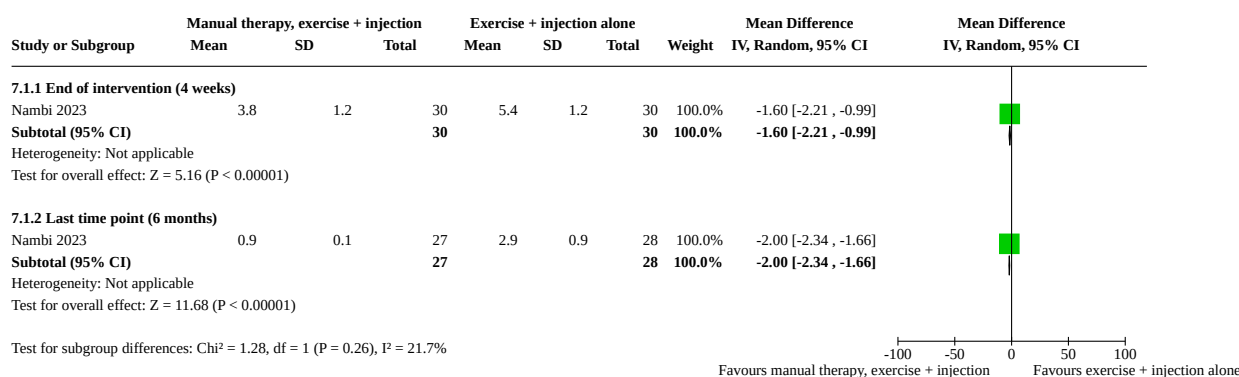
Analysis 6.2. Comparison 6: Manual therapy and prescribed exercises versus prescribed exercises alone, Outcome 2: Mean disability (0 to 100, 0 is no disability)**Analysis 6.3. Comparison 6: Manual therapy and prescribed exercises versus prescribed exercises alone, Outcome 3: Participant-reported treatment success**

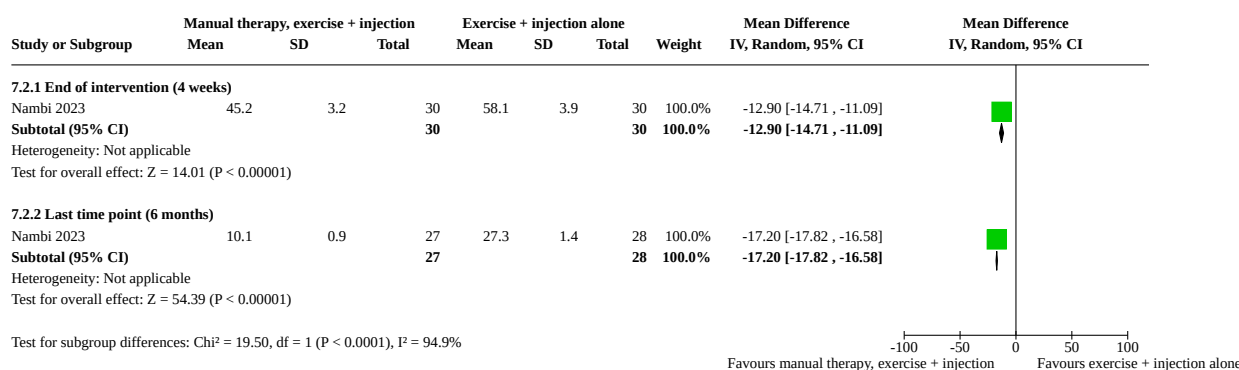
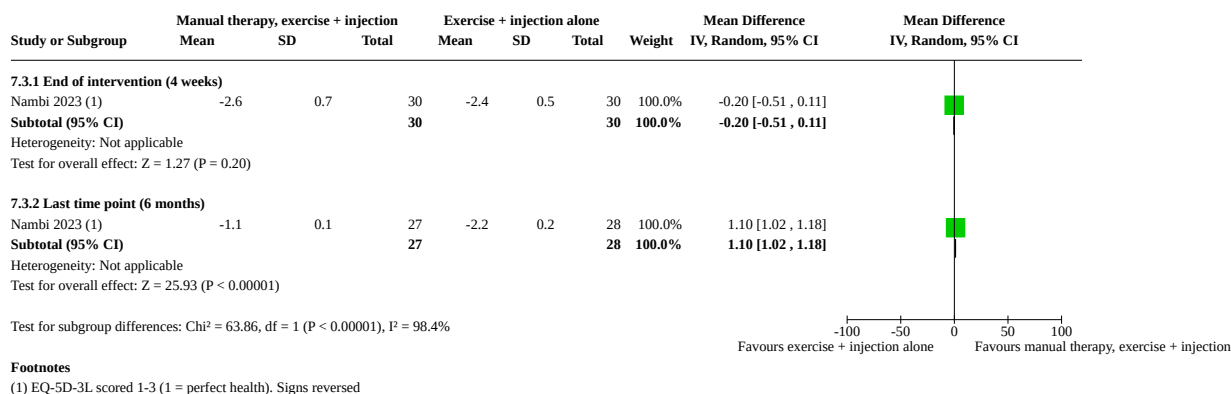
Analysis 6.4. Comparison 6: Manual therapy and prescribed exercises versus prescribed exercises alone, Outcome 4: Proportion of overall withdrawals**Analysis 6.5. Comparison 6: Manual therapy and prescribed exercises versus prescribed exercises alone, Outcome 5: Mean grip strength (kg)****Comparison 7. Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection alone**

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
7.1 Mean pain (0 to 10, 0 is no pain)	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
7.1.1 End of intervention (4 weeks)	1	60	Mean Difference (IV, Random, 95% CI)	-1.60 [-2.21, -0.99]
7.1.2 Last time point (6 months)	1	55	Mean Difference (IV, Random, 95% CI)	-2.00 [-2.34, -1.66]

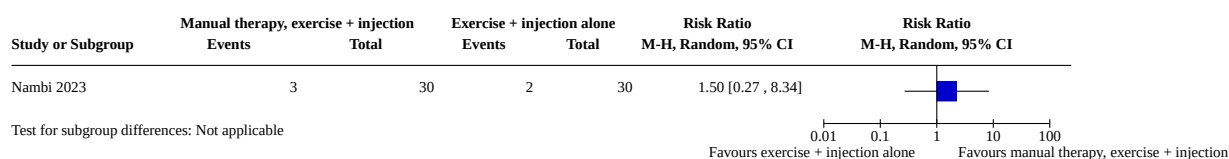
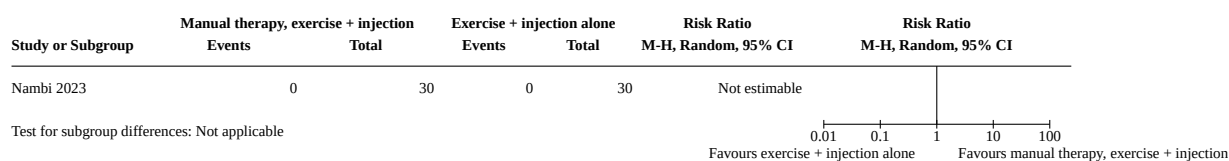
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
7.2 Mean disability (0 to 100, 0 is no disability)	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
7.2.1 End of intervention (4 weeks)	1	60	Mean Difference (IV, Random, 95% CI)	-12.90 [-14.71, -11.09]
7.2.2 Last time point (6 months)	1	55	Mean Difference (IV, Random, 95% CI)	-17.20 [-17.82, -16.58]
7.3 Mean health-related quality of life (1 to 3, 1 = perfect health)	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
7.3.1 End of intervention (4 weeks)	1	60	Mean Difference (IV, Random, 95% CI)	-0.20 [-0.51, 0.11]
7.3.2 Last time point (6 months)	1	55	Mean Difference (IV, Random, 95% CI)	1.10 [1.02, 1.18]
7.4 Proportion of overall withdrawals	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
7.5 Proportion with any adverse event	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
7.6 Mean grip strength (pain-free, kg)	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
7.6.1 End of intervention (4 weeks)	1	60	Mean Difference (IV, Random, 95% CI)	1.70 [0.38, 3.02]
7.6.2 Last time point (6 months)	1	55	Mean Difference (IV, Random, 95% CI)	5.40 [3.97, 6.83]

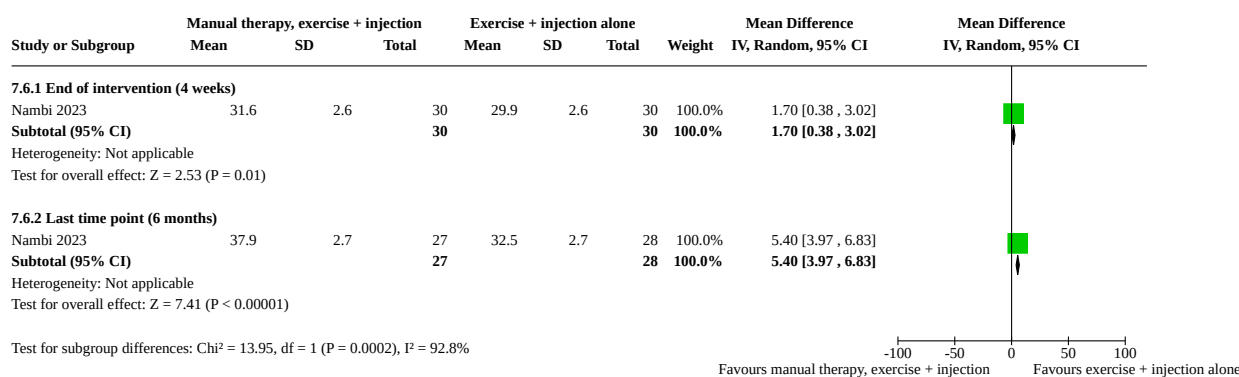
Analysis 7.1. Comparison 7: Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection alone, Outcome 1: Mean pain (0 to 10, 0 is no pain)



Analysis 7.2. Comparison 7: Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection alone, Outcome 2: Mean disability (0 to 100, 0 is no disability)**Analysis 7.3. Comparison 7: Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection alone, Outcome 3: Mean health-related quality of life (1 to 3, 1 = perfect health)****Footnotes**

(1) EQ-5D-3L scored 1-3 (1 = perfect health). Signs reversed

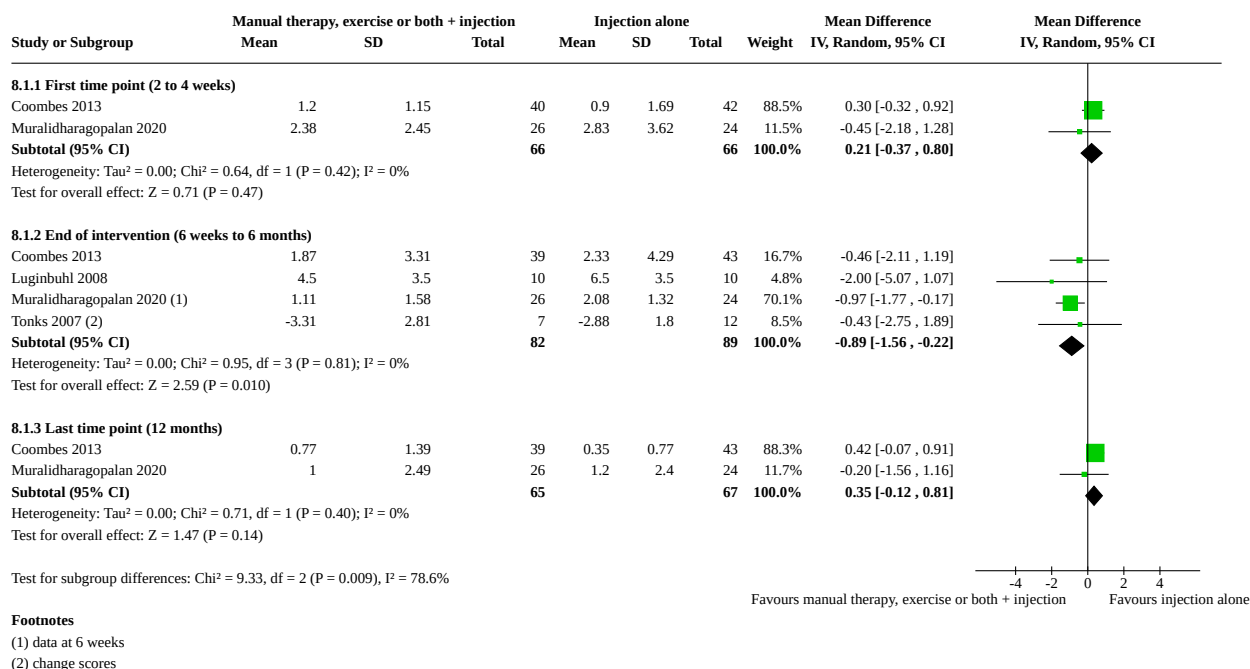
Analysis 7.4. Comparison 7: Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection alone, Outcome 4: Proportion of overall withdrawals**Analysis 7.5. Comparison 7: Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection alone, Outcome 5: Proportion with any adverse event**

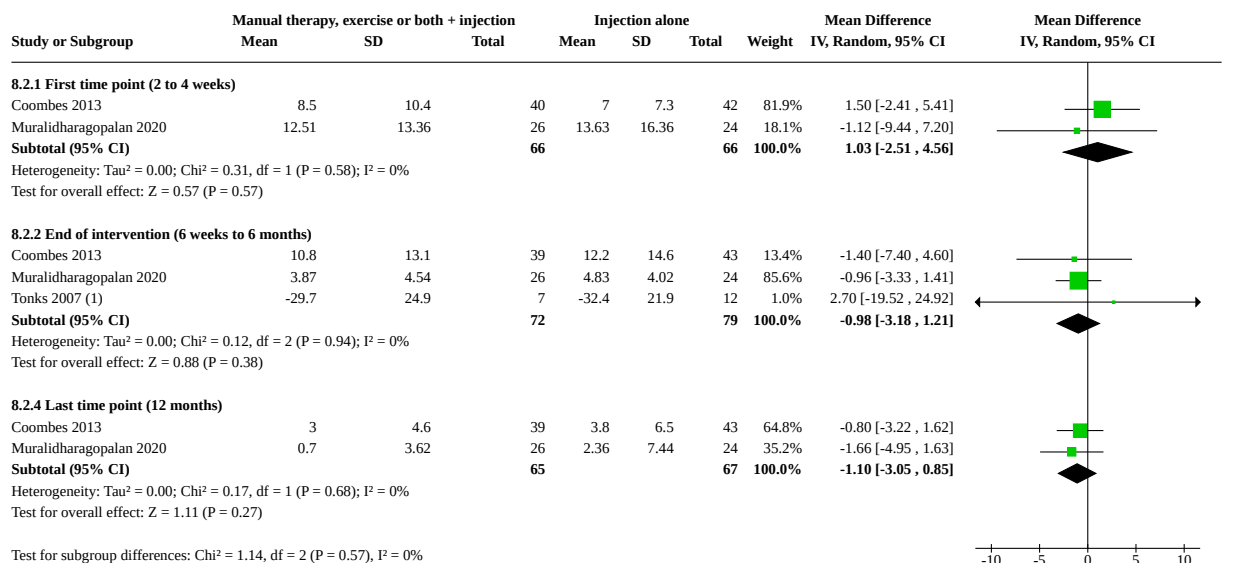
Analysis 7.6. Comparison 7: Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection alone, Outcome 6: Mean grip strength (pain-free, kg)**Comparison 8. Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection alone**

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
8.1 Mean pain (0 to 10, 0 is no pain)	4		Mean Difference (IV, Random, 95% CI)	Subtotals only
8.1.1 First time point (2 to 4 weeks)	2	132	Mean Difference (IV, Random, 95% CI)	0.21 [-0.37, 0.80]
8.1.2 End of intervention (6 weeks to 6 months)	4	171	Mean Difference (IV, Random, 95% CI)	-0.89 [-1.56, -0.22]
8.1.3 Last time point (12 months)	2	132	Mean Difference (IV, Random, 95% CI)	0.35 [-0.12, 0.81]
8.2 Mean elbow disability (0 to 100, 0 is no disability)	3		Mean Difference (IV, Random, 95% CI)	Subtotals only
8.2.1 First time point (2 to 4 weeks)	2	132	Mean Difference (IV, Random, 95% CI)	1.03 [-2.51, 4.56]
8.2.2 End of intervention (6 weeks to 6 months)	3	151	Mean Difference (IV, Random, 95% CI)	-0.98 [-3.18, 1.21]
8.2.4 Last time point (12 months)	2	132	Mean Difference (IV, Random, 95% CI)	-1.10 [-3.05, 0.85]
8.3 Mean health-related quality of life (EQ-5D utility score)	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
8.3.1 First time point (4 weeks)	1	82	Mean Difference (IV, Random, 95% CI)	-0.02 [-0.05, 0.01]
8.3.2 End of intervention (6 weeks to 6 months)	1	82	Mean Difference (IV, Random, 95% CI)	0.05 [0.02, 0.08]
8.3.3 Last time point (12 months)	1	82	Mean Difference (IV, Random, 95% CI)	-0.01 [-0.05, 0.03]

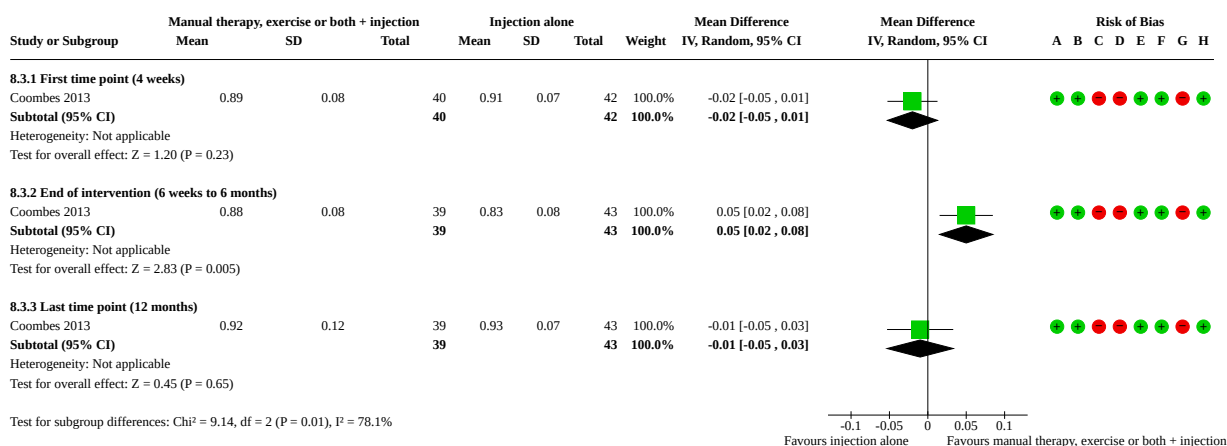
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
8.4 Participant-reported treatment success	2		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
8.4.1 First time point (4 weeks)	1	82	Risk Ratio (M-H, Random, 95% CI)	0.94 [0.71, 1.26]
8.4.2 End of intervention (3 to 6 months)	2	102	Risk Ratio (M-H, Random, 95% CI)	1.05 [0.74, 1.48]
8.4.3 Last time point (12 months)	1	82	Risk Ratio (M-H, Random, 95% CI)	0.98 [0.80, 1.19]
8.5 Proportion of overall withdrawals	2	107	Risk Ratio (M-H, Random, 95% CI)	2.78 [0.27, 28.79]
8.6 Proportion with any adverse event	3	157	Risk Ratio (M-H, Random, 95% CI)	0.78 [0.33, 1.83]
8.7 Mean grip strength (affected side/unaffected side x 100)	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
8.7.1 End of intervention (4 weeks to 3 months)	1	20	Mean Difference (IV, Random, 95% CI)	9.00 [-2.84, 20.84]

Analysis 8.1. Comparison 8: Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection alone, Outcome 1: Mean pain (0 to 10, 0 is no pain)

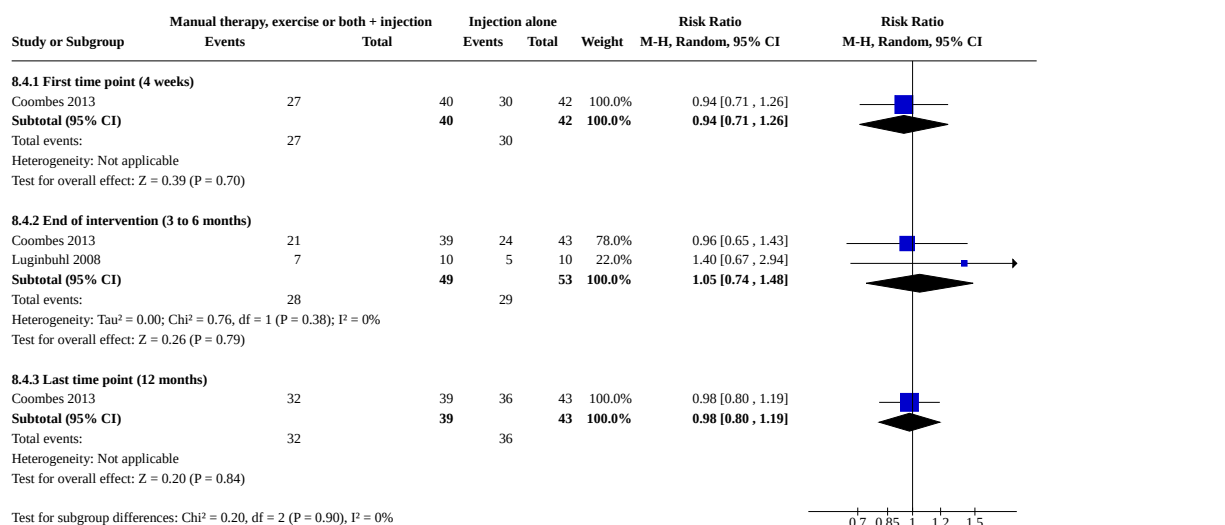
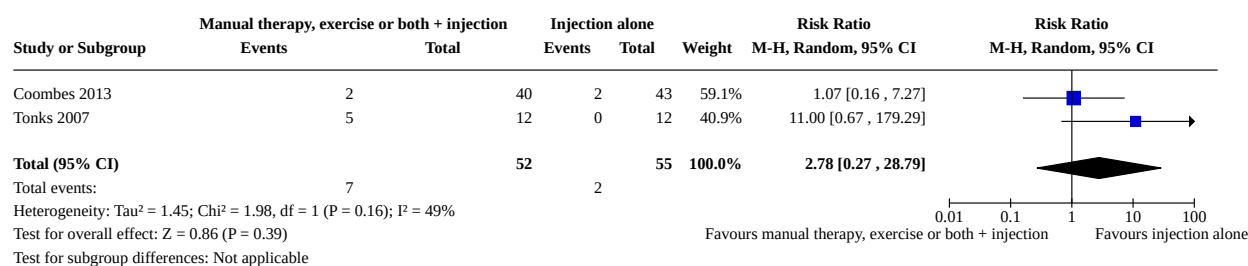
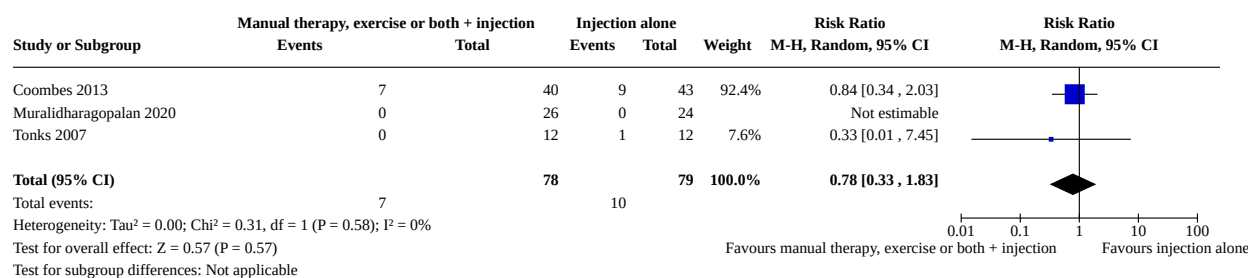


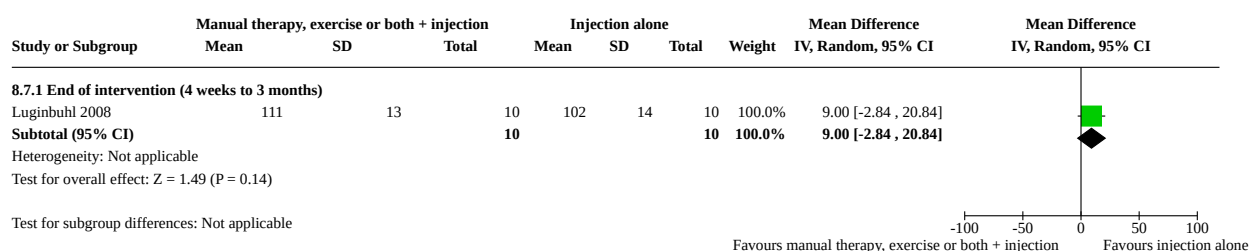
Analysis 8.2. Comparison 8: Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection alone, Outcome 2: Mean elbow disability (0 to 100, 0 is no disability)**Footnotes**

(1) change scores transformed to a 100-point scale

Analysis 8.3. Comparison 8: Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection alone, Outcome 3: Mean health-related quality of life (EQ-5D utility score)**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment - self reported (detection bias)
- (E) Blinding of outcome assessment - assessor reported outcomes
- (F) Incomplete outcome data (attrition bias)
- (G) Selective reporting (reporting bias)
- (H) Other bias

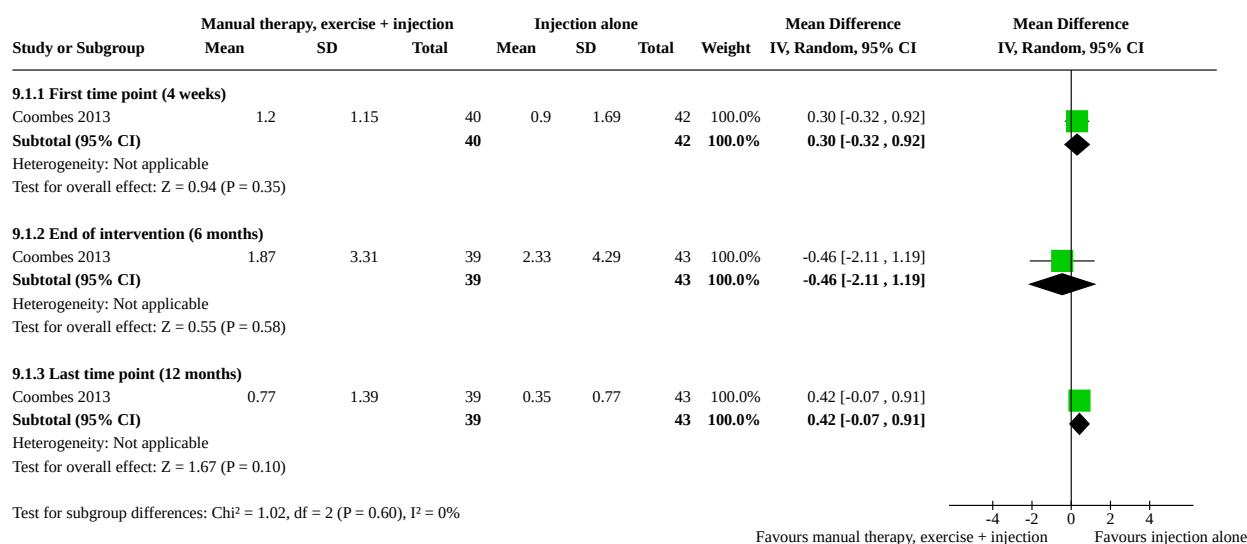
Analysis 8.4. Comparison 8: Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection alone, Outcome 4: Participant-reported treatment success**Analysis 8.5. Comparison 8: Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection alone, Outcome 5: Proportion of overall withdrawals****Analysis 8.6. Comparison 8: Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection alone, Outcome 6: Proportion with any adverse event**

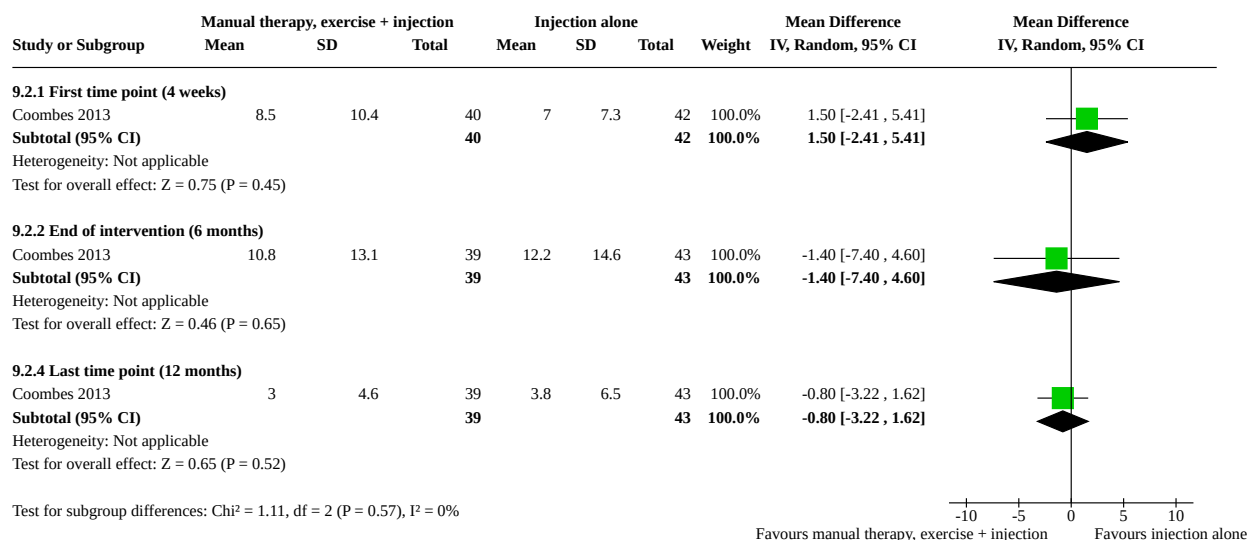
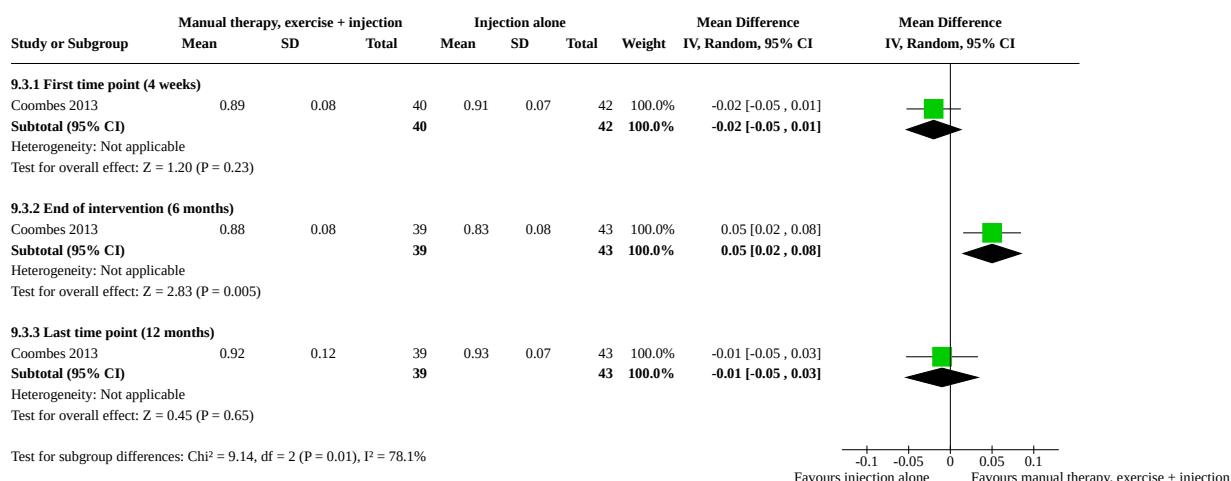
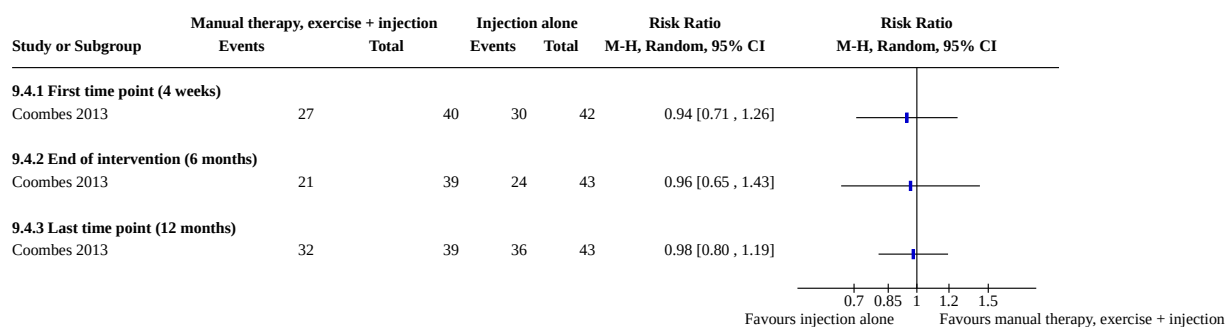
Analysis 8.7. Comparison 8: Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection alone, Outcome 7: Mean grip strength (affected side/unaffected side x 100)**Comparison 9. Manual therapy, prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone**

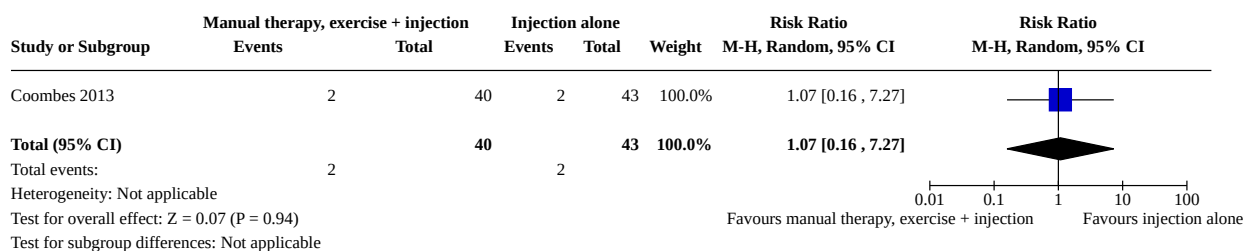
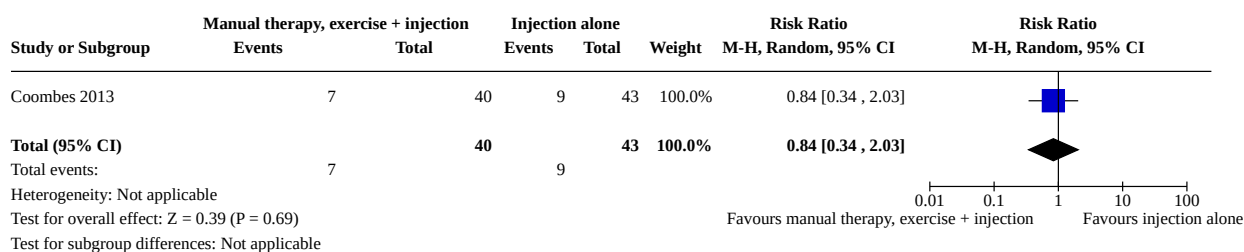
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
9.1 Mean pain (0 to 10, 0 is no pain)	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
9.1.1 First time point (4 weeks)	1	82	Mean Difference (IV, Random, 95% CI)	0.30 [-0.32, 0.92]
9.1.2 End of intervention (6 months)	1	82	Mean Difference (IV, Random, 95% CI)	-0.46 [-2.11, 1.19]
9.1.3 Last time point (12 months)	1	82	Mean Difference (IV, Random, 95% CI)	0.42 [-0.07, 0.91]
9.2 Mean elbow disability (0 to 100, 0 is no disability)	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
9.2.1 First time point (4 weeks)	1	82	Mean Difference (IV, Random, 95% CI)	1.50 [-2.41, 5.41]
9.2.2 End of intervention (6 months)	1	82	Mean Difference (IV, Random, 95% CI)	-1.40 [-7.40, 4.60]
9.2.4 Last time point (12 months)	1	82	Mean Difference (IV, Random, 95% CI)	-0.80 [-3.22, 1.62]
9.3 Mean health-related quality of life (EQ-5D utility score)	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
9.3.1 First time point (4 weeks)	1	82	Mean Difference (IV, Random, 95% CI)	-0.02 [-0.05, 0.01]
9.3.2 End of intervention (6 months)	1	82	Mean Difference (IV, Random, 95% CI)	0.05 [0.02, 0.08]
9.3.3 Last time point (12 months)	1	82	Mean Difference (IV, Random, 95% CI)	-0.01 [-0.05, 0.03]
9.4 Participant-reported treatment success	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
9.4.1 First time point (4 weeks)	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
9.4.2 End of intervention (6 months)	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
9.4.3 Last time point (12 months)	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
9.5 Proportion of overall withdrawals	1	83	Risk Ratio (M-H, Random, 95% CI)	1.08 [0.16, 7.27]
9.6 Proportion with any adverse event	1	83	Risk Ratio (M-H, Random, 95% CI)	0.84 [0.34, 2.03]

Analysis 9.1. Comparison 9: Manual therapy, prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 1: Mean pain (0 to 10, 0 is no pain)



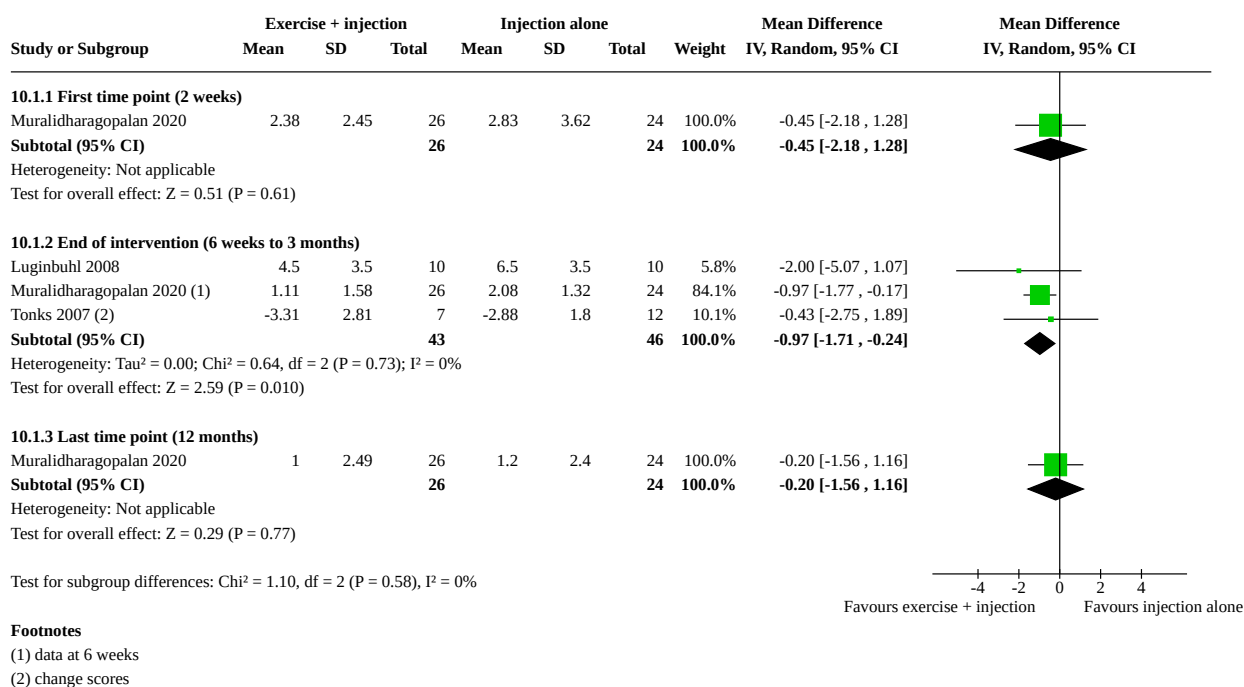
Analysis 9.2. Comparison 9: Manual therapy, prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 2: Mean elbow disability (0 to 100, 0 is no disability)**Analysis 9.3. Comparison 9: Manual therapy, prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 3: Mean health-related quality of life (EQ-5D utility score)****Analysis 9.4. Comparison 9: Manual therapy, prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 4: Participant-reported treatment success**

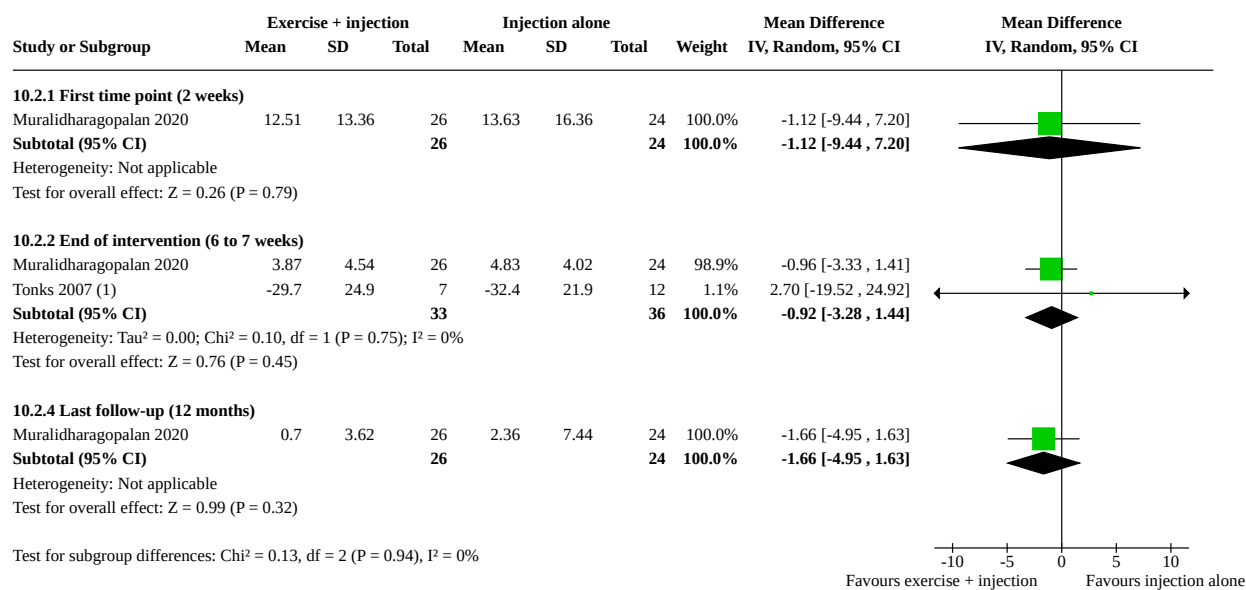
Analysis 9.5. Comparison 9: Manual therapy, prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 5: Proportion of overall withdrawals**Analysis 9.6. Comparison 9: Manual therapy, prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 6: Proportion with any adverse event****Comparison 10. Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone**

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
10.1 Mean pain (0 to 10, 0 is no pain)	3		Mean Difference (IV, Random, 95% CI)	Subtotals only
10.1.1 First time point (2 weeks)	1	50	Mean Difference (IV, Random, 95% CI)	-0.45 [-2.18, 1.28]
10.1.2 End of intervention (6 weeks to 3 months)	3	89	Mean Difference (IV, Random, 95% CI)	-0.97 [-1.71, -0.24]
10.1.3 Last time point (12 months)	1	50	Mean Difference (IV, Random, 95% CI)	-0.20 [-1.56, 1.16]
10.2 Mean elbow disability (0 to 100, 0 is no disability)	2		Mean Difference (IV, Random, 95% CI)	Subtotals only
10.2.1 First time point (2 weeks)	1	50	Mean Difference (IV, Random, 95% CI)	-1.12 [-9.44, 7.20]
10.2.2 End of intervention (6 to 7 weeks)	2	69	Mean Difference (IV, Random, 95% CI)	-0.92 [-3.28, 1.44]
10.2.4 Last follow-up (12 months)	1	50	Mean Difference (IV, Random, 95% CI)	-1.66 [-4.95, 1.63]

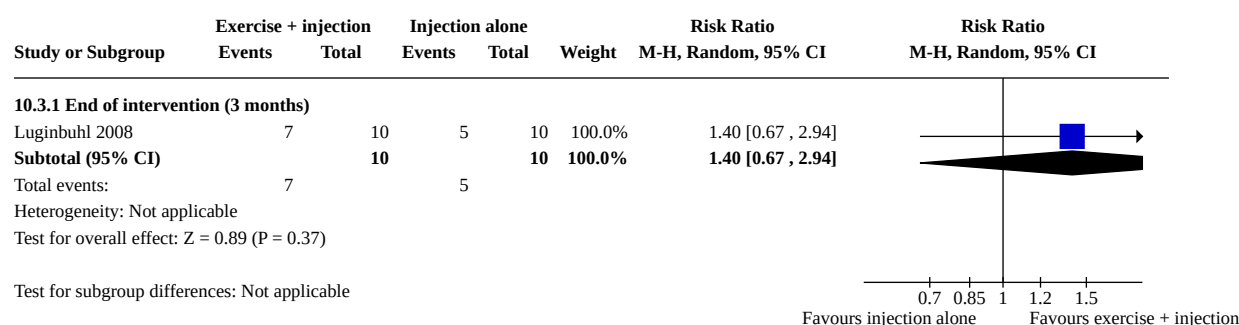
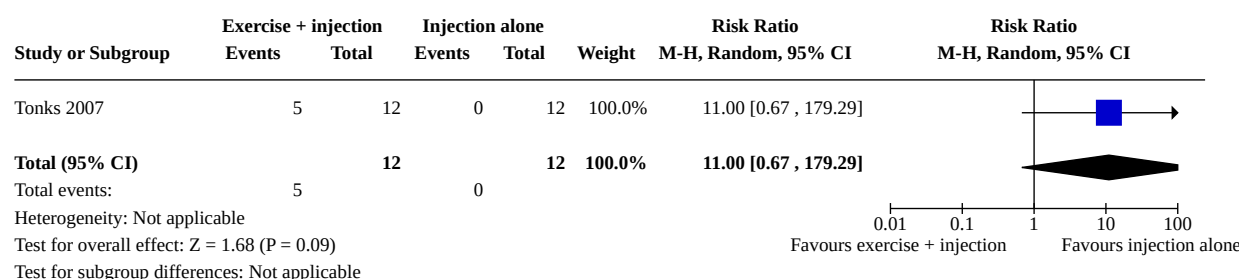
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
10.3 Participant-reported treatment success	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
10.3.1 End of intervention (3 months)	1	20	Risk Ratio (M-H, Random, 95% CI)	1.40 [0.67, 2.94]
10.4 Proportion of overall with-drawals	1	24	Risk Ratio (M-H, Random, 95% CI)	11.00 [0.67, 179.29]
10.5 Proportion with any adverse event	2	74	Risk Ratio (M-H, Random, 95% CI)	0.33 [0.01, 7.45]
10.6 Mean grip strength (affected side/unaffected side x 100)	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
10.6.1 End of intervention (3 months)	1	20	Mean Difference (IV, Random, 95% CI)	9.00 [-2.84, 20.84]

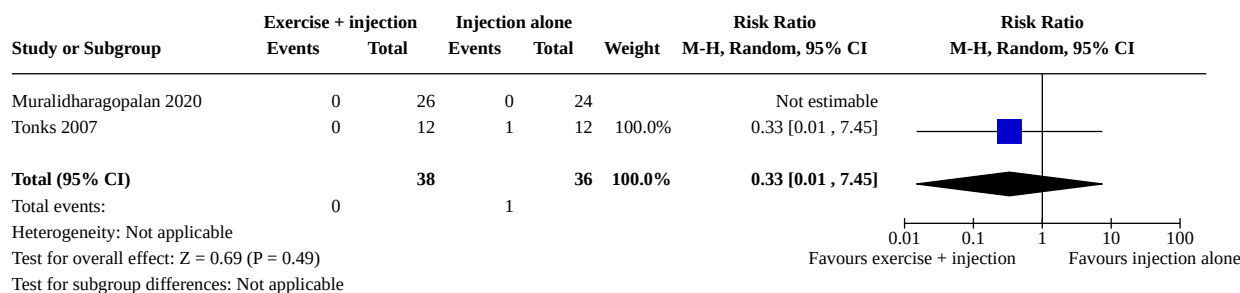
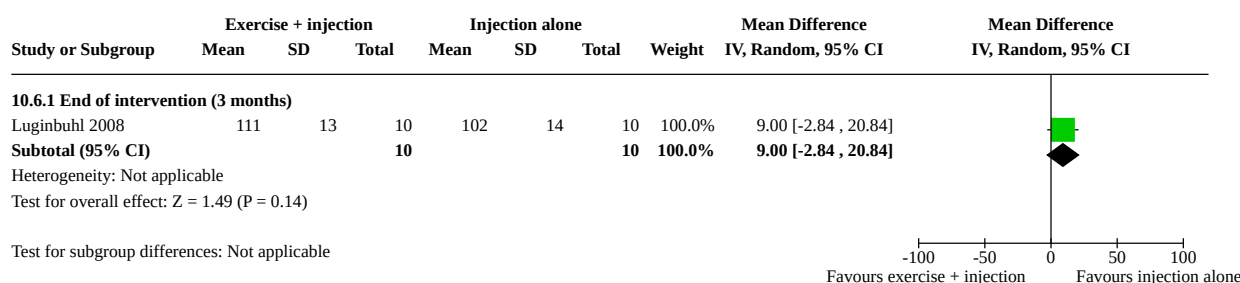
Analysis 10.1. Comparison 10: Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 1: Mean pain (0 to 10, 0 is no pain)



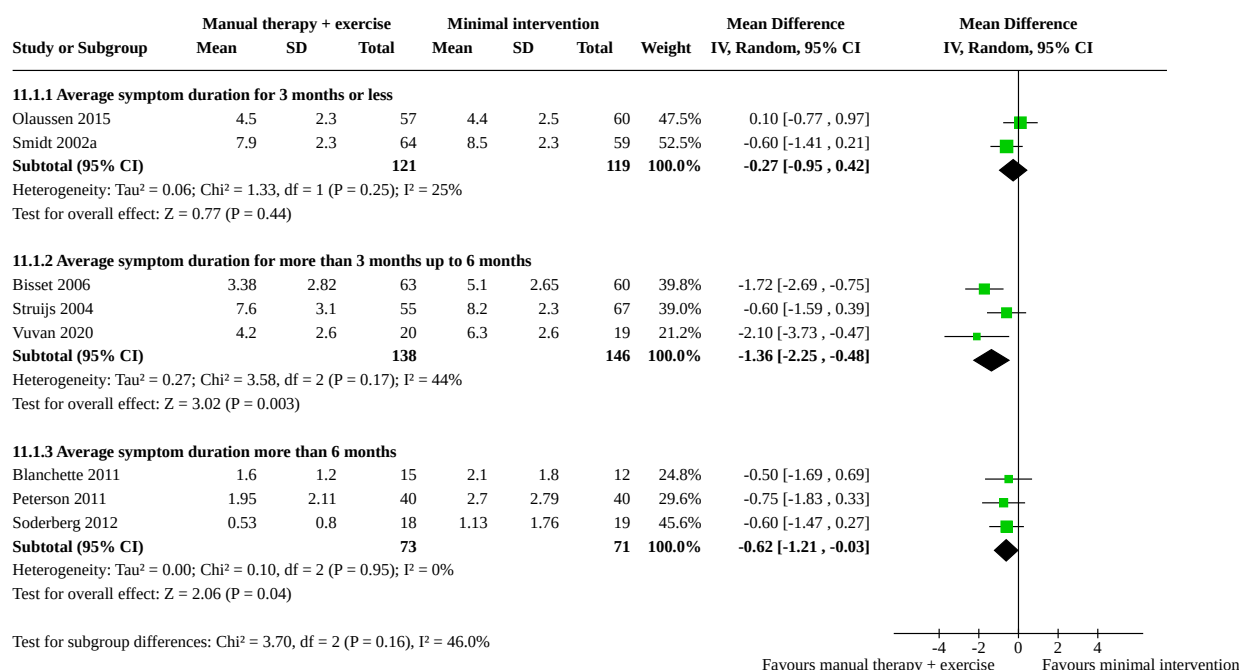
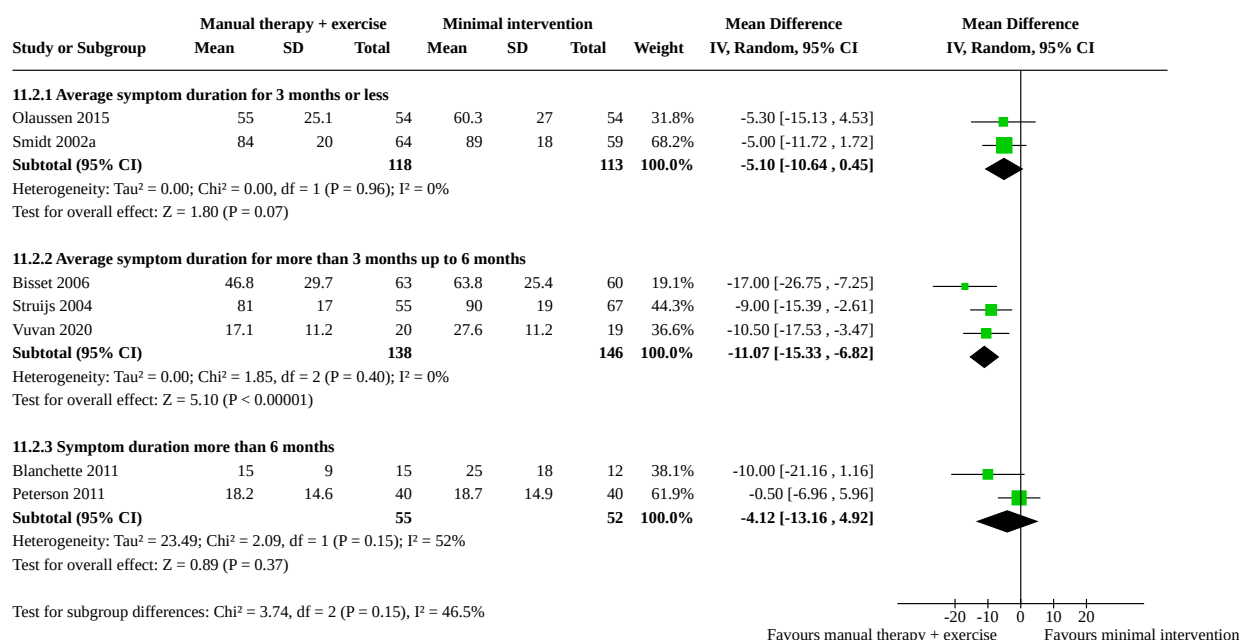
Analysis 10.2. Comparison 10: Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 2: Mean elbow disability (0 to 100, 0 is no disability)**Footnotes**

(1) change scores transformed to a 100-point scale

Analysis 10.3. Comparison 10: Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 3: Participant-reported treatment success**Analysis 10.4. Comparison 10: Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 4: Proportion of overall withdrawals**

Analysis 10.5. Comparison 10: Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 5: Proportion with any adverse event**Analysis 10.6. Comparison 10: Prescribed exercises and glucocorticoid injection versus glucocorticoid injection alone, Outcome 6: Mean grip strength (affected side/unaffected side x 100)****Comparison 11. Subgroup analysis short-term vs long-term symptoms**

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
11.1 Subgroup analysis: pain end of intervention	8		Mean Difference (IV, Random, 95% CI)	Subtotals only
11.1.1 Average symptom duration for 3 months or less	2	240	Mean Difference (IV, Random, 95% CI)	-0.27 [-0.95, 0.42]
11.1.2 Average symptom duration for more than 3 months up to 6 months	3	284	Mean Difference (IV, Random, 95% CI)	-1.36 [-2.25, -0.48]
11.1.3 Average symptom duration more than 6 months	3	144	Mean Difference (IV, Random, 95% CI)	-0.62 [-1.21, -0.03]
11.2 Subgroup analysis: disability end of intervention	7		Mean Difference (IV, Random, 95% CI)	Subtotals only
11.2.1 Average symptom duration for 3 months or less	2	231	Mean Difference (IV, Random, 95% CI)	-5.10 [-10.64, 0.45]
11.2.2 Average symptom duration for more than 3 months up to 6 months	3	284	Mean Difference (IV, Random, 95% CI)	-11.07 [-15.33, -6.82]
11.2.3 Symptom duration more than 6 months	2	107	Mean Difference (IV, Random, 95% CI)	-4.12 [-13.16, 4.92]

Analysis 11.1. Comparison 11: Subgroup analysis short-term vs long-term symptoms, Outcome 1: Subgroup analysis: pain end of intervention**Analysis 11.2. Comparison 11: Subgroup analysis short-term vs long-term symptoms, Outcome 2: Subgroup analysis: disability end of intervention**

ADDITIONAL TABLES

Table 1. Study characteristics of unpublished or awaiting classification, ongoing and terminated trials

Study reference Registration number	Principal investigator/s and country	Eligible comparison groups	Main selection criteria	Primary outcome/s	Registration date	Recruitment commenced	Trial status to 27 February 2024	Planned sample size	Actual sample size
Bartlett 2007 ISRCTN89551482	Alison Bartlett, United Kingdom	Manual therapy versus minimal or no treatment	Chronic lateral elbow pain; symptoms of greater than 6 weeks duration; age 18+ years	(1) Pain-free grip strength (2) Disability - PRFEQ	28 September 2007	No	Awaiting classification. Likely terminated trial with no response from primary contact	Unknown	Unknown
Prat 2018 Trial registration unknown	Pietro logna Prat, USA	Manual therapy and prescribed exercise versus prescribed exercises	Clinical diagnosis of lateral elbow pain	(1) Disability using DASH questionnaire (2) Pain-free and maximum grip strength (3) Pain pressure threshold	Unknown	Yes	Ongoing trial Abstract published	Unknown	29
Saheb-jamei 2019 IRC-T20180812040773N1	Afshin Saheb-jamei, Iran	Manual therapy and exercise versus minimal or no treatment	Aged between 25 and 55 years of age, lateral elbow pain for at least 4 to 6 weeks with at least 2 positive provocation tests	(1) Pain (2) Severity of specific activities (3) Severity of routine activities	29 January 2019	23 September 2017	Trial completion date: 21 April 2018	40	40
Sveinall 2022 NCT04803825	Håkon Sveinall, Norway	Prescribed exercises versus minimal or no treatment	Clinically diagnosed tennis elbow provoked by at least two pain provocation tests	(1) Disability - PRTEE (2) Disability - Quick-DASH	18 March 2021	23 August 2021	Ongoing trial	100	Unknown

Table 1. Study characteristics of unpublished or awaiting classification, ongoing and terminated trials (Continued)

				(3) EQ-5D-5L and EQ VAS					
				(4) Pain-free and maximum grip strength					
				(5) Global improvement					
Takamjani 2019	Ismail Ebrahimi Takamjani, Iran	Manual therapy and prescribed exercises versus prescribed exercises	Patients with chronic lateral epicondylitis symptoms lasting more than 3 months, aged between 35 and 55 years, referred by physician	(1) Pain (2) Disability (3) Grip strength	26 April 2019	Unknown	Ongoing trial	34	Unknown

DASH: Disabilities of the Arm, Shoulder and Hand

EQ-5D: EuroQol-5 Dimensions

PRFEQ: Patient-Rated Forearm Evaluation Questionnaire

PRTEE: Patient-Rated Tennis Elbow Evaluation

VAS: visual analogue scale

Table 2. Baseline trial and participant characteristics

Trial	Setting	Intervention groups (n)	Mean (SD) age, years	Female, n (%)	Dominant elbow, n (%)	Manual workers, n (%)	Mean (SD) duration of symp- toms, months	Mean (SD) pain at baseline (0 to 10)	Mean (SD) disability at base- line (0 to 100)
Manual therapy versus placebo (1 trial)									
Nour- bakhsh 2008	USA, commu- nity	Manual therapy (11)	51.7 (9.4)	5 (45)	Not re- ported	Not re- ported	17.1 (17.7)	5.1 (2.6)	52 (28)
		Placebo manual therapy (12)	52.4 (7.2)	4 (33)			20.3 (22.1)	4.6 (2.3)	61 (22)
Manual therapy, prescribed exercises or both versus minimal or no intervention (12 trials)									
Bisset 2006	Australia, community and primary care	Manual therapy, exercise (66)	47.9 (7.2)	21 (32)	42 (64)	21 (32)	3.7 (2.5 to 8.1) ¹	5.8 (2.5)	81 (16)

Table 2. Baseline trial and participant characteristics (Continued)

		Advice ³ (67)	47.3 (8.1)	24 (36)	42 (63)	27 (40)	6 (2.3 to 9.7) ¹	6.1 (2.3)	77 (21)
Blanchette 2011	Canada, community (University)	Manual therapy (15)	47 (10)	9 (60)	7 (46)	Not reported	22 (25)	4.6 (2.3)	37 (19)
		Advice (12)	46 (10)	6 (50)	6 (50)		43 (50)	3.9 (2.9)	30 (18)
Haahr 2003	Denmark, primary care	Prescribed exercises, ergonomic advice ² (141)	82 (58) ⁴	80 (57)	102 (72)	84 (60)	Not reported	6.1 (2.1)	Not reported
		Usual GP care (125)	81 (65) ⁴	64 (51)	83 (66)			6.0 (2.1)	Not reported
McQueen 2021	USA, outpatient clinic	Exercise (54)	Not reported	59 (63)	Not reported	20 (21)	Not reported	6.7 (not reported)	31 (not reported)
		Advice ³ , wrist orthosis ³ (40)						6.7 (not reported)	29 (not reported)
Olaussen 2015	Norway, primary care	Manual therapy, exercise, placebo injection ² (58)	48.8 (9.4)	20 (34)	42 (72)	29 (50)	1.8 (0.9-2.3) ¹	5.3 (1.9)	53 (18)
		Advice ³ (60)	44.0 (9.7)	25 (42)	41 (68)		1.7 (0.9-2.3) ¹	4.8 (2.2)	54 (16)
Peterson 2011	Sweden, community and primary care	Prescribed exercises (40)	49.1 (8.1)	16 (40)	Not reported	Not reported	25 (44)	4.2 (2.7)	29 (7.4)
		Waitlist control, advice ³ (41)	47.4 (8.6)	18 (44)			22 (27)	3.4 (2.9)	25 (15)
Rachold 2023	Germany, primary care	Manual therapy (18)	48.1 (9.7)	12 (67)	Not reported	Not reported	Not reported	5.2 (1.8)	41.0 (17.5)
		Forearm brace and electrotherapy (15)	45.7 (9.7)	10 (67)				5.9 (1.5)	39.7 (16.3)
		Manual therapy, forearm brace, electrotherapy (19)	45.9 (9.4)	9 (47)				5.5 (1.1)	42.3 (15.2)
Smidt 2002a	Netherlands, primary care	Manual therapy, prescribed exercises, ultrasound therapy ² (64)	48 (41 to 52) ¹	28 (44)	51 (80)	Not reported	2.5 (1.8 to 4.8) ¹	6 (4 to 7) ¹	48 (38 to 60) ¹
		Advice ³ (59)	46 (42 to 54) ¹	31 (53)	46 (78)		2.5 (1.8 to 4.8) ¹	6 (3 to 7) ¹	45 (35 to 55) ¹

Table 2. Baseline trial and participant characteristics (Continued)

Soderberg 2012	Sweden, community	Prescribed exercises (20)	48 (12.4)	9 (45)	Not reported	Not reported	12 (60) ⁵	2.4 (1.2 to 4.2) ¹	Not reported
		Advice ³ , forearm band ³ (22)	50 (10.8)	15 (68)			14 (64) ⁵	3.1 (1.5 to 4.0) ¹	Not reported
Struijs 2004	Netherlands, primary care	Manual therapy, exercise, ultrasound therapy ² (56)	47 (9)	28 (50)	40 (71)	Not reported	4.8 (8.5)	7.2 (1.5)	52 (16)
		Forearm brace ³ (68)	46 (11)	32 (47)	50 (74)		5.3 (6.9)	7.4 (1.8)	51 (17)
Tonks 2007 ⁶	UK, outpatient clinic	Exercise (12)	43.8 (7.5)	Not reported	Not reported	Not reported	Not reported	4.6 (1.8)	37 (20)
		Observation (12)	43.4 (7.1)					5.1 (1.2)	49 (13)
Vuvan 2020	Australia, community and primary care	Prescribed exercises (21)	48.0 (7.9)	7 (33)	14 (67)	17 (81)	6 (2 to 8) ¹	6.8 (1.8)	33 (12)
		Advice ³ (19)	49.0 (10.2)	4 (21)	10 (53)	16 (84)	3 (2 to 6) ¹	7.8 (1.4)	38 (14)
Manual therapy and prescribed exercises versus prescribed exercises (6 trials)									
Amro 2010	Palestine, community and primary care	Manual therapy, prescribed exercises ³ (17)	37.8 (unable to extract)	4 (24)	Not reported	Not reported	Not reported	6.3 (1.1)	52 (15)
		Prescribed exercises ³ , heat ³ , massage ³ , ultrasound therapy ³ (17)	36.8 (unable to extract)	6 (35)				6.6 (1.4)	58 (10)
Bagcaci 2023	Turkey, community (University Department of Physiotherapy and Rehabilitation)	Mulligans's and prescribed exercises ³ (15)	41.7 (4.4)	8 (53)	Not reported	Not reported	6.3 (2.7)	Not reported	
		Muscle energy techniques and prescribed exercises ³ (15)	44.5 (3.96)	8 (53)			7.3 (3.1)		
		Prescribed exercises ³ (15)	42.1 (4.3)	8 (53)			7.7 (2.98)		
Bostrøm 2019	Norway, primary care	Manual therapy, prescribed exercises ³ (12)	49 (2)	3 (25)	Not reported	2 (16)	Not reported	4.1 (1.3)	32 (15)
		Prescribed exercises ³ , advice ³ (11)	47 (3)	6 (55)		5 (45)		4.8 (1.3)	29 (14)

Table 2. Baseline trial and participant characteristics (Continued)

Kochar 2002	India, outpa- tient clinic	Manual therapy, prescribed exercises ³ (23)	40 (9)	11 (48)	16 (70)	Not re- ported	Not re- ported	6 (1.3)	Not re- ported
		Exercise, ultrasound therapy ³ (23)	45 (12)	10 (43)	18 (78)			5.8 (1.3)	Not re- ported
Reyhan 2020	Turkey, out- patient clinic	Manual therapy, prescribed exercises ³ (20)	43.4 (10.6)	15 (75)	Not re- ported	Not re- ported	4.05 (0.63)	5.1 (1.5)	Not re- ported
		Prescribed exercises ³ , advice ³ (20)	41.4 (9.4)	18 (90)	3.98 (0.91)		5.2 (2.1)	Not re- ported	
Yilmaz 2022	Turkey, out- patient clinic	Manual therapy, prescribed exercises ³ (20)	42.7 (7.57)	14 (70)	Not re- ported	Not re- ported	7.0 (8.0)	7.2 (1.7)	39 (14)
		Prescribed exercises ³ , advice ³ (20)	42.9 (10.3)	12 (60)	6.7 (7.5)		6.0 (1.9)	33 (14)	
Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection (1 trial)									
Nambi 2023	Saudi Arabia, University hospital out- patient clinic	Manual therapy, prescribed exercises and glucocorticoid injection (30)	48.4 (3.8)	0 (0)	21 (70)	21 (70)	6.2 (2.3)	7.6 (1.4)	74.1 (5.1)
		Sham manual therapy, prescribed exercis- es ³ and glucocorticoid injection ³ (30)	47.1 (3.5)	0 (0)	20 (67)	20 (67)	5.6 (2.7)	7.4 (1.3)	74.8 (4.9)
Manual therapy, prescribed exercises or both and glucocorticoid injection versus glucocorticoid injection (4 trials)									
Coombes 2013	Australia, community and primary care	Manual therapy, exercise, glucocorticoid injection ³ (40)	50.8 (8.5)	15 (38)	Not re- ported	Not re- ported	3.4 (2.3 to 6.0) ¹	5.9 (1.6)	38 (14)
		Glucocorticoid injection ³ , advice ³ (43)	49.3 (8.9)	16 (37)	3.7 (2.3 to 6.2) ¹		6.2 (2.0)	42 (14)	
Luginbuhl 2008	Switzerland, outpatient clinic	Prescribed exercises, glucocorticoid injec- tion ³ (10)	47 (9)	11 (55)	15 (75)	Not re- ported	10 (11)	3.1 (0.7)	Not re- ported
		Glucocorticoid injection ³ , forearm band ³ (10)						3.7 (0.7)	Not re- ported
Muralid- haragopalan 2020	India, outpa- tient clinic	Prescribed exercises, glucocorticoid injec- tion ³ (26)	37.6 (6.9)	15 (58)	Not re- ported	Not re- ported	7.3 (1.3)	7.1 (0.8)	35 (7.21)

Table 2. Baseline trial and participant characteristics (Continued)

		Glucocorticoid injection ³ (24)	37.5 (7.3)	14 (58)			7.3 (1.6)	7.1 (1.0)	33 (7.4)
Tonks 2007 ⁶	UK, outpa- tient clinic	Exercise, glucocorticoid injection ³ (12)	41.9 (7.4)	Not re- ported	Not re- ported	Not re- ported	Not re- ported	4.7 (2.3)	36 (23)
		Glucocorticoid injection ³ (12)	48.2 (6.5)					5.0 (1.4)	50 (27)

¹Median (interquartile range (IQR)).

²Additional intervention not in comparator group.

³Co-intervention delivered in all groups.

⁴Number (%) with age > 40 years.

⁵Number (%) with duration of symptoms > 6 months.

⁶Four-arm trial, with different arms included in the two comparisons.

GP: general practitioner; SD: standard deviation

Table 3. Characteristics of manual therapy interventions in included trials

Trial	Techniques	Provider (number in trial, qualifica- tions if reported)	Setting	Selection, modification and/or progression of techniques	Dosage of each tech- nique	Duration of session in min- utes	Number of ses- sions	Number of ses- sions per week	Number of weeks of treat- ment
Manual therapy versus placebo (1 trial)									
Nour- bakhsh 2008	Oscillating en- ergy (elbow)	Orthopaedic clin- ical specialist (1, trained in cran- iosacral therapy)	Not report- ed	Not reported	Duration from point to point ranged from 30 secs to 2 mins	20 to 30	6	2 to 3	2 to 3
Manual therapy, prescribed exercises or both versus minimal or no intervention (6 trials)									
Bisset 2006	Mobilisation with movement	Physiotherapist (6, postgraduate qual- ified)	Primary care	Selection of techniques was based on the therapist's classification of impair- ments and symptoms relat- ed to the elbow and cervi- cal spine. Decisions to con- tinue or change techniques were based on the thera-	6 to 10 reps	30	8 (max)	1 to 2	6

Table 3. Characteristics of manual therapy interventions in included trials (Continued)
pist's evaluation of treatment response.

Blanchette 2011	Augmented soft tissue mobilisation	Chiropractor (1, completing a master's degree in kinesiology)	University biomechanics laboratory	Not reported	Not reported	Not reported	10	2	5
Olaussen 2015	Deep transverse friction massage Mill's manipulation	Physiotherapist	Primary care	Not reported	15 mins 1 rep	Not reported	12	2	6
Rachold 2023	Manipulating the joints, muscles of fascia	Specialised therapist (1, unknown profession)	Primary care	To guide treatment, the therapist received correspondence from the primary physician regarding functional chains and trigger points	Not reported	20	6	2	3
Smidt 2002a	Deep friction massage	Physiotherapist (72 from 31 practices)	Primary care	Not reported	Not reported	Not reported	9	1 to 2	6
Struijs 2004	Deep friction massage	Physiotherapist	Outpatient department	Not reported	5 to 10 mins	5 to 10	9	1 to 3	6
Manual therapy and prescribed exercises versus prescribed exercises (6 trials)									
Amro 2010	Mobilisation with movement	Physiotherapist	Primary care	Not reported	36 reps	30 to 45	12	3	4
Bagcaci 2023	Mobilisation with movement	Physiotherapist (1, experienced and certified)	Department of Physiotherapy and Rehabilitation (University)	Not reported	3 sets, 10 reps. 15 to 20 sec pause between sets and 30 sec interval between reps	Not reported	12	3	4

Table 3. Characteristics of manual therapy interventions in included trials (Continued)

	Muscle energy techniques	Physiotherapist		Not reported	Not reported				
Boström 2019	Mobilisation with movement	Physiotherapist (2, specific manual therapy qualifications)	Outpatient interdisciplinary institute of sports medicine	The number of sessions depended on the patients' perceived pain intensity and the therapists' clinical evaluation	3 sets, 8 reps	Not reported	3 to 8	1	12
Kochar 2002	Mobilisation with movement	Physiotherapist (1)	Physiotherapy clinic in medical institute	Not reported	3 sets, 10 reps	Not reported	10	3 to 4	3
Reyhan 2020	Mobilisation with movement	Physiotherapist (1)	Outpatient department	Not reported	3 sets, 10 reps	Not reported	10	5	2
Yilmaz 2022	Radial nerve mobilisation	Physiotherapist	Outpatient department	Not reported	3 sets, 10 reps, with 3 sec holds and 2 mins rest between sets	Not reported	9	3	3
Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection (1 trial)									
Nambi 2023	Deep transverse friction massage	Physiotherapist	University hospital	Fixed physical therapy protocol	5 mins	30 to 40	12	3	4
	Mill's manipulation				1 repetition				
Manual therapy, prescribed exercises or both and glucocorticoid injection versus glucocorticoid injection (1 trial)									
Coombes 2013	Mobilisation with movement	Physiotherapist (11, postgraduate qualified)	Primary care	Selection of techniques was based on substantial improvement in pain-free grip force with the glide application	6 to 10 reps	30	8	1	8

Table 4. Characteristics of exercise interventions in included trials

Trial	Types	Pre-scriber (number and qualifications)	Equipment type	Starting levels, tailoring and/or progression strategies	Adherence strategies	Dosage	Number of sessions per day	Number of weeks of treatment	Number of therapist visits ¹
Manual therapy, prescribed exercises or both versus minimal or no intervention (10 trials)									
Bisset 2006	Resistance (wrist movements)	Physio-therapist (6, post-graduate qualified)	Resistance band	If symptoms were severe only one set of exercises was prescribed	Regular review by therapist, exercise instruction booklet	3 sets, 8 to 12 reps, 6 to 10 sec duration	2 to 3	6	8 ¹
	Stretching		Not reported			3 to 5 reps, 20 to 30 sec holds			
	Postural correction and functional retraining					If participant absent from the workforce, participant received work-specific tasks and activities incorporated in the exercise programme. Functional retraining involved progressively heavier and harder objects to grasp			
Haahr 2003	Resistance (isometric wrist extensors)	Ergonomist (1)	None reported (pressing back of hand for resistance)	Standardised programme	Not reported	10 sec hold	2	As long as symptoms persisted	12
	Stretching					15 to 20 sec, 3 to 5 reps			
McQueen 2021	Resistance (eccentric in wrist extension)	Occupational therapist (1)	Hand weight (1 to 2 pounds)	Eccentric exercises were initiated at the second therapy session with a one-pound hand weight at 10 repetitions. Participants were coached to progress to 30 repetitions, after which a two-pound weight was used for the same progression.	Therapist sessions	10 to 30 reps	1	12 to 16	4 to 6

Table 4. Characteristics of exercise interventions in included trials *(Continued)*

	Grip strengthening (in supination, neutral and pronation)		Theraput-ty	Began with gripping with the elbow flexed and progressing to elbow extension. Progressed from 3 to 5 reps at whichever level of the elbow position the patient was able to use while performing the exercise pain-free.		3 to 5 reps			
Olaussen 2015	Resistance (eccentric)	Physio-therapist	500 mL bottle	Progression using a 500 mL bottle filled with increasing volume of water or sand as participants improved. Exercises performed within tolerable pain level.	Not reported	3 sets, 30 reps	1	6	12 ¹
	Stretching		None reported	Not reported		40 sec holds	3		
Peterson 2011	Resistance (concentric and eccentric for wrist extensors)	Not reported	Water containers with handle	Initial load was standardised to 1 kg (1 litre of water) for women and 2 kg for men	Monitored at 4-week and 8-week trial follow-up	3 sets, 15 reps	1	12	12 ¹
Smidt 2002a	Resistance (isometric, concentric, wrist movements)	Physio-therapist (72 from 31 practices)	Stick, elastic band	Exercise programme started after pain reduction from manual therapy. Therapist sessions were used to prescribe more intensive programme over 4 phases	Exercise instruction book	2 to 3 sets, 10 reps with 8 sec duration	4 to 6	6	9 ¹
	Functional re-training		Soft ball, buttons, towel			Not reported			
	Stretching (flexion and extension at end of each session)		Not reported			30 sec holds			
Soderberg 2012	Resistance (eccentric wrist extensors)	Not reported	Bucket	Advised to adjust their resisted weight so that they were able to perform the prescribed repetitions during pain-free intervals (less than or equal to 2 on Borg's scale) throughout training period	Participant diary captured adherence but unclear if used to increase adherence	Week 1 to 3 = 2 sets, 8 to 12 reps Week 4 to 6 = 3 sets, 8 to 12 reps	1 to 2	6	2

Table 4. Characteristics of exercise interventions in included trials (Continued)

Struijs 2004	Resistance (isometric, concentric, wrist movements)	Physio-therapist (received training by re-searchers)	Stick, elastic band	Exercise programme started after pain reduction from manual therapy. Therapist sessions were used to prescribe more intensive programme over 4 phases.	Therapist sessions, exercise diary	2 to 3 sets, 10 reps with 8 sec duration	4 to 6	6	91
	Functional re-training		Soft ball, buttons, towel			Not reported			
	Stretching (flexion and extension at end of each session)		None			30 sec holds			
Tonks 2007	Resistance (isometric, concentric, wrist movements)	Physio-therapist	Stick, elastic band	Therapist sessions were used to prescribe more intensive programme over 4 phases	Not reported	2 to 3 sets, 10 reps with 8 sec duration	4 to 6	6	Not reported
	Functional re-training		Soft ball, buttons, towel			Not reported			
	Stretching (flexion and extension at end of each session)		None			30 sec holds			
Vuvan 2020	Resistance (isometric wrist extensors)	Physio-therapist (1, post-graduate Master's degree-level qualification)	Water container	Initial load was equivalent to 20% MVC of the unaffected side and increased to 35% by week 7. Participants were instructed to add the equivalent of 5% MVC more water to the container each fortnight (weeks 3, 5, 7). On alternate weeks, the time under tension was alternated (see dosage column)	Therapist monitored exercise diaries completed weekly by participants during and after exercise	Week 1, 3, 5, 7 = 30 sec holds, 4 reps, 120 sec total time under tension Week 2, 4, 6, 8 = 45 sec holds, 3 reps, 135 sec total time under tension	1	8	12

Table 4. Characteristics of exercise interventions in included trials (Continued)

Manual therapy and prescribed exercises versus prescribed exercises (6 trials)

Amro 2010	Resistance	Physio- therapist	Not reported					4	12 ¹
	Stretching								
Bagcaci 2023	Resistance (ec- centric for exten- sor carpi radialis brevis muscle)	Physio- therapist (number not report- ed)	Weight	Started with the weight that pa- tients could lift at 10 maximal reps	Phone call monitoring	10 reps	1	4	12 ¹
	Stretching		None	Not reported		10 secs			
Bostrøm 2019	Resistance (ec- centric)	Physio- therapist (2)	Not re- ported	Starting weight was standardised. Load increased once a week by 10% of the starting weight, or less if their pain intensified.	Weekly text message as a reminder to do their daily exer- cise	Not report- ed	1	12	5 ¹
Kochar 2002	Resistance (iso- metric, concen- tric and eccentric for wrist exten- sors)	Physio- therapist (1)	Not re- ported	For isometric exercises, progres- sion included forearm pronation as the starting position and increas- ing resistance. For concentric/ec- centric exercises, participants in- creased the range of movement and resistance by using heavier weights as pain allowed (without leading to aggravation). All exer- cises were progressed within pain- free limits.	Fortnightly review with therapist to progress the exercis- es and moti- vate them to comply with the exercise regime	Isometrics	3	9	10 ¹
	Stretching					- held for 5 to 10 secs Concen- tric/eccen- tric - 10 reps Held few secs			
Reyhan 2020	Resistance (con- centric, isomet- ric, wrist exten- sors and flexors)	Physio- therapist (1)	Tennis ball	Not reported	Not report- ed	Concentrics - 3 sets, 20 reps Isometrics - 20 sec, 20 reps	Not re- ported	2	10 ¹

Table 4. Characteristics of exercise interventions in included trials (Continued)

	Stretching		Not reported			30 sec, 3 reps			
	Active elbow movements					20 reps			
Yilmaz 2022	Resistance (eccentric wrist extensors)	Physio-therapist	Weight	Concept of 10 repetitions maximum was used to determine resistance load and progressions. Advised to stop the exercises if they felt pain at any time during the exercises or continue exercises without weights	Monitoring every second week by the therapist	3 sets, 10 reps	3	6	8 ¹
Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection (1 trial)									
Nambi 2023	Resistance (eccentric)	Physio-therapist (number not reported)	Therapist band	All wrist movements. Initially, the painful movements are trained with minimal resistance and then progressed. In the latter phase, activity specific rehabilitation. Intensity, frequency and duration in every treatment session are based on individual capacities, without exaggerating the symptoms	Monitored by a treating therapist before every session by checking the exercise log book	3 sets, 30 repetitions	1	4	12 ¹
	Stretching of radial extensors and supinators					30 seconds	3		
	Activity specific rehabilitation		Not reported			Not reported			
Manual therapy, prescribed exercises or both and glucocorticoid injection versus glucocorticoid injection (4 trials)									
Coombes 2013	Resistance (concentric, eccentric for wrist extensors)	Physio-therapist (11, post-graduate qualified)	Resistance bands with handle, free weights	Therapists selected exercises from the protocol and progressed the programme based on the patients' capabilities to allow for optimal exercise volume and load setting without exacerbating pain. Resistance band had varied resistance levels and also heavier weights to provide progressive resistance	Therapist monitored exercise diary at all sessions and emphasised importance of the exercise and self-management	Not reported	2	8	8 ¹
	Posture correction		Not reported						
	Functional re-training (gripping and forearm movements)			Progression included work or sport-specific exercises later in rehabilitation					

Table 4. Characteristics of exercise interventions in included trials (Continued)

Luginbuhl 2008	Resistance (isometric gripping and wrist extensors)	Not reported	Tennis ball	All exercises were instructed to be performed gently, and not cause relevant pain	Not reported	20 sec hold, 20 reps	2	9 to 12	3
Muralidharagopalan 2020	Resistance (eccentric, wrist extension)	Physiotherapist (number not reported)	Hand weight	Not reported	Monitored remotely for the 2 weeks	Not reported	Not reported	2	1 ²
Tonks 2007	Resistance (isometric, concentric, wrist movements)	Physiotherapist	Stick, elastic band	Therapist sessions were used to prescribe more intensive programme over 4 phases	Not reported	2 to 3 sets, 10 reps with 8 sec duration	4 to 6	6	Not reported
	Functional re-training		Soft ball, buttons, towel			Not reported			
	Stretching (flexion and extension at end of each session)		None			30 sec holds			

¹Therapist visits for manual therapy intervention.

²Prescribed at initial visit.

MVC: maximum voluntary contraction

Table 5. Orbit matrix for the major outcomes at the primary time point

Comparison	Time point	Pain	Disability	Quality of life	Participant success	Withdrawals	Adverse events	Serious adverse events
Study ID								
Manual therapy versus placebo (1 trial)								
Nourbakhsh 2008	2 to 3 weeks	0 to 10 NRS (0 = no pain)	0 to 10 NRS (10 = no disability)	#	#	#	#	#
Manual therapy, prescribed exercises or both versus minimal or no intervention (12 trials)								

Table 5. Orbit matrix for the major outcomes at the primary time point *(Continued)*

Bisset 2006	6 weeks	0 to 100 VAS (0 = no pain)	0 to 100 using Pain Free Function questionnaire (0 = no disability)	#	6-point scale measured as 'completely recovered' or 'much improved'	#	# ^a	# ^a
Blanchette 2011	6 weeks	0 to 100 VAS (0 = no pain)	0 to 100 using the PRTEE questionnaire (0 = no disability)	#	#	# ^a	# ^b	# ^b
Haahr 2003	3 months	0 to 36 (4 sub-scales 0 to 9, 9 = pain as bad as could be)	#	#	5-point scale measured as 'much better' or 'better')	#	#	#
McQueen 2021	3 months	0 to 10 VAS (0 = no pain)	0 to 100 using Quick-DASH (0 = no disability)	#	#	#	#	#
Olaussen 2015	6 weeks	0 to 100 VAS (0 = no pain)	0 to 100 using Pain Free Function questionnaire (0 = no disability)	#	6-point scale measured as 'completely recovered' or 'much improvement'	#	#	#
Peterson 2011	3 months	0 to 100 VAS (0 = no pain)	0 to 100 using DASH questionnaire (0 = no disability)	7-point Likert scale ranging from 'very bad' to 'excellent, could not be better' using the Gothenburg questionnaire	#	# ^a	#	#
Rachold 2023	4 weeks	0 to 10 VAS (0 = no pain)	0 to 100 using the PRTEE questionnaire (0 = no disability)	0 to 100 using SF-36 questionnaire (100 = perfect health)	#	#	#	#
Smidt 2002a	6 weeks	0 to 100 VAS (0 = no pain)	0 to 100 using Pain Free Function questionnaire (100 = no disability)	#	6-point scale measured as 'com-	# ^a	#	# ^a

Table 5. Orbit matrix for the major outcomes at the primary time point *(Continued)*

					pletely recovered' or 'much improved'			
Soderberg 2012	6 weeks	0 to 100 VAS (0 = no pain)	#	#	#	#	#	#
Struijs 2004	6 weeks	0 to 100 VAS (0 = no pain)	0 to 100 using Pain-Free Function Questionnaire (0 = no disability)	#	6-point scale measured as 'completely recovered' or 'much improved'	#	#	#
Tonks 2007	7 weeks	Pain sub-score of the Patient-rated Forearm Evaluation Questionnaire (0 = no pain) change from baseline scores	0 to 10 using the PRFEQ (0 = no disability)	#	#	#	# ^a	# ^a
Vuvan 2020	8 weeks	0 to 10 NRS (0 = no pain)	0 to 100 using the PRTEE questionnaire (0 = no disability)	#	6-point scale measured as 'completely recovered' or 'much improved'	# ^a	# ^a	# ^a
Manual therapy and prescribed exercises versus prescribed exercises (6 trials)								
Amro 2010	4 weeks	0 to 10 VAS (0 = no pain)	0 to 100 using PRTEE questionnaire (0 = no disability)	#	Number of participants who thought the treatment was effective	#	#	#
Bagcaci 2023	4 weeks	0 to 10 VAS (0 = no pain)	0 to 100 using the PRFEQ (0 = no disability)	#	#	# ^a	#	#
Bostrøm 2019	3 months	0 to 10 NRS (0 = no pain)	0 to 100 scale using Quick-DASH (0 = no disability)	#	#	#	# ^a	# ^a
Kochar 2002	3 months	0 to 10 VAS (0 = no pain)	#	#	#	#	#	#
Reyhan 2020	6 weeks	0 to 10 VAS (0 = no pain) ^c	0 to 100 using PRTEE questionnaire (0 = no disability)	#	6-point scale measured as 'complete recovery' or 'significant recovery'	#	# ^a	# ^a

Table 5. Orbit matrix for the major outcomes at the primary time point (Continued)

Yilmaz 2022	6 weeks	0 to 10 VAS (0 = no pain)	0 to 100 using DASH questionnaire (0 = no disability)	#	#	#	# ^a	# ^a
Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection (1 trial)								
Nambi 2023	4 weeks	0 to 10 VAS (0 = no pain)	0 to 100 using PRTEE questionnaire (0 = no disability)	1 to 3 using EQ-5D-3L (1 = perfect health)	6-point scale ^c	#	# ^a	#
Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection (4 trials)								
Coombes 2013	6 months	0 to 100 VAS (0 = no pain)	0 to 100 using PRTEE questionnaire (0 = no disability)	EQ-5D-3L utility score (a value of 1 = perfect health)	6-point scale measured as 'complete recovery' or 'much improvement'	#	#	#
Luginbuhl 2008	3 months ^d	0 to 4 scale measured by the Modified Nirschl/Pettrone score	#	#	4-point scale measured as 'much better' or 'better'	#	#	#
Muralidharagopalan 2020	6 weeks	0 to 10 VAS (0 = no pain)	0 to 100 using DASH questionnaire (0 = no disability)	#	#	#	# ^a	# ^a
Tonks 2007	7 weeks	Pain subscore of the PRFEQ (0 = no pain) change from baseline score	0 to 10 using the PRFEQ (0 = no disability)	#	#	# ^a	# ^a	# ^a

^aZero events in one or more of the groups.

^bStudy measured adverse events for one treatment group only.

^cInsufficient data reporting.

^dWe assumed 3 months follow-up represented 'end of intervention' as the majority of participants were lost to follow-up at 12 months.

DASH: Disability of the Arm, Shoulder and Hand

NRS: numeric rating scale

PRFEQ: Patient-rated Forearm Evaluation Questionnaire

PRTEE: Patient-Rated Tennis Elbow Evaluation questionnaire

VAS: visual analogue scale

Table 6. Summary of adverse events

Study ID	Included treatment groups(n)	Withdrawals	Totaladverse events	Serious ad-verse events
Manual therapy versus placebo (1 trial)				
Nourbakhsh 2008	Manual therapy (11)	Not reported		
	Placebo manual therapy (12)			
Manual therapy, prescribed exercises or both versus minimal or no intervention (12 trials)				
Bisset 2006	Manual therapy and exercise (66)	3 participants	7 participants: pain after treatment (n = 7), including pain lasting 7 days or longer (n = 1)	0 participants
	Advice (67)	5 participants	0 participants	
Blanchette 2011	Manual therapy (15)	0 participants	14 participants experienced aches and bruises following the treatment, and these effects had ceased at the 6-week follow-up	0 participants
	Advice (15)	3 participants	No data reported	
Haahr 2003	Prescribed exercises (148)	32 partici-pants	Not reported	
	Usual GP care (141)	25 partici-pants		
McQueen 2021	Prescribed exercises and brace (54)	16 partici-pants	Not reported	
	Brace (40)	19 partici-pants		
Olaussen 2015	Manual therapy, prescribed exercises and placebo injection (58)	5 participants	5 participants: gastrointestinal side effects (n = 3) and increased pain (n = 2)	0 participants
	Advice (60)	8 participants	9 participants: gastrointestinal side effects (n = 8) and increased pain (n = 2)	
Peterson 2011	Prescribed exercises (40)	0 participants	Not reported	
	Advice (41)	0 participants		
Rachold 2023	Manual therapy (30)	12 partici-pants	Not reported	
	Brace and electrotherapy (28)	13 partici-pants		
Smidt 2002a	Manual therapy and exercise (64)	0 participants	41 participants: radiating pain to forearm or upper arm (n = 27)	0 participants

Table 6. Summary of adverse events (Continued)

	Advice (59)	0 participants	10 participants: radiating pain to forearm or upper arm (n = 7)	
Soderberg 2012	Prescribed exercises and brace (20)	2 participants	Not reported	
	Brace (22)	3 participants		
Struijs 2004	Manual therapy, prescribed exercises and brace (56)	2 participants	Not reported	
	Brace (68)	5 participants		
Tonks 2007	Prescribed exercises (12)	4 participants	0 participants	
	Observation (12)	2 participants		
Vuvan 2020	Prescribed exercises (21)	1 participant	3 participants: reported feeling worse on the global rating of change scale and were classified as adverse events	0 participants
	Advice (19)	0 participants	0 participants	
Manual therapy and prescribed exercises versus prescribed exercises (6 trials)				
Amro 2010	Manual therapy and prescribed exercises (17)	Not reported		
	Prescribed exercises (17)			
Bagcaci 2023	Manual therapy (Mulligan's and prescribed exercises (15)	0 participants	Not reported	
	Manual therapy (muscle energy techniques) and prescribed exercises (15)	0 participants		
	Prescribed exercises (15)	0 participants		
Bostrøm 2019	Manual therapy and prescribed exercises (12)	0 participants		
	Prescribed exercises (11)			
Kochar 2002	Manual therapy and prescribed exercises (23)	Not reported		
	Prescribed exercises (23)			
Reyhan 2020	Manual therapy and prescribed exercises (20)	0 participants		
	Prescribed exercises (20)			
Yilmaz 2022	Manual therapy and prescribed exercises (20)	4 participants	1 participant: did not attend final follow-up due to pain	0 participants
	Prescribed exercises (20)	2 participants	0 participants	

Table 6. Summary of adverse events (Continued)

Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection (1 trial)

Nambi 2023	Manual therapy, prescribed exercises and glucocorticoid injection (30)	3 participants	0 participants
	Sham manual therapy, prescribed exercises and glucocorticoid injection (30)	2 participants	

Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection (4 trials)

Coombes 2013	Manual therapy, prescribed exercises and glucocorticoid injection (40)	2 participants	7 participants	Adverse events were reported as minor, transient and similar between injection and physiotherapy treatments. Skin depigmentation (4/83) and subcutaneous atrophy (3/83) occurred exclusively in patients receiving glucocorticoid injection, showed a delayed onset (evident on examinations at 8 weeks or 3 months) and were resolved by 6 months.	0 participants
	Glucocorticoid injection (43)	2 participants	9 participants		
Luginbuhl 2008	Prescribed exercises, brace and glucocorticoid injection (10)	6 participants without specifying groups	Not reported		
	Brace and glucocorticoid injection (10)				
Muralidharagopalan 2020	Prescribed exercises and glucocorticoid injection (26)	Not reported	0 participants		
	Glucocorticoid injection (24)				
Tonks 2007	Prescribed exercises and glucocorticoid injection (12)	5 participants	0 participants		0 participants
	Glucocorticoid injection (12)	0 participants	1 participant: depigmentation and atrophy		

GP: general practitioner

APPENDICES

Appendix 1. Cochrane Central Register of Controlled Trials search strategy

Database: EBM Reviews - Cochrane Central Register of Controlled Trials <December 2023>

Search Strategy:

Manual therapy and exercise for lateral elbow pain (Review)

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

```

1 exp Tendinopathy/ (1535)
2 elbow/ or elbow.tw. (5209)
3 1 and 2 (411)
4 Tennis Elbow/ (533)
5 (Lateral adj2 epicondylitis).tw. (691)
6 (tennis adj2 elbow).tw. (595)
7 (lateral adj2 epicondylalgia).tw. (101)
8 (lateral adj2 elbow).tw. (200)
9 (elbow adj2 pain).tw. (224)
10 (elbow adj3 tendinopathy).tw. (72)
11 (elbow adj3 tendinitis).tw. (8)
12 or/3-11 (1251)
13 exp rehabilitation/ (51034)
14 exp Physical Therapy Modalities/ (35852)
15 exp Musculoskeletal Manipulations/ (4061)
16 exp Exercise Movement Techniques/ (3181)
17 (rehabilitat$ or physiotherap$ or physical therap$ or manual therap$ or exercis$ or mobili$ or manipul$ or massag$).tw. (194626)
18 or/13-17 (218196)
19 12 and 18 (576)

```

Appendix 2. MEDLINE (Ovid) search strategy

Database: Ovid MEDLINE(R) <1946 to January 29, 2024>

Search Strategy:

```

1 exp Elbow Tendinopathy/ (1981)
2 (Lateral adj2 epicondylitis).tw. (981)
3 (tennis adj2 elbow).tw. (905)
4 (lateral adj2 epicondylalgia).tw. (151)
5 (lateral adj2 elbow).tw. (531)
6 (elbow adj2 pain).tw. (861)
7 (elbow adj3 tendinopathy).tw. (162)
8 (elbow adj3 tendinitis).tw. (25)
9 or/1-8 (3233)
10 exp rehabilitation/ (359093)
11 exp physical therapy techniques/ (181829)
12 exp musculoskeletal manipulations/ (18910)
13 exp exercise movement techniques/ (10569)
14 (rehabilitat$ or physiotherap$ or physical therap$ or manual therap$ or exercis$ or mobili$).mp. (1002300)
15 or/10-14 (1172382)
16 randomized controlled trial.pt. (607538)
17 controlled clinical trial.pt. (95519)
18 randomized.ab. (548034)
19 placebo.ab. (223109)
20 drug therapy.fs. (2662505)
21 randomly.ab. (357368)
22 trial.ab. (591455)
23 groups.ab. (2220294)
24 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 (5326899)
25 exp animals/ not humans.sh. (5191932)
26 24 not 25 (4588324)
27 9 and 15 and 26 (360)

```

Appendix 3. Embase search strategy

Database: Embase Classic+Embase <1947 to 2024 January 29>

Search Strategy:

```

1 exp epicondylitis/ (4859)
2 (Lateral adj2 epicondylitis).tw. (1612)
3 (tennis adj2 elbow).tw. (1520)
4 (lateral adj2 epicondylalgia).tw. (207)

```

5 (lateral adj2 elbow).tw. (833)
 6 (elbow adj2 pain).tw. (1466)
 7 (elbow adj3 tendinopathy).tw. (233)
 8 (elbow adj3 tendinitis).tw. (39)
 9 or/1-8 (6546)
 10 exp physiotherapy/ (121603)
 11 exp musculoskeletal manipulation/ (5131)
 12 exp kinesiotherapy/ (107732)
 13 exercise/ (380597)
 14 (rehabilitat\$ or physiotherap\$ or physical therap\$ or manual therap\$ or exercis\$ or mobili\$ or manipul\$ or massag\$).tw. (1465380)
 15 or/10-14 (1654198)
 16 Randomized controlled trial/ (808159)
 17 Controlled clinical study/ (472538)
 18 random\$.ti,ab. (2040133)
 19 randomization/ (99379)
 20 intermethod comparison/ (304453)
 21 placebo.ti,ab. (377233)
 22 (compare or compared or comparison).ti. (647981)
 23 ((evaluated or evaluate or evaluating or assessed or assess) and (compare or compared or comparing or comparison)).ab. (2867570)
 24 (open adj label).ti,ab. (112926)
 25 ((double or single or doubly or singly) adj (blind or blinded or blindly)).ti,ab. (284414)
 26 double blind procedure/ (218043)
 27 parallel group\$1.ti,ab. (33009)
 28 (crossover or cross over).ti,ab. (128068)
 29 ((assign\$ or match or matched or allocation) adj5 (alternate or group\$1 or intervention\$1 or patient\$1 or subject\$1 or participant \$1)).ti,ab. (427605)
 30 (assigned or allocated).ti,ab. (505523)
 31 (controlled adj7 (study or design or trial)).ti,ab. (466124)
 32 (volunteer or volunteers).ti,ab. (292795)
 33 human experiment/ (654234)
 34 trial.ti. (422640)
 35 or/16-34 (6558510)
 36 (random\$ adj sampl\$ adj7 (cross section\$ or questionnaire\$1 or survey\$ or database\$1)).ti,ab. not (comparative study/ or controlled study/ or randomi?ed controlled.ti,ab. or randomly assigned.ti,ab.) (9871)
 37 Cross-sectional study/ not (randomized controlled trial/ or controlled clinical study/ or controlled study/ or randomi?ed controlled.ti,ab. or control group\$1.ti,ab.) (377926)
 38 (((case adj control\$) and random\$) not randomi?ed controlled).ti,ab. (22020)
 39 (Systematic review not (trial or study)).ti. (273601)
 40 (nonrandom\$ not random\$).ti,ab. (19382)
 41 Random field\$.ti,ab. (3022)
 42 (random cluster adj3 sampl\$).ti,ab. (1628)
 43 (review.ab. and review.pt.) not trial.ti. (1166572)
 44 we searched.ab. and (review.ti. or review.pt.) (51340)
 45 update review.ab. (137)
 46 (databases adj4 searched).ab. (65588)
 47 (rat or rats or mouse or mice or swine or porcine or murine or sheep or lambs or pigs or piglets or rabbit or rabbits or cat or cats or dog or dogs or cattle or bovine or monkey or monkeys or trout or marmoset\$1).ti. and animal experiment/ (1239020)
 48 Animal experiment/ not (human experiment/ or human/) (2607198)
 49 or/36-48 (4452546)
 50 35 not 49 (5790669)
 51 9 and 15 and 50 (542)

Appendix 4. Orbit matrix for the major outcomes and all time points

Comparison	Major outcomes and time points														
Study ID	Pain			Disability			Quality of life			Participant success			With- drawals	Ad- verse events	Ser- ious ad- verse events
	First	End	Last	First	End	Last	First	End	Last	First	End	Last			
Manual therapy versus placebo															
Nourbakhsh 2008	#	2-3w	#	#	2-3w	#	#	#	#	#	#	#	#	#	#
Manual therapy, prescribed exercises or both versus minimal or no intervention															
Bisset 2006	3w	6w	12m	3w	6w	12m	#	#	#	3w	6w	12m	#	# ^a	# ^a
Blanchette 2011	#	6w	3m	#	6w	3m	#	#	#	#	#	#	# ^a	# ^b	# ^b
Haahr 2003	#	3m	12m	#	#	#	#	#	#	#	3m	12m	#	#	#
Olaussen 2015	#	6w	12m	#	6w	12m	#	#	#	#	6w	12m	#	#	#
McQueen 2021	6w	3m	12m	6w	3m	12m	#	#	#	#	#	#	#	#	#
Peterson 2011	4w	3m	#	#	3m	#	#	3m	#	#	#	#	# ^a	#	#
Rachold 2023	#	4w	8w	#	4w	8w	#	4w	8w	#	#	#	#	#	#
Smidt 2002a	3w	6w	12m	3w	6w	12m	#	#	12m	3w	6w	12m	# ^a	#	# ^a
Soderberg 2012	3w	6w	#	#	#	#	#	#	#	#	#	#	#	#	#
Struijs 2004	#	6w	12m	#	6w	12m	#	#	#	#	6w	12m	#	#	#
Tonks 2007	#	7w	#	#	7w	#	#	#	#	#	#	#	#	# ^a	# ^a
Vuvan 2020	#	8w	#	#	8w	#	#	#	#	#	8w	#	# ^a	# ^a	# ^a
Manual therapy and prescribed exercises versus prescribed exercises															



(Continued)

Amro 2010	#	4w	#	#	4w	#	#	#	#	#	4w	#	#	#	#
Bagcaci 2023	#	4w	#	#	4w	#	#	#	#	#	#	#	# ^a	#	#
Bostrøm 2019	3w	3m	12m	4w	3m	12m	#	#	#	#	#	#	#	# ^a	# ^a
Kochar 2002	3w	3m	#	#	#	#	#	#	#	#	#	#	#	#	#
Reyhan 2020	2w ^c	6w ^c	3m ^c	2w	6w	3m	#	#	#	2w	6w	3m	#	# ^a	# ^a
Yilmaz 2022	3w	6w	3m	3w	6w	3m	#	#	#	#	#	#	#	# ^a	# ^a
Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection (1 trial)															
Nambi 2023	#	4w	6m	#	4w	6m	#	4w	6m	#	4w ^c	6m ^c	#	# ^a	#
Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection															
Coombes 2013	4w	6m	12m	4w	6m	12m	4w	6m	12m	4w	6m	12m	#	#	#
Luginbuhl 2008	#	3m ^d	#	#	#	#	#	#	#	#	3m ^d	#	#	#	#
Muralidharagopalan 2020	2w	6w	12m	2w	6w	12m	#	#	#	#	#	#	#	# ^a	# ^a
Tonks 2007	#	7w	#	#	7w	#	#	#	#	#	#	#	# ^a	# ^a	# ^a

Footnotes

m = months; w = weeks

First = first follow-up (during intervention); end = end of intervention follow-up (primary time point); last = last follow-up.

^aZero events in one or more of the groups.

^bStudy measured adverse events for one treatment group only.

^cInsufficient data reporting.

^dWe assumed 3 months follow-up represented 'end of intervention' as the majority of participants were lost to follow-up at 12 months.

Appendix 5. Orbit matrix for the minor outcomes and all time points

Comparison	Grip strength			Work participa- tion
Study ID	First	End	Last	
Manual therapy versus placebo (1 trial)				
Nourbakhsh 2008	#	2-3w	#	#
Manual therapy, prescribed exercises or both versus minimal or no intervention (12 trials)				
Bisset 2006	3w	6w	12m	#
Blanchette 2011	#	6w	3m	#
Haahr 2003	#	#	#	#
Olaussen 2015	#	6w	12m	Sick leave
McQueen 2021	#	#	#	#
Peterson 2011	#	#	#	#
Rachold 2023	#	#	#	#
Smidt 2002a	3w	6w	12m	#
Soderberg 2012	3w	6w	#	#
Struijs 2004	#	6w	12m	#
Tonks 2007	#	7w	#	#
Vuvan 2020	#	8w	#	#
Manual therapy and prescribed exercises versus prescribed exercises (6 trials)				
Amro 2010	#	4w	#	#
Bagcaci 2023	#	4w	#	#
Bostrøm 2019	#	#	#	#

Manual therapy and exercise for lateral elbow pain (Review)

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

(Continued)

Kochar 2002	3w	3m	#	#
Reyhan 2020	2w	6w	3m	#
Yilmaz 2022	3w	6w	3m	#
Manual therapy, prescribed exercises and glucocorticoid injection versus prescribed exercises and glucocorticoid injection (1 trial)				
Nambi 2023	#	4w	6m	#
Manual therapy, prescribed exercises or both, and glucocorticoid injection versus glucocorticoid injection (4 trials)				
Coombes 2013	#	#	#	#
Luginbuhl 2008	#	3m	#	#
Muralidharagopalan 2020	#	#	#	#
Tonks 2007	#	7w	#	#

Footnotes

m = months; w = weeks

First = first follow-up (during intervention); end = end of intervention follow-up (primary time point); last = last follow-up.

WHAT'S NEW

Date	Event	Description
28 June 2024	Amended	Wording errors in Plain Language Summary corrected.

HISTORY

Protocol first published: Issue 6, 2018

Review first published: Issue 5, 2024

Date	Event	Description
27 June 2017	Amended	We added new review authors and updated the methods to conform with the latest Cochrane recommended standards. We amended and split the original protocol, 'Physiotherapy and physiotherapeutic modalities for lateral epicondylitis', into two protocols: 'Manual therapy and exercise for lateral elbow pain' and 'Electrotherapy modalities for lateral elbow pain'.

CONTRIBUTIONS OF AUTHORS

Nynke Smidt (NS) and Rachelle Buchbinder (RB) conceived the idea for the review, and contributed to drafting the protocol.

Jason Wallis (JW), Allison Bourne (AB), Rebecca Jessup (RLJ), Renea Johnston (RJ) and Aviva Frydman (AF) contributed to study selection.

JW, AB, SC and RJ assessed the risk of bias of studies.

JW and RJ undertook data analysis, GRADE assessment and presentation, and wrote the first draft of the manuscript.

All authors contributed to the writing of the protocol and critically revising the drafts for important intellectual content.

All authors approved the final version of the manuscript and agree to be accountable for its content.

DECLARATIONS OF INTEREST

Wallis J: none known

Bourne A: none known

Jessup R: none known

Johnston R: none known

Frydman A: none known

Cyril S: none known

Buchbinder R: none known

SOURCES OF SUPPORT

Internal sources

- School of Public Health and Preventive Medicine, Monash University, Australia

In kind support

External sources

- National Health and Medical Research Council, Australia

R Buchbinder is supported by an Australian National Health and Medical Research Council Senior Principal Research Fellowship

DIFFERENCES BETWEEN PROTOCOL AND REVIEW

In the [Criteria for considering studies for this review](#), we made the following changes to the types of interventions eligible for inclusion: (1) we excluded trials that compared any manual therapy or exercise intervention, or both delivered together, to a different type of manual therapy or exercise, or another active intervention, and (2) we excluded trials testing a single application of an intervention.

We expanded the [Types of outcome measures](#) from pain, global improvement, elbow specific functional status, grip strength, generic functional status or sick leave to:

- Major outcomes: pain, disability, participant-reported treatment success, health-related quality of life, overall withdrawals, adverse events.
- Minor outcomes: grip strength, return to work.

For data collection and analysis, we changed the grouping of outcomes from five time points (up to 3 weeks; > 3 weeks to 6 weeks (primary time point); last follow up > 6 weeks to 3 months; > 3 months to 6 months; > 6 months) to three time points (first time point, end of intervention and last time point). We also selected overall score for disability measures in preference to sub-scores. We extracted overall withdrawals (i.e. for any reason rather than due to adverse events).

For trials that reported multiple measures of pain, we had planned to extract the measures in the following hierarchy: overall pain, unspecified pain, pain at rest, pain with activity or daytime pain. We changed the hierarchy to overall pain (or similar, such as average pain in the last 24 hours or other timespan), followed by unspecified pain, pain with activity or worst pain, pain at night or at rest. We made this change because trials reported measures that we did not anticipate and we considered pain with activity more relevant than pain at rest for lateral elbow pain.

In data synthesis, the questions and comparisons reflect the changes made to the types of interventions eligible for inclusion. The questions that could not be answered in this review included (1) whether manual therapy and/or exercise is more effective than another intervention,

and (2) whether one type of manual therapy or exercise intervention is more effective than another type of exercise or manual therapy intervention.

We would have restricted the subgroup and sensitivity analyses to placebo-controlled trials. However, as there was only one placebo-controlled trial we performed the analyses on the following comparison: manual therapy or exercise or both delivered together versus minimal or no intervention for the outcomes pain and disability. We altered the subgroups from two to three subgroups to assess whether pain and function differ between participants who have acute symptoms (defined as three months or less) compared to those with symptoms for more than three months and up to six months, and to those with symptoms for more than six months. This is because we expected pain to be the most intense in the first three months ([Mallen 2009](#)), and to have diminished up to 12 months ([Ikonen 2022](#)).

NOTES

The original protocol was published as, 'Physiotherapy and physiotherapeutical modalities for lateral epicondylitis', and has been split into two amended protocols, 'Manual therapy and exercise for lateral elbow pain' and 'Electrotherapy modalities for lateral elbow pain'.

INDEX TERMS

Medical Subject Headings (MeSH)

*Bias; Combined Modality Therapy [methods]; *Exercise Therapy [methods]; *Glucocorticoids [therapeutic use]; Injections, Intra-Articular; *Musculoskeletal Manipulations [methods]; Quality of Life; *Randomized Controlled Trials as Topic; *Tennis Elbow [therapy]

MeSH check words

Adult; Female; Humans; Middle Aged